

CS 598 Practical Statistical Learning

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2025-09-10

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Course Information

Course Description

This course provides an introduction to modern techniques for statistical analysis of complex and massive data. Examples of these include model selection for regression, classification, nonparametric models such as splines and kernel models, regularization, dimension reduction, and clustering analysis. Applications are discussed as well as computation and theoretical foundations.

Course Prerequisites

Knowledge of basic multivariate calculus, statistical inference, and linear algebra. You should be comfortable with the following concepts: probability distribution functions, expectations, conditional distributions, likelihood functions, random samples, estimators, and linear regression models.

Course Learning Outcomes

By the end of the course, you will be able to:

- Use a broad range of methods and techniques in machine learning.
- Have a deeper understanding of major algorithms and techniques in machine learning.
- Build analytics pipelines for regression problems.
- Build analytics pipelines for classification problems.
- Build analytics pipelines for recommendation problems.

Textbook and Readings

There is no required textbook for this course.

Recommended resources for a statistical approach to machine learning are the following textbooks:

- An Introduction to Statistical Learning with Applications in R (basic)
- An Introduction to Statistical Learning with Applications in Python (basic)

- The Elements of Statistical Learning: Data Mining, Inference, and Prediction (more advanced)
- You may also want to view the Data School YouTube videos associated with the first book, as an additional resource.

The detailed syllabus for the course can be found on Coursera.

Chapter 1

Introduction to Statistical Learning

Statistical Learning is a realm in statistics, computer science, and data science focused on developing and understanding models that can *learn* from the data. While Statistics has long been dealing with model building, model validation, and prediction across a wide range of applications, the fairly recent increase in the volume, complexity and multidimensional nature of data has challenged traditional statistical techniques.

To this end, the introduction of sophisticated computer science algorithms, data management tools and optimization techniques has enabled us to design more powerful methods that are suited to handle the ever expanding nature of modern data.

Typically, *learning from data* means that given an **outcome/measurement** variable that may be quantitative or categorical, we are interested in making **predictions** based on a (typically large) set of features. More specifically, a *learner* or **prediction model** enables us to predict the outcome for **new unseen** objects.

We can separate statistical learning problems in two types:

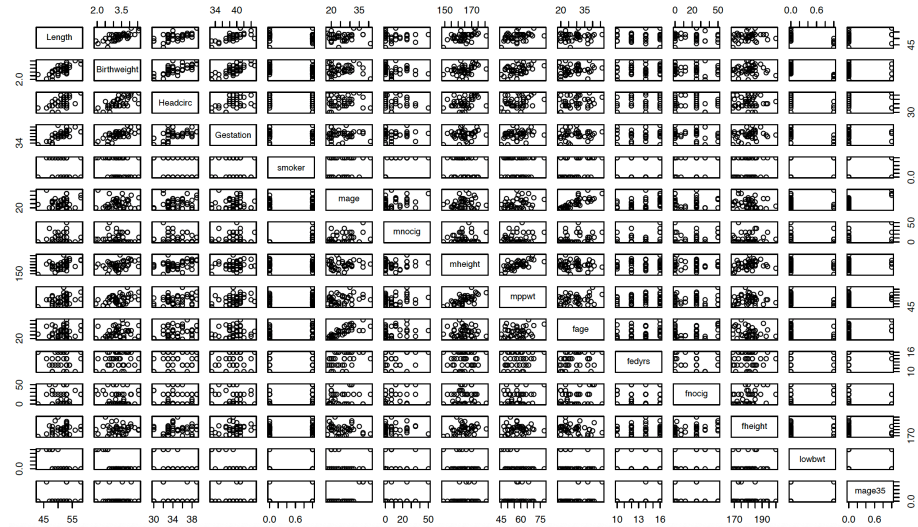
- **Supervised Learning:** when an outcome variable is present to guide the learning process. Examples of supervised learning are:
 - *Regression:* Response is quantitative, i.e. a *number*.
 - *Classification:* Response is qualitative, i.e. categorical, discrete, or a factor – a *label* (binary or multi-class).
- **Unsupervised Learning:** when an outcome variable is not present to guide the learning process. In this case, our goal is to identify latent structures in the data, e.g. clustering, association rules, HMM, etc.

1.1 Examples of Statistical Learning Problems

Birthweight Data (Regression)

This is an example in which the researchers are interested in predicting the weight of a newborn based on a set of baby and parent characteristics, such as gestation period, length, head circumference, mother's and father's height, mother's weight, parents' smoking habits, etc. The study particularly focuses on babies born prematurely, i.e. before 40 weeks of gestation.

The scatterplots below illustrate the pairwise relationship of each variable with the other variables in the data set:



Note in the plots that three of the predictors, `smoker`, `lowbwt`, and `mage35` are categorical.

This is a typical example where a multiple linear regression model is a good starting point. If the fit of the model is good and all key model assumptions are satisfied, then the predictive power of this model will be strong.

Trees and Shrubs Data (Binary Classification)

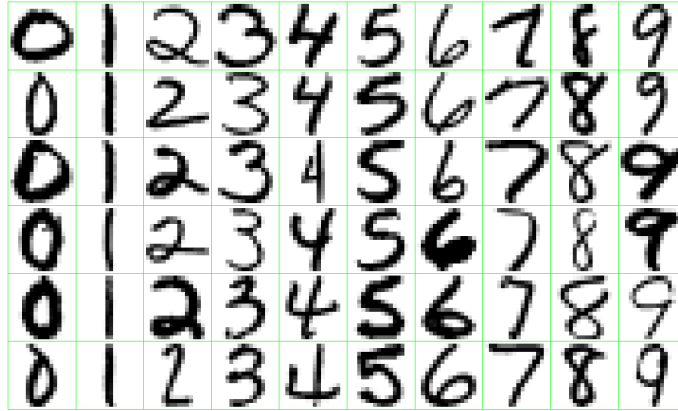
In this problem, the goal is to understand whether a specific woody plant, common in the Black Forest region in Southwestern Germany (Ref: Lederer) is a tree or a shrub. A sample of the data set is shown below:

Name	Height	Diameter	GrowthRate	Longevity
Silver Fir	55.0	1.50	0.5	500
European Beech	40.0	1.00	0.5	200
Common Hazel	5.0	0.10	0.5	70
Red Raspberry	1.5	0.01	0.4	10
European Spruce	50.0	1.20	0.5	600

This is a binary classification problem in which we want to use a set of characteristics to predict whether a specific plant is a tree or a shrub.

Handwritten Digits (Multiclass Classification)

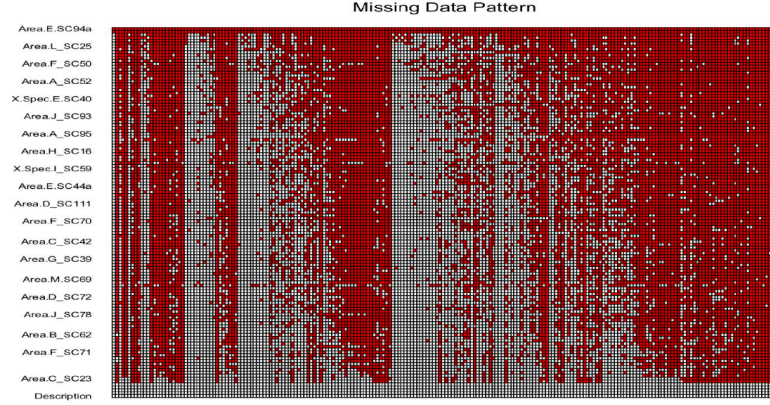
This is a classic example of multiclass classification where the goal is to predict the handwritten digit (0, ..., 9) from the envelope of a letter, based on a 16×16 eight-bit grayscale map (Ref: ESL). The challenge in this example is to keep the error rate below a desirable threshold to avoid misdirection of mail.



Proteomics Data (Clustering)

Proteomics is the field of biology and chemistry that is focused on studying proteins in large-scale. Typically, proteomics data sets are obtained as a result of an experiment and through specific processes, such as protein purification or through mass spectrometry, a technique that is used to measure mass-to-charge ratio of ions.

In the data set below, we show the expression matrix of 437 proteins only 101 of which are of interest (Ref: Romanova et al. – STAT 427 dataset). The challenge in these data sets is the amount of missing data (grey pixels in the plot) and the goal is to cluster the observations into various protein groups for further analysis.



1.2 Supervised Learning Framework

In this class, our main focus is *supervised learning*. This means that we have a **target/outcome/response** variable Y that we need to predict using a set of *features/predictors*, X . We can assume that the relationship between the response Y and the set of features are linked via a *parametric* function f as follows

$$\mathbf{y} \leftarrow f(\mathbf{x}, w)$$

If we know the function f up to a parameter w , e.g. we know it is a linear, then our problem reduces to **learning** w .

For this reason, we collect (*training*) data $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^n$ that will help us to *minimize an objective function* with respect to w :

$$F(w) = \sum_{i=1}^n \text{loss}(y_i, f(\mathbf{x}_i, w))$$

The *loss* function quantifies the distance between the model, i.e. $f(\mathbf{x}_i, w)$ and the respective outcome $\mathbf{y}_i$. It is up to the practitioner to choose the *loss* function. The most popular examples are:

- the Squared Error Loss

$$\text{loss} = [y_i - f(x_i; w)]^2$$

- the Absolute Error Loss

$$\text{loss} = |y_i - f(x_i; w)|$$

- the Log Loss

$$\text{loss} = \begin{cases} -\log f(x_i; w), & \text{if } y_i = 1 \\ -\log(1 - f(x_i; w)), & \text{if } y_i = 0 \end{cases}$$

- the 0-1 Loss

$$loss = \mathbf{1}_{\{y_i \neq f(x_i; w)\}} = \begin{cases} 1, & \text{if } y_i \neq f(x_i; w) \\ 0, & \text{if } y_i = f(x_i; w) \end{cases}$$

Depending on the choice of *loss* function, the minimizer of the objective function may or may not be in closed form. If not, we can try optimization algorithms that can guarantee convergence to the global minimizer. In the worst case scenario, one can always use gradient descent.

1.3 Why is Statistical Learning is Challenging

In our framework, we reduced the problem of learning, to estimating a vector of parameters. However, we assumed that we already know the true function f up to this parameter. In practice, we typically need to estimate or better *approximate* the function f . This approximation, denoted by $\hat{f}$ introduces an *additional source of error*.

Furthermore, in our framework we worked with a set of *training* data that will be used to learn the model. The objective function quantifies the distance between the *true* f parametrized by w and the observation in the training data set. However, to assess the actual performance and the predictive power of the fitted model, we need to *test* it to new, unobserved data. Thus, in the statistical learning context, our goal is to minimize the **test** or **generalization error**, not the training error.

Therefore, statistical learning is a difficult task because

- the training error typically underestimates the test/generalization error.
- the model performance might be good for training data, but poor for future (test) data due to overfitting.
- the number of parameters needed to learn the underlying regression or classification function f might be large; and in some cases larger than the available data.
- The test error increases significantly when the number of model parameters becomes larger.

Let's consider the following simple cases:

- (1) In *classification*, the one-nearest-neighbor approach predicts perfectly on the training data, but poorly on testing data. An interesting illustration on how dimensionality changes the performance of linear classifiers can be found here: <https://www.visiondummy.com/2014/04/curse-dimensionality-affect-classification/>

- (2) On the other hand, in *regression*, if we take $\mathbf{y}_{n \times 1}$ to be the regression vector, where n is the sample size, and $\mathbf{X}_{n \times p}$ is the design matrix (i.e. matrix of predictors) with p predictors. The underlying regression model is

$$\mathbf{y}_{n \times 1} = \mathbf{X}_{n \times p} \mathbf{w}_{p \times 1} + \text{error}$$

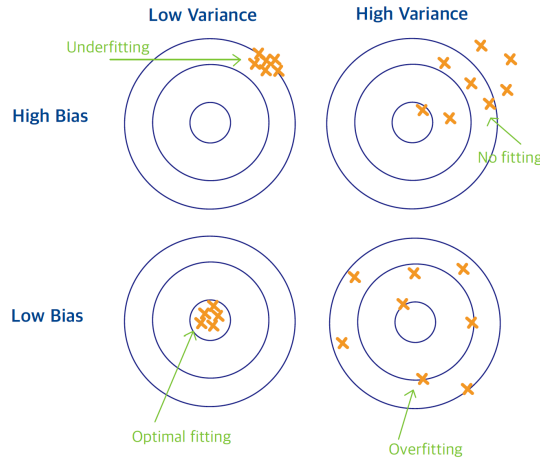
where $w_{p \times 1}$ are the model coefficients (i.e. parameters to estimate). If

- $p < n$: we have n (more) equations than the p parameters; this is when typically regression methods can be successful.
- $p = n$: we have n equations and n parameters; this leads to a perfect fit on training data.
- $p > n$: we have fewer equations n than parameters p : this is a scenario when classical regression methods do not perform well.

1.4 Bias-Variance Trade-Off

We mentioned before that we have **two sources of error** that affect the quality of the predictions we make. Specifically, we try to balance between:

- **Bias**: The bias is the difference between the estimated parameter or function and the true underlying parameter or function. High bias leads to under-fitting and an inaccurate model.
- **Variance** (of an estimated function): The variance quantifies the ability of the function to “adapt” to small changes in the data. High variance leads to over-fitting and an unreliable model.

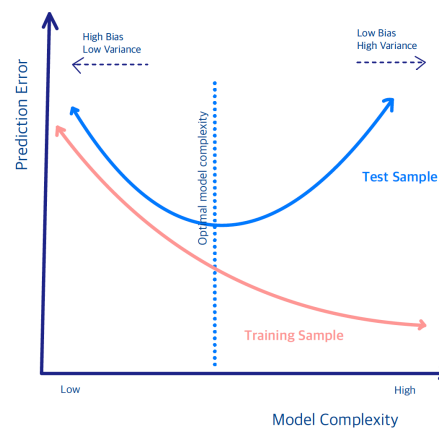


In general, one can think of our learning problem as a **target** in which the true model lies in the center. *How close to the center the fitted values are is captured by the bias*. If most of the points “miss” the target, then it means that our

model does not approximate the true f . If the bias is large, then increasing the sample size will not help us to “hit the target”.

On the other hand, *the spread of the data is measured by the variance*. If all data points are all close to each other, then the variance is low, while if they are widely spread, then the variance is high.

Ideally, we would like to minimize both variance and bias at the same time. In our context, this is not feasible which means that we need to balance between the two. Keeping in mind that in statistical learning our goal is to *minimize the generalization error*—not the training error—, we can say that:



- When the number of model parameters is large, or in other words when the model is complex, the prediction error in the *training data* tends to **reduce**. However, when we overfit, we end up with a fitted model that follows precisely the pattern of the data and that won't generalize well, leading to a higher variance in the *testing data*.
- When the model is very simple, i.e. when we have very few parameters, then it is very likely that we underfit the data, leading to larger bias and a poor generalization error.

In this class, we will discuss:

- (i) Flexible modeling techniques to reduce bias.
- (ii) Useful strategies to achieve the trade-off between bias and variance.

As an example, two successful approaches that balance the bias-variance trade-off that we will study are.

- **Regularization:** Restrict the parameters to a low-dimensional space, which is adaptively determined by the data.
- **Ensemble:** Average many low-bias high-variance models $\rightarrow$ averaging reduces variance.

1.5 Two Toy Examples: k NN vs. Linear Regression

Before we wrap up the introduction, we will review two simple supervised learning examples (i) k -Nearest Neighbors (k NN) (ii) Linear Regression and will examine their performance and understand the bias-variance trade-off.

1.5.1 k -Nearest Neighbors

In the k -Nearest Neighbors (k NN) method, we use observations in the training set that are closest to $\mathbf{x}$ to form $\hat{\mathbf{y}}$. Specifically, the k -Nearest Neighbor fit for $\hat{y}$ is

$$\hat{y}(\mathbf{x}) = \frac{1}{k} \sum_{x_i \in N_k(\mathbf{x})} y_i$$

where $N_k(\mathbf{x})$ is the neighborhood of $\mathbf{x}$ defined by the k closest points x_i in the training sample. In a regression context the k NN fitted $\hat{\mathbf{y}}$ predicts $\mathbf{y}$ via a **local average**, while in the classification context k NN returns the **majority vote** in $N_k(\mathbf{x})$ or a probability calculated on the frequencies in $N_k(\mathbf{x})$. What can be challenging in the k NN approach is *tuning* k , the neighborhood size, and determining the metric to define the neighborhood.

- The choice of k is directly linked to the complexity of the method which is roughly equal to n/k .
 - When $k = 1$, the prediction at x_i is **exactly** y_i which means that we have zero training error.
 - When $k = n$, every neighborhood contains all the n training samples, so the prediction is the same no matter x .
- The default metric to define the neighborhood is the *Euclidean distance*:

$$d(\mathbf{x}, \tilde{\mathbf{x}}) = \sum_{j=1}^p w_j (x_j - \tilde{x}_j)^2,$$

where we would like to learn the w_j 's from the data.

1.5.2 Linear Regression

Given a vector of inputs $\mathbf{x}^T = (x_1, x_2, \dots, x_p)$, we **approximate** Y via a *linear* function

$$f(\mathbf{x}) \approx \beta_0 + \sum_{j=1}^p x_j \beta_j$$

Our goal is to *estimate* the parameters β_j using the Least-Squares (LS) method by minimizing the Residual Sum of Squares (*objective function*)

$$\min_{\beta_0, \dots, \beta_p} \sum_{i=1}^n \left(y_i - \beta_0 - x_{i1}\beta_1 - \dots - x_{ip}\beta_p \right)^2$$

The solution is easy to obtain (both in R/Python and analytically under certain assumptions) and the fitted value for the i th input x_i is given by

$$\hat{y}_i = \hat{y}(x_i) = x_i^T \hat{\beta}$$

We leave the details to be discussed in Week 2.

Linear Regression in a Classification Context

We can apply linear regression on classification problems with $Y = 0$ or 1 . In this case, we predict Y to be 1 if the LS prediction $f(x)$ is bigger than 0.5, and 0 otherwise. This approach has drawbacks. First of all, the squared difference $p(\mathbf{x}) = \left(y_i - f(\mathbf{x}_i)\right)^2$ is not a good evaluation metric, since considering a linear function $f(\mathbf{x})$ may result in values outside $[0, 1]$. Therefore, when we are in this context a Logistic regression is the gold standard according to which

$$\log \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} \approx \beta_0 + \sum_{j=1}^p x_j \beta_j$$

More on this in Week 8.

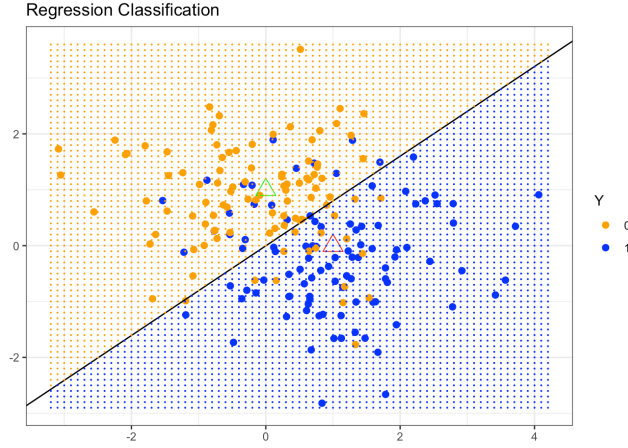
1.5.3 Simulated Binary Classification Example

Consider a response variable G that takes two values (0 – BLUE or 1 – ORANGE). In our simulation, we generate 200 such values; 100 in each class, and our goal is to use both regression and k NN to classify the data. The code behind the simulation and the plots can be found [here](#).

We start by fitting a linear regression model to the simulated data. In a naive approach, we treat the response is a continuous variable. Hence, the continuous fitted values $\hat{Y}$ are converted to a fitted class variable $\hat{G}$ according to the following rule:

$$\hat{G} = \begin{cases} \text{Blue, if } \hat{Y} > 0.5 \\ \text{Orange, otherwise} \end{cases}$$

Our classification example is in two dimensions which means that the decision boundary (the boundary that separates the orange from the blue region) is a straight line.



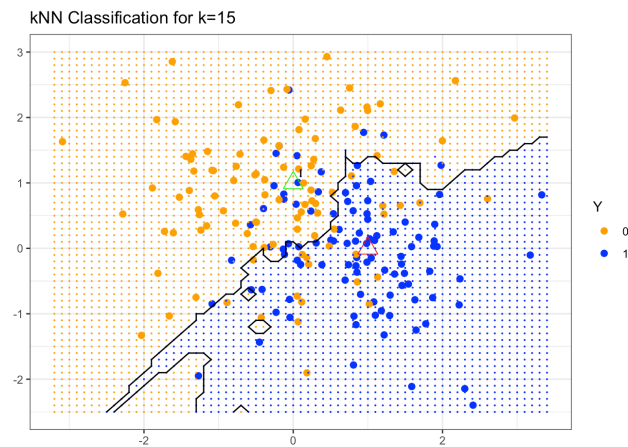
Specifically, the fitted decision boundary is a straight line (the black line in the plot) defined by $\mathbf{x}^T \hat{\beta} = 0.5$. Based on our simulation, we know that the blue region should be above the black fitted regression line, while the orange region should be below the black fitted regression line. We observe that there are many misclassifications on both sides of the decision boundary.

The regression line seems to be very smooth and too rigid when it comes to classifying the data. On the other end of the spectrum, we have the k Nearest-Neighbor approach. So, for the same simulated data, the nearest neighbor method will use the observations in the training set closest in input space to X to form $\hat{Y}$.

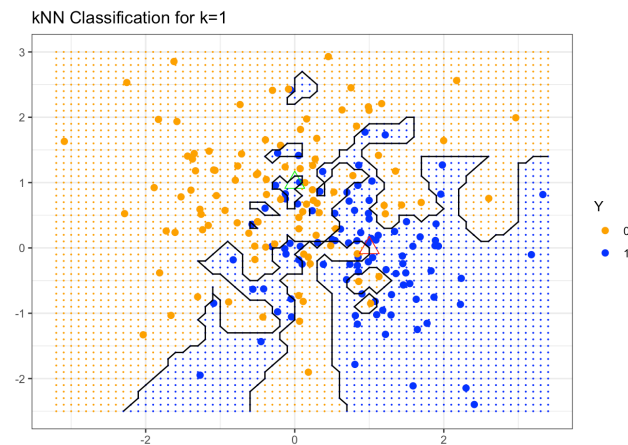
Using a 15-nearest-neighbor averaging of the binary coded response such that $\hat{Y}$ is the proportion of blue's in the neighborhood, then we assign

$$\hat{G} = \begin{cases} \text{Blue, if } \hat{Y} > 0.5 \\ \text{Orange, otherwise} \end{cases}$$

In this case, the predicted class is chosen by majority vote among the 15 nearest neighbors.



We observe that the decision boundary separating the blue from the orange region is far more irregular than before and sensitive to local clusters of blue and orange dots. As a result, we have fewer misclassified observations than before. Remember that we can tune the neighborhood size. So, if we take the extreme scenario in which $k = 1$ and we only consider **one** neighbor, we have

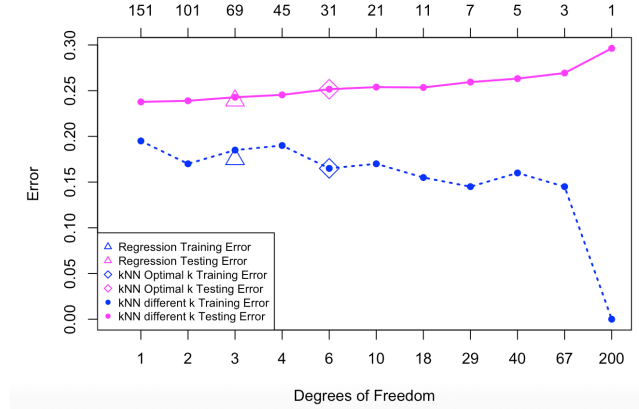


This results in a much rougher decision boundary with hardly any misclassified data. However, is this the ideal choice? The answer is no, but in order to understand the reason we need to consider the *generalization error*.

Up to now, we have used the same data for training and comparison purposes. As a result, a method like k NN has seemingly 0 error for $k = 1$. In order to make fair comparisons, we should consider another data set, independent of the one used to fit the data, so that we can compute the test/generalization error for both methods.

In the plot below, we compare the misclassification error on the testing data

set as a function of the degrees of freedom. In other words, we compare several k NN fits (for different k s) and the regression fit. The testing set here contains 10,000 observations.



The magenta curve is the test error and the blue curve is the training error for the k NN classification for different k s. In our simulated example, we used a 5-fold cross validation method to identify the optimal k . The results for the optimal k in k NN are denoted with a diamond. The results for linear regression are the magenta and blue triangles at 3 DFs (the DFs were determined based on the dimension of the linear model we fitted).

The regression method has the advantage of it being linear with only $p = 3$ parameters to estimate which means that it has a relatively *low variance*. However, the linearity assumption seems to be quite restrictive for the classification problem under consideration, which means that we expect to have high bias.

On the other hand, the k NN approach has no assumption of the shape of the underlying f , maybe except some local smoothness. This flexibility results in overfitting and a low bias (we saw that in the extreme case of $k = 1$). It can be shown that as $k, n \rightarrow \infty$ such that $k/n \rightarrow 0$, k NN is consistent. At the same time, the method has higher variance with the extreme case of when the number of parameters for k NN is roughly $n = k$, which goes to ∞ in order to achieve consistency.

1.5.4 Code for the Examples in the Lectures

We replicate one of the examples in the ESL book to illustrate the differences between the two simplest prediction methods for binary outcomes: the *Regression method* and the *k-Nearest-Neighbors method*.

The python version of the example below was written by our TA Jim Yici Yan and can be found here: [Python Version](#)

1.5.4.1 Simulation

- The *generated data* consist of a **binary classification** with two classes, labeled as 0 and 1.
- The *features* are two-dimensional.
- **Class 1** data points are generated from a Gaussian distribution with mean μ_1 and variance σ^2 , i.e. $X_{1,1}, \dots, X_{1,n_1} \sim \mathcal{N}(\mu_1, \sigma^2)$.
- **Class 0** data points are generated from a Gaussian distribution with mean μ_2 and variance σ^2 , i.e. $X_{0,1}, \dots, X_{0,n_2} \sim \mathcal{N}(\mu_0, \sigma^2)$.
- In total, we generate *200 training samples* ($n = 100$ for each class), and we assign labels to the training data (100 **Class 1** and 100 **Class 0**).
- Similarly, we generate *10,000 test samples*.

1.5.4.2 Code for the Simulation

Set the model parameters for the simulation:

```
p = 2;          ## No. of parameters
sigma = 1;      ## St. Dev for the Normals (common)
mu1 = c(1, 0);  ## Vector of means for the first Normal
mu0 = c(0, 1);  ## Vector of means for the second Normal
```

Generate n i.i.d. (independent and identically distributed) samples from each normal to create the *training* data set.

```
n = 100;        ## Training Sample Size for each Normal

## rnorm(2*n*p) generates 2*n*p N(0,1) random variables.
## matrix(rnorm(2*n*p), 2*n, p)*sigma generates a 2n-by-p matrix of two N(0, sigma^2) each of len
## Adding matrix(rep(mu1, n), nrow=n,byrow=TRUE) to each column of the previous matrix
## shifts each of the columns to generate the Normals with means mu1 or mu0.
## Note that both mu0, mu1 are 2-dimensional.

traindata = matrix(rnorm(2*n*p), 2*n, p)*sigma +
              rbind(matrix(rep(mu1, n), nrow=n,byrow=TRUE),
                    matrix(rep(mu0, n), nrow=n, byrow=TRUE))

# dim(traindata)

## We generate the 0 or 1 labels.
Ytrain = factor(c(rep(1,n), rep(0,n)))
```

Generate N *test* samples in a similar way:

```
N=10000;

testdata = matrix(rnorm(2*N*p), 2*N, p)*sigma+
            rbind(matrix(rep(mu1, N), nrow=N,byrow=TRUE),
```

```
matrix(rep(mu0, N), nrow=N, byrow=TRUE))

Ytest = factor(c(rep(1,N), rep(0,N)))
```

1.5.4.3 Visualization of the Simulated Data

This section also serves as a review of plotting in R.

We visualize the data we generated – those in the `traindata` matrix.

In the figure generated by the code below, points from two groups are colored in orange and blue, respectively; the two centers are plotted as +, and a legend is added to explain the association of each color.

```
## Create an empty plotting area: The axes are the two vectors of normals generated,
## each one saved in a column of the `traindata` matrix.
## The following line creates an empty plot, since we used the option type="n"
## We do this so that we can color-code the data.
plot(traindata[,1], traindata[,2], type="n", xlab="", ylab="");

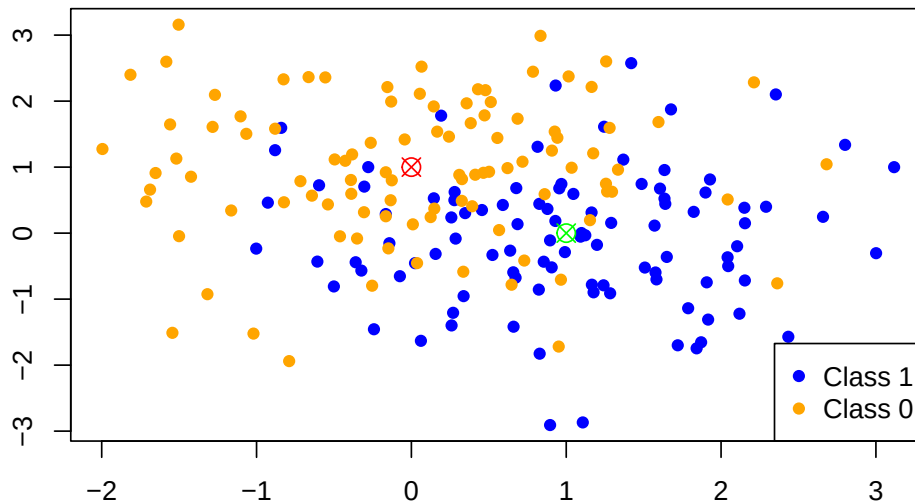
# Add the "Class 1" points - in blue color.
points(traindata[1:n, 1], traindata[1:n,2], pch=16, col="blue");

# Add the "Class 0" points - in orange color
points(traindata[(n+1):(2*n),1], traindata[(n+1):(2*n),2], pch=16, col="orange");

# Add the centers for class 1
points(mu1[1], mu1[2], pch=13, cex=1.5, col="green");

# Add the centers for class 0
points(mu0[1], mu0[2], pch=13, cex=1.5, col="red");

legend("bottomright", pch = c(16,16), col = c("blue", "orange"),
      legend = c("Class 1", "Class 0"))
```



1.5.4.5 Using the ggplot2 R library

In this section, we present an alternative way to construct the plot above via the `ggplot2` package. The `ggplot2` package allows us to create elaborate plots. More information can be found here: <http://ggplot2.org/>

```
# install.package("ggplot2")
library("ggplot2")

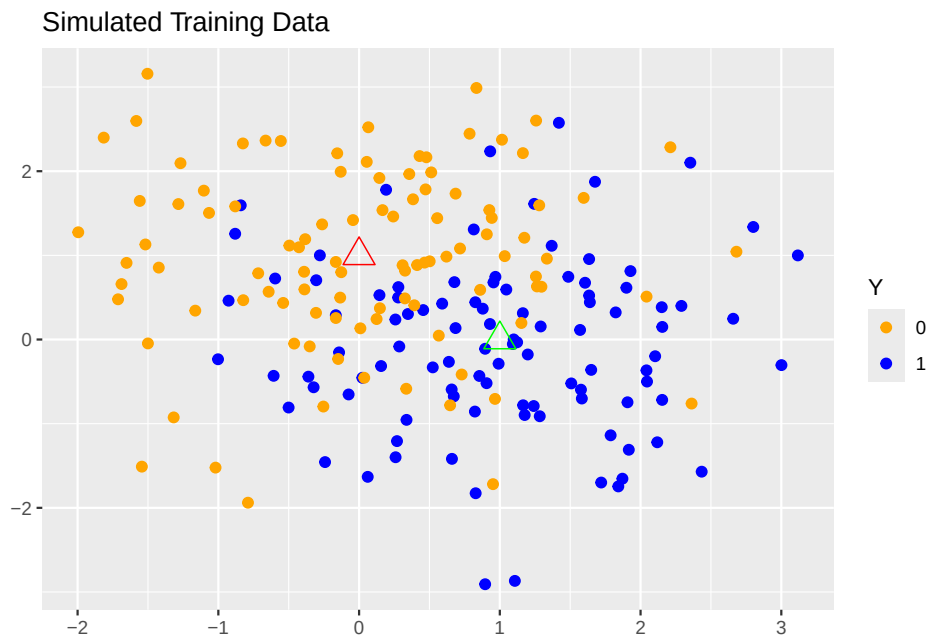
## The input in a `ggplot` function can only be a data.frame

## In our case the data is a matrix, so we convert them to data.frames here:
mytraindata = data.frame(X1=traindata[,1], X2=traindata[,2], Y=Ytrain)

## The ggplot output is an object which is saved in training.scatter.
## In this object, we can later add --and plot-- additional features.

training.scatter = ggplot(mytraindata, aes(X1, X2)) + ## creates the empty plot
  geom_point(aes(colour=Y), size=2) + ## adds the points color-coded by the labels in Y
  scale_color_manual(values = c("orange", "blue")) + ## change the default colors
  ## Use geom_point to add the centers (as before)
  geom_point(data=data.frame(X1=mu1[1], X2=mu1[2]), aes(X1, X2), colour="green", shape=2, size=100) +
  geom_point(data=data.frame(X1=mu0[1], X2=mu0[2]), aes(X1, X2), colour="red", shape=2, size=100) +
  ggtitle("Simulated Training Data") + ## add a title
  labs(x = "", y="") ## remove axes labels

plot(training.scatter)
```



1.5.4.6 k Nearest-Neighbors Method

To apply the k -NN method, we need to choose k . In the example below, we use the neighborhood sizes suggested from the textbook (ESL). We also apply and plot the results for $k = 1$ and $k = 15$.

```
library("class")

neighbor_size = c(151, 101, 69, 45, 31, 21, 11, 7, 5, 3, 1); ## These are different k
m = length(neighbor_size); ## needed to run our for-loop below.

train.err.knn = rep(0,m); ## vector to store training error
test.err.knn = rep(0, m); ## vector to store testing error

## knn is the R function that runs the kNN method. Output is a factor.

for( j in 1:m){
  Ytrain.pred = knn(traindata, traindata, Ytrain, k=neighbor_size[j]) ## predictions
  train.err.knn[j]= sum(Ytrain != Ytrain.pred)/(2*n) ## mis-classification training
  Ytest.pred = knn(traindata, testdata, Ytrain,k=neighbor_size[j]) ## predictions
  test.err.knn[j] = sum(Ytest != Ytest.pred)/(2*N) ## mis-classification testing
}

cbind(train.err.knn, test.err.knn) ## matrix containing the train and test errors

##      train.err.knn test.err.knn
```

```
## [1,]      0.220      0.23800
## [2,]      0.225      0.23985
## [3,]      0.210      0.24000
## [4,]      0.215      0.23800
## [5,]      0.205      0.23830
## [6,]      0.230      0.24010
## [7,]      0.200      0.24390
## [8,]      0.195      0.24685
## [9,]      0.185      0.25230
## [10,]     0.150      0.26160
## [11,]     0.000      0.32395
```

1.5.4.7 5-Fold Cross-Validation for Choosing optimal k

A systematic way to determine the *optimal* value for the neighborhood size, k , in a range of values is the so-called 5-fold Cross-Validation (CV) method. Essentially, the method selects the k value that minimizes the CV error. In a nutshell, the 5-fold CV error for each k is a *sum of 5 prediction errors*, one corresponding to each fold.

In the code below, we have an outside loop from 1 to 5 (the folds), and an inside loop from 1 to m (all possible values for k). Inside the loop, we use 80% (i.e., four folds) of the data as training and predict on the 20% (i.e., one fold) holdout set.

```
## In this chunk of code, we use the same vector of k's as above - the neighbor_size vector of s

## Initialize a vector cv.error to store the CV error for each k.
cv.error = rep(0,m);

id = sample(1:(2*n),(2*n), replace=FALSE);
fold = c(0, 40, 80, 120, 160, 200)

for(i in 1:5)
  for(j in 1:m){

    ## ith.fold = rows which are in the i-th fold
    ith.fold = id[(fold[i]+1):fold[i+1]];
    tmp = knn(traindata[-ith.fold,], traindata[ith.fold,], Ytrain[-ith.fold], k=neighbor_size[j])
    cv.error[j]=cv.error[j] + sum(tmp != Ytrain[ith.fold])
  }

## Find the optimal k value based 5-fold CV
k.optimal = neighbor_size[order(cv.error)[1]]

## Error of KNN for the k is chosen by 5-fold CV
```

```

Ytrain.pred = knn(traindata, traindata, Ytrain, k=k.optimal)
train.err.knn.CV = sum(Ytrain != Ytrain.pred)/(2*n)
Ytest.pred = knn(traindata, testdata, Ytrain, k=k.optimal)
test.err.knn.CV = sum(Ytest != Ytest.pred)/(2*N)

```

1.5.4.8 Least Squares Method

We run a regression of Ytrain vs. the traindata, and we classify the results as follows:

$$\hat{Y} = \begin{cases} 1, & \text{if } fitted(Y) > 0.5 \\ 0, & \text{if } fitted(Y) \leq 0.5 \end{cases}$$

```

## Run a regression using the lm function
## Ytrain is a factor, so we need to convert it to a numeric vector to run the lm
RegModel = lm(as.numeric(Ytrain)-1 ~ traindata)

## Compute the \hat{Y} for training
Ytrain_pred_LS = as.numeric(RegModel$fitted > 0.5)

## Compute the predicted values for testing data and then the \hat{Y}
Ytest_pred_LS = RegModel$coef[1] + RegModel$coef[2] * testdata[,1] + RegModel$coef[3] *
Ytest_pred_LS = as.numeric(Ytest_pred_LS > 0.5 )

## Cross-tab for training data and training error
table(Ytrain, Ytrain_pred_LS);

##      Ytrain_pred_LS
## Ytrain  0  1
##      0 80 20
##      1 23 77

train.err.LS = sum(Ytrain != Ytrain_pred_LS) / (2*n);

## Cross-tab for test data and test error
table(Ytest, Ytest_pred_LS);

##      Ytest_pred_LS
## Ytest    0    1
##      0 7741 2259
##      1 2465 7535

test.err.LS = sum(Ytest != Ytest_pred_LS) / (2*N);

```

1.5.4.9 Illustration of the Results

First, we illustrate the classification achieved by each method.

1.5.4.10 kNN Classification for two k's: $k = 15$ and $k = 1$

```
## Grid Using kNN Classification: We first define the boundaries for the grid
```

```
x.min = round(min(mytraindata$X1), digits=1)-0.1
x.max = round(max(mytraindata$X1), digits=1)+0.1
y.min = round(min(mytraindata$X2), digits=1)-0.1
y.max = round(max(mytraindata$X2), digits=1)+0.1
x.range = seq(from=x.min, to=x.max, by=0.1)
y.range = seq(from=y.min, to=y.max, by=0.1)
x.new = expand.grid(x.range, y.range)
names(x.new) = names(mytraindata[,-3])
```

```
## Basic scatterplot (same as before):
```

```
## this is used as a base to add the boundary and shaded areas
```

```
grid.plot = ggplot(mytraindata, aes(X1, X2)) +
  geom_point(aes(colour=Y), size=2) +
  scale_color_manual(values = c("orange", "blue")) +
  geom_point(data=data.frame(X1=mu1[1], X2=mu1[2]), aes(X1, X2), colour="red", shape=2,
  geom_point(data=data.frame(X1=mu0[1], X2=mu0[2]), aes(X1, X2), colour="green", shape=
  labs(x = "", y="")
```

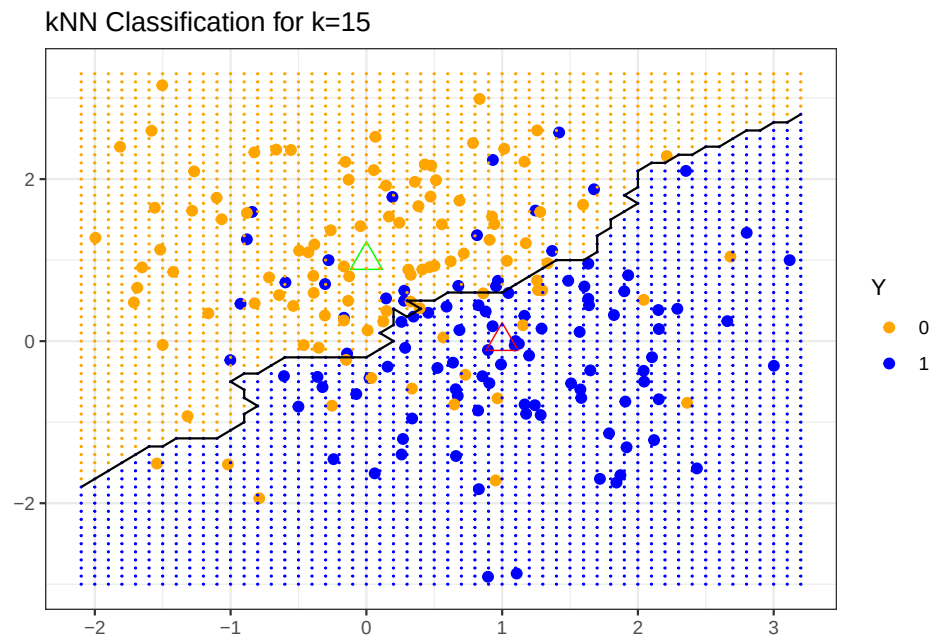
```
## Plot for k=15
```

```
knn.yhat.15 = knn(traindata, x.new, Ytrain, k=15)
knn.pred.15 = ifelse(as.numeric(knn.yhat.15)>0.5,"1","0")
```

```
knn.plotdata.15 = as.data.frame(cbind(x.new, knn.yhat.15))
```

```
knn.plot.15 = grid.plot +
  theme_bw() + #remove grey background
  geom_point(data=knn.plotdata.15, aes(x=X1, y=X2, colour=knn.yhat.15), size=0.05) +
  geom_contour(data=knn.plotdata.15, aes(x=X1, y=X2, z=as.numeric(knn.yhat.15)), bins=1
  ggtitle("kNN Classification for k=15")
```

```
plot(knn.plot.15)
```



```
## Plot for k=1
```

```
knn.yhat.1 = knn(traindata, x.new, Ytrain, k=1)
```

```
knn.pred.1 = ifelse(as.numeric(knn.yhat.1)>0.5, "1", "0")
```

```
knn.plotdata.1 = as.data.frame(cbind(x.new, knn.yhat.1))
```

```
knn.plot.1 = grid.plot +
```

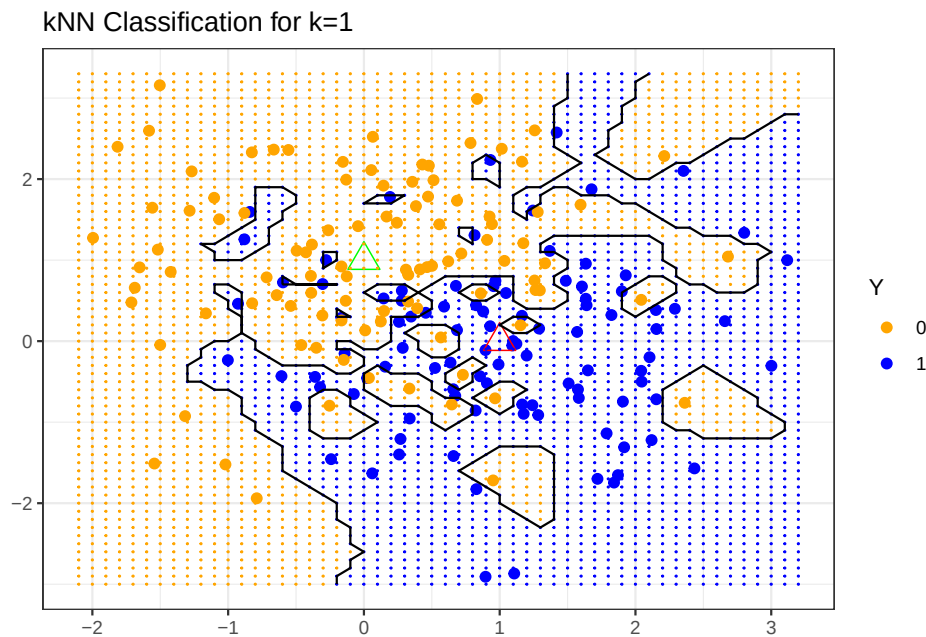
```
  theme_bw() +
```

```
  geom_point(data=knn.plotdata.1, aes(x=X1, y=X2, colour=knn.yhat.1), size=0)
```

```
  geom_contour(data=knn.plotdata.1, aes(x=X1, y=X2, z=as.numeric(knn.yhat.1)))
```

```
  ggtitle("kNN Classification for k=1")
```

```
plot(knn.plot.1)
```



```
## Grid Using Linear Model Classification

RegModel = lm(as.numeric(Ytrain)-1 ~ traindata)
beta.hat = coef(RegModel)

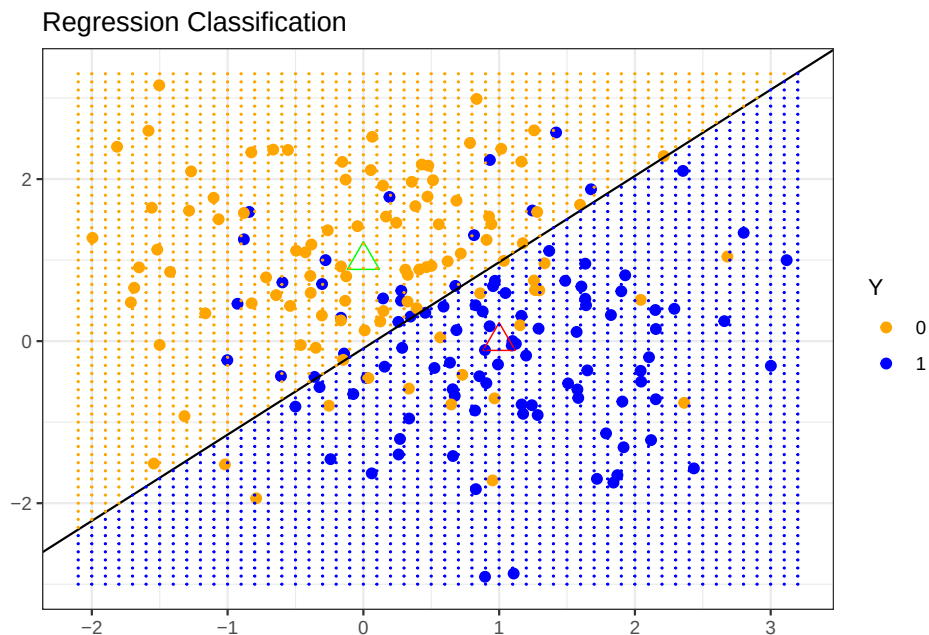
## We use the same ranges for the grid as before

Reg.yhat = as.matrix(cbind(rep(1, nrow(x.new)), x.new)) %*% beta.hat ## predicted Y values
Reg.pred = ifelse(Reg.yhat > 0.5, "1", "0") ## convert numeric Y to factor

Reg.plot.data = cbind(x.new, Reg.pred)

## Basic plot is same as before, so now we add the regression boundary and classification result
reg.plot = grid.plot +
  theme_bw() +
  geom_abline(slope=-beta.hat[2]/beta.hat[3], intercept=(.5-beta.hat[1])/beta.hat[3], color="black") +
  geom_point(data = Reg.plot.data, aes(x=X1, y=X2, color=Reg.pred), size=0.05) +
  ggtitle("Regression Classification")

plot(reg.plot)
```



1.5.4.12 Plot the Performance and Compare the two Methods

Test errors are in magenta and training errors are in blue. The upper x -coordinate indicates the k values, and the lower x -coordinate indicates the degrees-of-freedom of the k NN procedures so that the labels are reciprocally related to k .

The training and test errors for linear regression are plotted at $df = 3$ (corresponding to $k = (2n)/3$), since the linear model has 3 parameters, i.e., 3 dfs.

The training and test errors for KNN with k chosen by CV are plotted at the chose k values.

```
plot(c(0.5,m), range(test.err.LS, train.err.LS, test.err.knn, train.err.knn),
     type="n", xlab="Degrees of Freedom", ylab="Error", xaxt="n")

df = round((2*n)/neighbor_size)
axis(1, at=1:m, labels=df)
axis(3, at=1:m, labels=neighbor_size)

points(1:m, test.err.knn, col="magenta", pch=16)
lines(1:m, test.err.knn, col="magenta", lty=1, lwd=2)
points(1:m, train.err.knn, col="blue", pch=16)
lines(1:m, train.err.knn, col="blue", lty=3, lwd=2)

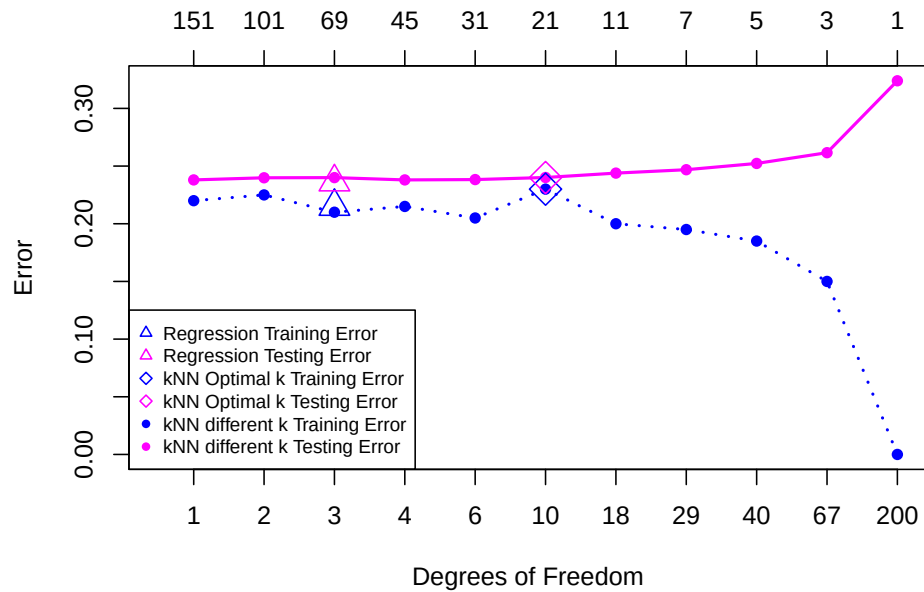
points(3, train.err.LS, pch=2, cex=2, col="blue")
points(3, test.err.LS, pch=2, cex=2, col="magenta")
```

```

points((1:m)[neighbor_size == k.optimal], train.err.knn.CV, col="blue", pch=5, cex=2)
points((1:m)[neighbor_size == k.optimal], test.err.knn.CV, col="magenta", pch=5, cex=2)

legend("bottomleft", pch = c(2, 2, 5, 5, 16, 16), col = c("blue", "magenta"),
      legend = c("Regression Training Error", "Regression Testing Error", "kNN Optimal k Training Error", "kNN Optimal k Testing Error", "kNN different k Training Error", "kNN different k Testing Error"))

```



Chapter 2

Linear Regression Models

A regression model is a model for the **conditional expectation** of the response/ outcome variable given a set of predictors/features. A linear regression model specifically, is *assumes* that $E(Y|\mathbf{X} = \mathbf{x})$ is a linear function of the inputs $X_1, X_2, \dots, X_p$, i.e.

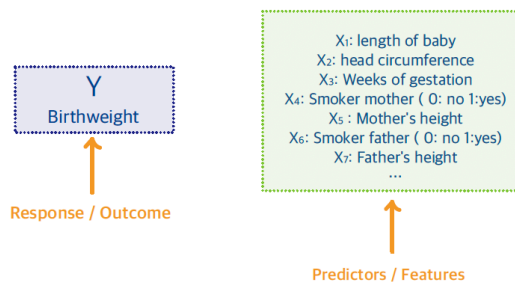
$$E(Y|\mathbf{X} = \mathbf{x}) \sim \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

To describe the properties and characteristics of the MLR model, we introduce the **Birthweight** example.

Introduction to the Birthweight Example

The goal of the birthweight study is to **predict** the birthweight (*response/ outcome*) of a newborn baby given a set of *predictors/features*, specifically for babies born prematurely.

The features include baby-related characteristics, such as length, weight, head circumference, gestation period, and parent characteristics, such as mother's/father's height, smoking habits, mother's pre-pregnancy weight, mother's/father's age, etc.



The **predictors** we consider may be:

- *quantitative*, such as length, weight, gestation period, etc.
- *qualitative/categorical*, such as smoking habits (yes/no), or
- *transformations of quantitative inputs*: For example, power transformation of variables, e.g. X_3^3 ; continuous functions of variables, e.g. $\log X_5$; basic expansions leading to a polynomial representation, e.g. $\sum_{k=1}^n c_k X_{7k}$; interactions, e.g. $X_1 \cdot X_2$, etc.

2.1 Multiple Linear Regression (MLR) Model

The model formulation is given by:

Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$$

where

- β_0 is the intercept
- β_j is the regression coefficient associated with predictor X_j
- ε is the error term. Usual assumptions for the error terms are $\varepsilon \sim IID(0, \sigma^2 \mathbf{I})$

Given the **training data** $\{x_{i1}, x_{i2}, \dots, x_{ip}; y_i\}_{i=1}^n$, we can re-write the model as

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n$$

where n is the sample size. Using the following matrix representation for the response and the features:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \quad \mathbf{X} = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix}$$

the model equation can be written in as

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

which leads to the concise form

$$\begin{array}{ccccccc}
 \mathbf{y}_{n \times 1} & = & \mathbf{X}_{n \times (p+1)} & & (p+1) \times 1 & + & n \times 1 \\
 \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 \text{Response} & & \text{Design} & & \text{Coefficients} & & \text{Error} \\
 & & \text{Matrix} & & & & \text{Term}
 \end{array}$$

where n is the sample size, and $p + 1$ is number of predictors or columns of $\mathbf{X}$ plus the intercept (the “+1”).

2.2 MLR Model Fitting

Given the training data $\{x_{i1}, x_{i2}, \dots, x_{ip}; y_i\}_{i=1}^n$, we want to estimate β , i.e. express:

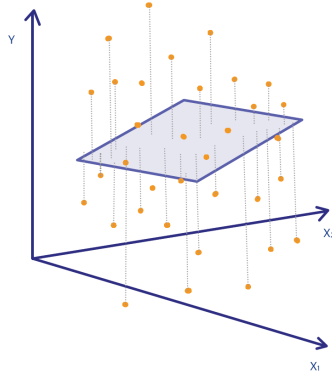
$$\hat{\mathbf{y}} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)^T$$

as a **function of the data**.

The estimated regression coefficients β are obtained by minimizing the *Residual Sum of Squares* (RSS):

$$RSS = \|\mathbf{y} - \mathbf{X}\beta\|^2 = (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)$$

The RSS minimizes the (Euclidean) distance of the points from the regression surface:



This approach makes minimal assumptions, since it only requires that (x_i, y_i) are *random draws from their population*. Specifically, it makes **no assumptions** about the underlying distribution of the response. It simply finds the **best linear fit**, making no assumptions about model validity. In fact, the underlying (true) model does not even need to be linear for this approach to provide us with estimators. Therefore, it often gives good results, no matter how the data were obtained. If a linear model is a good approximation for the underlying non-linear (true) model, then the regression surface can also be thought as a criterion that measures the **lack-of-fit**.

Least-Squares Method

We want to estimate the **vector** of coefficients, by minimizing the *sum of squared residuals*:

$$RSS = \|y - \mathbf{X}\beta\|^2 = (y - \mathbf{X}\beta)^T(y - \mathbf{X}\beta)$$

Therefore, we take derivatives with respect to β 's and set to zero:

$$\begin{aligned} \frac{\partial RSS}{\partial \beta} &= \mathbf{0}_{p \times 1} \Leftrightarrow \\ -2 \mathbf{X}_{p \times n}^T (y - \mathbf{X}\beta)_{n \times 1} &= \mathbf{0}_{p \times 1} \end{aligned}$$

This leads to the so-called **Normal Equations**

$$\mathbf{X}^T(y - \mathbf{X}\beta) = \mathbf{0}$$

Solving the Normal Equations

$$(\mathbf{X}^T \mathbf{X}) \beta = \mathbf{X}^T y$$

leads to the

Least Squares Estimators in MLR

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y$$

We assume that **the rank of $\mathbf{X}$ is $p + 1$** , i.e. no columns of $\mathbf{X}$ are a linear combinations of the other columns of $\mathbf{X}$. *Since $\mathbf{X}$ has rank p , the inverse of $(\mathbf{X}^T \mathbf{X})$ exists.*

Fitted, Predicted Values & Residuals

The **fitted value** of y_i at $x_i = (x_{i1}, x_{i2}, \dots, x_{ip})$ is computed as:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_p x_{ip}$$

More generally, using matrix formulation, we can compute the **fitted values** of $\mathbf{y}$ based on the model as follows:

$$\begin{aligned} \hat{\mathbf{y}}_{n \times 1} &= \mathbf{X} \hat{\beta} \\ &= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y \\ &= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y := \mathbf{H}_{n \times n} y_{n \times 1} \end{aligned}$$

The Hat Matrix

We define

$$\mathbf{H}_{n \times n} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

to be the **hat matrix**, since it returns the “*y-hat*” values.

The **predicted** value of y_i at $x_i^* = (x_{i1}^*, x_{i2}^*, \dots, x_{ip}^*)$ is given by

$$\hat{\mathbf{y}}^* = \hat{\beta}_0 + \hat{\beta}_1 x_{i1}^* + \hat{\beta}_2 x_{i2}^* + \dots + \hat{\beta}_p x_{ip}^*$$

More generally, using matrix formulation, the **predicted values** of $\mathbf{y}$ are given by

$$\hat{\mathbf{y}}^* = \mathbf{X}^* \hat{\beta}$$

Note that the difference between the fitted and predicted values is that the $\mathbf{x}$ is a vector of features that we have already observed (it is part of our training data), while $\mathbf{x}^*$ is an **unobserved** vector of features that is **independent** of the training data.

The **residual** of y_i at $x_i = (x_{i1}, x_{i2}, \dots, x_{ip})$ is obtained by

$$\begin{aligned} r_i &= y_i - \hat{y}_i \\ &= y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \dots - \hat{\beta}_p x_{ip} \end{aligned}$$

Using matrix formulation, the residuals can be computed as

$$\begin{aligned} \mathbf{r}_{n \times 1} &= \mathbf{y} - \hat{\mathbf{y}} \\ &= \mathbf{y} - \mathbf{X} \hat{\beta} = \mathbf{y} - \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \\ &= \mathbf{y} - \mathbf{H} \mathbf{y} = (\mathbf{I} - \mathbf{H}) \mathbf{y} \end{aligned}$$

The residuals $\mathbf{r}$ are used to estimate the **error variance**:

$$\hat{\sigma}^2 = \frac{1}{n - p - 1} \sum_i r_i^2 = \frac{RSS}{n - p - 1}$$

Note that the denominator in the formula is equal to the *degrees of freedom of the residuals* ($n - p - 1$).

Properties of the Residuals

The LS estimator is the β vector that satisfies the **normal equations**, that is

$$\mathbf{X}^T (\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{X}^T (\mathbf{y} - \mathbf{X} \hat{\beta}) = \mathbf{0}$$

Based on this, we can derive the following properties for the residuals $\mathbf{r}_{n \times 1}$:

- The cross-products between the residual vector $\mathbf{r}$ and *each column of $\mathbf{X}$* are zero, i.e.

$$\begin{aligned} \mathbf{X}^T \mathbf{r} &= \mathbf{X}^T (\mathbf{y} - \mathbf{X} \hat{\beta}) \\ &= \mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \hat{\beta} \\ &= \mathbf{X}^T \mathbf{y} - (\mathbf{X}^T \mathbf{X}) (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{0} \end{aligned}$$

- The cross-product between the fitted value $\hat{y}$ and the residual vector r is zero, i.e.

$$\hat{\mathbf{y}}^T \mathbf{r} = \hat{\beta}^T \mathbf{X}^T \mathbf{r} = \mathbf{0}$$

This implies that the residual vector $\mathbf{r}$ is **orthogonal to each column of $\mathbf{X}$ and to $\hat{\mathbf{y}}$** .

2.3 Least-Squares & Normal Equations

In this section, we focus on the system of linear equations that we solve to obtain the least squares estimators for β . Recall, that we want to find a vector $\hat{\beta}$ that minimizes:

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|^2$$

Any vector that provides a minimum value for this expression is called a least-squares solution.

The set of all least squares solutions is precisely the **set of solutions** to

$$(\mathbf{X}^T \mathbf{X})\beta = \mathbf{X}^T \mathbf{y}$$

There is a *unique* solution **if and only if** $\text{rank}(\mathbf{X}) = p+1$ in which case $(\mathbf{X}^T \mathbf{X})$ is invertible.

System of Linear Equations

Linear Algebra Review Let us review a few facts from linear algebra related to solutions to a system of linear equations. For simplicity in the notation, we use the *generic* system of equations:

$$\mathbf{A}z = c$$

where

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{pmatrix}, \quad z = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{pmatrix}$$

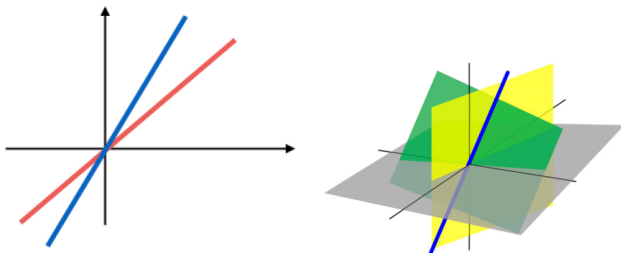
Denote by $\mathbf{A}_i = (a_{1i}, a_{2i}, \dots, a_{mi})^T$ the vector representing the *columns* of $\mathbf{A}$. Therefore, $\mathcal{C}(\mathbf{A})$ is the space generated by the columns of $\mathbf{A}$, or in other words the $\text{span}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k)$.

Definition

The **span** of a collection of vectors $(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k)$ is the set of all linear combinations of these vectors:

$$\begin{aligned} & \text{span}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k) \\ &= \left\{ d_1 \mathbf{A}_1 + d_2 \mathbf{A}_2 + \dots + d_k \mathbf{A}_k, \text{ for any constants } d_1, \dots, d_k \in \mathbb{R} \right\} \end{aligned}$$

The column space of $\mathbf{A}$, $\mathcal{C}(\mathbf{A})$, is a **subspace** in $\mathbb{R}^n$. A linear subspace of $\mathbb{R}^n$ is thought of as a “flat” surface within $\mathbb{R}^n$, and it is a collection of vectors that is closed under linear combinations. Illustration of subspaces of $\mathbb{R}^n$ are shown below:



Solving the System of Equations $\mathbf{A}z = c$

For a system of equations, $\mathbf{A}z = c$, to have a solution, c must be a linear combination of the columns of $\mathbf{A}$, i.e. $c \in \mathcal{C}(\mathbf{A})$. This is simply obtained by the definition of *matrix multiplication* and *equality*:

$$\mathbf{A}z = c \Leftrightarrow c = z_1 \mathbf{A}_1 + \dots + z_k \mathbf{A}_k$$

When $c \notin \mathcal{C}(\mathbf{A})$, we need to **find $\hat{c}$ living in $\mathcal{C}(\mathbf{A})$ that is closest to c** . If this is the case, $\mathbf{A}z = \hat{c}$ has a unique solution, and $\hat{c}$ comes as close to the original data as possible. To do so, we need to **project c orthogonally** onto $\mathcal{C}(\mathbf{A})$, by multiplying both sides by $\mathbf{A}^T$:

$$\mathbf{A}^T \mathbf{A}z = \mathbf{A}^T c$$

* These are the normal equations.

In the sketch to the right, $\mathcal{C}(\mathbf{A})$, the space spanned by the columns of $\mathbf{A}$, is a *flat* surface, and c is a point that exists off of that flat surface. The **shortest distance** from the point c to the plane $\mathcal{C}(\mathbf{A})$ is the one *orthogonal* to the plane.

The *normal equations* essentially help us find the closest point to c that belongs to $\mathcal{C}(\mathbf{A})$ by means of an orthogonal projection.

Geometric Representation of LS

Taking the previous discussion to the linear regression framework:

- $\mathcal{C}(\mathbf{X})$ is the space that is *spanned* by the predictors $X_1, X_2, \dots, X_p$, or in other words the columns of the design matrix. It is a *flat* subspace of $\mathbb{R}^n$.
- The response $\mathbf{y}$ is a vector **off** the subspace, and the fitted value $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$.

- The residuals, on the other hand, $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$ are **orthogonal** to $\hat{\mathbf{y}}$ and to $\mathcal{C}(\mathbf{X})$.

Essentially, what the least-squares method does is that it decomposes the data vector $\mathbf{y}$ into two orthogonal components:

$$\mathbf{y}_{n \times 1} = \hat{\mathbf{y}}_{n \times 1} + \mathbf{r}_{n \times 1}$$

2.4 Goodness-Of-Fit: R -Square

A measure of how well the model fits the data is the **R -square** or the so-called **coefficient of determination** or *percentage of variance explained*. It is defined as

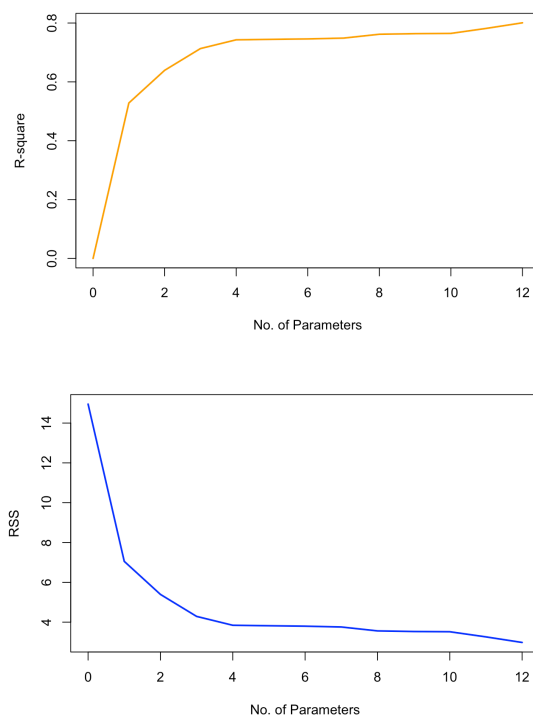
$$\begin{aligned} R^2 &= \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2} = \frac{\text{distance of model from grand mean}}{\text{distance of observations from grand mean}} \\ &= \frac{\|\hat{\mathbf{y}} - \bar{\mathbf{y}}\|^2}{\|\mathbf{y} - \bar{\mathbf{y}}\|^2} \\ &= \frac{\|\mathbf{y} - \bar{\mathbf{y}}\|^2 - \|\hat{\mathbf{y}} - \mathbf{y}\|^2}{\|\mathbf{y} - \bar{\mathbf{y}}\|^2} \\ &= 1 - \frac{\|\hat{\mathbf{y}} - \mathbf{y}\|^2}{\|\mathbf{y} - \bar{\mathbf{y}}\|^2} := 1 - \frac{RSS}{TSS} \end{aligned}$$

where we used orthogonality to get

$$\underbrace{\|\mathbf{y} - \bar{\mathbf{y}}\|^2}_{\text{total variation}} = \underbrace{\|\hat{\mathbf{y}} - \bar{\mathbf{y}}\|^2}_{\text{model variation from mean}} + \underbrace{\|\hat{\mathbf{y}} - \mathbf{y}\|^2}_{\text{error}}$$

Properties of R^2

- $0 \leq R^2 \leq 1$
- R^2 is invariant of any location and/or scale change of Y or X .
- R^2 alone does not tell us much about the effectiveness of the LS method.
- A small R^2 does not imply that the LS model is bad.
- Adding a new predictor, even if it is randomly generated and has nothing to do with Y will decrease RSS and therefore increase R^2 . These relationships are illustrated in the two plots below:



2.5 Linear Transformations on X

Suppose we have a linear regression model of Y on X . If we **scale** or **shift** a predictor, then the fitted values, $\hat{y}$, and R^2 stay the same, but LS estimators, $\hat{\beta}$, change!

Some Examples

- As an example, consider scaling a predictor, i.e. $\tilde{x}_{i2} = 2x_{i2}$ or $\tilde{x}_{i3} = x_{i3}/4$. This transformation will not impact $\hat{y}$, or R^2 .
- In the `birthweight` example, the length of a baby is given in cm. If we want to change the unit to inches by dividing by 2.54 the column that corresponds to length, i.e. $\tilde{X}_1 = X_1/2.54$, then this won't affect our predicted values.

These statements hold true, if we apply any linear transformation on the p predictors **as long as the transformation does not change the rank of X** .

2.6 Rank deficiency

The design matrix $\mathbf{X}$ is an $n \times p$ matrix. If this matrix is **not of full rank** (i.e., its columns are not linearly independent), the matrix $\mathbf{X}^T \mathbf{X}$ can not be inverted. This implies that there is some redundancy in the columns of $\mathbf{X}$, i.e. one column can be written as a linear combination of other columns. As a result, the matrix $\mathbf{X}^T \mathbf{X}$ is *singular* which means that the LS solution is not unique (*identifiability problem*).

However, $\mathcal{C}(\mathbf{X})$ is well defined and thus $\hat{y}$ is well-defined and can be computed. R/ Python can cope well with this problem, and the returned model can be used for prediction.

2.7 Hypothesis Testing in MLR

When fitting a regression model, we often want to understand which predictors are statistically significant or which model (a larger or a smaller) is more appropriate –either for prediction or for estimation purposes–.

Testing for the statistical significance of one (or more) predictor(s), can be formulated as testing whether the corresponding β is zero, or as a model comparison test. In this class, we summarize all testing questions as the following hypothesis test:

$$\begin{cases} H_0 : \text{Reduced Model with } p_0 \text{ coefficients} \\ H_\alpha : \text{Full/ Larger Model with } p_\alpha \text{ coefficients} \end{cases}$$

Obviously, $p_\alpha > p_0$, since the reduced model is a subset/special case of the full model. Therefore, we expect the reduced model to have RSS_0 smaller than RSS_α , the full model's RSS , since $p_\alpha > p_0$. In this context, the main tool for testing is the so-called **partial F test** and is formulated as follows:

Partial F test

Hypothesis Test:

$$\begin{cases} H_0 : \text{Reduced Model with } p_0 \text{ coefficients} \\ H_\alpha : \text{Full/ Larger Model with } p_\alpha \text{ coefficients} \end{cases}$$

Test Statistic:

$$F = \frac{(RSS_0 - RSS_\alpha)/(p_\alpha - p_0)}{RSS_\alpha/(n - p_\alpha)} \sim F_{p_\alpha - p_0, n - p_\alpha}$$

Decision Rule: Reject H_0 , if the F test statistic is large (or larger than $F_{p_\alpha - p_0, n - p_\alpha}$), i.e. the variation missed by the reduced model, when being compared with the error variance, is significantly large.

- The **numerator** in the partial F test quantifies the variation in the data not explained by the reduced model, but explained by the full model.
- On the other hand, the **denominator** is equal to the variation in the data not explained by the full model (i.e., not explained by either model), which is used to estimate the error variance.

Birthweight Example (cont'd)

We want to decide whether the group of the predictors that refer to father's characteristics (i.e. variables $X_9 - X_{12}$) are significant –as a group.

The hypothesis test is formulated as

$$\begin{cases} H_0 : Y \sim 1 + \beta_1 X_1 + \dots + \beta_8 X_8 \\ H_\alpha : Y \sim 1 + \beta_1 X_1 + \dots + \beta_8 X_8 + \dots + \beta_{12} X_{12} \end{cases}$$

In this case $p_0 = 8+1$ is the number of β 's in the reduced model, and $p_\alpha = 12+1$ is the number of β 's in the full model. Using **R/Python**, we extract RSS_0 , RSS_α to perform the partial F test and make a decision.

It turns out that $F = 0.7225$ is smaller than $F_{p_\alpha - p_0, n - p_\alpha} = 2.7$ meaning that the model under the null (reduced) is preferred.

Partial F Test: Special Cases

- **Test for a Single Predictor β_j**

$$\begin{cases} H_0 : Y \sim 1 + \beta_1 X_1 + \dots + \beta_{j-1} X_{j-1} + \beta_{j+1} X_{j+1} + \dots + \beta_p X_p \\ H_\alpha : Y \sim 1 + \beta_1 X_1 + \dots + \beta_{j-1} X_{j-1} + \beta_j X_j + \beta_{j+1} X_{j+1} + \dots + \beta_p X_p \end{cases}$$

is equivalent to the so-called **t -test for each regression parameter**.

- **Test for all Predictors**

$$\begin{cases} H_0 : Y \sim 1 & \text{[intercept-only model]} \\ H_\alpha : Y \sim 1 + \beta_1 X_1 + \dots + \beta_{j-1} X_{j-1} + \beta_j X_j + \beta_{j+1} X_{j+1} + \dots + \beta_p X_p \end{cases}$$

This is the **overall F test** that can be thought of as a *goodness-of-fit test*.

2.8 Categorical Variables in MLR

As we discussed some of the predictors may be *categorical*. Assume a predictor that is categorical with k levels. When added in a linear regression model, it is coded as $k - 1$ *numerical predictors*. The default coding is of the form:

$$D_i = \begin{cases} 0, & \text{if not level } i \\ 1, & \text{if level } i \end{cases}$$

where Level 1 is the *reference level*.

An Example with Categorical Variables

Consider a categorical predictor with three levels (a, b, c):

$$\begin{pmatrix} a \\ a \\ b \\ b \\ b \\ c \\ c \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

In this setup,

- Col 1 corresponds to level a , i.e. $X_1 = 1$ if in column 1, and 0 otherwise.
- Col 2 corresponds to level b , i.e. $X_2 = 1$ if in column 2, and 0 otherwise.
- Col 3 corresponds to level c : this is the reference level, since its value is absorbed by the intercept.

In general, any level can be chosen to be the reference level; this is more a choice that depends on the context rather than the statistics behind. There are also alternative ways to code the variables. If we choose a different coding, then this will affect the estimated regression coefficients (and the interpretation), but it won't affect the predicted/fitted values.

2.9 Collinearity

In practice, we often encounter problems in which many of the predictors are **highly correlated** with each other. In such cases, the fitted values and sampling variance of the regression coefficients can be highly dependent on the particular predictors chosen for the model. This results in sensitive and unreliable estimators.

Possible *symptoms* of collinearity include :

- high pair-wise (sample) correlation between predictors,
- relatively large R^2 ,
- statistically significant F test without *any* statistically significant predictors.

There are also specific tests and metrics that allow us to assess the severity of collinearity in our model, but this is beyond the scope of this course.

The easiest way to **remedy** collinearity is to remove some predictors from highly correlated groups of predictors. Other (more advanced) approaches include PCA, or regularization using penalized Least Squares.

2.10 Model Diagnostics

We made several assumptions when defining the linear regression model:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \text{ where } \varepsilon \sim IID(0, \sigma^2)$$

This means that we assumed:

- *Linearity*, i.e. the model is linear with respect to the predictors.
- *Constant Variance*, i.e. all Y 's have common, constant variance.
- *Uncorrelated errors*, i.e. Y_i and Y_j are uncorrelated for $i \neq j$.
- (*Normality*): this is not part of the assumptions above, and it is not required for the derivation of the least-squares estimators. However, it is needed when one wants to discuss statistical properties of the estimators and the fitted values and residuals. It is, in particular, one of the most critical assumptions when it comes to hypothesis testing.

If we are interested in using the model for understanding the effect of the predictors on the response (i.e. not purely for prediction purposes), then we need to make sure that all assumptions are satisfied (via diagnostic test/plots), and if not take appropriate actions (remedial measures).

Since the focus of this class is prediction, we refer the students to a regression course or textbook to review diagnostics and remedial measures for a MLR model. As an example, you can refer to ...

Outliers

Finally, outliers might be a concern both in large and smaller data sets. If the data set is not very large, it is suggested to test for outliers, e.g. using outlier tests based on the leave-one-out prediction error. Otherwise, if the data set is really large, then this test is not useful and it is suggested to adjust for multiple comparisons or consider alternative approaches.

If outliers seem to be an issue, then it is important to (i) know the range of each variable, (ii) apply log, square-root or other transformations on right skewed predictors and Y , or (iii) apply winsorization to remove the effect of extreme values.

2.11 The Birthweight Data Set Example

The `birthweight.csv` data set contains data from a study on the birth weight of 42 babies. The variables in the data set are the following:

- **ID**: Unique Identification number of a baby
- **Length**: Length of the baby at time of birth in cm (X_1)
- **Birthweight**: Weight of the baby at time of birth in kg (Y)
- **Headcirc**: Head circumference of the baby at time of birth in cm (X_2)
- **Gestation**: Completed weeks of gestation (X_3)
- **smoker**: Mother is/is not a smoker {0: No, 1: Yes} (X_4)
- **mage**: Mother's age at time of birth (X_5)
- **mnocig**: Mother's number of cigarettes smoked per day (X_6)
- **mheight**: Mother's height in cm (X_7)
- **mppwt**: Mother's pre-pregnancy weight (X_8)
- **fage**: Father's age at time of birth (X_9)
- **fedyrs**: Father's years of education (X_{10})
- **fnocig**: Father's number of cigarettes smoked per day (X_{11})
- **fheight**: Father's height (X_{12})

The python version of the example below was written by our TA Peng Xu and can be found here: [Python Version](#)

You can also download the birthweight and seatpos here: [Birthweight.csv](#) and [seatpos.csv](#)

```
birthweight <- read.csv("data/week2/Birthweight.csv", header=TRUE)
dim(birthweight)
```

```
## [1] 42 14
```

```
head(birthweight)
```

```
##      ID Length Birthweight Headcirc Gestation smoker mage mnocig mheight mppwt
## 1 1360     56      4.55      34      44      0    20      0     162     57
## 2 1016     53      4.32      36      40      0    19      0     171     62
## 3  462     58      4.10      39      41      0    35      0     172     58
## 4 1187     53      4.07      38      44      0    20      0     174     68
## 5  553     54      3.94      37      42      0    24      0     175     66
## 6 1636     51      3.93      38      38      0    29      0     165     61
##      fage fedyrs fnocig fheight
## 1    23     10     35     179
## 2    19     12      0     183
## 3    31     16     25     185
## 4    26     14     25     189
## 5    30     12      0     184
## 6    31     16      0     180
```

Since the ID variable is not of interest in our analysis, we won't use it in our

analysis. For convenience can remove it from the data frame we are working with:

```
birthweight = birthweight[, -1]
```

Fitting a Multiple Linear Regression

We start by fitting a linear regression model using all the predictors and we print the summary of the LS results:

```
birthweight.mlr.full = lm(Birthweight ~ ., data=birthweight)
summary(birthweight.mlr.full)
```

```
##
## Call:
## lm(formula = Birthweight ~ ., data = birthweight)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.37925 -0.25558 -0.05541  0.21700  0.60329
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.812311   2.080150  -1.833  0.07713 .
## Length      0.034152   0.031456   1.086  0.28656
## Headcirc    0.086594   0.029489   2.937  0.00644 **
## Gestation   0.096565   0.031323   3.083  0.00447 **
## smoker      -0.243895   0.168516  -1.447  0.15854
## mage        -0.019004   0.018537  -1.025  0.31374
## mnocig      0.001064   0.006854   0.155  0.87770
## mheight     0.004187   0.014153   0.296  0.76944
## mppwt       0.010346   0.011228   0.921  0.36443
## fage        0.007187   0.016327   0.440  0.66305
## fedys       0.003220   0.030939   0.104  0.91783
## fnocig      0.003445   0.004013   0.858  0.39769
## fheight    -0.013293   0.009614  -1.383  0.17731
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3342 on 29 degrees of freedom
## Multiple R-squared:  0.7834, Adjusted R-squared:  0.6938
## F-statistic: 8.742 on 12 and 29 DF,  p-value: 1.011e-06
```

If we want to check the objects returned by `lm` and `summary(lm...)`:

```
names(birthweight.mlr.full)
```

```
## [1] "coefficients" "residuals"    "effects"       "rank"
```

```
## [5] "fitted.values" "assign"      "qr"      "df.residual"
## [9] "xlevels"       "call"      "terms"   "model"
```

```
names(summary(birthweight.mlr.full))
```

```
## [1] "call"      "terms"      "residuals"  "coefficients"
## [5] "aliased"   "sigma"      "df"         "r.squared"
## [9] "adj.r.squared" "fstatistic" "cov.unscaled"
```

The design matrix in this example can be extracted using:

```
design.birthweight = model.matrix(birthweight.mlr.full);
```

If we want to extract the Regression coefficients including the intercept, i.e. $\hat{\beta}$, we have

```
birthweight.mlr.full$coef
```

```
## (Intercept)      Length      Headcirc      Gestation      smoker      mage
## -3.812311183  0.034151922  0.086594329  0.096565435 -0.243895136 -0.019003806
##      mnocig      mheight      mppwt      fage      fedyrs      fnocig
##  0.001064074  0.004187315  0.010346096  0.007187428  0.003219952  0.003444674
##      fheight
## -0.013293205
```

The fitted values and residuals are obtained by:

```
full.model.residuals = birthweight.mlr.full$resid
full.model.fitted = birthweight.mlr.full$fitted
```

Check the calculation for Residual standard error and Multiple R-squared in the summary output:

The residuals $\mathbf{r}$ are used to estimate the *error variance*:

$$\hat{\sigma}^2 = \frac{1}{n - p - 1} \sum_i r_i^2 = \frac{RSS}{n - p - 1}$$

To obtain an estimator for $\hat{\sigma}$, we can either use the formula (from the notes):

```
n = dim(birthweight)[1]
p = dim(birthweight)[2] - 1
sqrt(sum(birthweight.mlr.full$residuals^2)/(n - p - 1))
```

```
## [1] 0.3341631
```

or we can extract it from the `lm` object:

```
summary(birthweight.mlr.full)$sigma
```

```
## [1] 0.3341631
```

The coefficient of determination computes as:

$$\frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2} = 1 - \frac{RSS}{TSS}$$

We can extract R^2 from the summary output or compute it:

```
summary(birthweight.mlr.full)$r.squared

## [1] 0.783425
1 - sum(birthweight.mlr.full$residuals^2)/(var(birthweight$Birthweight)*(n-1))

## [1] 0.783425
1 - var(birthweight.mlr.full$residuals)/var(birthweight$Birthweight)

## [1] 0.783425
```

Invariance of R^2 & Fitted Values

We can also check the invariance property of R^2 by rescaling one of the variables:

```
## Center the Length variable
birthweight.new = birthweight
birthweight.new$Length.new = birthweight.new$Length - rep(mean(birthweight.new$Length), 42)

## Model with original `Length`
summary(lm(Birthweight~., data=birthweight.new[-14]))$r.squared

## [1] 0.783425
## Model with centered `Length`
summary(lm(Birthweight~., data=birthweight.new[-1]))$r.squared

## [1] 0.783425
```

If we want, we can also check that the fitted values from both models are identical:

```
## Difference in fitted values between Model w `Length` - Model w `centered-Length`
### The values are added to make the output cleaner.

sum( fitted(lm(Birthweight~., data=birthweight.new[-14])) - fitted(lm(Birthweight~., data=birthweight.new[-1])) )

## [1] -8.881784e-16
```

Note that the difference is not exactly 0, since there are rounding errors that accumulated when the various R functions are applied.

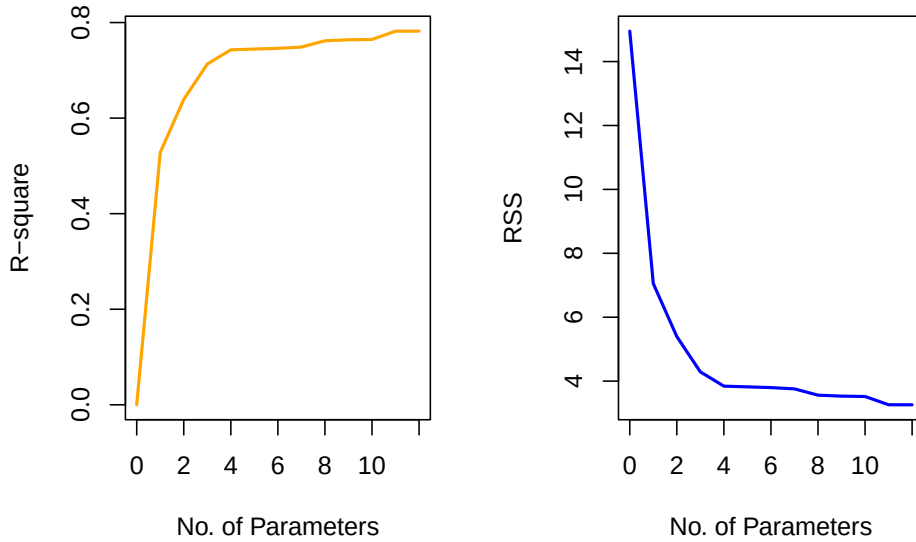
R^2 and RSS behavior when adding more X s

We can also visualize how the R^2 and RSS change as a function of the number of variables in the model to observe that adding a new predictor, even if it is randomly generated and has nothing to do with Y will decrease RSS and therefore increase R^2 .

So, start with the null model that contains only the intercept and then we add variables one at a time, keeping the R^2 and RSS that we will plot in the end. We will also randomly generate a new variable `randomvar` to illustrate that even an irrelevant variable has an effect on both R^2 and RSS :

```
birthweight.new = birthweight
birthweight.new$randomvar = rnorm(42)

no.var = 0:12
par(mfrow=c(1,2))
plot(no.var, rsq, xlab="No. of Parameters", ylab = "R-square", type='n');
lines(no.var, rsq, lwd=2, col="orange")
plot(no.var, rss, xlab="No. of Parameters", ylab = "RSS", type='n');
lines(no.var, rss, lwd=2, col="blue")
```



This is to illustrate why R^2 and RSS are not good metrics when it comes to model selection. They are useful, because they help us understand the quality of the model or RSS is a key ingredient in constructing hypothesis tests, but they are never used for model selection.

Rank Deficiency

We have rank deficiency when $\mathbf{X}$ is not of full rank which leads to the $(\mathbf{X}^T \mathbf{X})$ matrix not being invertible.

Here we illustrate a scenario where it is easy to identify the cause of the problem.

```
## Create a new variable: the average height of both parents
birthweight.new = birthweight
birthweight.new$aveheight = (birthweight.new$mheight+birthweight.new$fheight)/2

## Regress birthweight against all predictors
summary(lm(Birthweight ~ ., data=birthweight.new))

##
## Call:
## lm(formula = Birthweight ~ ., data = birthweight.new)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.37925 -0.25558 -0.05541  0.21700  0.60329
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.812311   2.080150  -1.833  0.07713 .
## Length      0.034152   0.031456   1.086  0.28656
## Headcirc     0.086594   0.029489   2.937  0.00644 **
## Gestation    0.096565   0.031323   3.083  0.00447 **
## smoker       -0.243895   0.168516  -1.447  0.15854
## mage         -0.019004   0.018537  -1.025  0.31374
## mnocig        0.001064   0.006854   0.155  0.87770
## mheight       0.004187   0.014153   0.296  0.76944
## mppwt         0.010346   0.011228   0.921  0.36443
## fage          0.007187   0.016327   0.440  0.66305
## fedyrs        0.003220   0.030939   0.104  0.91783
## fnocig        0.003445   0.004013   0.858  0.39769
## fheight      -0.013293   0.009614  -1.383  0.17731
## aveheight           NA           NA      NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3342 on 29 degrees of freedom
## Multiple R-squared:  0.7834, Adjusted R-squared:  0.6938
## F-statistic: 8.742 on 12 and 29 DF,  p-value: 1.011e-06
```

As you see the β_j that corresponds to the new variable cannot be calculated (hence the NA), since `aveheight` is linearly dependent on both `mheight` and `fheight`. At the same time, although the matrix $\mathbf{X}$ is not of full rank, R produces solutions for the rest of the β s and the fitted model is valid for prediction purposes. The predictions derived from this model will align with those obtained from fitting a linear regression model excluding the columns with NA coefficient.

This issue can be easily fixed, by removing one or more variables. Which variable will be removed depends on the context. For example, we need to (conceptually) answer the question on whether we prefer to have the `aveheight` or each parent's height individually.

A harder issue to resolve is when two predictors are not exactly linearly dependent, but approximately - this means that they are highly correlated, but not an exact function of one another. This is more challenging to deal with and we will discuss it later.

Test for the Significance of Predictors

We use the partial F test

$$F = \frac{(RSS_0 - RSS_\alpha)/p_\alpha - p_0}{RSS_\alpha/(n - p_\alpha)} \sim F_{p_\alpha - p_0, n - p_\alpha}$$

to compare the full model with all available predictors versus the model that exclude father-related variables. That is, the hypothesis test is formulated as

$$\begin{cases} H_0 : Y \sim 1 + \beta_1 X_1 + \dots + \beta_8 X_8 \\ H_\alpha : Y \sim 1 + \beta_1 X_1 + \dots + \beta_8 X_8 + \dots + \beta_{12} X_{12} \end{cases}$$

```
## Full Model was fitted before
## and is stored in birthweight.mlr.full
```

```
## Reduced Model
```

```
birthweight.mlr.reduced1 = lm(Birthweight ~ Length + Headcirc + Gestation + smoker + mage + mnocig)
```

We can use one function to perform the test:

```
anova(birthweight.mlr.reduced1, birthweight.mlr.full)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: Birthweight ~ Length + Headcirc + Gestation + smoker + mage +
##      mnocig + mheight + mppwt
```

```
## Model 2: Birthweight ~ Length + Headcirc + Gestation + smoker + mage +
##      mnocig + mheight + mppwt + fage + fedys + fnocig + fheight
```

```
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      33 3.5610
## 2      29 3.2383  4    0.32271 0.7225 0.5837
```

The F statistic value is 0.7225, and the corresponding p -value is 0.5837 which means that we fail to reject the null and conclude that the model under the H_0 is preferred. In other words, we prefer the smaller model which implies that the father-related predictors can be removed from the model.

We can also build the F statistic, by extracting all necessary components:

```

RSS0 = deviance(birthweight.mlr.reduced1)
RSS0

## [1] 3.560996

RSSalpha = deviance(birthweight.mlr.full)
RSSalpha

## [1] 3.238285
p0 = 9;
palpha = 13;
n = 42;

## Partial F test
F = ((RSS0 - RSSalpha)/(palpha-p0))/(RSSalpha/(n-palpha))
F

## [1] 0.7224983
## Critical value use to compare with the F statistic.
qf(0.95, palpha-p0, n-palpha)

## [1] 2.701399

```

the conclusion is the same since $F < F_{p_\alpha - p_0, n - p_\alpha}$.

Testing for a Single β_j

Let's test for the significance of variable **Length** (X_1), i.e.

$$\begin{cases} H_0 : Y \sim 1 + \beta_2 X_2 + \dots + \beta_{12} X_{12} \\ H_\alpha : Y \sim 1 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_{12} X_{12} \end{cases}$$

We fit the full model and then compare by means of the partial F test:

```

## Full Model was fitted before
## and is stored in birthweight.mlr.full

## Reduced Model
birthweight.mlr.reduced2 = lm(Birthweight ~ Headcirc + Gestation + smoker + mage + mnocig + mheight + mppwt + fage + fedyrs + fnocig + fheight)

## Partial F test
anova(birthweight.mlr.reduced2, birthweight.mlr.full)

## Analysis of Variance Table
##
## Model 1: Birthweight ~ Headcirc + Gestation + smoker + mage + mnocig +
##      mheight + mppwt + fage + fedyrs + fnocig + fheight

```

```
## Model 2: Birthweight ~ Length + Headcirc + Gestation + smoker + mage +
##      mnocig + mheight + mppwt + fage + fedys + fnocig + fheight
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      30 3.3699
## 2      29 3.2383  1   0.13162 1.1787 0.2866
```

```
## Compare the  $F$  value above with the square of the  $t$  test
## for the length variable in the output below:
summary(birthweight.mlr.full)
```

```
##
## Call:
## lm(formula = Birthweight ~ ., data = birthweight)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.37925 -0.25558 -0.05541  0.21700  0.60329
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.812311    2.080150  -1.833  0.07713 .
## Length       0.034152    0.031456   1.086  0.28656
## Headcirc     0.086594    0.029489   2.937  0.00644 **
## Gestation    0.096565    0.031323   3.083  0.00447 **
## smoker       -0.243895    0.168516  -1.447  0.15854
## mage        -0.019004    0.018537  -1.025  0.31374
## mnocig       0.001064    0.006854   0.155  0.87770
## mheight      0.004187    0.014153   0.296  0.76944
## mppwt        0.010346    0.011228   0.921  0.36443
## fage         0.007187    0.016327   0.440  0.66305
## fedys        0.003220    0.030939   0.104  0.91783
## fnocig       0.003445    0.004013   0.858  0.39769
## fheight     -0.013293    0.009614  -1.383  0.17731
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3342 on 29 degrees of freedom
## Multiple R-squared:  0.7834, Adjusted R-squared:  0.6938
## F-statistic: 8.742 on 12 and 29 DF,  p-value: 1.011e-06
## The square of the t-value is obtained by
summary(birthweight.mlr.full)$coefficients[2,3]^2
```

```
## [1] 1.178721
```

Testing for all the predictors

The F test statistic and corresponding p -value in the `summary` output correspond to the following hypothesis:

$$\begin{cases} H_0 : Y \sim 1 \\ H_\alpha : Y \sim 1 + \beta_1 X_1 + \dots + \beta_{j-1} X_{j-1} + \beta_j X_j + \beta_{j+1} X_{j+1} + \dots + \beta_p X_p \end{cases}$$

which is the so-called **overall F test**.

```
summary(birthweight.mlr.full)$fstatistic
```

```
##      value      numdf      dendif
## 8.741902 12.000000 29.000000
```

In this case, the F value is equal to 8.74 and it follows an $F_{12,29}$ distribution. The corresponding p -value is equal to 1.011e-06 which is much less than $\alpha = 0.05$. This implies that we reject the null and conclude that the model with all the predictors is more adequate compared to the intercept-only model.

Categorical Variables

In the `birthweight` data set we only have binary variables, so for illustration purposes, we will create a new categorical variable, `mage.group` with 3 levels: a =below 20, b =20-30, and c =above 30:

```
birthweight.new = birthweight
birthweight.new$mage.group <- as.factor(ifelse(birthweight.new$mage < 20, 'a', ifelse(birthweight.new$mage < 30, 'b', 'c')))
```

Let's fit the model with the age group variable and look at the design matrix:

```
birthweight.mlr.categorical = lm(Birthweight ~ ., data=birthweight.new[, -6])
model.matrix(birthweight.mlr.categorical)
```

```
##      (Intercept) Length Headcirc Gestation smoker mnocig mheight mppwt fage
## 1             1      56        34         44      0      0      162    57    23
## 2             1      53        36         40      0      0      171    62    19
## 3             1      58        39         41      0      0      172    58    31
## 4             1      53        38         44      0      0      174    68    26
## 5             1      54        37         42      0      0      175    66    30
## 6             1      51        38         38      0      0      165    61    31
## 7             1      52        34         40      0      0      157    50    31
## 8             1      53        33         42      0      0      165    61    21
## 9             1      54        38         38      0      0      172    50    20
## 10            1      50        35         38      0      0      157    48    22
## 11            1      53        33         41      0      0      164    62    37
## 12            1      51        36         40      0      0      168    53    29
## 13            1      52        36         38      0      0      164    57    35
## 14            1      53        33         41      0      0      155    55    25
```

## 15	1	53	34	40	0	0	167	60	30
## 16	1	48	33	37	0	0	158	54	39
## 17	1	48	35	39	0	0	162	62	27
## 18	1	48	33	34	0	0	167	64	25
## 19	1	53	34	39	0	0	165	57	23
## 20	1	43	32	33	0	0	149	45	26
## 21	1	53	38	40	1	2	170	59	24
## 22	1	51	33	41	1	7	160	53	24
## 23	1	50	30	37	1	7	165	60	20
## 24	1	50	35	39	1	7	159	52	23
## 25	1	48	30	37	1	7	163	47	20
## 26	1	46	32	35	1	7	166	57	37
## 27	1	48	30	33	1	7	161	50	20
## 28	1	58	39	41	1	12	173	70	38
## 29	1	53	34	40	1	12	163	49	41
## 30	1	49	36	40	1	12	152	48	37
## 31	1	52	35	38	1	12	165	64	38
## 32	1	51	38	39	1	17	157	48	32
## 33	1	50	33	39	1	17	156	53	24
## 34	1	50	33	45	1	25	163	54	30
## 35	1	52	36	39	1	25	170	78	40
## 36	1	53	37	41	1	25	161	66	46
## 37	1	52	37	40	1	25	170	62	30
## 38	1	52	33	39	1	25	181	69	23
## 39	1	49	34	38	1	25	162	57	32
## 40	1	53	34	41	1	35	163	51	31
## 41	1	47	33	35	1	35	170	57	23
## 42	1	53	32	40	1	50	168	61	31
##	fedyrs	fnocig	fheight	mage.groupb	mage.groupc				
## 1	10	35	179	1	0				
## 2	12	0	183	0	0				
## 3	16	25	185	0	1				
## 4	14	25	189	1	0				
## 5	12	0	184	1	0				
## 6	16	0	180	1	0				
## 7	16	0	173	1	0				
## 8	10	25	185	1	0				
## 9	12	7	172	0	0				
## 10	14	0	179	1	0				
## 11	14	0	170	1	0				
## 12	16	0	181	1	0				
## 13	16	0	183	0	1				
## 14	14	25	183	1	0				
## 15	16	25	182	1	0				
## 16	10	0	171	1	0				
## 17	14	0	178	1	0				

```
## 18      12      25      175          1          0
## 19      14       2      193          0          0
## 20      16       0      169          1          0
## 21      12      12      185          1          0
## 22      16      12      176          1          0
## 23      14       0      183          0          0
## 24      14      25      200          1          0
## 25      10      35      185          1          0
## 26      14      25      173          0          1
## 27      10      35      180          1          0
## 28      14      25      180          0          1
## 29      12      50      191          0          1
## 30      12      25      170          1          0
## 31      14      50      180          1          0
## 32      12      25      169          1          0
## 33      12       7      179          1          0
## 34      16       0      183          1          0
## 35      16      50      178          1          0
## 36      16       0      175          0          1
## 37      10      25      181          1          0
## 38      16       2      181          1          0
## 39      16      50      194          0          1
## 40      16      25      185          1          0
## 41      12      50      186          1          0
## 42      16       0      173          0          1
## attr("assign")
## [1] 0 1 2 3 4 5 6 7 8 9 10 11 12 12
## attr("contrasts")
## attr("contrasts")$mage.group
## [1] "contr.treatment"
```

When fitting the regression model, R automatically converted the factor to the appropriate number of indicator/variables which were included in the design matrix. We observe that level *a* is the baseline and the two generated columns are levels *b* and *c*.

In general, the reference level can be changed as well as the coding. Those changes will affect the estimated β coefficients, but they will not affect the fitted or predicted values.

Mean Response Estimation & Prediction

In practice, we have target values for which we need to estimate the mean response of values for which we want to do prediction.

Since this is not the case here, in order to illustrate both methods, we use as inputs the mean value of every predictor. This is done as follows:

```
meanvalue = apply(birthweight[, -2], 2, mean)
```

Now, we create the X `data.frame` to use as an *input* in the `predict()` function:

```
x=data.frame(t(meanvalue))
```

Using the `predict` function we have:

```
predict.lm(birthweight.mlr.full, x)
```

```
##          1
## 3.312857
```

Training vs. Testing Error

In this example, and in linear regression models, as we introduce more variables, training error (measured by RSS or MSE) consistently decreases. However, this reduction in training error does not guarantee a corresponding decrease in test error, which measures prediction accuracy on independent test data. To demonstrate this, we randomly split our dataset into 60% training and 40% test portions, progressively adding predictors. This approach highlights how additional predictors can lower training error while test error may not follow the same trend!

```
n = dim(birthweight)[1]      ## sample size
p = dim(birthweight)[2]-1    ## number of non-intercept predictors

ntrain = round(n*0.6)
set.seed(598)
train.id = sample(1:n, ntrain)
rm(.Random.seed, envir=globalenv())
train.id

## [1] 22 21 33 2 23 31 41 29 1 38 19 42 36 18 13 30 12 10 17 5 34 40 35 16 39

train.MSE = rep(0, p)
test.MSE = rep(0, p)

train.Y = birthweight[train.id, 2]
train.Y

## [1] 3.14 3.64 2.74 4.32 2.78 3.00 2.66 3.59 4.55 3.41 2.90 2.75 3.55 2.92 3.23
## [16] 3.32 3.27 3.42 3.03 3.94 3.87 3.19 3.86 3.11 3.18

test.Y = birthweight[-train.id, 2]
test.Y

## [1] 4.10 4.07 3.93 3.77 3.65 3.63 3.35 3.20 3.15 2.65 2.51 2.37 2.05 1.92 4.57
## [16] 3.32 3.53
```



```

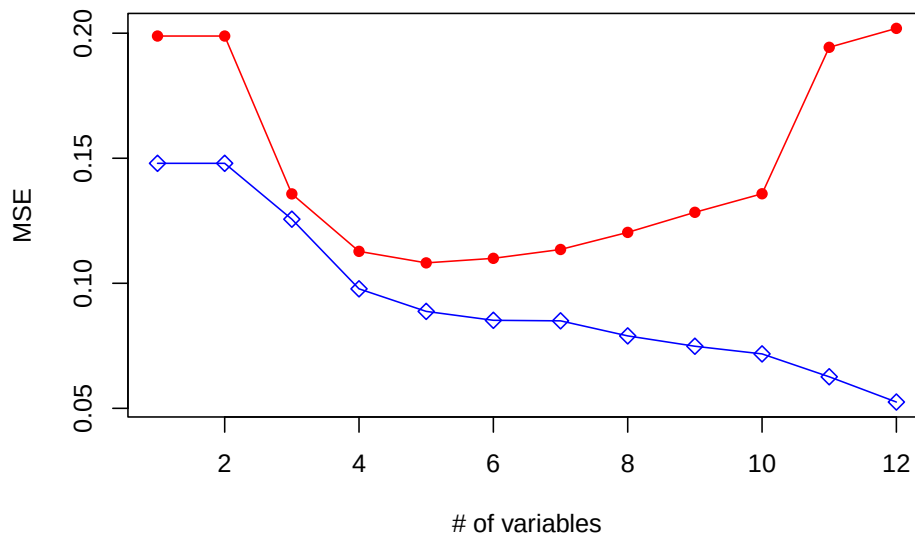
for(i in 1:p){
  var_id = c(1:i, (p+1))
  var_id = var_id[var_id != 2] # remove the response variable
  myfit = lm( birthweight[train.id, 2]~., birthweight[ train.id, var_id ] )

  train.Y.pred = myfit$fitted
  train.MSE[i] = mean((train.Y - train.Y.pred)^2)

  test.Y.pred = predict(myfit, newdata = birthweight[-train.id, ])
  test.MSE[i] = mean((test.Y - test.Y.pred)^2)
}

plot(c(1, p), range(train.MSE, test.MSE), type="n", xlab="# of variables", ylab="MSE")
points(train.MSE, col = "blue", pch = 5)
lines(train.MSE, col = "blue", pch = 5)
points(test.MSE, col = "red", pch = 16)
lines(test.MSE, col = "red", pch = 16)

```



In this example, the test errors initially decreased but began to rise again as more variables were added. Feel free to use this code in other examples to observe more patterns between training/testing errors.

In general, we expect that in each iteration, the blue line (representing the training error) consistently exhibits a monotonically decreasing trend, signifying the reduction of training error as more predictors are incorporated. On the other hand, the red line's (testing error) trajectory may not uniformly decrease.

Test for Outliers

In this example, we check the data set for unusual observations. Specifically, we will check for:

To identify outliers, we use the Bonferroni test. For that purpose, we need to compute the studentized residuals t_i (an externally standardized version of residuals). Under the null hypothesis H_0 , $t_i \sim t_{n-p}$, where n is the sample size and p is the number of predictors without the intercept. We perform this test for all n observations testing case at level $\frac{\alpha}{n}$:

```
## We work with the full model again.
n = dim(birthweight)[1];
p = length(variable.names(birthweight.mlr.full));
```

We first compute the studentized residuals using the `rstudent` R function and the Bonferroni critical value using Student's distribution:

```
## Computing Studentized Residuals #
birthweight.resid = rstudent(birthweight.mlr.full);

## Critical value WITH Bonferroni correction #
bonferroni_cv = qt(.05/(2*n), n-p-1)
bonferroni_cv
```

```
## [1] -3.60755
```

Now, we need to find which (if any) studentized residuals exceed the Bonferroni critical value:

```
## Sorting the residuals in descending order to find outliers (if any)
birthweight.resid.sorted = sort(abs(birthweight.resid), decreasing=TRUE)[1:10]
print(birthweight.resid.sorted)
```

```
##          2          29          34          1          28          20          42          24
## 2.068621 1.890299 1.830714 1.682375 1.645611 1.613154 1.435773 1.411769
##          19          39
## 1.347956 1.324123
```

```
birthweight.outliers = birthweight.resid.sorted[abs(birthweight.resid.sorted) > abs(bonferroni_cv)]
print(birthweight.outliers)
```

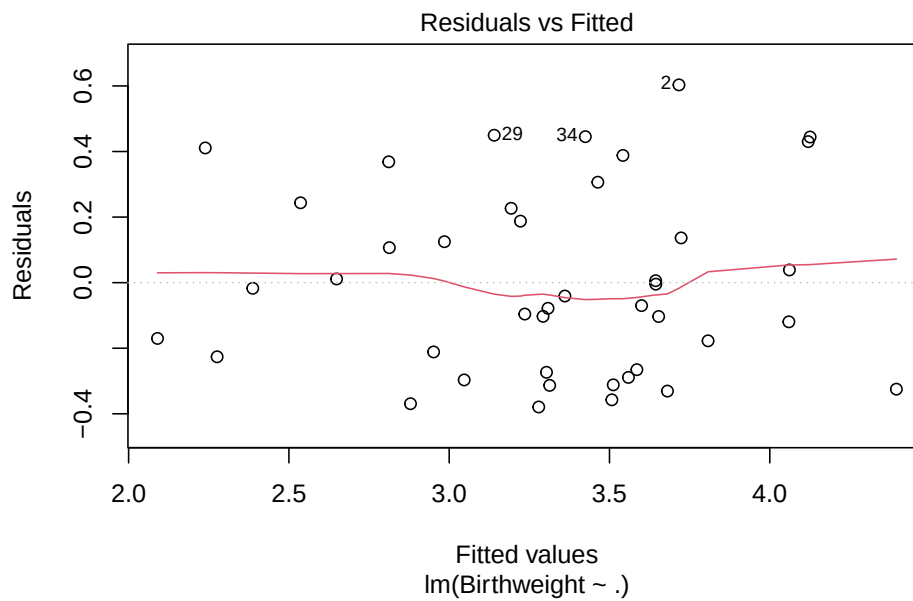
```
## named numeric(0)
```

Above, we computed a t-value of $|-3.61|$ at $\alpha = 0.05$. If an observation's studentized residual is higher (in absolute value) than the critical value of the T distribution with Bonferroni correction, then this observation will be considered an outlier. According to this criterion, we can see that **we don't have any outliers in the data set, since none of the studentized residuals is higher than $|-3.61|$.**

Checking Model Assumptions

We start by using residual plots and specifically the residuals against fitted values:

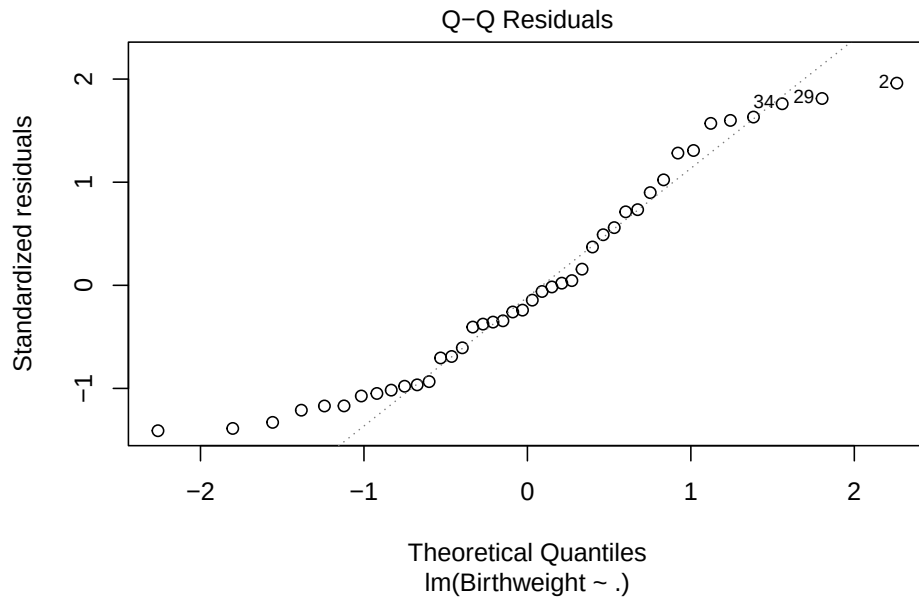
```
plot(birthweight.mlr.full, which=1)
```



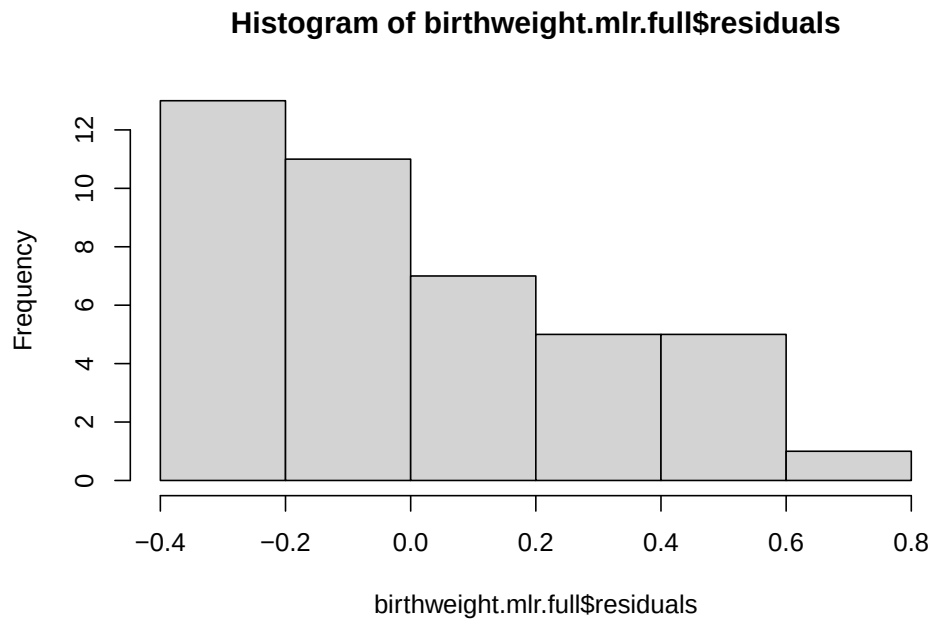
The points on the plot are randomly scattered around the zero line, so we conclude that the constant variance assumption is satisfied.

We continue by checking the normality assumption:

```
plot(birthweight.mlr.full, which=2)
```



```
hist(birthweight.mlr.full$residuals)
```



We seem to have some departures from the normality assumption in this case.

Collinearity: Car Seat Position Data (a Faraway data set)

Car drivers like to adjust the seat position for their own comfort. Car designers would find it helpful to know how different drivers will position the seat depending on their *size* and *age*. Researchers at the HuMoSim laboratory at the University of Michigan collected the following data on **38** drivers:

Age: Drivers age in years

Weight: Drivers weight in lbs

HtShoes: height with shoes in cm

Ht: height without shoes in cm

Seated: seated height in cm

Arm: lower arm length in cm

Thigh: thigh length in cm

Leg: lower leg length in cm

hipcenter: horizontal distance of the midpoint of the hips from a fixed location in the car in mm

To read the data we need to load the **faraway** library and then open the **seatpos** data set.

```
library(faraway)
data(seatpos)
attach(seatpos)
```

Let's start by fitting the **full** model with **hipcenter** as a response and everything else as a predictor:

```
# Fit the FULL model
position.full=lm(hipcenter ~ ., seatpos)
summary(position.full)
```

```
##
## Call:
## lm(formula = hipcenter ~ ., data = seatpos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -73.827 -22.833  -3.678  25.017  62.337
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 436.43213  166.57162   2.620   0.0138 *
## Age          0.77572    0.57033   1.360   0.1843
```

```
## Weight      0.02631    0.33097    0.080    0.9372
## HtShoes     -2.69241    9.75304   -0.276    0.7845
## Ht          0.60134   10.12987    0.059    0.9531
## Seated      0.53375    3.76189    0.142    0.8882
## Arm         -1.32807    3.90020   -0.341    0.7359
## Thigh       -1.14312    2.66002   -0.430    0.6706
## Leg         -6.43905    4.71386   -1.366    0.1824
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 37.72 on 29 degrees of freedom
## Multiple R-squared:  0.6866, Adjusted R-squared:  0.6001
## F-statistic:  7.94 on 8 and 29 DF,  p-value: 1.306e-05
```

Note that the p -value of the overall F test is very small ($1.306e-05$), which means that *at least one of the predictors is statistically significant*, but at the same time none of the predictors is statistically significant according to the individual t tests. This is strange and definitely a *red flag* that we may have collinearity issues.

To investigate the presence of high correlated predictors as well as the linear correlation of the response with each of the predictors, we start by checking the **correlation matrix**:

```
## We use the round function with 2 digits, to round the numbers in the output
## This only affects the printed numbers, not the ones R has stored.
round(cor(seatpos), dig=2)
```

```
##      Age Weight HtShoes    Ht Seated    Arm Thigh    Leg hipcenter
## Age      1.00  0.08  -0.08 -0.09  -0.17  0.36  0.09 -0.04    0.21
## Weight   0.08  1.00  0.83  0.83   0.78  0.70  0.57  0.78   -0.64
## HtShoes  -0.08  0.83  1.00  1.00   0.93  0.75  0.72  0.91   -0.80
## Ht       -0.09  0.83  1.00  1.00   0.93  0.75  0.73  0.91   -0.80
## Seated   -0.17  0.78  0.93  0.93   1.00  0.63  0.61  0.81   -0.73
## Arm       0.36  0.70  0.75  0.75   0.63  1.00  0.67  0.75   -0.59
## Thigh     0.09  0.57  0.72  0.73   0.61  0.67  1.00  0.65   -0.59
## Leg      -0.04  0.78  0.91  0.91   0.81  0.75  0.65  1.00   -0.79
## hipcenter 0.21 -0.64 -0.80 -0.80  -0.73 -0.59 -0.59 -0.79    1.00
```

We observe that the response is *highly correlated with most of the variables* which justifies the low p -value in the overall F test. However, we have **highly correlated predictors** (e.g. $\text{Corr}(\text{Leg}, \text{HtShoes})=0.91$) which means that we will have *collinearity*.

Let's look at the **condition number** of the $\mathbf{X}^T\mathbf{X}$ matrix. Recall that the condition number is *not scale invariant*. This implies that calculating it before standardizing the matrix might lead to incorrect conclusions. So, let's standardize it first:

```
## Extract the design matrix and remove the column of 1s that corresponds to the intercept.
x = model.matrix(position.full)[,-1]
```

```
## Standardize the matrix
```

```
x = x - matrix(apply(x,2, mean), 38,8, byrow=TRUE)
x = x / matrix(apply(x, 2, sd), 38,8, byrow=TRUE)
apply(x,2,mean)
```

```
##      Age      Weight      HtShoes      Ht      Seated
## -5.843279e-18  2.534522e-16  9.524545e-16  1.577685e-16 -1.079546e-15
##      Arm      Thigh      Leg
## -1.285521e-16  9.860533e-17 -1.029878e-16
```

```
apply(x,2,var)
```

```
##      Age Weight HtShoes      Ht Seated      Arm      Thigh      Leg
##      1      1      1      1      1      1      1      1
```

Compute the condition number using the standardized matrix:

```
## Compute the eigenvalues of the matrix
```

```
eigenvalues.x = eigen(t(x) %*% x)
eigenvalues.x$val
```

```
## [1] 209.90786979  45.76108236  17.15850736   8.91545889   7.18612386
## [6]   5.14944541   1.86274750   0.05876483
```

```
## Compute the condition number:
```

```
sqrt(eigenvalues.x$val[1]/eigenvalues.x$val[8])
```

```
## [1] 59.7662
```

The condition number is **59.77**, larger than 30, so we conclude that *collinearity is present*.

Let's also check the **Variance Inflation Factor (VIF)**:

```
## Variance Inflation Factor (VIF)
```

```
round(vif(x), dig=2)
```

```
##      Age Weight HtShoes      Ht Seated      Arm      Thigh      Leg
##      2.00   3.65  307.43  333.14   8.95   4.50   2.76   6.69
```

```
sqrt(307.43)
```

```
## [1] 17.53368
```

Note that the *standard error* for the coefficient associated with HtShoes is **17.5 times larger** than it would have been without collinearity.

The next step is to **investigate which variable to remove**. To do so, we study the *pairwise correlations* and perform *partial F-tests* as follows:

```

cor(Seated+Thigh, Ht)

## [1] 0.9389819
cor(Seated+Leg, Ht)

## [1] 0.965607
cor(Seated+Arm, Ht)

## [1] 0.9465523
position.red1 = lm(hipcenter ~ Age + Weight + Ht + Seated, data=seatpos)
summary(position.red1)

##
## Call:
## lm(formula = hipcenter ~ Age + Weight + Ht + Seated, data = seatpos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -90.869 -21.163  -3.144   26.773   59.423
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  478.65890   159.73362    2.997  0.00515 **
## Age           0.58396    0.42573    1.372  0.17943
## Weight       -0.01535    0.31640   -0.049  0.96159
## Ht           -4.99025    1.64389   -3.036  0.00466 **
## Seated        2.04632    3.41283    0.600  0.55287
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 36.83 on 33 degrees of freedom
## Multiple R-squared:  0.6599, Adjusted R-squared:  0.6186
## F-statistic: 16.01 on 4 and 33 DF, p-value: 2.224e-07
position.red2 = lm(hipcenter ~ Ht, data=seatpos)
summary(position.red2)

##
## Call:
## lm(formula = hipcenter ~ Ht, data = seatpos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -99.956 -27.850   5.656   20.883   72.066
##

```



```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 556.2553    90.6704   6.135 4.59e-07 ***
## Ht          -4.2650     0.5351  -7.970 1.83e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 36.37 on 36 degrees of freedom
## Multiple R-squared:  0.6383, Adjusted R-squared:  0.6282
## F-statistic: 63.53 on 1 and 36 DF,  p-value: 1.831e-09
anova(position.red2, position.red1)

## Analysis of Variance Table
##
## Model 1: hipcenter ~ Ht
## Model 2: hipcenter ~ Age + Weight + Ht + Seated
##   Res.Df  RSS Df Sum of Sq    F Pr(>F)
## 1      36 47616
## 2      33 44774   3    2841.6 0.6981 0.5599
```

Based on the F test provided in the ANOVA table, we conclude that *the reduced model with Ht as the only variable is better than the model that includes Age, Weight, Ht and Seated.*

Chapter 3

Variable Selection & Regularization

Let's consider the multiple linear regression model with p predictors **plus** the intercept, i.e.

$$Y \sim 1 + X_1 + X_2 + \dots + X_p$$

In many applications, the number of explanatory variables, i.e., p is large while in some cases we could even have $p \gg n$. However, this does not necessarily mean that all the variables are relevant to the response Y . In fact, *only a small portion* of the p variables are believed to be relevant to Y .

Our **goal** in this chapter is to develop methods that will allow us to efficiently identify the set of predictors that are useful in estimating or predicting the response. Since the least squares estimator $\hat{\beta}$ is *unbiased*, this implies that irrelevant estimators $\hat{\beta}_j$ will eventually go to zero. So, if our task is to do well on prediction, *then we need to reflect on whether it is important to remove unnecessary variables from the model*.

To better understand the implications of unnecessary parameters in a MLR model, we further discuss and quantify the **Training** and **Testing Errors**.

3.1 Training vs. Testing Errors

Consider that we *split our data in two parts*:

- **Training data** $\{\mathbf{x}_i, y_i\}_{i=1}^n$ are used to fit our model
- **Testing data** $\{\mathbf{x}_i, y_i^*\}_{i=1}^n$ are an **independent** data set collected at locations $\mathbf{x}_i$

Remark: In practice, we are given a full data set to analyze, which we then need to split it in two parts (typically in a random fashion) - a *training* part and a *testing* part with a higher percentage allocated to the training data (usually ~80%).

We assume that both *testing* and *training* data come from the same population or are collected at the same locations $\mathbf{x}_i$. Then, *statistically* we can write both models as follows:

$$\mathbf{y}_{n \times 1}, \mathbf{y}_{n \times 1}^* \sim^{iid} N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n) \text{ and } \mathbf{X} = \mathbf{X}$$

or equivalently,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim^{iid} \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n) \quad \text{and} \quad \mathbf{y}^* = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}^*, \quad \boldsymbol{\epsilon}^* \sim^{iid} \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

with $\boldsymbol{\epsilon}, \boldsymbol{\epsilon}^*$ independent.

Having these models in mind, we compute the **MSE for train and testing data**:

Note that for simplicity in the calculations below, we have suppressed the intercept (so there is no β_0). If we have an intercept, then instead of p , we will have $p + 1$.

$$\begin{aligned} \mathbb{E}(\text{Train Error}) &= \mathbb{E} \left\| \mathbf{y} - \hat{\mathbf{y}} \right\|^2 = \mathbb{E} \left\| (\mathbf{I} - \mathbf{H})\mathbf{y} \right\|^2 \\ &= \text{tr} \left((\mathbf{I} - \mathbf{H}) \text{Cov}(\mathbf{y}) (\mathbf{I} - \mathbf{H})^T \right) \\ &= \sigma^2 \text{tr} \left((\mathbf{I} - \mathbf{H}) \right) = (n - p) \sigma^2 \\ &= n\sigma^2 - p\sigma^2 \end{aligned}$$

$$\begin{aligned} \mathbb{E}(\text{Test Error}) &= \mathbb{E} \left\| \mathbf{y}^* - \hat{\mathbf{X}} \right\|^2 \\ &= \mathbb{E} \left\| (\mathbf{y}^* - \mathbf{X}) + (\mathbf{X} - \hat{\mathbf{X}}) \right\|^2 \\ &= \mathbb{E} \left\| \mathbf{y}^* - \right\|^2 + \mathbb{E} \left\| \mathbf{X} - \hat{\mathbf{X}} \right\|^2 \\ &= \mathbb{E} \left\| \boldsymbol{\epsilon}^* \right\|^2 + \text{tr} \left(\mathbf{X} \text{Cov}(\boldsymbol{\epsilon}) \mathbf{X}^T \right) \\ &= n\sigma^2 + \sigma^2 \text{tr} \mathbf{H} \\ &= n\sigma^2 + p\sigma^2 \end{aligned}$$

From the previous equations we can conclude that:

- the *training error decreases with p* .
- the *testing error increases with p* .

This implies that if our goal is *pure prediction*, adding more variables to matrix $\mathbf{X}$ is not the best option. But, *does this imply that the intercept-only model with $p = 0$, i.e. the one with the smallest expected test error is the best?* **No!**

The previous analysis is based on the following **Assumptions**:

1. The mean of $\mathbf{y}$, i.e. $E(\mathbf{Y}|\mathbf{X})$, is in $\mathcal{C}(\mathbf{X})$, i.e., there exists some coefficient vector such that $E(\mathbf{Y}|\mathbf{X}) = \mathbf{X}$.
2. The design matrix $\mathbf{X}$ above contains all available predictors. But, when we run a linear regression model using only a **subset** of the columns of $\mathbf{X}$, there will be *an additional **Bias** term*.

Even more generally, we may have a *model*:

$$Y = f(X) + \varepsilon$$

in which f is not even linear. In such cases, we approximate f with a linear function f^* which means that we introduce bias. (Intuitively, this means that usually when we do not know the underlying true model, a linear approximation is often a good first step.) In the same spirit, even when the underlying function is truly linear, we may still introduce bias when we use a subset of the design matrix $\mathbf{X}$, i.e. when the model misses some relevant variables.

In such cases, the training and testing errors compute as:

$$\begin{aligned}\mathbb{E}(\text{Test Error})^2 &= n\sigma^2 + p\sigma^2 + \text{Bias} \\ \mathbb{E}(\text{Training Error})^2 &= n\sigma^2 - p\sigma^2 + \text{Bias}\end{aligned}$$

where $p\sigma^2$ is the unavoidable error – the ε – and Bias is the model error.

- Bigger model (i.e., p large) $\rightarrow$ small Bias, but large Variance ($p\sigma^2$)
- Smaller model (i.e., p small) $\rightarrow$ large Bias, but small Variance ($p\sigma^2$).

To reduce the *test error* (i.e., prediction error), the key is to find the best **trade-off** between Bias and Variance.

3.2 Subset Selection

The idea behind *subset selection* is to score each model according to an information criterion and then use a search algorithm to find the optimal model. In this way, we take into account

Training Error + Complexity Penalty

In the context of linear regression models, the *complexity of a model increases with the number of predictor variables* (i.e., p).

- **Training Error:** an increasing function of RSS .
- **Complexity Penalty:** an increasing function of p .

Why don't we just use R^2 or RSS ? The main reason is that R^2 always increases when we introduce variables in the model, while RSS always reduces. Therefore, R^2 and RSS do not penalize for introducing unnecessary variables in the model.

3.2.1 Information Criteria-based procedures

Akaike Information Criterion & Bayesian Information Criterion

AIC and BIC are defined as

$$\begin{aligned} AIC &= -2 \cdot \loglik + 2 p \\ BIC &= -2 \cdot \loglik + \log(n) p \end{aligned}$$

where p is the number of predictors included in model under consideration. $\loglik$ denotes the log-likelihood of the model under consideration. For the normal-error linear regression model, the first term computes:

$$-2 \cdot \loglik = n \log \frac{RSS}{n}$$

which means that

$$\begin{aligned} AIC &= n \log \frac{RSS}{n} + 2 p \\ BIC &= n \log \frac{RSS}{n} + \log(n) p \end{aligned}$$

The **lower** the AIC/BIC the **better**. Note that when n is large, *adding an additional predictor costs a lot more in BIC than AIC*. So, AIC tends to pick a bigger model than BIC.

Adjusted- R^2 for model with p predictors

$$\begin{aligned}
R_a^2 &= 1 - \frac{RSS/(n-p-1)}{TSS/(n-1)} \\
&= 1 - (1 - R^2) \left(\frac{n-1}{n-p-1} \right) \\
&= 1 - \frac{\hat{\sigma}^2}{\hat{\sigma}_0^2}
\end{aligned}$$

where $\hat{\sigma}$ is the estimated error variance for the current fitted model, and $\hat{\sigma}_0^2$ is the estimated error variance for the intercept-only fitted model.

The **higher** the R_a^2 the **better**.

Mallow's C_p

$$C_p = \frac{RSS_{\mathbf{p}}}{\hat{\sigma}_0^2} + 2p - n$$

where $\hat{\sigma}_0^2$ is the estimate of the error variance for the full model. Mallow's C_p behaves very similar to AIC and the **lower** the C_p the **better**.

Illustration of Complexity Criteria vs. p

As an illustration, the complexity penalties R_{adjusted}^2 , Mallow's C_p , AIC , and BIC are plotted as a function of p (the number of parameters in the model) for the **student-grades** data set (the example in the last section):

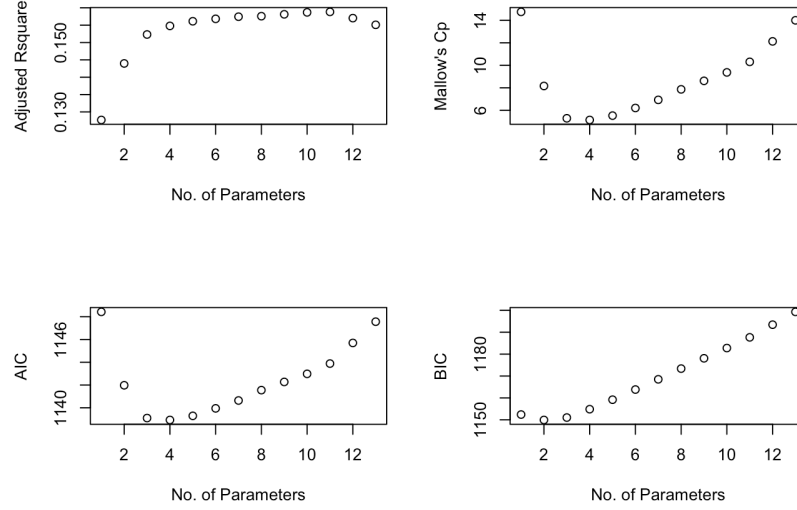
3.2.2 Search Algorithms

In the beginning of this section, we discussed that to select the optimal model, we need to penalize the models using the preferred penalty. If we have (in total) p available predictors, then there are 2^p potential models. Ideally, we would like to search and score all of these models. If p is large, then we need to employ systematic algorithms that search and score potential models.

Level-wise search algorithms: These are algorithms return the global optimal solution among *all possible models* and work only for less than 40 variables.

The idea is that they finds m different models (default m in R is 8) of up to size p with the smallest RSS among all the models of the same size. Then, they evaluate the score on the p models and report the optimal one. It is important to note that the algorithm does not need to visit every model. For example, if we know that $RSS(X1, X2) < RSS(X3, X4, X5, X6)$, then it is not necessary to search any size 2 or 3 *sub-models* of set $(X3, X4, X5, X6)$ meaning that these models may be leaped over.

Greedy algorithms: These algorithms add and/or remove variables based on the score given by a criterion such as AIC/BIC. They can move :

Figure 3.1: Subset selection criteria vs. p .

- *Backwards*: start with the full model and sequentially delete predictors until the score does not improve.
- *Forward*: start with the null model and sequentially add predictors until the score does not improve.
- *Stepwise*: consider both deleting and adding one predictor at each stage.

These algorithms are computationally efficient, but they only return a *locally optimal* solution which usually good enough in practice.

Some considerations

In some cases, specially when $p \gg n$ starting with the full model and using the stepwise procedure is not feasible. Therefore, we need to screen some of the variables and remove them, before employing the search algorithms. A common variable screening approach is the following:

- Pick a *smaller initial model* as a starting point by **ranking** the p predictors by the absolute value of their (marginal) correlation with Y .
- Keep the **top** K predictors (e.g., $K = n/3$).
- Use the stepwise procedure so that you can add removed variables back to the model.

3.3 Shrinkage Methods

The variable selection methods we already discussed work quite well in practice when it comes to data sets where we do not have too many predictors. For example, recall that the *level-wise* algorithms are good when $p < 40$.

In this section, we are going to study two different methods we can use to **shrink** the number of predictors in order to select the optimal model that balances the *trade-off between model bias and prediction error (variance)*.

Let us then re-frame what we have been doing so far:

Regression with Penalization

$$\begin{aligned}\hat{\beta} &= \arg \min_{\beta} ||\mathbf{y} - \mathbf{X}\beta||^2 + \underbrace{\lambda \sum_{j=1}^p \mathbf{1}_{\beta_j \neq 0}}_{\text{penalty}} \\ &= \arg \min_{\beta} ||\mathbf{y} - \mathbf{X}\beta||^2 + \underbrace{\lambda ||\beta||_0}_{\text{penalty}}\end{aligned}$$

where $||\beta||_0 = \sum_{j=1}^p \mathbf{1}_{\beta_j \neq 0}$. The *vector norm* $||\beta||_0$ can be thought of as a function that decides whether a variable is **in** or **out** of the model. Then, the different criteria we discussed before rise by choosing λ properly.

Taking this idea one step further (remembering our linear algebra tools), we can replace the ℓ_0 -norm above with a different one, e.g. the L^1 or the L^2 norm. Exactly this observation, lead to two extremely popular *penalized regression* methods:

- the **Ridge Regression**

$$\hat{\beta} = \arg \min_{\beta} ||\mathbf{y} - \mathbf{X}\beta||^2 + \underbrace{\lambda ||\beta||^2}_{\text{penalty}}, \text{ where } ||\beta||^2 = \sum_{j=1}^p \beta_j^2$$

- the **Lasso Regression**

$$\hat{\beta} = \arg \min_{\beta} ||\mathbf{y} - \mathbf{X}\beta||^2 + \underbrace{\lambda |\beta|}_{\text{penalty}}, \text{ where } |\beta| = \sum_{j=1}^p |\beta_j|$$

In the next sections, we discuss the mathematics and the implementation of these methods. However, before proceeding, we need to make two consider the following:

Implementation Considerations

When the penalization norm changes, we observe that the penalty term that arises (in both cases) is *not location or scale invariant*. This implies that if

we re-scale a variable then the penalized β estimators will be sensitive to these values. Therefore, it is suggested that we **center** and **scale** each column of the design matrix $\mathbf{X}$. Specifically, if $\mathbf{x}_j$, $j = 1, \dots, p$ is a column of $\mathbf{X}$, then

$$\tilde{\mathbf{x}}_j = \frac{\mathbf{x}_j - \bar{\mathbf{x}}_j}{sd_{\mathbf{x}_j}}$$

In addition to that we **center** $\mathbf{y}$ to *suppress the intercept*, i.e.

$$\tilde{\mathbf{y}}_i = \mathbf{y}_i - \bar{\mathbf{y}}$$

After the model selection/fitting, we can always transform the variables back to their original values so that we continue with interpretation and/or prediction. We can also *estimate back the intercept* as follows:

$$\hat{\beta}_0 = \bar{y} - \sum_{j=1}^n \hat{\beta}_j \bar{x}_j$$

Note that in R the **glmnet** package handles the centering and scaling (and transformation back to original) automatically.

3.3.1 Ridge Regression

Ridge regression assumes that after normalization, some of the regression coefficients should not be very large. Ridge regression is very useful when you have collinearity and the LS regression coefficients are unstable. In fact, it was initially introduced by A. Tikhonov to remedy multi-collinearity problems by adding a non-negative constant to the diagonal of the design matrix.

Ridge Regression

The idea of the method is to add a **penalty** term to the LS minimization problem :

$$\text{minimize}_{\beta} (y - X\beta)^T (y - X\beta) + \lambda \sum_j \beta_j^2$$

for some $\lambda \geq 0$. The penalty term is $\sum_j \beta_j^2$.

As discussed before, for the method to be more effective, we prefer to *standardize* the predictors first (centered by their means and scaled by their standard deviations) and center the response y .

One of the main advantages of the ridge regression is that it provides us with closed-form solutions for the β coefficients. Indeed, solving the minimization problem we obtain:

$$\hat{\beta}_{\text{Ridge}} = (X^T X + \underbrace{\lambda \mathbf{I}}_{\text{ridge}})^{-1} X^T y$$

It is easy to see that when $\lambda = 0$ the ridge regression estimation problem reduces to the standard least squares problem, while when $\lambda \rightarrow \infty$, the ridge coefficients $\hat{\beta} \rightarrow \mathbf{0}$. As a result, during implementation, one of the main considerations is how to choose λ . Choosing a very small (relatively speaking) value for λ leads back to the usual LS estimators, while choosing a very large value for λ makes most estimators zero.

To balance this trade-off in practice, a common approach is to use automated methods such as *Generalized Cross-Validation* (GCV). The main *disadvantage* of the ridge regression estimators is that they are **biased**, that is $\mathbb{E}(\hat{\beta}) = \beta + \text{bias}$.

To better understand the structure of the ridge regression LS Coefficients, assume that $\mathbf{X}^T \mathbf{X} = \mathbf{I}_p$, that is the columns of the design matrix are orthogonal. Then, the general formula above reduces to

$$\hat{\beta}_{ridge} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} = \frac{1}{1 + \lambda} \mathbf{X}^T \mathbf{y}$$

Similarly, the fitted values

$$\hat{\mathbf{y}}_{ridge} = \mathbf{X} \hat{\beta}_{ridge} = \frac{1}{1 + \lambda} \hat{\mathbf{y}}_{LS}$$

If the columns of the design matrix are not orthogonal, then we can run the regression against an orthonormal version of $\mathbf{X}$, known as *principal components analysis*, or *singular value decomposition*.

Singular Value Decomposition (SVD)

The Singular Value Decomposition (SVD) is one of the most important concepts in applied mathematics. It is used for a number of application including dimension reduction and data analysis. Principal Components Analysis (PCA) is also a special case of the SVD.

SVD of $\mathbf{X}_{n \times p}$

Consider the design matrix $\mathbf{X}_{n \times p}$. Then, $\mathbf{X}$ can be written as

$$\mathbf{X} = \mathbf{U}_{n \times p} \mathbf{D}_{p \times p} \mathbf{V}_{p \times p}^T$$

where

- $\mathbf{U}_{n \times p}$ orthogonal matrix with columns that are spanning $\mathcal{C}(\mathbf{X})$.
- $\mathbf{V}_{p \times p}$ orthogonal matrix with columns that are spanning $\mathbb{R}^p$.
- $\mathbf{D}_{p \times p}$ diagonal values with $d_1 \geq \dots \geq d_p \geq 0$ the singular values of $\mathbf{X}$. If one or more $d_j = 0$, then $\mathbf{X}$ is singular, i.e. not full-rank.

This factorization of the matrix $\mathbf{X}$ is called the **singular value decomposition of $\mathbf{X}$** , and the columns of $\mathbf{U}$ and $\mathbf{V}$ are called the left- and right-hand **singular vectors** of $\mathbf{X}$.

For simplicity in the discussion, let $n > p$ and $\text{rank}(\mathbf{X}) = p$, which also implies that $d_p > 0$.

Some Useful Properties of the SVD

1. The left-hand singular vectors are a set of orthonormal eigenvectors for $\mathbf{X}^T\mathbf{X}$, i.e. $\mathbf{U}^T\mathbf{U} = \mathbf{I}$.
2. The right-hand singular vectors are a set of orthonormal eigenvectors for $\mathbf{X}\mathbf{X}^T$, i.e. $\mathbf{V}^T\mathbf{V} = \mathbf{I}$.
3. The eigenvectors $\mathbf{v}_j$ are also called the *principal components directions* of the columns of $\mathbf{X}$.
4. The first principal component direction v_1 has the property that $z_1 = \mathbf{X}\mathbf{v}_1 = \mathbf{u}_1 d_1$ has the largest sample variance among all normalized linear combinations of the columns of $\mathbf{X}$.
5. The singular values are the square roots of the eigenvalues for $\mathbf{X}^T\mathbf{X}$ and $\mathbf{X}\mathbf{X}^T$, since these matrices have the same eigenvalues.
6. The first singular value is equal to

$$\sigma_1 = \max_{\|x\|=1} \|\mathbf{X}x\|_2$$

One of the goals of *principal components analysis* is to find the new coordinates, or *scores*, of the data in the principal components basis. If the original (centered or standardized) data was contained in the matrix $\mathbf{X}$ and the eigenvectors of the covariance/correlation matrix ($\mathbf{X}^T\mathbf{X}$) were columns of a matrix $\mathbf{V}$, then to find the scores ($\mathbf{S}$) of the observations on the eigenvectors we can use the following equation:

$$\mathbf{X} = \mathbf{S}\mathbf{V}^T$$

where each columns of $\mathbf{S}_{n \times p} = \mathbf{U}\mathbf{D}$ is the so-called **principal component** and each column of $\mathbf{V}$ is the **principal component direction** of $\mathbf{X}$.

SVD of Fitted Values: LS vs. Ridge

To observe how the *ridge* affects the structure of the fitted β , we have the following.

- In *least-squares* regression, the fitted values compute as:

$$\begin{aligned}
\hat{\mathbf{y}}_{LS} &= \mathbf{X} \hat{\beta}_{LS} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} \mathbf{V}^T (\mathbf{V} \mathbf{D}^2 \mathbf{V}^T)^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} \mathbf{V}^T (\mathbf{V}^T)^{-1} \mathbf{D}^{-2} \mathbf{V}^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} \mathbf{D}^{-2} \mathbf{D} \mathbf{U}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{U}^T \mathbf{y} \\
&= \sum_{j=1}^p (\mathbf{u}_j^T \mathbf{y}) \mathbf{u}_j
\end{aligned}$$

where we used the facts that $\mathbf{U}^T \mathbf{U} = \mathbf{I}$, and that

$$\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{D}^T \mathbf{U}^T \mathbf{U} \mathbf{D} \mathbf{V}^T = \mathbf{V} \mathbf{D}^2 \mathbf{V}^T$$

The last expression is the *eigen-decomposition* of $\mathbf{X}^T \mathbf{X}$.

Similarly, for the ridge regression coefficients, we have:

$$\begin{aligned}
\hat{\mathbf{y}}_{ridge} &= \mathbf{X} \hat{\beta}_{ridge} \\
&= \mathbf{X}(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} \mathbf{V}^T (\mathbf{V} \mathbf{D}^2 \mathbf{V}^T + \lambda \mathbf{I})^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} \mathbf{V}^T (\mathbf{V} \mathbf{D}^2 \mathbf{V}^T + \lambda \mathbf{V} \mathbf{V}^T)^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} \mathbf{V}^T (\mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I}) \mathbf{V}^T)^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} \mathbf{V}^T (\mathbf{V}^T)^{-1} (\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{V}^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\
&= \mathbf{U} \mathbf{D} (\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{D} \mathbf{U}^T \mathbf{y}
\end{aligned}$$

In this last expression, note that

- $\mathbf{D} (\mathbf{D}^2 + \lambda \mathbf{I})^{-1}$ is a diagonal matrix with elements given by $\frac{d_j^2}{d_j^2 + \lambda}$.
- the vector $\mathbf{U}^T \mathbf{y}$ is the coordinates of the vector $\mathbf{y}$ in the basis spanned by the p columns of $\mathbf{U}$.

Therefore, $\hat{\mathbf{y}}_{ridge}$ simplifies to

$$\hat{\mathbf{y}}_{ridge} = \sum_{j=1}^p \mathbf{u}_j \frac{d_j^2}{d_j^2 + \lambda} \mathbf{u}_j^T \mathbf{y}$$

As we can observe, the inner products $\mathbf{u}_j^T \mathbf{y}$ are scaled by the factors $\frac{d_j^2}{d_j^2 + \lambda}$, or in other words the ridge estimate $\hat{\beta}_{ridge}$ **shrinks the LS estimate $\hat{\beta}_{LS}$** by

a factor of $\frac{d_j^2}{d_j^2 + \lambda}$ where *the smaller the eigenvalues the more the shrinkage*. Essentially, ridge regression projects $\mathbf{y}$ onto the principal components, and then shrinks the coefficients of the low-variance components more than the high-variance components.

Complexity of Ridge Regression

To quantify the complexity of a model, heuristically we need to understand the number of coefficients that need to be estimated. So, a linear regression model is a model with p covariates and p β coefficients, so it has p degrees of freedom. In ridge regression however, the estimated β s are still p -dimensional, however the method does not use all strength of the p covariates due to shrinkage. If λ is extremely large, there will be no effective covariates left in the model, meaning that they should all be close to zero. On the other hand, if λ is 0, then we go back to linear regression with p covariates and p degrees of freedom. So, in the case of the ridge regression, the truth lies somewhere between 0 and p .

Let's formalize the intuition, by recalling that one method to compute the degrees of freedom of a model is to relate them to the correlation between the observed and fitted values as follows:

$$df = \sum_{i=1}^n \text{Cor}(y_i, \hat{y}_i)$$

- **Linear regression:** We have already shown that the fitted values can be expressed as $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, in which case

$$df = \sum_{i=1}^n \text{Cor}(y_i, \hat{y}_i) = \sum_{i=1}^n H_{ii} = \text{tr}(\mathbf{H}) = p$$

where (recall) $\mathbf{H}$ is the hat (projection) matrix equal to $\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$.

- **Ridge regression:** We have shown that $\hat{\mathbf{y}} = \underbrace{\mathbf{X}(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T}_{:=\mathbf{S}_\lambda}\mathbf{y} = \mathbf{S}_\lambda\mathbf{y}$. So, the effective degrees of freedom of the ridge regression are

$$\begin{aligned} df(\lambda) &= \sum_{i=1}^n \text{Cor}(y_i, \hat{y}_i) = \sum_{i=1}^n [S_\lambda]_{ii} \\ &= \text{tr}(\mathbf{S}_\lambda) \\ &= \text{tr}(\mathbf{X}(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T) \\ &= \text{tr}\left(\sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda} \mathbf{u}_j \mathbf{u}_j^T\right) = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda} \end{aligned}$$

Based on the expression above, we can see that the degrees of freedom of the ridge regression are a *decreasing* function of λ , and reduce to p when $\lambda = 0$. One important consequence of this expression is that we can use it to determine the values of λ to use when applying cross validation. This can be done by thinking $df(\lambda)$ as a function of λ and then setting $df(\lambda) = k$, with $k = 1, \dots, p$ representing all possible dfs. Then, we can solve with respect to λ using a numerical approach.

3.3.2 Lasso Regression

Lasso Regression is similar to the Ridge regression in the sense that it minimizes the least squares criterion *subject to a penalty term*. However, the penalty term is different in the case of lasso.

Lasso Regression

$\hat{\beta}_{\text{LASSO}}$ minimizes:

$$\text{minimize } (y - X\beta)^T(y - X\beta) + \lambda \sum_j |\beta_j|$$

for some $\lambda \geq 0$. The penalty term is $\sum_j |\beta_j|$ (L_1 constraint).

In two-dimensions the constraint defines a square, while in higher dimensions it defines a polytope. Lasso is useful when the response can be explained by *few* predictors with zero effect on the remaining predictors (Lasso is similar to a variable selection method). When $\beta_j = 0$ the corresponding predictor is eliminated which is not the case for ridge regression. Therefore, we use lasso when the effect of predictors is **sparse**. This means that only few predictors will have an effect on the response (e.g. gene expression data) or when number of predictors is large ($p > n$).

The lasso solution is defined as

$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta \in \mathbb{R}^p} \left((y - X\beta)^\top (y - X\beta) + \lambda \sum_j |\beta_j| \right)$$

and does not have a closed form expression.

If we assume that $\mathbf{X}^T \mathbf{X} = \mathbf{I}_p$, then

$$\begin{aligned} \|\mathbf{y} - \mathbf{X}\beta\|^2 &= \|\mathbf{y} - \mathbf{X}\hat{\beta}_{LS} + \mathbf{X}\hat{\beta}_{LS} - \mathbf{X}\beta\|^2 \\ &= \|\mathbf{y} - \mathbf{X}\hat{\beta}_{LS}\|^2 + \|\mathbf{X}\hat{\beta}_{LS} - \mathbf{X}\beta\|^2 \end{aligned}$$

where

$$2(\mathbf{y} - \mathbf{X}\hat{\beta}_{LS})^T (\mathbf{X}\hat{\beta}_{LS} - \mathbf{X}\beta) = 2 r^T (\mathbf{X}\hat{\beta}_{LS} - \mathbf{X}\beta) = 0$$

since the second term is a linear combination of columns of $\mathbf{X}$ no matter what value β takes, and thus is in $\mathcal{C}(\mathbf{X})$, therefore orthogonal to the residual vector r .

Obtaining the Lasso Solution

Although we do not have a closed form solution in the case of lasso, the minimization problem we have to solve is not very challenging.

The lasso solution can be expressed as:

$$\begin{aligned}
\hat{\beta}_{lasso} &= \arg \min_{\beta \in \mathbb{R}^p} \left(\|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda|\beta| \right) \\
&= \arg \min_{\beta \in \mathbb{R}^p} \left(\|\mathbf{X}\hat{\beta}_{LS} - \mathbf{X}\beta\|^2 + \lambda|\beta| \right) \\
&= \arg \min_{\beta \in \mathbb{R}^p} \left((\hat{\beta}_{LS} - \beta)^T \mathbf{X}^T \mathbf{X} (\hat{\beta}_{LS} - \beta) + \lambda|\beta| \right) \\
&= \arg \min_{\beta \in \mathbb{R}^p} \left((\hat{\beta}_{LS} - \beta)^T (\hat{\beta}_{LS} - \beta) + \lambda|\beta| \right) \\
&= \arg \min_{\beta_1, \dots, \beta_p} \sum_{i=1}^p \left((\beta_j - \hat{\beta}_{(LS)j})^2 + \lambda|\beta_j| \right)
\end{aligned}$$

We can find the *optimal* β_j for each of $j = 1, \dots, p$ **separately** by solving the following generic problem for each dimension:

$$\arg \min_x (x - a)^2 + \lambda|x|, \quad \lambda > 0$$

Therefore, to solve the one-dim lasso above, define

$$f(x) = (x - a)^2 + \lambda|x|, \quad a \in \mathbb{R}, \quad \lambda > 0$$

The value x^* that minimizes $f(x)$ must satisfy:

$$\begin{aligned}
\frac{\partial}{\partial x} (x^* - a)^2 + \lambda \frac{\partial}{\partial x} |x^*| &= 0 \\
2(x^* - a) + \lambda z^* &= 0
\end{aligned}$$

where z^* is the *sub-gradient* of the absolute value function evaluated at x^* , which equals to $sign(x^*)$ if $x^* \neq 0$ and any number in $[-1, 1]$, if $x^* = 0$.

Therefore, the minimizer of $f(x)$ is given by

$$x^* = S_{\lambda/2}(a) = sign(a)(|a| - \lambda/2)_+ = \begin{cases} a - \lambda/2, & \text{if } a > \lambda/2 \\ 0, & \text{if } |a| \leq \lambda/2 \\ a + \lambda/2, & \text{if } a < -\lambda/2 \end{cases}$$

where $S_{\lambda/2}(\cdot)$ is often referred to as the **soft-thresholding operator**.

When the design matrix $\mathbf{X}$ is orthogonal, the lasso solution simplifies to

$$\hat{\beta}_j^{lasso} = \begin{cases} \text{sign}(\hat{\beta}_j^{ls})(|\hat{\beta}_j^{ls}| - \lambda/2)_+, & \text{if } |\hat{\beta}_{(LS)j}| > \lambda/2 \\ 0, & \text{if } |\hat{\beta}_{(LS)j}| \leq \lambda/2 \end{cases}$$

A large λ will cause some of the coefficients to be exactly zero. So, lasso does both variable (subset) selection and (soft) shrinkage.

Remarks

- It is suggested to use lasso when the effect of predictors is *sparse*, since lasso will “make” some of the β coefficients zero keeping the coefficients that will have an effect on $\mathbf{y}$. This is the reason why this method also works when the number of predictors is larger than the sample size ($p > n$). These are scenarios often encountered in genomic or proteomic data where the design matrices tend to have a lot of zeros and too many predictors.
- In lasso as in ridge regression, we can select λ using Cross-Validation (CV). When λ increases, the number of predictors decreases.

Comparing Ridge Regression and Lasso

Lasso selects a sub-set of predictors (some coefficients will equal to zero), while ridge regression performs better when the response is a function of many predictors with coefficients around the same size. Lasso will perform better when a relatively small number of predictors have large coefficients and the rest are very small or equal to zero. Since the number of predictors is never known *a priori*, cross-validation can be used to decide which approach is better for a particular data set.

Ridge regression does a proportional shrinkage. Lasso translates each coefficient by a constant factor, truncating at zero.

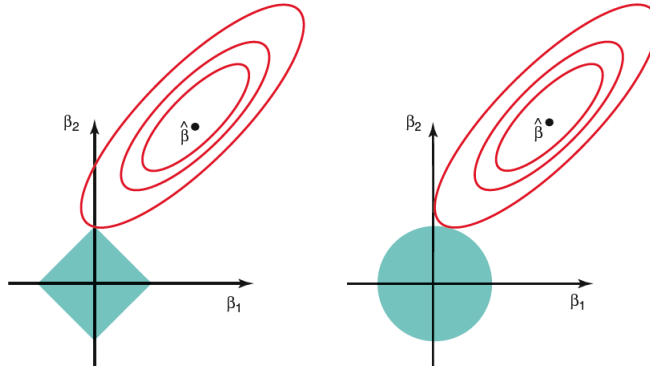


Figure 3.2: Contour of the optimization for Lasso (left) and Ridge (right).

In the picture below, we can see the difference in the contours of optimization for lasso (left) and Ridge (right) when there are only two parameters. The residual sum of squares has elliptical contours, centered at the full least squares estimate, $\hat{\beta}_{LS}$. The constraint region for ridge regression is the disk $\beta_1^2 + \beta_2^2 \leq t$, while that for lasso is the diamond $|\beta_1| + |\beta_2| \leq t$. Both methods find the first point where the elliptical contours hit the constraint region. Unlike the disk, the diamond has corners which means that if the solution occurs at a corner, then it has one parameter β_j equal to zero. When $p > 2$, the diamond becomes a rhomboid, and has many corners, flat edges and faces; there are many more opportunities for the estimated parameters to be zero.

Lasso with $p > n$

When $\mathbf{X}$ is of full rank:

- the lasso solution is the minimizer of a convex function over a convex set
- the lasso solution is unique since the first term is a strictly convex function.

When $\mathbf{X}$ is not of full rank, or when $p > n$:

- The lasso criterion is no longer convex which means that it may not have a unique minimizer. first term is no longer strictly convex. In the $p > n$ case, the lasso selects at most n variables before it saturates, because of the nature of the convex optimization problem which is a limiting feature for a variable selection method.

3.4 The Student Performance Example

The `student-mat.csv` data set contains data from a study on student achievement in secondary education of two Portuguese schools. The data are based on the following publication:

“Using data mining to predict secondary school student performance”
By P. Cortez, A. M. G. Silva. 2008. Published in Proceedings of 5th
Annual Future Business Technology Conference.

and can be found in the UCI data repository: [here](#)

The python version of the example below was written by our TA Tommy Tang and can be found here: [Python Version](#)

The variables in the data set are the following:

- **school** Student’s school (*binary*: GP - Gabriel Pereira or MS - Mousinho da Silveira)
- **sex** Sex student’s sex (*binary*: F - female or M - male)
- **age** Student’s age (*numeric*: from 15 to 22)
- **address** Student’s home address type (*binary*: U - urban or R - rural)

- **famsize** Family size (*binary*: LE3 - less or equal to 3 or GT3 - greater than 3)
- **Pstatus** Parent's cohabitation status (*binary*: T - living together or A - apart)
- **Medu** Mother's education level (*numeric*: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)
- **Fedu** Father's education level (*numeric*: 0 - none, 1 - primary education (4th grade), 2 5th to 9th grade, 3 - secondary education or 4 higher education)
- **Mjob** Mother's occupation (*nominal*: **teacher**, **healthcare** related, **civil services** (e.g. administrative or police), **at_home** or **other**)
- **Fjob** Father's occupation (*nominal*: **teacher**, **healthcare** related, **civil services** (e.g. administrative or police), **at_home** or **other**)
- **reason** Reason to choose this school (*nominal*: **close to home**, **school reputation**, **course preference** or **other**)
- **guardian** Student's guardian (*nominal*: **mother**, **father** or **other**)
- **traveltime** Home to school travel time (*numeric*: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)
- **studytime** Weekly study time (*numeric*: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours)
- **failures** Number of past class failures (*numeric*: **n** if $1 \leq n < 3$, else 4)
- **schoolsup** Extra educational support (*binary*: **yes** or **no**)
- **famsup** Family educational support (*binary*: **yes** or **no**)
- **paid** Extra paid classes within the course subject (Math or Portuguese) (*binary*: **yes** or **no**)
- **activities** Extra-curricular activities (*binary*: **yes** or **no**)
- **nursery** Attended nursery school (*binary*: **yes** or **no**)
- **higher** Wants to take higher education (*binary*: **yes** or **no**)
- **internet** Internet access at home (*binary*: **yes** or **no**)
- **romantic** With a romantic relationship (*binary*: **yes** or **no**)
- **famrel** Quality of family relationships (*numeric*: from 1 - very bad to 5 - excellent)
- **freetime** Free time after school (*numeric*: from 1 - very low to 5 - very high)
- **goout** Going out with friends (*numeric*: from 1 - very low to 5 - very high)
- **Dalc** Workday alcohol consumption (*numeric*: from 1 - very low to 5 - very high)
- **Walc** Weekend alcohol consumption (*numeric*: from 1 - very low to 5 - very high)
- **health** Current health status (*numeric*: from 1 - very bad to 5 - very good)

- `absences` Number of school absences (*numeric*: from 0 to 93)
- `G1` First period grade (*numeric*: from 0 to 20)
- `G2` Second period grade (*numeric*: from 0 to 20)
- `G3` Final grade (*numeric*: from 0 to 20, output target)

```
stu_performance0 = read.csv2("data/week3/student-mat.csv", header=TRUE)
```

In our example, we want to focus on predicting the final grade (column `G3`) using penalized regression methods. A special variation of penalized regression is required when using categorical variables (e.g. *group lasso*), thus in this example we only focus on numerical predictors.

```
## Remove categorical predictors
stu_performance = stu_performance0[, -c(1, 2, 4, 5, 6, 9, 10, 11, 12, 16, 17, 18, 19, 20, 21,

## Rename the response to "Grades"
names(stu_performance)[14] = 'Grades'

head(stu_performance)
```

```
##   age Medu Fedu traveltime studytime failures famrel freetime goout Dalc Walc
## 1  18   4   4         2         2         0    4         3    4    1    1
## 2  17   1   1         1         2         0    5         3    3    1    1
## 3  15   1   1         1         2         3    4         3    2    2    3
## 4  15   4   2         1         3         0    3         2    2    1    1
## 5  16   3   3         1         2         0    4         3    2    1    2
## 6  16   4   3         1         2         0    5         4    2    1    2
##   health absences Grades
## 1      3         6      6
## 2      3         4      6
## 3      3        10     10
## 4      5         2     15
## 5      5         4     10
## 6      5        10     15
```

```
n = dim(stu_performance)[1] ## sample size
p = dim(stu_performance)[2] - 1 ## number of non-intercept predictors
```

We start by fitting the **full linear model** including all the predictors:

```
performance.full = lm(Grades ~ ., data=stu_performance)
summary(performance.full)
```

```
##
## Call:
## lm(formula = Grades ~ ., data = stu_performance)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.2769  -2.1753   0.4053   2.7927   8.7122
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.53702    3.33724   4.056 6.05e-05 ***
## age         -0.25623    0.17880  -1.433  0.15268
## Medu         0.58494    0.25650   2.280  0.02313 *
## Fedu        -0.09135    0.25465  -0.359  0.72000
## traveltime  -0.40184    0.31439  -1.278  0.20197
## studytime    0.26583    0.26903   0.988  0.32373
## failures    -1.83146    0.31201  -5.870 9.49e-09 ***
## famrel       0.28018    0.24537   1.142  0.25422
## freetime     0.34139    0.23096   1.478  0.14021
## goout       -0.62638    0.22179  -2.824  0.00499 **
## Dalc        -0.13723    0.32034  -0.428  0.66862
## Walc         0.35312    0.23834   1.482  0.13928
## health      -0.17429    0.15620  -1.116  0.26521
## absences     0.03197    0.02760   1.158  0.24745
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.211 on 381 degrees of freedom
## Multiple R-squared:  0.183, Adjusted R-squared:  0.1551
## F-statistic: 6.564 on 13 and 381 DF,  p-value: 2.525e-11
```

We may want to do model selection using the *testing approach* (either via a backward/forward selection) removing or adding one variable at a time, or we can even compare the full model with a reduced containing only the variables that seem to be statistically significant:

```
performance.red = lm(Grades ~ failures + goout + Medu, data=stu_performance)
```

```
## Model comparison based on a partial F test
anova(performance.red, performance.full)
```

```
## Analysis of Variance Table
##
## Model 1: Grades ~ failures + goout + Medu
## Model 2: Grades ~ age + Medu + Fedu + traveltime + studytime + failures +
##          famrel + freetime + goout + Dalc + Walc + health + absences
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      391 6956.9
## 2      381 6756.7 10      200.22 1.129  0.339
```

In this case, we fail to reject the null so the reduced model is preferred. However,

this approach is not efficient nor optimal. Therefore, we use algorithms to search the space of models and a penalty to penalize the inclusion of terms that are not as useful.

3.4.0.1 Subset Selection via Level-wise algorithms & the leaps package

```
library(leaps)
regsubsets_model = regsubsets(Grades ~ ., data=stu_performance, nvmax = p)
rs = summary(regsubsets_model)
```

The `rs` object above contains all the different criteria for all the models that have been evaluated. The default maximum size of models is `8` and this is changed above with the `nvmax` option to $p = 13$.

Below we extract the R^2 , Adjusted R^2 , AIC , BIC and C_p -Mallows criteria for the 13 models that have been evaluated:

```
## Extract the models' R^2
rs$rsq
```

```
## [1] 0.1298989 0.1483134 0.1587670 0.1633698 0.1668323 0.1696634 0.1724079
## [8] 0.1746913 0.1773575 0.1800384 0.1823149 0.1827012 0.1829771
```

The best model is the 13th model (as expected).

```
## Extract the models' Adjusted R^2
rs$adjr2
```

```
## [1] 0.1276849 0.1439681 0.1523125 0.1547890 0.1561232 0.1568231 0.1574385
## [8] 0.1575865 0.1581269 0.1586853 0.1588305 0.1570269 0.1550997
```

The best model is the 11th model.

```
## Extract the models' Cp Mallows
rs$cp
```

```
## [1] 14.751803 8.164618 5.289832 5.143431 5.528772 6.208548 6.928720
## [8] 7.863899 8.620578 9.370388 10.308796 12.128682 14.000000
```

The best model is the 4th model.

```
## Extract the models' BIC
rs$bic
```

```
## [1] -43.004859 -45.475340 -44.374670 -40.562949 -36.222213 -31.587827
## [7] -26.916684 -22.029154 -17.328400 -12.638892 -7.758187 -1.965927
## [13] 3.879571
```

The best model is the 2nd model.

If we want to identify which variables are included in each model, then we use the `rs$which` command, where the rows correspond to the model and the columns to the variable. TRUE means that the variable is included in the model and FALSE means otherwise.

```
rs$which
```

```
##      (Intercept)  age  Medu  Fedu traveltime studytime failures famrel freetime
## 1      TRUE FALSE FALSE FALSE      FALSE      FALSE      TRUE  FALSE  FALSE
## 2      TRUE FALSE  TRUE FALSE      FALSE      FALSE      TRUE  FALSE  FALSE
## 3      TRUE FALSE  TRUE FALSE      FALSE      FALSE      TRUE  FALSE  FALSE
## 4      TRUE FALSE  TRUE FALSE      FALSE      FALSE      TRUE  FALSE   TRUE
## 5      TRUE FALSE  TRUE FALSE      TRUE      FALSE      TRUE  FALSE   TRUE
## 6      TRUE FALSE  TRUE FALSE      TRUE      FALSE      TRUE  FALSE   TRUE
## 7      TRUE  TRUE  TRUE FALSE      TRUE      FALSE      TRUE  FALSE   TRUE
## 8      TRUE  TRUE  TRUE FALSE      TRUE      FALSE      TRUE  FALSE   TRUE
## 9      TRUE  TRUE  TRUE FALSE      TRUE      FALSE      TRUE   TRUE   TRUE
## 10     TRUE  TRUE  TRUE FALSE      TRUE      FALSE      TRUE   TRUE   TRUE
## 11     TRUE  TRUE  TRUE FALSE      TRUE       TRUE      TRUE   TRUE   TRUE
## 12     TRUE  TRUE  TRUE FALSE      TRUE       TRUE      TRUE   TRUE   TRUE
## 13     TRUE  TRUE  TRUE  TRUE      TRUE       TRUE      TRUE   TRUE   TRUE
##      goout  Dalc  Walc health absences
## 1  FALSE FALSE FALSE  FALSE  FALSE
## 2  FALSE FALSE FALSE  FALSE  FALSE
## 3   TRUE FALSE FALSE  FALSE  FALSE
## 4   TRUE FALSE FALSE  FALSE  FALSE
## 5   TRUE FALSE FALSE  FALSE  FALSE
## 6   TRUE FALSE  TRUE  FALSE  FALSE
## 7   TRUE FALSE FALSE  FALSE   TRUE
## 8   TRUE FALSE  TRUE  FALSE   TRUE
## 9   TRUE FALSE  TRUE   TRUE  FALSE
## 10  TRUE FALSE  TRUE   TRUE   TRUE
## 11  TRUE FALSE  TRUE   TRUE   TRUE
## 12  TRUE  TRUE  TRUE   TRUE   TRUE
## 13  TRUE  TRUE  TRUE   TRUE   TRUE
```

For example, Model 1 only contains `failures`, and the `intercept`.

Next, compute “by hand” the AIC/BIC for those p models and find the one that achieves the the smallest score:

```
msize = 1:p;
par(mfrow=c(1,2))
Aic = n*log(rs$rss/n) + 2*msize;
Bic = n*log(rs$rss/n) + msize*log(n);
```

```
Aic
```

```
## [1] 1148.427 1141.978 1139.100 1138.932 1139.294 1139.950 1140.642 1141.551
```

```
## [9] 1142.272 1142.983 1143.885 1145.698 1147.565
```

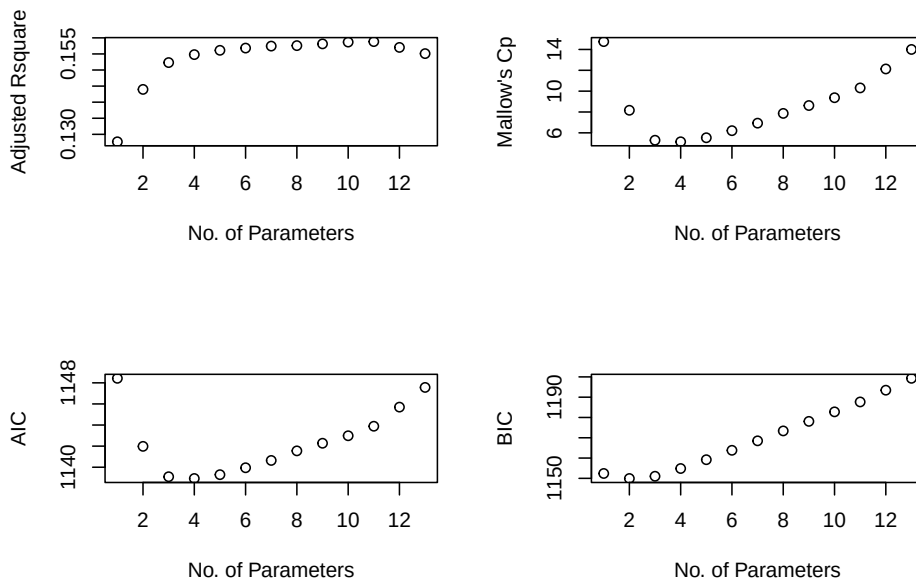
AIC chooses the 4th model.

```
Bic
```

```
## [1] 1152.406 1149.936 1151.036 1154.848 1159.189 1163.823 1168.494 1173.382
## [9] 1178.082 1182.772 1187.653 1193.445 1199.290
```

As we saw before, BIC selects the 2nd model. We can also plot the results for the various models. So, we have

```
par(mfrow=c(2,2))
plot(msize, rs$adjr2, xlab="No. of Parameters", ylab = "Adjusted Rsquare");
plot(msize, rs$cp, xlab="No. of Parameters", ylab = "Mallow's Cp");
plot(msize, Aic, xlab="No. of Parameters", ylab = "AIC");
plot(msize, Bic, xlab="No. of Parameters", ylab = "BIC")
```



For this particular data set, AIC and BIC end up selecting different models. **The model selected by AIC is larger than the one selected by BIC**, which is common. AIC favors larger models while BIC favors smaller models. Although the model selected by BIC does not have the smallest AIC score, its AIC score is very close to the smallest one.

```
cbind(rs$which[which.min(Aic),], rs$which[which.min(Bic), ])
```

```
##           [,1] [,2]
## (Intercept) TRUE  TRUE
## age        FALSE FALSE
## Medu       TRUE  TRUE
## Fedu       FALSE FALSE
```



```
## traveltime FALSE FALSE
## studytime  FALSE FALSE
## failures   TRUE  TRUE
## famrel     FALSE FALSE
## freetime   TRUE FALSE
## goout      TRUE FALSE
## Dalc       FALSE FALSE
## Walc       FALSE FALSE
## health     FALSE FALSE
## absences   FALSE FALSE
```

leaps does not return AIC, but BIC. Its BIC differs from what has been computed above, but the difference is a constant, so the two BIC formulas (ours and the one used by leaps) are essentially the same.

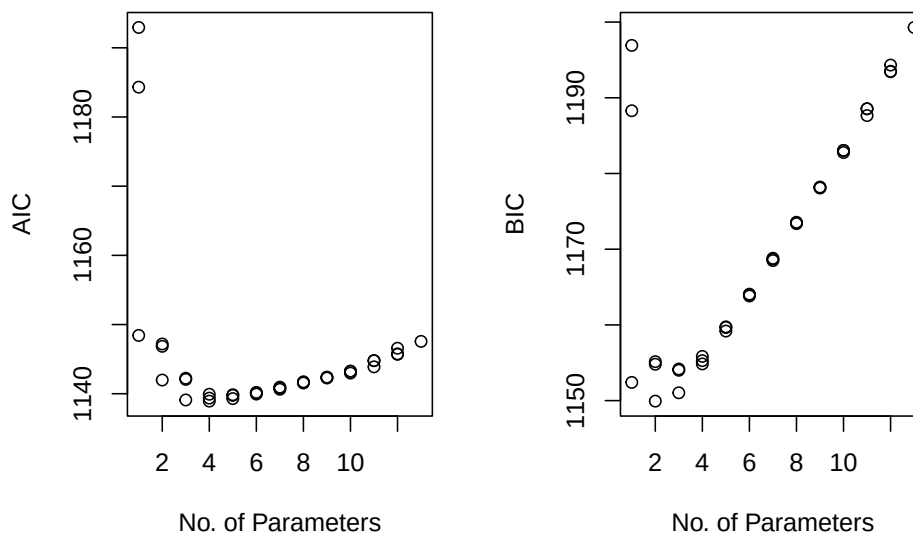
```
cbind(rs$bic, Bic, rs$bic - Bic)
```

```
##           Bic
## [1,] -43.004859 1152.406 -1195.411
## [2,] -45.475340 1149.936 -1195.411
## [3,] -44.374670 1151.036 -1195.411
## [4,] -40.562949 1154.848 -1195.411
## [5,] -36.222213 1159.189 -1195.411
## [6,] -31.587827 1163.823 -1195.411
## [7,] -26.916684 1168.494 -1195.411
## [8,] -22.029154 1173.382 -1195.411
## [9,] -17.328400 1178.082 -1195.411
## [10,] -12.638892 1182.772 -1195.411
## [11,] -7.758187 1187.653 -1195.411
## [12,] -1.965927 1193.445 -1195.411
## [13,]  3.879571 1199.290 -1195.411
```

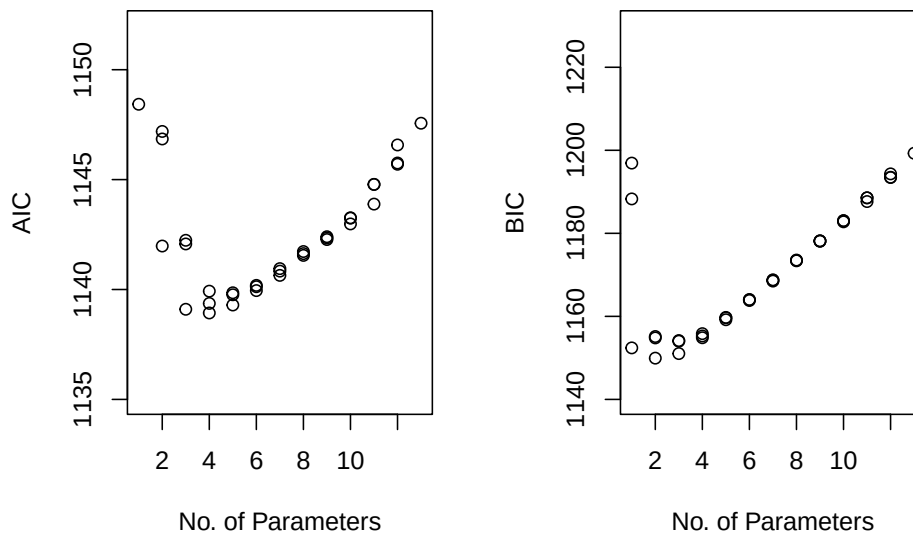
What are the 2nd and 3rd best models in terms of AIC/BIC? This cannot be answered by looking at the AIC/BIC plots shown above. Instead, we need to run the following code.

```
regsubsets_model.1 = regsubsets(Grades ~ ., data=stu_performance, nbest = 3, nvmax = p)
rs.1 = summary(regsubsets_model.1)
#rs.1$which
msize.1 = apply(rs.1$which, 1, sum) - 1
```

```
par(mfrow=c(1,2))
Aic.1 = n*log(rs.1$rss/n) + 2*msize.1
Bic.1 = n*log(rs.1$rss/n) + msize.1*log(n)
plot(msize.1, Aic.1, xlab="No. of Parameters", ylab = "AIC")
plot(msize.1, Bic.1, xlab="No. of Parameters", ylab = "BIC")
```



```
par(mfrow=c(1,2))
plot(msize.1, Aic.1, ylim = c(1135, 1152), xlab="No. of Parameters", ylab = "AIC")
plot(msize.1, Bic.1, ylim = c(1140, 1230), xlab="No. of Parameters", ylab = "BIC")
```



```
# top three models by AIC
rs.1$which[order(Aic.1)[1:3],]
```

```
## (Intercept) age Medu Fedu traveltime studytime failures famrel freetime
## 4 TRUE FALSE TRUE FALSE FALSE FALSE TRUE FALSE TRUE
## 3 TRUE FALSE TRUE FALSE FALSE FALSE TRUE FALSE FALSE
## 5 TRUE FALSE TRUE FALSE TRUE FALSE TRUE FALSE TRUE
## goout Dalc Walc health absences
```

```
## 4 TRUE FALSE FALSE FALSE FALSE
## 3 TRUE FALSE FALSE FALSE FALSE
## 5 TRUE FALSE FALSE FALSE FALSE

# top three models by BIC
rs.1$which[order(Bic.1)[1:3],]

## (Intercept) age Medu Fedu traveltime studytime failures famrel freetime
## 2 TRUE FALSE TRUE FALSE FALSE FALSE TRUE FALSE FALSE
## 3 TRUE FALSE TRUE FALSE FALSE FALSE TRUE FALSE FALSE
## 1 TRUE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE
## goout Dalc Walc health absences
## 2 FALSE FALSE FALSE FALSE FALSE
## 3 TRUE FALSE FALSE FALSE FALSE
## 1 FALSE FALSE FALSE FALSE FALSE
```

3.4.0.2 Subset Selection via the step function

We can use the **stepwise** approach to search possible models moving forward, backwards or in both directions (stepwise).

The R function to do this is the **step** function. The default is the stepwise method, so if we want to change this, we choose the **direction** option. The method by default outputs **all** the steps of the algorithm, so we can set **trace=0** if we do not want to see the intermediate results. Also, the default criterion for model selection is the AIC. However, we can select the BIC criterion by selecting **k = long(n)**.

```
## Stepwise AIC

stepAIC = step(performance.full, trace=0, direction="both")
#stepAIC = step(performance.full, direction="both")

## Stepwise BIC

n = dim(stu_performance)[1]
stepBIC = step(performance.full, trace=0, direction="both", k=log(n))
```

If we want to retrieve the output from the **step** function, then we can use the following code:

```
sel.var.AIC = attr(stepAIC$terms, "term.labels")
sel.var.BIC = attr(stepBIC$terms, "term.labels")
sel.var.AIC

## [1] "Medu" "failures" "freetime" "goout"

length(sel.var.AIC)

## [1] 4
```

```
length(sel.var.BIC)

## [1] 2
c("Medu", "failures", "freetime", "goout") %in% sel.var.AIC

## [1] TRUE TRUE TRUE TRUE
sel.var.BIC %in% sel.var.AIC

## [1] TRUE TRUE
```

3.4.0.3 LASSO and RIDGE Regression

glmnet is the library needed for both lasso ($\alpha = 1$) and ridge ($\alpha = 0$). Check more glmnet examples at https://web.stanford.edu/~hastie/glmnet/glmnet_beta.html

```
library(glmnet)

## Loading required package: Matrix
## Loaded glmnet 4.1-10
## The input in the glmnet function should be in matrix format:

X = as.matrix(stu_performance[, names(stu_performance) != "Grades"])
Y = stu_performance$Grades
```

We are going to split the data in training (80%) and testing (20%) data sets. Then, we will select the variables in the training set and estimate the coefficients and compute the average MSE on the testing data.

```
## Training/ Testing Data Sets

ntest = round(n*0.2); ## testing set sample size
ntrain = n - ntest; ## training set sample size
test.id = sample(1:n, ntest); ## sampling indices for the test data

Ytest = Y[test.id]; ## Response in the Testing data
```

Training the **full model** in the *training* data set:

```
## Fit the Full model in the training data
full.model = lm(Y[-test.id]~., data = stu_performance[-test.id, names(stu_performance)
```

We then use the full model to **predict** values and *compute the MSE on the testing data set*:

```
## Predicted values on the testing data set
Ytest.pred = predict(full.model, newdata= stu_performance[test.id, names(stu_performance)
```

```
## Averaged MSE on the test set
sum((Ytest - Ytest.pred)^2)/ntest

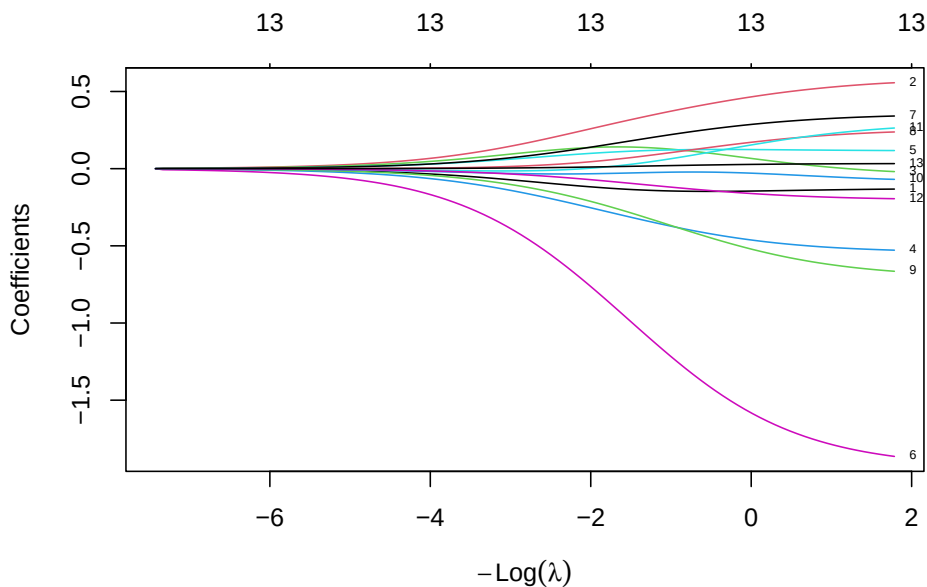
## [1] 20.08458
```

3.4.1 Ridge Regression

Fit a set of ridge regression models using `glmnet` with the default λ sequence. Selecting $\alpha = 0$ to run a **ridge**

We plot the fitted output of the ridge regression. We can use the `xvar` argument to decide what is plotted on the x-axis. We can also label the curves with the variable index numbers simply by setting `label = TRUE`.

```
ridge.lm = glmnet(X[-test.id, ], Y[-test.id], alpha = 0)
plot(ridge.lm , label = TRUE, xvar = "lambda")
```



We can obtain the λ values

```
length(ridge.lm$lambda)
```

```
## [1] 100
```

```
## remove the length function to see the lambda values
```

and the coefficients :

```
dim(ridge.lm $beta)      ## coefficients for non-intercept predictors
```

```
## [1] 13 100
```

```
length(ridge.lm $a0)      ## intercept

## [1] 100
## You can also retrieve coefficients (including intercept) using coef(myridge)
dim(coef(ridge.lm ))

## [1] 14 100
## The two coefficient matrices should be the same
sum((coef(ridge.lm ) - rbind(ridge.lm $a0, ridge.lm $beta))^2)

## [1] 0
```

How are the intercepts computed?

```
k = 2;

## Compute the "intercept" by hand:
mean.ridge = apply(X[-test.id, ], 2, mean) ## mean vector for training X
mean(Y[-test.id]) - sum(mean.ridge * ridge.lm $beta[, k])

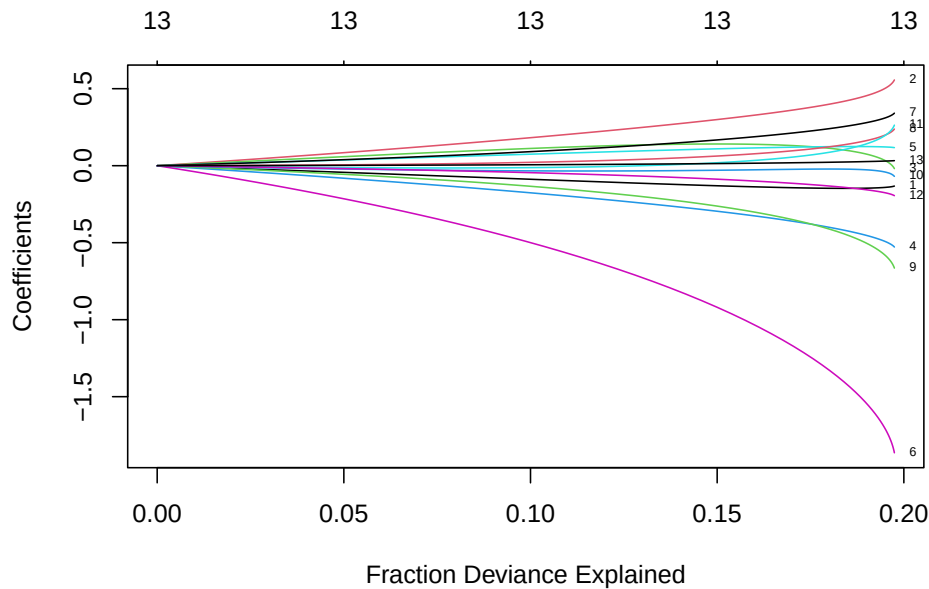
## [1] 10.33926
## Extract intercept from the estimated coefficients
ridge.lm$a0[k]      ## intercept for lambda = myridge$lambda[k]

##          s1
## 10.33926
## Both approaches give the same value
sum((mean(Y[-test.id]) - mean.ridge %*% ridge.lm $beta - ridge.lm $a0)^2)

## [1] 1.388395e-28
```

We can also choose `xvar="dev"` to *plot the fitted coefficients against the percent deviance*. This is the percent of deviance explained on the training data, and is a measure of complexity of the model. We can see that the coefficients do not change much after a point the but the coefficients start blowing up a bit.

```
plot(ridge.lm, xvar = "dev", label = TRUE)
```



3.4.1.1 Cross-Validation & Path Plot

If we want to use cross-validation to choose λ , then we can use the function `cv.glmnet`. This function is using a K -fold cross validation method to obtain the “optimal lambda”. The `cv.glmnet` has the attribute `nfolds` to select the number of folds we prefer, the option `foldid` to specify our own folds, and `type.measure` to choose the loss function (deviance or `mse` for squared loss, and `mae` for mean absolute error).

```
ridge.cv = cv.glmnet(X[-test.id, ], Y[-test.id], alpha = 0)
## the default in nfolds is 10
```

Some basic output information can be obtained

```
print(ridge.cv)
```

```
##
## Call:  cv.glmnet(x = X[-test.id, ], y = Y[-test.id], alpha = 0)
##
## Measure: Mean-Squared Error
##
##      Lambda Index Measure      SE Nonzero
## min  1.718    75  17.86 1.0397      13
## 1se 14.595    52  18.89 0.9698      13
```

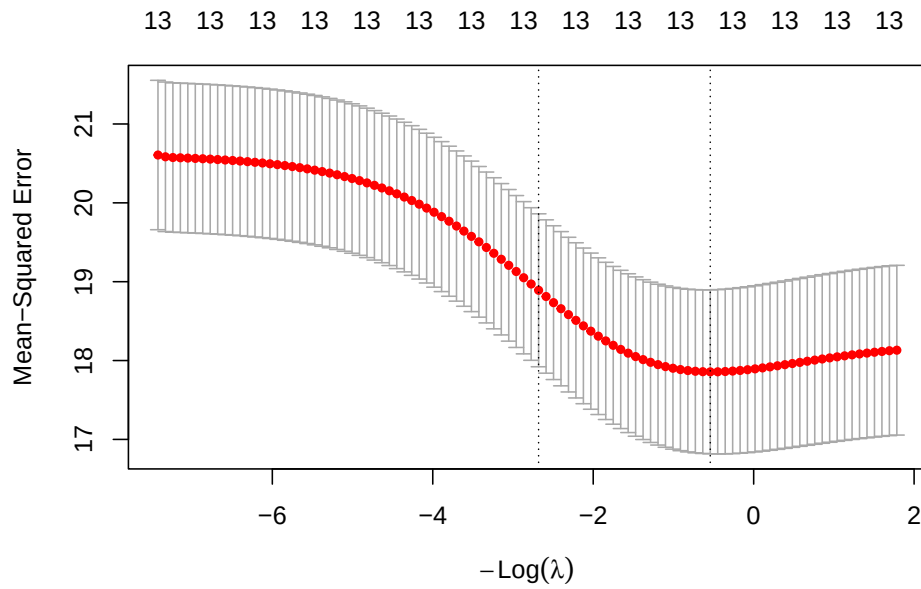
As a side note, the `cv.glmnet` function allows parallel computing for large scale problems.

The results are stored in

`ridge.cv$cvm` mean CV errors `ridge.cv$cvstd`: standard errors of CVm

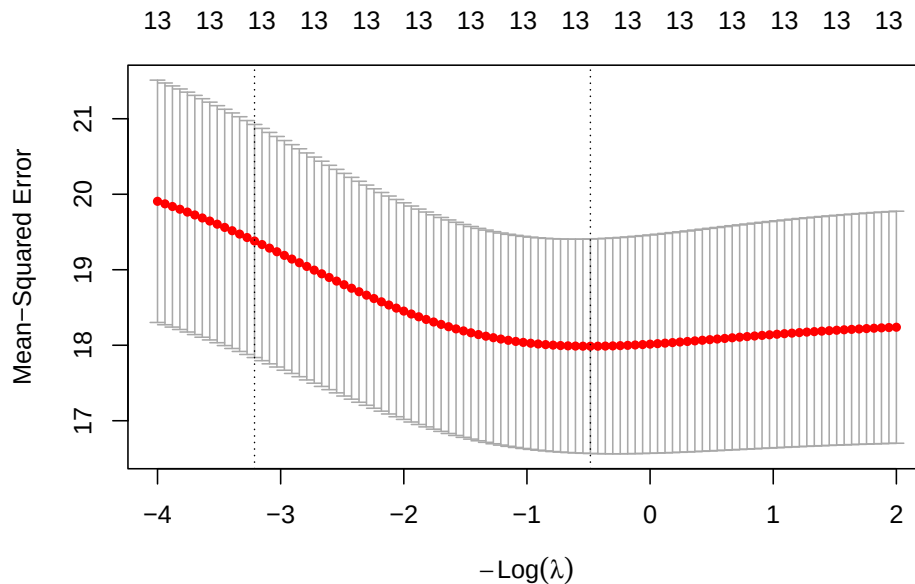
We can also plot the result of the `cv.glmnet`:

```
plot(ridge.cv)
```



The CV performance continues to drop when λ gets smaller. So, we provide R with a λ sequence dense in $[-2, 2]$:

```
lambda.seq = exp(seq(-2, 4, length=100))
ridge.cv.seq = cv.glmnet(X[-test.id,], Y[-test.id], alpha=0, lambda=lambda.seq)
plot(ridge.cv.seq)
```

We can still use the `coef` and `predict` methods for `cv.glmnet` objects in the same way as for the `glmnet`. In this case, we also have two special strings supported by `s` (the values of λ requested):

- `lambda.min`: the value of `lambda` that achieves the minimum MSE (i.e. `cvm` – the left dashed vertical line)
- `lambda.1se`: the largest value of `lambda` (i.e., the largest regularization, the smallest `df`) whose MSE (i.e. `cvm`) is within one-standard-error of the `cvm` of `lambda.min` – i.e. the smallest MSE – the left dashed vertical line).

Note: by definition `lambda.1se` (right dashed vertical line) is always bigger than or equal to `lambda.min` (left dashed vertical line).

Check how `lambda.min` and `lambda.1se` are computed.

```
names(ridge.cv.seq)
```

```
## [1] "lambda"      "cvm"         "cvstd"       "cvup"       "cvlo"
## [6] "nzero"      "call"       "name"       "glmnet.fit" "lambda.min"
## [11] "lambda.1se" "index"
```

```
## Find lambda.min
```

```
ridge.cv.seq$lambda.min
```

```
## [1] 1.623929
```

```
ridge.cv.seq$lambda[which.min(ridge.cv.seq$cvm)]
```

```
## [1] 1.623929
```

```
## Find lambda.1se
ridge.cv.seq$lambda.1se

## [1] 24.8317
tmp.id = which.min(ridge.cv.seq$cvm)
max(ridge.cv.seq$lambda[ridge.cv.seq$cvm < ridge.cv.seq$cvm[tmp.id] + ridge.cv.seq$cvs

## [1] 24.8317
```

3.4.1.1.1 Prediction Next we apply the two ridge regression models—one with `lambda.min` and one with `lambda.1se` on a new *test* data set and report the corresponding **mean squared prediction errors**:

```
ridge.lambda.pred = glmnet(X[-test.id,], Y[-test.id], alpha=0, lambda = lambda.seq)

## Using the lambda.min
Ytest.ridge.min.pred = predict(ridge.lambda.pred, s = ridge.cv.seq$lambda.min, newx=X[
mean((Ytest.ridge.min.pred - Y[test.id])^2)

## [1] 19.85768

## Using lambda.1se
Ytest.ridge.1se.pred = predict(ridge.lambda.pred, s = ridge.cv.seq$lambda.1se, newx=X[
mean((Ytest.ridge.1se.pred - Y[test.id])^2)

## [1] 21.30005
```

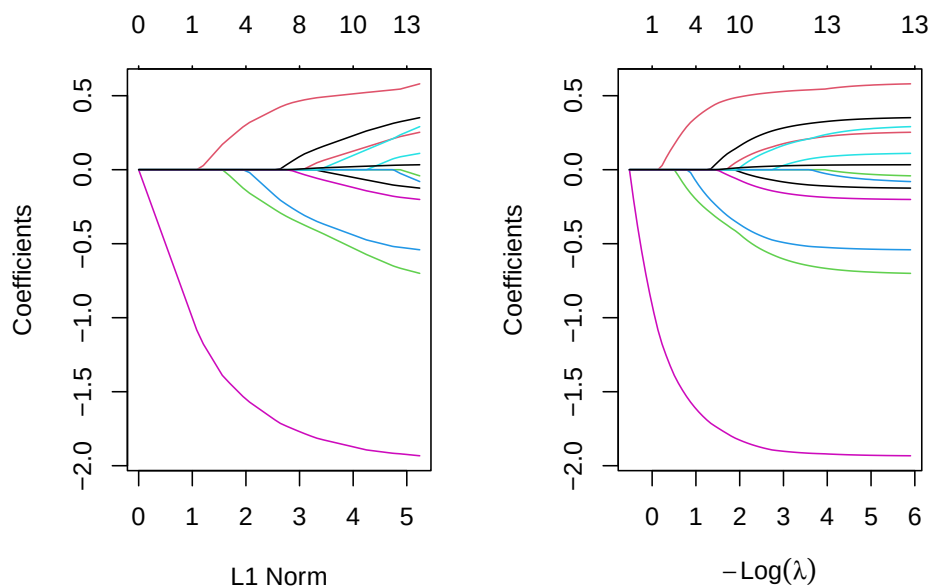
3.4.2 LASSO Regression

If we want to use LASSO instead of Ridge, we need to choose $\alpha = 1$ in the `glmnet` function. In this case, we have

```
lasso.lm = glmnet(X[-test.id,], Y[-test.id], alpha = 1)
```

Using cross validation, we can see the CV path plot:

```
par(mfrow = c(1, 2))
plot(lasso.lm, label=TRUE, xvar = "norm") ## norm is the default value for xvar
plot(lasso.lm, label=TRUE, xvar = "lambda")
```



```
par(mfrow=c(1,1))
```

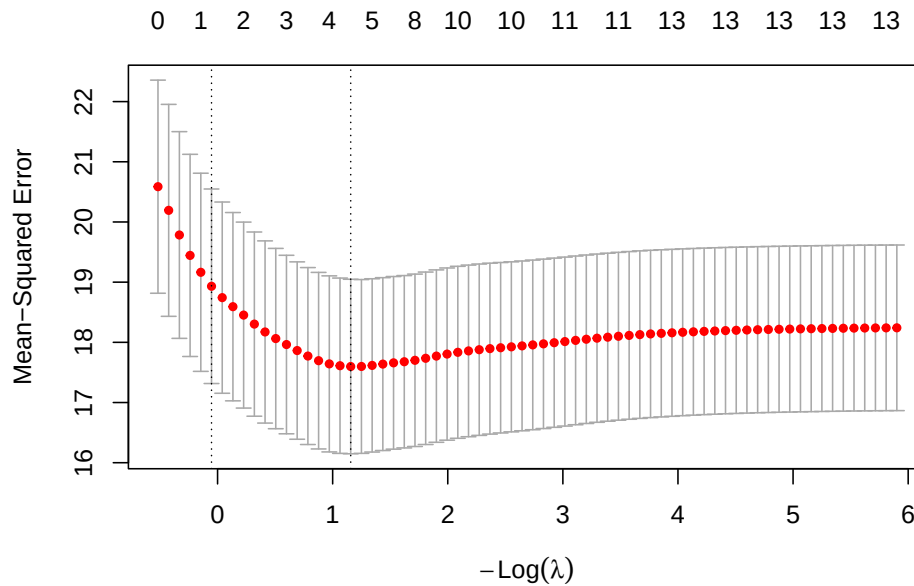
We can extract the number of non-zero coefficients for each value of λ .

```
lasso.lm$df
```

```
## [1] 0 1 1 1 1 1 1 1 2 2 2 2 3 3 3 4 4 4 4 4 5 5 7 7 8
## [26] 8 9 10 10 10 10 10 10 10 10 10 11 11 11 11 11 11 11 11 12 12 12 13 13
## [51] 13 13 13 13 13 13 13 13 13 13 13 13 13 13 13 13 13 13 13
```

As with the ridge regression example, we can still use cross-validation by calling the `cv.glmnet`:

```
lasso.cv = cv.glmnet(X[-test.id, ], Y[-test.id], alpha = 1)
plot(lasso.cv)
```



From the `cv.glmnet` object we can still compute `lambda.min` and `lambda.1se` in the same way as before:

```
## Computing lambda.min
lasso.cv$lambda.min
```

```
## [1] 0.3144502
```

```
##Computing lambda.1se
lasso.cv$lambda.1se
```

```
## [1] 1.053912
```

We can also extract the LASSO coefficients:

```
lasso.min.pred.coef = predict(lasso.lm, s=lasso.cv$lambda.min, type="coefficients");
lasso.1se.pred.coef = predict(lasso.lm, s=lasso.cv$lambda.1se, type="coefficients");
cbind(lasso.min.pred.coef, lasso.1se.pred.coef);
```

```
## 14 x 2 sparse Matrix of class "dgCMatrix"
##              s=0.3144502 s=1.053912
## (Intercept)  10.7544315 10.5966544
## age          .          .
## Medu         0.3880521 .
## Fedu         .          .
## traveltime  -0.1419488 .
## studytime   .          .
## failures    -1.6650296 -0.8417575
## famrel      .          .
## freetime    .          .
```

```
## goout          -0.2461255  .
## Dalc           .          .
## Walc           .          .
## health         .          .
## absences       .          .

## number of variables selected (including the intercept)
sum(lasso.1se.pred.coef != 0)

## [1] 2

sum(lasso.min.pred.coef != 0)

## [1] 5

## names of selected non-intercept variables
row.names(lasso.1se.pred.coef)[which(lasso.1se.pred.coef != 0)[-1]]

## [1] "failures"

row.names(lasso.min.pred.coef)[which(lasso.min.pred.coef != 0)[-1]]

## [1] "Medu"          "traveltime" "failures"  "goout"
```

3.4.2.0.1 Prediction Apply the two fitted models for prediction on test data:

```
lasso.lambda.pred = glmnet(X[-test.id, ], Y[-test.id], alpha = 1)

## Using the lambda.min
Ytest.lasso.min.pred = predict(lasso.lambda.pred, s = lasso.cv$lambda.min, newx = X[test.id,])
mean((Ytest.lasso.min.pred - Y[test.id])^2)

## [1] 20.12897

## Using lambda.1se
Ytest.lasso.1se.pred = predict(lasso.lambda.pred, s = lasso.cv$lambda.1se, newx=X[test.id,])
mean((Ytest.lasso.1se.pred - Y[test.id])^2)

## [1] 21.06227
```

3.4.3 Refit to Reduce Bias

We refit an ordinary linear regression model using the variables selected by `lambda.1se` and then use it for prediction.

```
lasso.coef.1se = predict(lasso.lambda.pred, s = lasso.cv$lambda.1se, type="coefficients")
variables.lasso = row.names(lasso.coef.1se)[which(lasso.coef.1se != 0)[-1]]
variables.lasso

## [1] "failures"
```

```
tmp.X = X[ , colnames(X) %in% variables.lasso]
lasso.refit = coef( lm( Y[-test.id] ~ tmp.X[-test.id ] ) )
Ytest.pred = lasso.refit[1] + tmp.X[test.id] * lasso.refit[-1]
mean((Ytest.pred - Y[test.id])^2)

## [1] 20.27528
```

Chapter 4

NonLinear Regression

As we discussed in the previous sections, a Multiple Linear Regression model has a set of assumptions on the error terms (normality, homoscedasticity, independence) as well as limitations, such as the linearity assumption.

If one or more of these assumptions are not satisfied, then the applicability of our model is restricted. In this section, we will introduce models that lift the linearity assumption. Therefore, we are going to discuss the *three* most popular *nonlinear* regression models:

- (i) Polynomial Regression
- (ii) Splines Regression
- (iii) Smoothing Splines

4.0.1 A Note on Nonlinearity

Assume that the true underlying model is of the following form:

$$Y = f(\mathbf{X}) + \varepsilon$$

where ε satisfies the usual assumptions. So far we have been **approximating** the unknown function f via a multiple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} + \varepsilon_i$$

The MLR model above is **linear both** with respect to the predictors (X_{ij}) **and** with respect to the coefficients (β_j).

Consider now the following models:

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i1}^2 + \beta_3 X_{i2} + \beta_4 X_{i2}^2 + \beta_5 X_{i1} X_{i2} + \varepsilon_i$$

or

$$\log_{10} Y_i = \beta_1 X_{i1} + \beta_2 \sqrt{X_{i1}} + \beta_3 e^{X_{i3}} + \varepsilon_i$$

These two models are both *non-linear* with respect to the predictors but **linear** with respect to the coefficients β_j .

On the other side, the model below

$$Y_i = \frac{\gamma_0}{1 + \gamma_1 e^{\gamma_2 X_i}} + \varepsilon_i$$

is non-linear both with respect to the parameters *and* the predictors.

During the course of this week, we are going to approximate the general non-linear model, using nonlinear models with respect to the predictors (so that we describe the underlying nonlinear relationship between response and features), but all the models that we will introduce will be *linear with respect to the coefficients/parameters*. In this way, one can think of the models we will work with as models in which the features have been transformed to a new predictor in a nonlinear fashion.

The main advantage of this approach is that we manage to describe (quite efficiently most of the time) a nonlinear relationship between predictor and features, but at the same time we are able to estimate the model parameters easily.

To be more specific, the coefficients will still be obtained via a least-squares approach which (due to the linearity wrt predictor assumption) will result in a system of linear equations as before.

4.1 Polynomial Regression

The simplest form of nonlinear regression is the polynomial regression which is an extension of the linear model by adding higher order terms of the predictor(s). To study this type of regression, we need to first define the polynomial basis functions.

4.1.1 Polynomial Basis Functions

If $b_j(x)$ is the j th basis function, then f has the following representation

$$f(x) = \sum_{j=0}^d b_j(x) \beta_j$$

for some values β_j . Therefore, we can write the nonlinear model $y_i = f(x_i) + \varepsilon_i$ as a linear model (with respect to the coefficients)

$$y_i = \beta_0 + \sum_{j=1}^d b_j(x_i) \beta_j + \varepsilon_i$$

Suppose that f is believed to be a 4th order polynomial, so *the space of polynomials of order 4 and below contains f* .

A basis for this space is

$$\begin{aligned}b_0(x) &= 1 \\b_1(x) &= x \\b_2(x) &= x^2 \\b_3(x) &= x^3 \\b_4(x) &= x^4\end{aligned}$$

so that the model becomes

$$y_i = \underbrace{\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \beta_4 x_i^4}_{=\beta_0 + \sum_{j=1}^d b_j(x_i)\beta_j} + \varepsilon_i$$

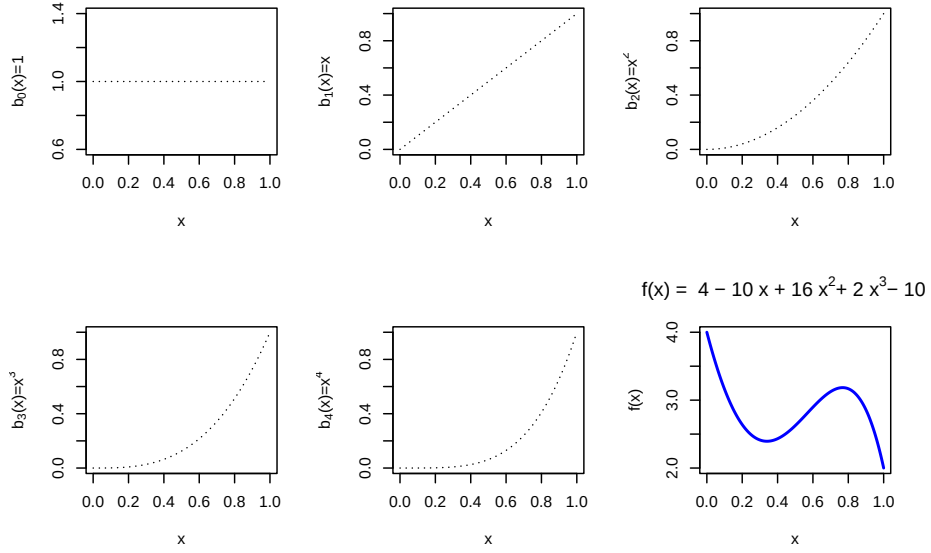
Illustration of the Polynomial Basis Functions

Representation of a function in terms of basis functions using a polynomial basis. The following code creates the plots the polynomial basis function up to order 4

```
x=seq(0, 1, by=0.001)
b0 = rep(1, length(x))
b1 = x
b2 = x^2
b3 = x^3
b4 = x^4

fun1 = 4*b0 -10* b1 + 16*b2 + 2*b3 -10*b4

par(mfrow = c(2,3))
plot(x, b0, type='l', lty=3, ylab=expression("b"[0]*(x)=1"))
plot(x, b1, type='l', lty=3, ylab=expression("b"[1]*(x)=x"))
plot(x, b2, type='l', lty=3, ylab=expression("b"[2]*(x)=x^2))
plot(x, b3, type='l', lty=3, ylab=expression("b"[3]*(x)=x^3))
plot(x, b4, type='l', lty=3, ylab=expression("b"[4]*(x)=x^4))
plot(x, fun1, type='l', ylab="f(x)", main=expression("f(x) = 4 - 10 x + 16 x^2 + 2 x^3 - 10 x^4"))
```



4.1.2 Polynomial Regression

From now on, assume $x \in \mathbb{R}$ is one-dimensional, and extensions to multi-dimensional cases will be discussed later. So, for $x_i \in \mathbb{R}$,

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d + \varepsilon_i$$

Then, we create the new variables $X_2 = X^2, \dots, X_d = X^d$, and treat this as a multiple linear regression model:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1} = \begin{pmatrix} 1 & x_1 & x_1^2 & \dots & x_1^d \\ 1 & x_2 & x_2^2 & \dots & x_2^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^d \end{pmatrix}_{n \times (d+1)} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_d \end{pmatrix}_{(d+1) \times 1} + \varepsilon$$

Therefore, we can say that a polynomial regression model is defined as follows:

Polynomial Regression Model

A *non-linear* model can be represented using a basis of polynomial functions as follows:

$$y_i = f(x_i) + \varepsilon_i \longrightarrow y_i = \beta_0 + \sum_{j=1}^d b_j(x_i) \beta_j + \varepsilon_i$$

where d is the degree of the polynomial component.

How do we choose d ?

1. *Forward Approach*: Keep *adding* terms until the *last* added term is not significant.

2. *Backward Approach*: Start with a large d , and keep *eliminating* the terms that are not statistically significant, starting with the highest order term.

Once we pick a value of d , then we usually do **not** test the significance of the lower-order terms. Therefore, when we decide to use a polynomial of degree d , by default, we include *all the lower-order terms in our model*.

Reasoning In regression analysis, we do not want our results to be affected by a change of location/scale of the data. Consider the following example: Suppose the data $\{y_i, x_i\}_{i=1}^n$ are generated by the model:

$$y_i = x_i^2 + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

But, they are instead recorded as $\{z_i, x_i\}_{i=1}^n$, where $z_i = x_i + 2$, that is,

$$y_i = (z_i - 2)^2 + \varepsilon_i = 4 - 4z_i + z_i^2 + \varepsilon_i$$

The linear term could become significant, if we shift the x values.

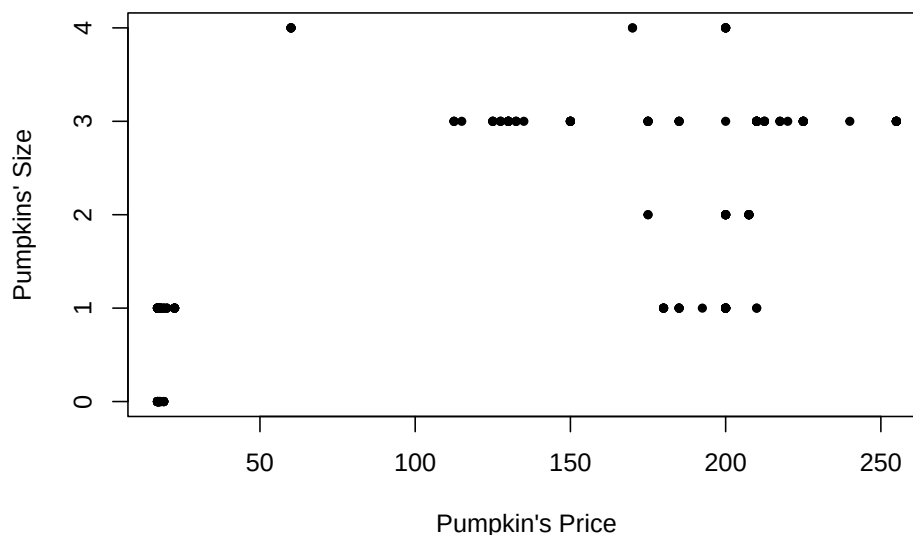
Exception: When we have a particular polynomial function in mind, e.g. the data are collected to test a particular physics formula $Y \approx X^2 + \text{constant}$, then you should test whether you can drop the linear term.

The Chicago Pumpkins Example

The `pumpkins.csv` data set contains information regarding the **size** and **price** of pumpkins sold in the Chicago area (data can be found [here](#)). Our goal in this example is to *predict* the **size** of the pumpkin (response) based on its **price** (predictor).

The scatter plot of the data is shown below:

```
pumpkins = read.csv("data/week4/chicagopumpkins.csv",header=TRUE)
plot(pumpkins$price, pumpkins$size, pch=20, xlab="Pumpkin's Price", ylab="Pumpkins' Size")
```



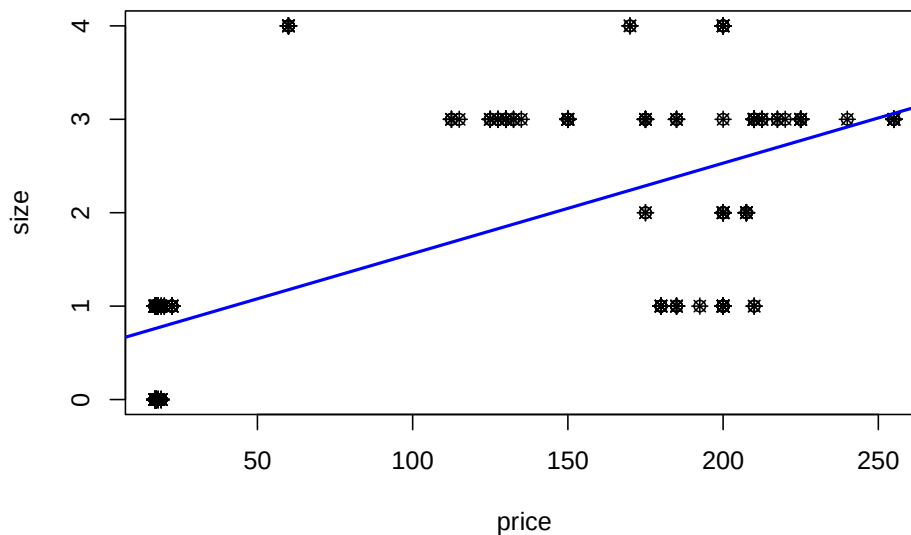
We see that a linear fit is probably *not* a good idea. Indeed,

```
lm.pumpkins = lm(size ~ price, data=pumpkins)
summary(lm.pumpkins)
```

```
##
## Call:
## lm(formula = size ~ price, data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6272 -0.7594  0.2244  0.3728  2.8245
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.5948315  0.0988339   6.018 6.35e-09 ***
## price        0.0096781  0.0006856  14.117 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9477 on 246 degrees of freedom
## Multiple R-squared:  0.4475, Adjusted R-squared:  0.4453
## F-statistic: 199.3 on 1 and 246 DF,  p-value: < 2.2e-16
```

Although the predictor is significant at explaining the response, the R^2 is on the lower end and the scatter plot does not support a straight line as a good fit.

```
plot(size ~ price, data=pumpkins)
points(size ~ price, data=pumpkins, pch=8)
abline(lm.pumpkins, col="blue", lwd=2)
```



We want to select a *higher order* model and we will do so following a Forward Selection approach first and a Backward Selection method second:

- (1) We start with a Forward Selection approach, i.e. we start by a linear model, and we keep **adding** higher order terms until the added term becomes *statistically insignificant*.

```
# Forward Selection
lm.pumpkins = lm(size ~ price , data=pumpkins)
summary(lm.pumpkins)

##
## Call:
## lm(formula = size ~ price, data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6272 -0.7594  0.2244  0.3728  2.8245
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.5948315  0.0988339   6.018 6.35e-09 ***
## price        0.0096781  0.0006856  14.117 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9477 on 246 degrees of freedom
## Multiple R-squared:  0.4475, Adjusted R-squared:  0.4453
## F-statistic: 199.3 on 1 and 246 DF,  p-value: < 2.2e-16

lm.pumpkins2 = lm(size ~ price + I(price^2), data=pumpkins)
summary(lm.pumpkins2)

##
## Call:
## lm(formula = size ~ price + I(price^2), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6438 -0.6029  0.3695  0.4895  2.3941
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.093e-01  1.174e-01   0.932   0.353
## price        3.037e-02  3.208e-03   9.469 < 2e-16 ***
## I(price^2)  -9.051e-05  1.375e-05  -6.582  2.8e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.8754 on 245 degrees of freedom
## Multiple R-squared:  0.5305, Adjusted R-squared:  0.5267
## F-statistic: 138.4 on 2 and 245 DF,  p-value: < 2.2e-16
lm.pumpkins3 = lm(size ~ price + I(price^2) + I(price^3), data=pumpkins)
summary(lm.pumpkins3)

##
## Call:
## lm(formula = size ~ price + I(price^2) + I(price^3), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.2786 -0.4937 -0.1368  0.5531  1.8633
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.414e+00  1.691e-01   -8.36 4.83e-15 ***
## price        1.256e-01  9.079e-03   13.83 < 2e-16 ***
## I(price^2)   -9.845e-04  8.239e-05  -11.95 < 2e-16 ***
## I(price^3)    2.228e-06  2.034e-07   10.95 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7182 on 244 degrees of freedom
## Multiple R-squared:  0.6853, Adjusted R-squared:  0.6814
## F-statistic: 177.1 on 3 and 244 DF,  p-value: < 2.2e-16
lm.pumpkins4 = lm(size ~ price + I(price^2) + I(price^3) + I(price^4), data=pumpkins)
summary(lm.pumpkins4)

##
## Call:
## lm(formula = size ~ price + I(price^2) + I(price^3) + I(price^4),
##     data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3200 -0.4497 -0.1241  0.5539  1.7925
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.871e+00  3.782e-01  -7.590 6.83e-13 ***
## price        2.314e-01  2.630e-02   8.800 2.60e-16 ***
## I(price^2)   -2.565e-03  3.785e-04  -6.776 9.25e-11 ***
## I(price^3)    1.061e-05  1.973e-06   5.378 1.76e-07 ***
```

```
## I(price^4) -1.470e-08 3.443e-09 -4.271 2.80e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6941 on 243 degrees of freedom
## Multiple R-squared:  0.7073, Adjusted R-squared:  0.7025
## F-statistic: 146.8 on 4 and 243 DF,  p-value: < 2.2e-16

lm.pumpkins5 = lm(size ~ price + I(price^2) + I(price^3) + I(price^4) + I(price^5), data=pumpkins)
summary(lm.pumpkins5)

##
## Call:
## lm(formula = size ~ price + I(price^2) + I(price^3) + I(price^4) +
##     I(price^5), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.38082 -0.46537 -0.08211  0.53463  1.91954
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.691e+00  7.112e-01  -2.378   0.0182 *
## price        1.305e-01  5.791e-02   2.253   0.0251 *
## I(price^2)   -2.479e-04  1.244e-03  -0.199   0.8422
## I(price^3)   -1.078e-05  1.113e-05  -0.969   0.3333
## I(price^4)    7.118e-08  4.409e-08   1.615   0.1077
## I(price^5)   -1.248e-10  6.386e-11  -1.954   0.0519 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6901 on 242 degrees of freedom
## Multiple R-squared:  0.7118, Adjusted R-squared:  0.7059
## F-statistic: 119.6 on 5 and 242 DF,  p-value: < 2.2e-16
```

We see that the *5th order model* has a 5th order term with *p-value equal to 0.0519*, which is **higher than 5%**, so we conclude that the optimal order for the polynomial, according to the forward selection method is $d = 4$. So, the fitted model is:

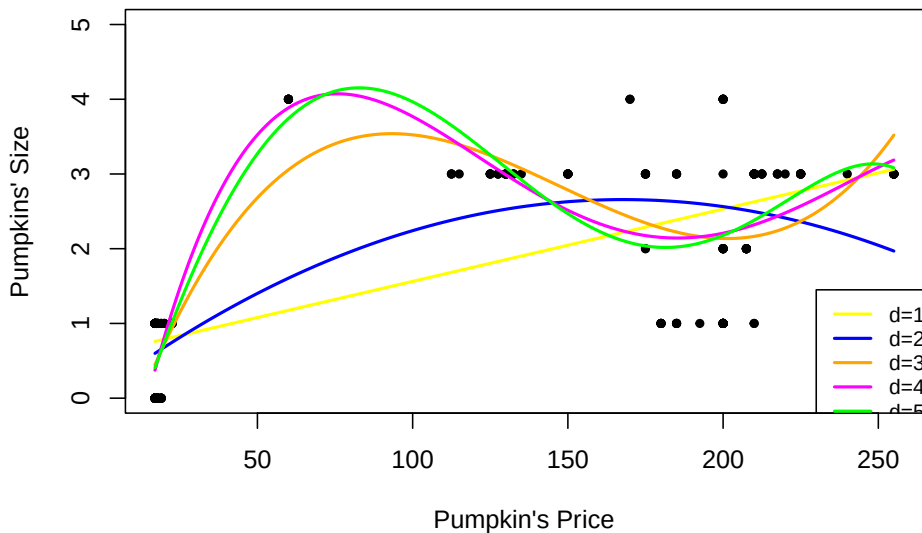
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 + \hat{\beta}_3 x^3 + \hat{\beta}_4 x^4$$

If we plot all the fitted models, we have:

```
newprice = data.frame(price=seq(17, 255, 1))
plot(pumpkins$price, pumpkins$size, pch=20, ylim=c(0,5), xlab="Pumpkin's Price", ylab="Pumpkins'
lines(newprice$price, predict(lm.pumpkins, newprice), col="yellow", lty=1, lwd=2);
lines(newprice$price, predict(lm.pumpkins2, newprice), col="blue", lty=1, lwd=2);
```

```
lines(newprice$price, predict(lm.pumpkins3, newprice), col="orange", lty=1, lwd=2);
lines(newprice$price, predict(lm.pumpkins4, newprice), col="magenta", lty=1, lwd=2);
lines(newprice$price, predict(lm.pumpkins5, newprice), col="green", lty=1, lwd=2);
legend(230, 1.45, legend=c("d=1", "d=2", "d=3", "d=4", "d=5"),
      col=c("yellow", "blue", "orange", "magenta", "green"), lty=c(1,1,1,1,1), cex=0.8)
```

Forward Selection Models



The magenta line is the one that corresponds to the 4th order model.

- (2) We can also select d using the Backward Elimination approach, that is we start with a large value for d and we eliminate terms *until the highest order term in the model is statistically significant*:

```
lm.pumpkins10 = lm(size ~ price + I(price^2) + I(price^3) + I(price^4) + I(price^5) + I(price^6) + I(price^7) + I(price^8) + I(price^9) + I(price^10), data = pumpkins)
summary(lm.pumpkins10)
```

```
##
## Call:
## lm(formula = size ~ price + I(price^2) + I(price^3) + I(price^4) +
##      I(price^5) + I(price^6) + I(price^7) + I(price^8) + I(price^9) +
##      I(price^10), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.44082 -0.45530 -0.01853  0.53535  2.12546
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.331e+01  2.671e+01   0.498   0.619
```



```
## price      -2.191e+00  4.274e+00 -0.513   0.609
## I(price^2)  1.396e-01  2.625e-01  0.532   0.595
## I(price^3) -4.389e-03  8.140e-03 -0.539   0.590
## I(price^4)  8.198e-05  1.458e-04  0.562   0.575
## I(price^5) -9.818e-07  1.624e-06 -0.605   0.546
## I(price^6)  7.731e-09  1.163e-08  0.664   0.507
## I(price^7) -3.970e-11  5.374e-11 -0.739   0.461
## I(price^8)  1.276e-13  1.549e-13  0.824   0.411
## I(price^9) -2.321e-16  2.535e-16 -0.916   0.361
## I(price^10) 1.821e-19  1.800e-19  1.012   0.313
##
## Residual standard error: 0.6483 on 237 degrees of freedom
## Multiple R-squared:  0.7509, Adjusted R-squared:  0.7404
## F-statistic: 71.45 on 10 and 237 DF,  p-value: < 2.2e-16

lm.pumpkins9 = lm(size ~ price + I(price^2) + I(price^3) + I(price^4)+ I(price^5)+ I(price^6) +
summary(lm.pumpkins9)

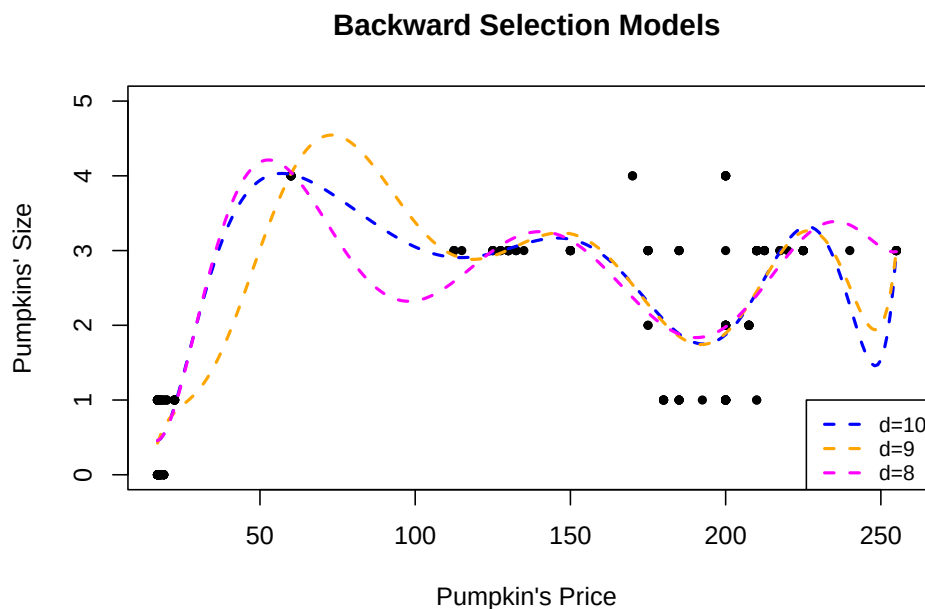
##
## Call:
## lm(formula = size ~ price + I(price^2) + I(price^3) + I(price^4) +
##     I(price^5) + I(price^6) + I(price^7) + I(price^8) + I(price^9),
##     data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.4514 -0.4230 -0.0091  0.5177  2.1072
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.191e+01  9.589e+00  -1.242   0.2154
## price        1.877e+00  1.450e+00   1.295   0.1967
## I(price^2)   -1.125e-01  8.271e-02  -1.360   0.1751
## I(price^3)    3.506e-03  2.316e-03   1.514   0.1314
## I(price^4)   -6.094e-05  3.623e-05  -1.682   0.0939 .
## I(price^5)    6.252e-07  3.391e-07   1.844   0.0664 .
## I(price^6)   -3.875e-09  1.945e-09  -1.993   0.0475 *
## I(price^7)    1.425e-11  6.704e-12   2.125   0.0346 *
## I(price^8)   -2.859e-14  1.276e-14  -2.241   0.0259 *
## I(price^9)    2.411e-17  1.030e-17   2.340   0.0201 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6483 on 238 degrees of freedom
## Multiple R-squared:  0.7499, Adjusted R-squared:  0.7404
## F-statistic: 79.27 on 9 and 238 DF,  p-value: < 2.2e-16
```

```
lm.pumpkins8 = lm(size ~ price + I(price^2) + I(price^3) + I(price^4) + I(price^5) + I(p
summary(lm.pumpkins8)
```

```
##
## Call:
## lm(formula = size ~ price + I(price^2) + I(price^3) + I(price^4) +
##      I(price^5) + I(price^6) + I(price^7) + I(price^8), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.39988 -0.45678 -0.05827  0.53309  2.02119
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.140e+00  3.347e+00   2.731 0.006791 **
## price       -1.368e+00  4.257e-01  -3.215 0.001486 **
## I(price^2)    7.522e-02  2.032e-02   3.702 0.000266 ***
## I(price^3)   -1.798e-03  4.783e-04  -3.759 0.000214 ***
## I(price^4)    2.262e-05  6.154e-06   3.675 0.000293 ***
## I(price^5)   -1.611e-07  4.541e-08  -3.549 0.000466 ***
## I(price^6)    6.535e-10  1.918e-10   3.407 0.000771 ***
## I(price^7)   -1.406e-12  4.316e-13  -3.259 0.001282 **
## I(price^8)    1.246e-15  4.008e-16   3.108 0.002112 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6544 on 239 degrees of freedom
## Multiple R-squared:  0.7441, Adjusted R-squared:  0.7355
## F-statistic: 86.87 on 8 and 239 DF,  p-value: < 2.2e-16
```

Starting with an order 10 model, we identify that an 9th order model is *optimal* according to the backward elimination criterion. If we plot all the fitted models, we have:

```
newprice = data.frame(price=seq(17, 255, 1))
plot(pumpkins$price, pumpkins$size, ylim=c(0,5), pch=20, xlab="Pumpkin's Price", ylab=
lines(newprice$price, predict(lm.pumpkins10, newprice), col="blue", lty=2, lwd=2);
lines(newprice$price, predict(lm.pumpkins9, newprice), col="orange", lty=2, lwd=2);
lines(newprice$price, predict(lm.pumpkins8, newprice), col="magenta", lty=2, lwd=2);
legend(226, 1, legend=c("d=10", "d=9", "d=8"),
      col=c("blue", "orange", "magenta"), lty=c(2,2,2), cex=0.8, lwd=2)
```



The magenta line is the one that corresponds to the *9th* order model.

For the fitted model we finally choose, we should (as always) perform diagnostic tests.

4.1.3 Orthogonal Polynomials

Fitting high order polynomials is generally **not recommended**, since they are very *unstable* and *difficult to interpret*. In addition, successive predictors x^j are *highly correlated* introducing multicollinearity problems. One way around this is to fit **orthogonal polynomials** of the form:

$$y_i = \beta_0 + \beta_1 z_1 + \dots + \beta_d z_d + \varepsilon_i$$

where each $z_j = a_1 + b_2 x + \dots + \kappa_j x^j$ is a polynomial of order j with coefficients chosen such that $z_i^\top z_j = 0$ (i.e. the inner product of any two polynomials is zero).

The Chicago Pumpkins Example

In R, we can fit orthogonal polynomials using the `poly` function. In the code below, we repeat the same process as before (for choosing d) using the orthogonal polynomials.

```
# Forward Selection
lm.pumpkins02 = lm(size ~ poly(price,2), data=pumpkins)
summary(lm.pumpkins02)
```

```
##
```

```
## Call:
## lm(formula = size ~ poly(price, 2), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6438 -0.6029  0.3695  0.4895  2.3941
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.70161    0.05559   30.612 < 2e-16 ***
## poly(price, 2)1  13.37846    0.87538   15.283 < 2e-16 ***
## poly(price, 2)2  -5.76139    0.87538   -6.582 2.8e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8754 on 245 degrees of freedom
## Multiple R-squared:  0.5305, Adjusted R-squared:  0.5267
## F-statistic: 138.4 on 2 and 245 DF,  p-value: < 2.2e-16
lm.pumpkins03 = lm(size ~ poly(price,3), data=pumpkins)
summary(lm.pumpkins03)

##
## Call:
## lm(formula = size ~ poly(price, 3), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.2786 -0.4937 -0.1368  0.5531  1.8633
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.7016    0.0456   37.313 < 2e-16 ***
## poly(price, 3)1  13.3785    0.7182   18.628 < 2e-16 ***
## poly(price, 3)2  -5.7614    0.7182   -8.022 4.35e-14 ***
## poly(price, 3)3   7.8672    0.7182   10.954 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7182 on 244 degrees of freedom
## Multiple R-squared:  0.6853, Adjusted R-squared:  0.6814
## F-statistic: 177.1 on 3 and 244 DF,  p-value: < 2.2e-16
lm.pumpkins04 = lm(size ~ poly(price,4), data=pumpkins)
summary(lm.pumpkins04)

##
```

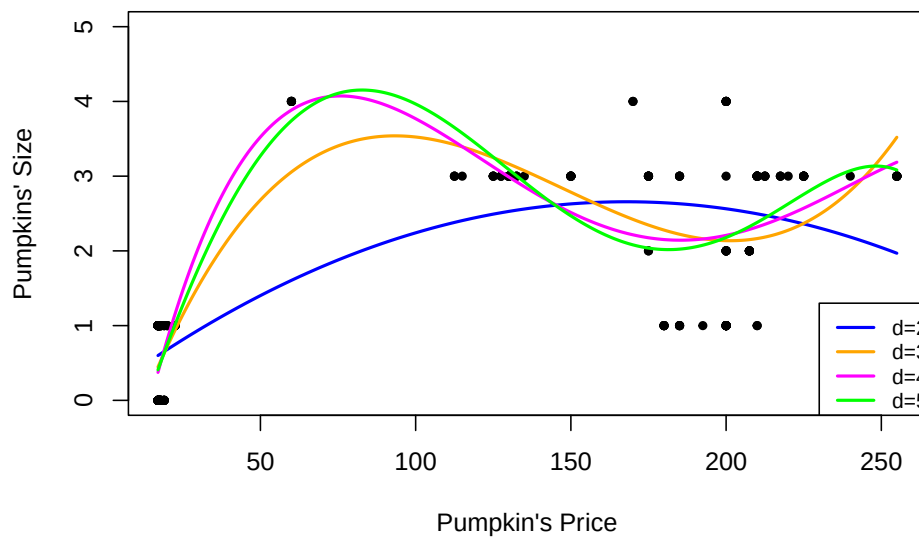
```
## Call:
## lm(formula = size ~ poly(price, 4), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3200 -0.4497 -0.1241  0.5539  1.7925
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.70161    0.04407  38.608 < 2e-16 ***
## poly(price, 4)1 13.37846    0.69408  19.275 < 2e-16 ***
## poly(price, 4)2 -5.76139    0.69408  -8.301 7.22e-15 ***
## poly(price, 4)3  7.86718    0.69408  11.335 < 2e-16 ***
## poly(price, 4)4 -2.96423    0.69408  -4.271 2.80e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6941 on 243 degrees of freedom
## Multiple R-squared:  0.7073, Adjusted R-squared:  0.7025
## F-statistic: 146.8 on 4 and 243 DF,  p-value: < 2.2e-16

lm.pumpkins05 = lm(size ~ poly(price,5), data=pumpkins)
summary(lm.pumpkins05)

##
## Call:
## lm(formula = size ~ poly(price, 5), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.38082 -0.46537 -0.08211  0.53463  1.91954
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.70161    0.04382  38.831 < 2e-16 ***
## poly(price, 5)1 13.37846    0.69009  19.387 < 2e-16 ***
## poly(price, 5)2 -5.76139    0.69009  -8.349 5.35e-15 ***
## poly(price, 5)3  7.86718    0.69009  11.400 < 2e-16 ***
## poly(price, 5)4 -2.96423    0.69009  -4.295 2.53e-05 ***
## poly(price, 5)5 -1.34841    0.69009  -1.954  0.0519 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6901 on 242 degrees of freedom
## Multiple R-squared:  0.7118, Adjusted R-squared:  0.7059
## F-statistic: 119.6 on 5 and 242 DF,  p-value: < 2.2e-16
```

```
newprice = data.frame(price=seq(17, 255, 1))
plot(pumpkins$price, pumpkins$size, pch=20, ylim=c(0,5), xlab="Pumpkin's Price", ylab="Pumpkins' Size",
     lines(newprice$price, predict(lm.pumpkins02, newprice), col="blue", lty=1, lwd=2);
     lines(newprice$price, predict(lm.pumpkins03, newprice), col="orange", lty=1, lwd=2);
     lines(newprice$price, predict(lm.pumpkins04, newprice), col="magenta", lty=1, lwd=2);
     lines(newprice$price, predict(lm.pumpkins05, newprice), col="green", lty=1, lwd=2);
     legend(230, 1.3, legend=c("d=2", "d=3", "d=4", "d=5"),
           col=c("blue", "orange", "magenta", "green"), lty=c(1,1,1,1), cex=0.8, lwd=c(2,2,2,2))
```

Forward Selection Models: Orthogonal Polynomials



```
# Backward Selection
lm.pumpkins010 = lm(size ~ poly(price,10), data=pumpkins)
summary(lm.pumpkins010)
```

```
##
## Call:
## lm(formula = size ~ poly(price, 10), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.44082 -0.45530 -0.01853  0.53535  2.12546
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.70161    0.04117  41.335 < 2e-16 ***
## poly(price, 10)1  13.37846    0.64829  20.636 < 2e-16 ***
## poly(price, 10)2  -5.76139    0.64829  -8.887 < 2e-16 ***
```

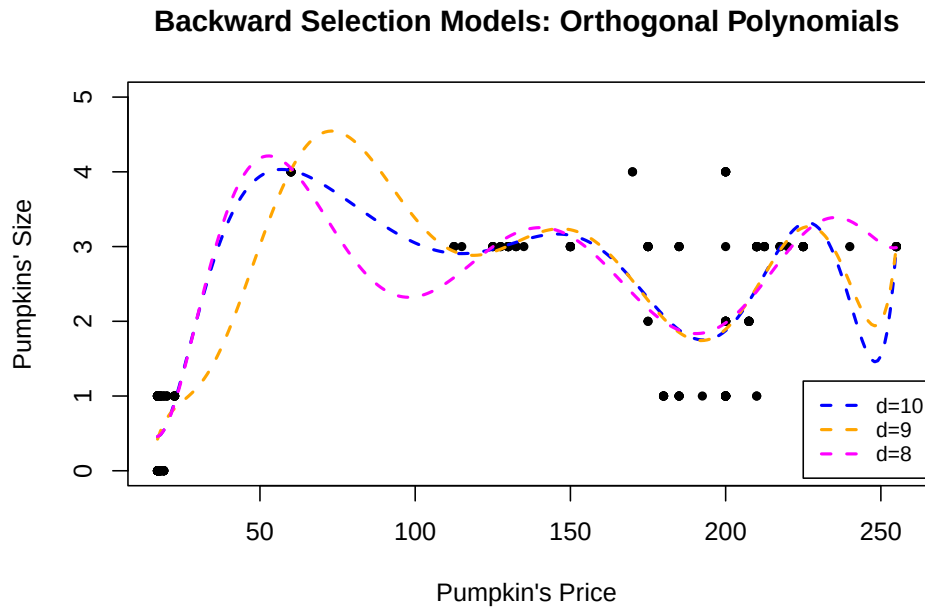
```
## poly(price, 10)3    7.86718    0.64829    12.135    < 2e-16 ***
## poly(price, 10)4   -2.96423    0.64829    -4.572    7.77e-06 ***
## poly(price, 10)5   -1.34841    0.64829    -2.080    0.03861 *
## poly(price, 10)6   -1.45456    0.64829    -2.244    0.02578 *
## poly(price, 10)7   -2.57955    0.64829    -3.979    9.20e-05 ***
## poly(price, 10)8    2.03378    0.64829     3.137    0.00192 **
## poly(price, 10)9    1.51698    0.64829     2.340    0.02012 *
## poly(price, 10)10  0.65591     0.64829     1.012    0.31269
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6483 on 237 degrees of freedom
## Multiple R-squared:  0.7509, Adjusted R-squared:  0.7404
## F-statistic: 71.45 on 10 and 237 DF,  p-value: < 2.2e-16

lm.pumpkins09 = lm(size ~ poly(price,9), data=pumpkins)
summary(lm.pumpkins09)

##
## Call:
## lm(formula = size ~ poly(price, 9), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.4514 -0.4230 -0.0091  0.5177  2.1072
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.70161    0.04117   41.333 < 2e-16 ***
## poly(price, 9)1 13.37846    0.64833   20.635 < 2e-16 ***
## poly(price, 9)2 -5.76139    0.64833   -8.887 < 2e-16 ***
## poly(price, 9)3  7.86718    0.64833   12.135 < 2e-16 ***
## poly(price, 9)4 -2.96423    0.64833   -4.572 7.76e-06 ***
## poly(price, 9)5 -1.34841    0.64833   -2.080  0.03861 *
## poly(price, 9)6 -1.45456    0.64833   -2.244  0.02578 *
## poly(price, 9)7 -2.57955    0.64833   -3.979 9.20e-05 ***
## poly(price, 9)8  2.03378    0.64833     3.137  0.00192 **
## poly(price, 9)9  1.51698    0.64833     2.340  0.02012 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6483 on 238 degrees of freedom
## Multiple R-squared:  0.7499, Adjusted R-squared:  0.7404
## F-statistic: 79.27 on 9 and 238 DF,  p-value: < 2.2e-16

lm.pumpkins08 = lm(size ~ poly(price,8), data=pumpkins)
summary(lm.pumpkins08)
```

```
##
## Call:
## lm(formula = size ~ poly(price, 8), data = pumpkins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.39988 -0.45678 -0.05827  0.53309  2.02119
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.70161    0.04155  40.951 < 2e-16 ***
## poly(price, 8)1 13.37846    0.65437  20.445 < 2e-16 ***
## poly(price, 8)2 -5.76139    0.65437  -8.805 2.71e-16 ***
## poly(price, 8)3  7.86718    0.65437  12.023 < 2e-16 ***
## poly(price, 8)4 -2.96423    0.65437  -4.530 9.32e-06 ***
## poly(price, 8)5 -1.34841    0.65437  -2.061 0.040421 *
## poly(price, 8)6 -1.45456    0.65437  -2.223 0.027162 *
## poly(price, 8)7 -2.57955    0.65437  -3.942 0.000106 ***
## poly(price, 8)8  2.03378    0.65437   3.108 0.002112 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6544 on 239 degrees of freedom
## Multiple R-squared:  0.7441, Adjusted R-squared:  0.7355
## F-statistic: 86.87 on 8 and 239 DF, p-value: < 2.2e-16
plot(pumpkins$price, pumpkins$size, pch=20, ylim=c(0,5), xlab="Pumpkin's Price", ylab="
lines(newprice$price, predict(lm.pumpkins010, newprice), col="blue", lty=2, lwd=2);
lines(newprice$price, predict(lm.pumpkins09, newprice), col="orange", lty=2, lwd=2);
lines(newprice$price, predict(lm.pumpkins08, newprice), col="magenta", lty=2, lwd=2);
legend(225, 1.2, legend=c("d=10", "d=9", "d=8"), col=c("blue", "orange", "magenta"),
```

4.1.4 Piece-wise Polynomials

If the true mean of $\mathbb{E}(Y|X = x) = f(x)$ is *too wiggly*, we might need to fit a higher order polynomial, which is not always a good idea. Instead we consider **piece-wise polynomials**:

1. we divide the range of x into several intervals, and
2. within each interval, $f(x)$ is a low-order polynomial, e.g., cubic or quadratic, but the polynomial coefficients will be different from interval to interval
3. we require the overall $f(x)$ to be continuous up to certain derivatives.

This method is also called “*broken-stick regression*”. Its benefit is that it localizes the influence of each data point to a particular segment, but overall it is not a very smooth line as the one we obtain by fitting a single polynomial for the whole data set.

4.2 Splines Regression

Splines are piecewise polynomials that are constructed so that they can be both sensitive and smooth, but also capture local features of the data.

A **Cubic Spline** is a curve constructed from sections of cubic polynomials, joined together so that the curve is *continuous up to second derivative*. The points at which the sections join are called the **knots** of the spline. For a conventional spline, the knots occur wherever there is a datum, but for the

regression splines the locations of the knots must be chosen. Typically, the knots would either be *evenly spaced* through the range of observed x values, or placed at the *quantiles* of the distribution of unique x values. Each section of cubic has different coefficients, but at the knots *it will match its neighboring sections in value and first two derivatives*.

Cubic Splines (Mathematical) Definition

A function g defined on $[a, b]$ is a **cubic spline** with respect to **knots** $\{\xi_i\}_{i=1}^m$ (specifically $a < \xi_1 < \xi_2 < \dots < \xi_m < b$) if:

- g is a cubic polynomial in each of the $m + 1$ intervals, i.e.

$$g(x) = d_i x^3 + c_i x^2 + b_i x + a_i, \quad x \in [\xi_i, \xi_{i+1}]$$

where $i = 0, \dots, m$, $\xi_0 = a$ and $\xi_{m+1} = b$.

- g is continuous up to the 2nd derivative. Since g is continuous up to the 2nd derivative *for any point inside an interval*, it suffices to check the following conditions:

$$g^{(0,1,2)}(\xi_i^+) = g^{(0,1,2)}(\xi_i^-), \quad i = 1, \dots, m$$

This expression indicates that the function and the first and second order derivatives are continuous at the knots.

How many free parameters do we need to represent a cubic spline?

(+) 4 parameters (d_i, c_i, b_i, a_i) for each of the $(m + 1)$ intervals.

(-) 3 constraints at each of the m knots (continuity constraints).

The total number of free parameters (similar to the number of degrees of freedom) is:

$$4(m + 1) - 3m = m + 4$$

A property of the cubic splines

Suppose the knots $\{\xi_i\}_{i=1}^m$ are given.

- If $g_1(x)$ and $g_2(x)$ are cubic splines, the linear combination

$$a_1 g_1(x) + a_2 g_2(x)$$

is also a cubic spline, where a_1 and a_2 are known constants.

That is, for a set of given knots, the corresponding *cubic splines form a linear space (of functions) with dim $(m + 4)$* .

4.2.1 Examples of Cubic Splines Basis

1. A set of basis functions for cubic splines (w.r.t knots $\{\xi_i\}_{i=1}^m$) is given by:

$$\begin{aligned} h_0(x) &= 1 \\ h_1(x) &= x \\ h_2(x) &= x^2 \\ h_3(x) &= x^3; \\ h_{i+3}(x) &= (x - \xi_i)_+^3, \quad i = 1, 2, \dots, m \end{aligned}$$

That is, any cubic spline can be uniquely expressed as:

$$\beta_0 + \sum_{j=1}^{m+3} \beta_j h_j(x)$$

2. Given knot locations, there are many alternative, but equivalent ways of writing down a basis for cubic splines. For example, *another* basis for cubic splines can be written as:

$$\begin{aligned} h_0(x) &= 1 \\ h_1(x) &= x \\ h_{i+1}(x) &= R(x, \xi_i^*), \quad i = 1, \dots, q-1 \end{aligned}$$

where

$$\begin{aligned} R(x, z) &= [(z - 1/2)^2 - 1/12] [(x - 1/2)^2 - 1/12] / 4 \\ &\quad - [(|x - z| - 1/2)^4 - 1/2(|x - z| - 1/2)^2 + 7/240] / 24 \end{aligned}$$

An Example of a Cubic Splines Basis

We first define the function $R(x, z)$ as above in R:

```
R_xz <- function(x,z){
  ((z-0.5)^2-1/12)*((x-0.5)^2-1/12)/4-((abs(x-z)-0.5)^4-(abs(x-z)-0.5)^2/2+7/240)/24
}
```

We then need to define the knots

```
new.knots= c(1/6, 3/6, 5/6)
```

In regression examples the x variable will be the predictor (there is no need to generate any x values). Here, we need a “generic” x for illustration purposes.

```
x=seq(0, 1, by=0.01)
```

Finally, we defined the splines functions, based on the definition given above:

```

spline1 = rep(1, length(x)) # First basis function
spline2 = x # Second basis function
spline3 = R_xz(x,new.knots[1]) # Third basis function
spline4 = R_xz(x,new.knots[2]) # Fourth basis function
spline5 = R_xz(x,new.knots[3]) # Fifth basis function

```

If we want to represent a function using the basis above, we have:

```

fun1s = 4*spline1 - 0.05*spline2 - 6*spline3 +2*spline4 +10*spline5

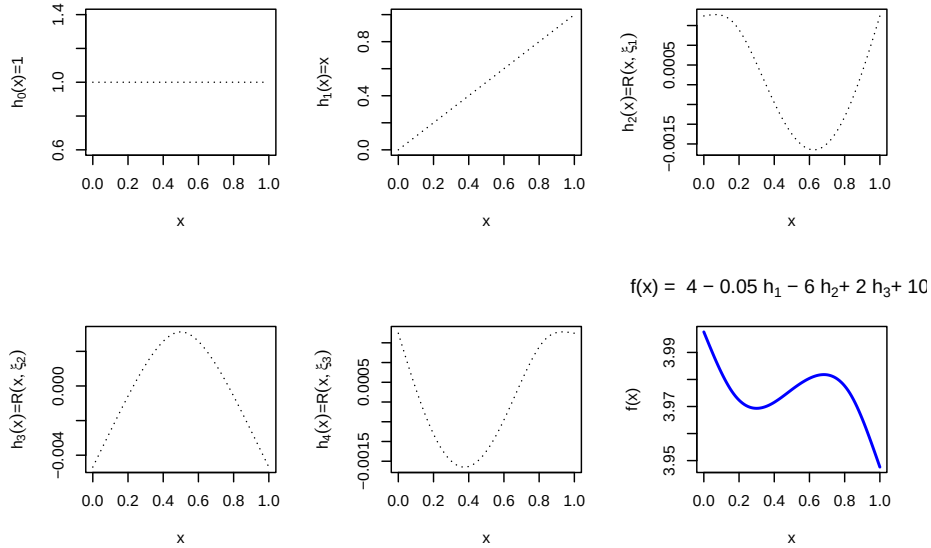
```

For illustration purposes, we plot the basis functions defined above as well as the function f :

```

par(mfrow = c(2,3))
plot(x, spline1, type='l', lty=3, ylab=expression("h"[0]*"(x)=1"))
plot(x, spline2, type='l', lty=3, ylab=expression("h"[1]*"(x)=x"))
plot(x, spline3, type='l', lty=3, ylab=expression(paste(h[2](x))*"="*paste(R(x,xi[1]))))
plot(x, spline4, type='l', lty=3, ylab=expression(paste(h[3](x))*"="*paste(R(x,xi[2]))))
plot(x, spline5, type='l', lty=3, ylab=expression(paste(h[4](x))*"="*paste(R(x,xi[3]))))
plot(x, fun1s, type='l', ylab="f(x)", main=expression("f(x) = 4 - 0.05 h"[1]*" - 6 h"

```



4.2.2 B-Splines Basis Functions in R

In this class, one of the cubic splines basis that we are going to use is called **B-Splines**.

Cubic B Splines Definition

The cubic B Splines basis is defined on an interval $[a, b]$ by the following requirements on the interior basis functions with knotpoints at $\{\xi_i\}_{i=1}^m$:

1. A given basis function is nonzero on an interval defined by *four successive knots and zero elsewhere*.
2. The basis function is a cubic polynomial for each subinterval between successive knots.
3. The basis function is continuous and is also continuous in its first and second derivatives at each knotpoint.
4. The basis function integrates to 1 over its support.
5. The boundary function definitions are adjusted to account for continuity in derivatives at the boundaries of the interval.

These are part of the `splines` library in R and the function to generate then is called `bs`. The arguments of the `bs` function are:

$$bs(x, df, knots, degree, intercept, Boundary.knots)$$

The R documentation can be found [here](#).

Arguments in the `bs` function:

- `x` is the predictor, i.e. the variable to which we want to apply the splines function.
- `df` are the degrees of freedom for the splines functions.
- `knots` are the *internal* breakpoints/knots that define the spline, i.e. location where the knots should be placed.
- `degree` is the degree of the piecewise polynomial. The default is `degree=3` for cubic splines.
- `intercept` is whether we want to add an intercept in the splines basis or not. *This is not the intercept in the regression we perform*. The default is `intercept=FALSE`.
- `Boundary.knots` are the boundary points at which we “anchor” the B-splines basis. The default values are obtained from the range of non-NA values of `x`, i.e. `Boundary.knots=FALSE`.

So, *in our examples*, we will need to feed the appropriate predictor `x` and then the desired `knots` **or** degrees of freedom, `df`, *not both*. The rest we leave them at their default values.

The **output** of the `bs()` function is a *matrix* of dimension `c(length(x), df)`, if `df` was supplied or of dimension `df = length(knots) + degree (+1 if intercept=TRUE)` if `knots` were supplied.

Knots or Degrees of Freedom in `bs()`

As we mentioned, we need to supply the degrees of freedom or the location of the knots in the `bs()` function, and of course there is an equivalence between

the two. Some details to keep in mind on how you can correctly specify these arguments:

- If you know where the knots should be placed, then you need to define a vector of the **locations** of the knots and input that in the **knots** argument in the **bs()** function. For example,

Defining knots location in **bs()**

```
library(splines)
x=seq(0, 1, by=0.01) # generic `x` for illustration purposes
new.knots= c(1/6, 3/6, 5/6) # define three knots at locations 1/6, 3/6, 5/6.
Bsplines.basis1 = bs(x, knots=new.knots)
head(Bsplines.basis1)
```

```
##           1           2           3 4 5 6
## [1,] 0.000000 0.000000 0.000000 0 0 0
## [2,] 0.165912 0.0034896 0.0000144 0 0 0
## [3,] 0.304896 0.0135168 0.0001152 0 0 0
## [4,] 0.418824 0.0294192 0.0003888 0 0 0
## [5,] 0.509568 0.0505344 0.0009216 0 0 0
## [6,] 0.579000 0.0762000 0.0018000 0 0 0
```

In this case, R calculates the degrees of freedom using the following formula:

$$df = \text{length}(\text{knots}) + \text{degree}$$

or if **intercept=TRUE** then

$$df = \text{length}(\text{knots}) + \text{degree} + 1$$

- If you prefer to specify the degrees of freedom, then the function **bs()** chooses **df-degree** many knots at suitable quantiles of x (ignoring any missing values). If you specified **intercept=TRUE**, then the number of knots will be **df-degree-1**. For example,

Defining knots location in **bs()**

```
x=seq(0, 1, by=0.01) # generic `x` for illustration purposes
Bsplines.basis2 = bs(x, df=4)
head(Bsplines.basis2)
```

```
##           1           2           3 4
## [1,] 0.000000 0.000000 0.000000 0
## [2,] 0.058214 0.000592 0.000002 0
## [3,] 0.112912 0.002336 0.000016 0
## [4,] 0.164178 0.005184 0.000054 0
## [5,] 0.212096 0.009088 0.000128 0
## [6,] 0.256750 0.014000 0.000250 0
```

As an example, below we plot the first 7 B Splines basis functions:

Illustration of B Splines Basis Function

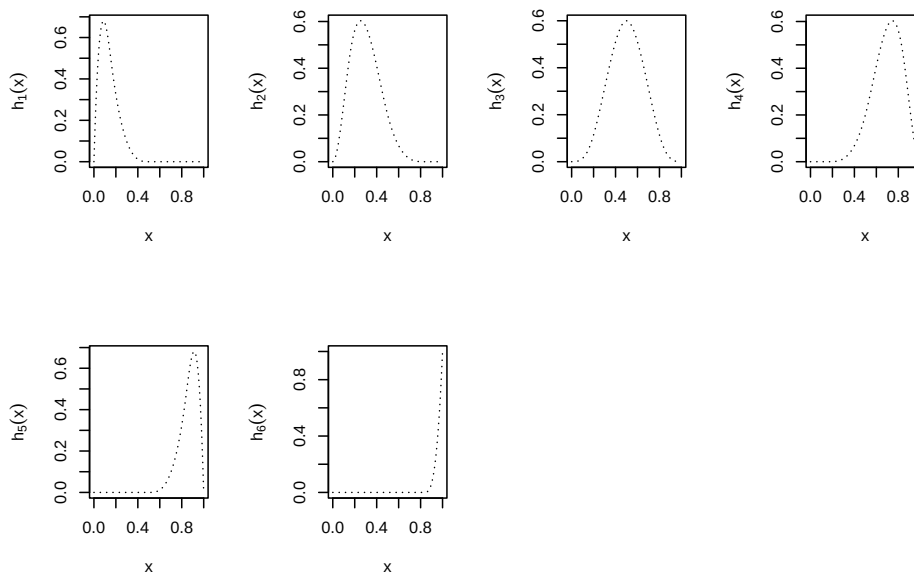
```
Bsplines.basis = bs(x, knots=new.knots)
dim(Bsplines.basis)
```

```
## [1] 101 6
```

```
head(Bsplines.basis)
```

```
##           1           2           3 4 5 6
## [1,] 0.000000 0.0000000 0.0000000 0 0 0
## [2,] 0.165912 0.0034896 0.0000144 0 0 0
## [3,] 0.304896 0.0135168 0.0001152 0 0 0
## [4,] 0.418824 0.0294192 0.0003888 0 0 0
## [5,] 0.509568 0.0505344 0.0009216 0 0 0
## [6,] 0.579000 0.0762000 0.0018000 0 0 0
```

```
par(mfrow = c(2,4))
plot(x, Bsplines.basis[,1], type='l', lty=3, ylab=expression(paste(h[1](x))))
plot(x, Bsplines.basis[,2], type='l', lty=3, ylab=expression(paste(h[2](x))))
plot(x, Bsplines.basis[,3], type='l', lty=3, ylab=expression(paste(h[3](x))))
plot(x, Bsplines.basis[,4], type='l', lty=3, ylab=expression(paste(h[4](x))))
plot(x, Bsplines.basis[,5], type='l', lty=3, ylab=expression(paste(h[5](x))))
plot(x, Bsplines.basis[,6], type='l', lty=3, ylab=expression(paste(h[6](x))))
```



4.2.3 Natural Cubic Splines (NCS)

A cubic spline on $[a, b]$ is a **Natural Cubic Spline** if its *second* and *third* derivatives are *zero at a and b* . This condition implies that NCS is a linear function in the two extreme intervals $[a, \xi_1]$ and $[\xi_m, b]$. The linear functions in the two extreme intervals are completely determined by their neighboring intervals. The degrees of freedom of NCS's with m knots are:

$$4(m+1) - 3m - 4 = m$$

since we have 4 additional constraints (the ones on the boundary points).

Remark: For a curve estimation problem with data $(x_i, y_i)_{i=1}^n$, if we put n knots at the n data points (assumed to be unique), then using NCS we obtain a *smooth* curve passing through **all** y 's.

Natural Cubic Spline

A Natural Cubic Spline with m knots is represented by m basis functions, for example one such basis is given by

$$\begin{aligned} N_1(x) &= 1 \\ N_2(x) &= x \\ N_{k+2}(x) &= d_k(x) - d_{k-1}(x) \end{aligned}$$

where

$$d_k(x) = \frac{(x - \xi_k)_+^3 - (x - \xi_m)_+^3}{\xi_m - \xi_k}$$

Each of these derivatives can be seen to have zero second and third derivative for $x \geq \xi_m$.

In **R** we can find the NCS as part of the **splines** library, by calling the **ns()** function. Specifically, we have that

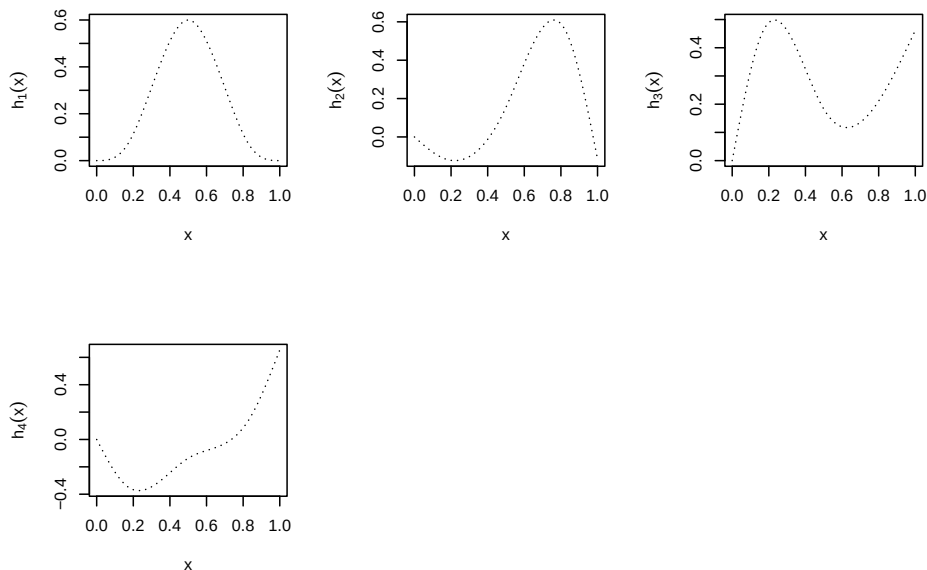
```
ns(x, df, knots, intercept = TRUE, Boundary.knots)
```

Natural Cubic Splines in R

```
NCS.basis = ns(x, knots=new.knots, Boundary.knots=c(0,1))
dim(NCS.basis)
```

```
## [1] 101 4
```

```
par(mfrow = c(2,3))
plot(x, NCS.basis[,1], type='l', lty=3, ylab=expression(paste(h[1](x))))
plot(x, NCS.basis[,2], type='l', lty=3, ylab=expression(paste(h[2](x))))
plot(x, NCS.basis[,3], type='l', lty=3, ylab=expression(paste(h[3](x))))
plot(x, NCS.basis[,4], type='l', lty=3, ylab=expression(paste(h[4](x))))
```

Recall that the linear functions in the two extreme intervals are completely determined by the other cubic splines. This means that data points in the two extreme intervals (i.e., outside the two boundary knots) are wasted since they do not affect the fitting. Therefore, by default, R puts the two boundary knots as the *min* and *max* of the x_i 's.

As in the case of the `bs()` function:

- You can tell R the location of knots, which are the interior knots. Recall that a NCS with m knots has m df. So, the df is equal to the number of (interior) knots plus 2, where 2 means the two boundary knots.
- Or you can tell R the df. If `intercept = TRUE`, then we need $m = df - 2$ knots, otherwise we need $m = df - 1$ knots. Again, by default, R puts knots at the $1/(m+1), \dots, m/(m+1)$ quantiles of $x_{1:n}$.

4.2.4 Regression Splines

Recall that for a given set of knots, the corresponding cubic splines form a *linear space* of functions with dimension $(m+4)$. So, the **Regression Splines** use a basis expansion approach:

$$g(x) = \beta_1 h_1(x) + \beta_2 h_2(x) + \dots + \beta_p h_p(x)$$

* If Cubic Splines are used as basis functions $p = m + 4$.

- If Natural Cubic Splines (NCS) are used as basis functions $p = m$.

We can represent the model on the observed n data points using matrix notation:

$$\begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{pmatrix}_{n \times 1} = \begin{pmatrix} h_1(x_1) & \dots & h_p(x_1) \\ h_1(x_2) & \dots & h_p(x_2) \\ \dots & \dots & \dots \\ h_1(x_n) & \dots & h_p(x_n) \end{pmatrix}_{n \times p} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \dots \\ \beta_p \end{pmatrix}_{p \times 1}$$

where our “*design*” matrix is the matrix $\mathbf{F}$ of basis functions. Therefore, we can estimate the coefficients by solving the following Least-Squares problem:

$$\hat{\beta} = \arg \min ||\mathbf{y} - \mathbf{F}\beta||^2$$

4.2.5 K-Fold Cross-Validation

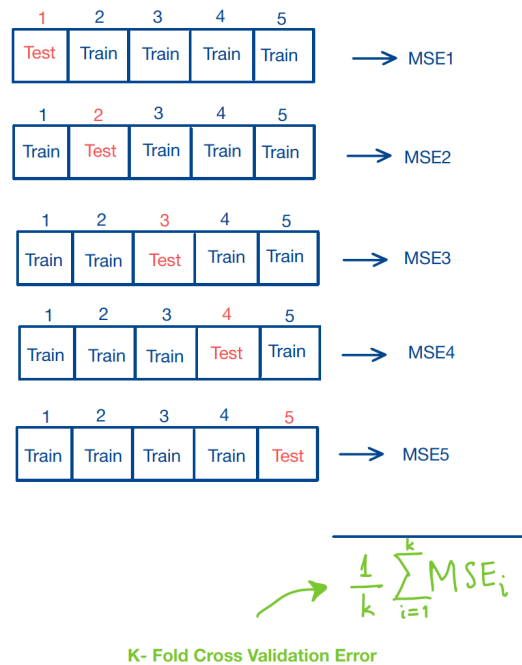
One of the challenges in using splines regression is the choice of the degrees of freedom/ knots. One way to optimally select the number of knots or degrees of freedom is the so-called K-fold cross-validation approach. This is outlined in the steps below:

1. Set a fixed number of knots (or df).
2. Divide the set of observations into k groups (or *folds*).
3. Leave the first fold as a validation set (not used to fit the model). Fit the Regression Spline with a fixed number of knots using the remaining $k - 1$ folds.
4. Calculate the Mean Square Error for fold 1: MSE_1 .
5. Repeat the previous steps k times. Each time a new validation set is used to calculate MSE_i .
6. Calculate the average k -fold Cross-Validation error:

$$CV(k) = \frac{1}{k} \sum_{i=1}^k MSE_i$$

7. Repeat 2 to 6 with a new number of knots (or df).
8. Select the number of knots that minimizes the k -fold CV error or $CV(k)$.

Then, we repeat the same process for a different number of knots and in the end we select the one that minimizes the $CV(k)$. A sketch of this approach is shown below:



4.3 Smoothing Splines

In Regression Splines, we need to choose the number and the location of knots (or the degrees of freedom). As we discussed, B-splines and NCS are both methods that construct a $n \times p$ basis matrix $\mathbf{F}$, and then model the outcome using a linear regression on $\mathbf{F}$. Inevitably, we need to select the order of the spline, the number of knots (AIC, BIC, CV) and even the location of knots, which is a quite challenging task. So, we need to consider whether there is an alternative approach that we can use to select the number and location of knots automatically.

A Smoothing Spline is a spline designed to balance fit with smoothness, and it starts by suggesting a very bad solution to our problem: by putting knots at *all* the observed data points $(x_1, \dots, x_n)$:

$$\mathbf{y}_{n \times 1} = \mathbf{F}_{n \times p} \beta_{p \times 1}$$

Then, we can construct n NCS basis. However, we know that with this approach we are going to run into overfitting problems. So, instead of selecting the knots, we recall last week's discussion on penalizing models with many parameters. This leads to minimizing the following objective functions with a ridge-type shrinkage:

$$\min_{\beta} \left\{ \|\mathbf{y} - \mathbf{F}\beta\|^2 + \lambda \beta^T \Omega \beta \right\}$$

where the tuning parameter λ is often chosen by CV (Ω will be defined later).

So, now we want to understand whether solving such a minimization problem can provide us with a good solution with desirable properties.

4.3.1 The Roughness Penalty Approach

Let $S[a, b]$ be the space of all “smooth” functions defined on $[a, b]$. This is a second order Sobolev space where global polynomial functions and cubic splines functions live ($S[a, b]$ is an infinite-dimensional function space). Our goal is to find the best function in $S[a, b]$ to approximate f .

Let us consider solving the following Penalized Residual Sum of Squares problem:

$$RSS_{\lambda}(g) = \sum_{i=1}^n [y_i - g(x_i)]^2 + \lambda \int_a^b [g''(x)]^2 dx.$$

where λ is a smoothing parameter. The first term measures the closeness of the model to the data, while the second term penalizes the roughness/curvature of the function. We solve this problem on $[a, b] = [\min x_i, \max x_i]$ for functions with finite roughness penalty $\int_a^b [g''(x)]^2 dx < \infty$. This is known as a second order Sobolev space.

Note that $\int_a^b [g''(x)]^2 dx$ is called the **roughness penalty**.

From the expression above, we can see that λ is the smoothing parameter that controls the bias-variance trade-off. Therefore, when $\lambda = 0$, we interpolate the data and that leads us to overfitting. When $\lambda = \infty$, then we return to linear least-squares regression. It turns out that the solution to the penalized residual sum of squares has to be a NCS. Indeed,

Theorem

$$\min_g RSS_{\lambda}(g) = \min_{\tilde{g}} RSS_{\lambda}(\tilde{g})$$

} where $\tilde{g}$ is a NCS with knots at the n data points.

Let's call $g(x)$ the as the optimal solution. Since the loss part in $RSS_{\lambda}(g)$ only involves n data points, we can find define a natural cubic spline (NCS) fit that we can call $\tilde{g}(x)$ such that it matches $g(x)$ at the observations $x_i, i = 1, \dots, n$, i.e.

$$g(x_i) = \tilde{g}(x_i), i = 1, \dots, n$$

We can always find such $\tilde{g}$ since our space consists of n basis. Then, we can show that

$$\int g''^2 dx \geq \int \tilde{g}''^2 dx$$

meaning that we will *always* prefer the $\tilde{g}$, the NCS “representation” of g , since the *penalty* is smaller, and the *loss* doesn't change.

4.3.2 Proof

To establish the Theorem, we essentially need to show that

$$\int g''^2 dx \geq \int \tilde{g}''^2 dx$$

Thus, we define $h(x) = g(x) - \tilde{g}(x)$ for which we know that $h(x_i) = 0$ for $i = 1, \dots, n$. Then

$$\int g''^2 dx = \int \tilde{g}''^2 dx + \int h''^2 dx + 2 \int \tilde{g}'' h'' dx$$

and without loss of generality assuming that the x_i s are ordered, we obtain:

$$\begin{aligned} \int \tilde{g}'' h'' dx &= \tilde{g}'' h' \Big|_a^b - \int_a^b h' \tilde{g}^{(3)} dx \\ &= - \sum_{i=1}^{n-1} \tilde{g}^{(3)}(x_j^+) \int_{x_j}^{x_{j+1}} h' dx \quad (\tilde{g}^{(3)} \text{ constant piecewise}) \\ &= - \sum_{i=1}^{n-1} \tilde{g}^{(3)}(x_j^+) (h(x_{j+1}) - h(x_j)) \end{aligned}$$

The second equation is because $\tilde{g}$ is a NCS and therefore we know that it has zero second derivative on the two boundaries a and b . The third equation is true, because $\tilde{g}$ is at most a 3rd order polynomial on any region and as a consequence has constant third derivatives, which we can pull out of the integration. The last equation holds because we said that $h(x) = 0$ on all the observation points x_i . Therefore, this shows that the roughness penalty of our NCS solution is no larger than the best solution g . If we also take into account that $\tilde{g}$ is also in the space $S[a, b]$, then g must be our NCS solution.

4.4 Fitting Smoothing Splines

From the previous discussion, we conclude that g has a **finite sample representation**

$$\hat{g}(x) = \sum_i \beta_i N_i(x)$$

where N_i are a set of NCS basis functions with knots at each of the x *unique* values. Therefore, the penalty function becomes

$$\begin{aligned} \int_a^b g''^2 dx &= \int \left(\sum_i \beta_i N_i''(x) \right)^2 dx \\ &= \sum_{i,j} \beta_i \beta_j \int N_i''(x) N_j''(x) dx \\ &= \beta^T \Omega \beta \end{aligned}$$

where $\Omega_{n \times n}$ with $\Omega_{ij} = \int N_i''(x)N_j''(x)dx$.

Hence our goal is to find β that minimizes

$$RSS_{\lambda}(\beta) = (\mathbf{y} - \mathbf{F}\beta)^T(\mathbf{y} - \mathbf{F}\beta) + \lambda\beta^T\Omega\beta$$

This is a ridge penalized function and the solution is

$$\begin{aligned}\hat{\beta} &= \arg \min_{\beta} RSS_{\lambda}(\beta) \\ &= (\mathbf{F}^T\mathbf{F} + \lambda\Omega)^{-1}\mathbf{F}^T\mathbf{y}\end{aligned}$$

The smoothing spline version of the “hat” matrix is called the **smoother matrix**

$$\hat{\mathbf{y}} = \mathbf{F}(\mathbf{F}^T\mathbf{F} + \lambda\Omega)^{-1}\mathbf{F}^T\mathbf{y} = S_{\lambda}\mathbf{y}$$

Remarks

We have done the analysis of degrees of freedom for ridge type regression. The degrees of freedom of a smoothing spline are

$$df = tr(S_{\lambda})$$

which ranges between 0 and n . Under some special constructions (Demmler and Reinsch, 1975), a basis with double orthogonality property, can lead to

$$\mathbf{F}^T\mathbf{F} = \mathbf{I}, \Omega = diag(d_i)$$

where d_i s are arranged in an increasing order and in addition $d_2 = d_1 = 0$. Using this basis we have

$$\hat{\beta} = (\mathbf{F}^T\mathbf{F} + \lambda\Omega)^{-1}\mathbf{F}^T\mathbf{y} = \left(\mathbf{I} + \lambda diag(d_i)\right)^{-1}\mathbf{F}^T\mathbf{y}$$

i.e.

$$\hat{\beta}_i = \frac{1}{1 + \lambda d_i} \hat{\beta}_i^{LS}$$

Choosing λ

Choosing the penalty λ is the same as in ridge regression.

– Leave-one-out CV:

$$CV(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{g}(x_i)}{1 - S_{\lambda}(i, i)} \right)^2.$$

– Generalized CV:

$$GCV(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{g}(x_i)}{1 - \frac{1}{n}tr(S_{\lambda})} \right)^2.$$

4.5 The Birthrates Example in R

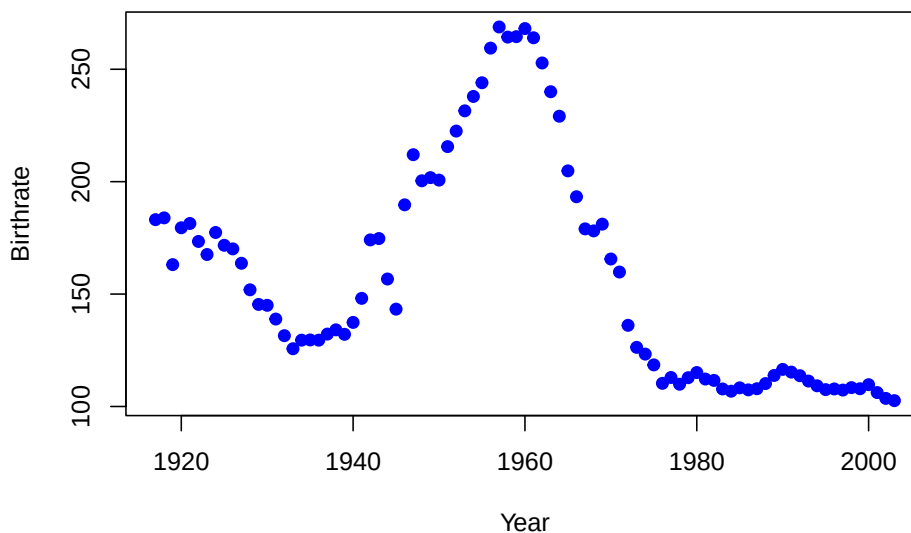
The Birthrate data set contains U.S. birth rate data from 1917 to 2003.

```
birthrates = read.csv("data/week4/birthrates.csv")
head(birthrates)
```

```
##   Year Birthrate
## 1 1917      183.1
## 2 1918      183.9
## 3 1919      163.1
## 4 1920      179.5
## 5 1921      181.4
## 6 1922      173.4
```

Plotting the data we have

```
plot(birthrates, pch = 19, col = "blue")
```



which is an highly nonlinear trend.

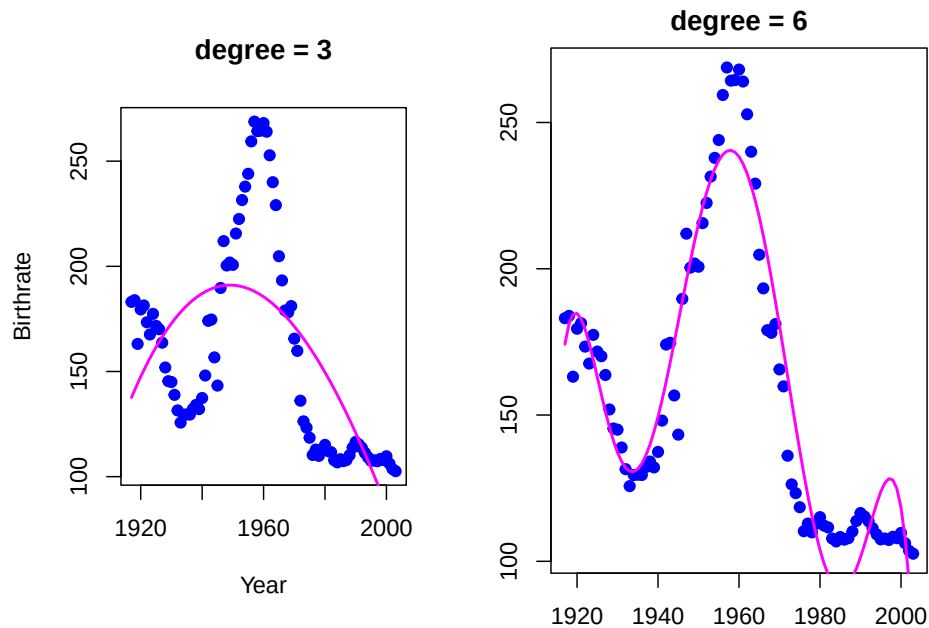
We first try to fit a polynomial regression with orthogonal polynomials for polynomials with degrees 3, and 6. The fitted lines can be shown below:

```
par(mfrow=c(1,2))

poly3.fit <- lm(Birthrate ~ poly(Year, 3), data = birthrates)
plot(birthrates, pch = 19, col = "blue")
lines(birthrates$Year, poly3.fit$fitted.values, lty = 1, col = "magenta", lwd = 2)
title("degree = 3")

par(mar = c(2,3,2,0))
```

```
poly6.fit <- lm(Birthrate ~ poly(Year, 6), data = birthrates)
plot(birthrates, pch = 19, col = "blue")
lines(birthrates$Year, poly6.fit$fitted.values, lty = 1, col = "magenta", lwd = 2)
title("degree = 6")
```



We can see that the 6th order polynomial is better, but it is not very good. We can try fitting local polynomials - piecewise polynomials of lower order (we try constant functions and linear function in the example below):

```
myknots = c(1936, 1960, 1978)

bounds = c(1917, myknots, 2003)

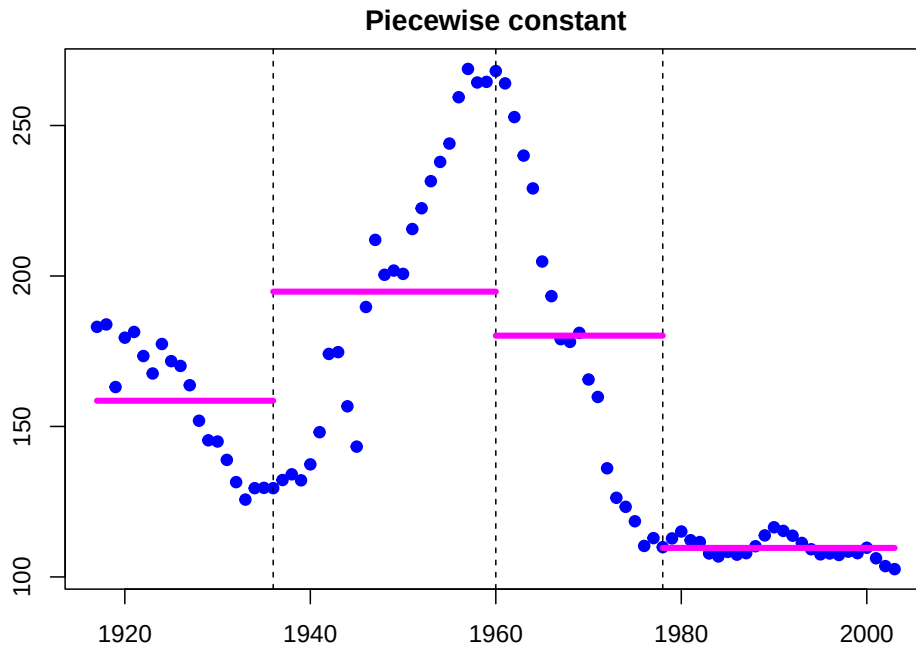
# Piecewise Constant Polynomials
mybasis = cbind("x_1" = (birthrates$Year < myknots[1]),
                "x_2" = (birthrates$Year >= myknots[1])*(birthrates$Year < myknots[2]),
                "x_3" = (birthrates$Year >= myknots[2])*(birthrates$Year < myknots[3]),
                "x_4" = (birthrates$Year >= myknots[3]))

const.fit <- lm(birthrates$Birthrate ~ . -1, data = data.frame(mybasis))
par(mar = c(2,3,2,0))
plot(birthrates, pch = 19, col = "blue")
abline(v = myknots, lty = 2)
title("Piecewise constant")

for (k in 1:4)
```



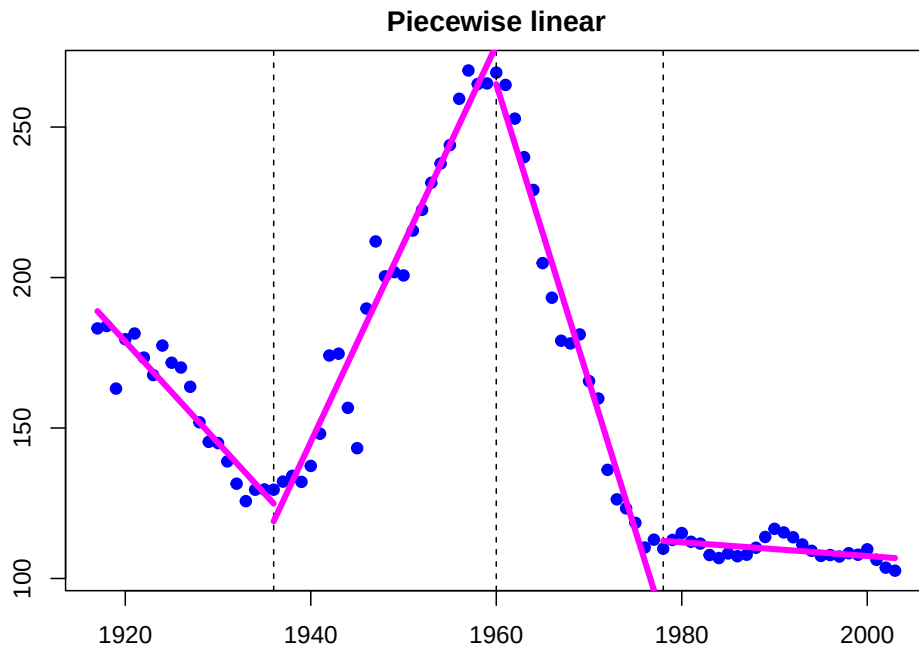
```
points(c(bounds[k], bounds[k+1]), rep(const.fit$coefficients[k], 2), type = "l", lty = 1,
```



```
# Piecewise Linear Polynomials
mybasis = cbind("x_1" = (birthrates$Year < myknots[1]),
               "x_2" = (birthrates$Year >= myknots[1])*(birthrates$Year < myknots[2]),
               "x_3" = (birthrates$Year >= myknots[2])*(birthrates$Year < myknots[3]),
               "x_4" = (birthrates$Year >= myknots[3]),
               "x_11" = birthrates$Year*(birthrates$Year < myknots[1]),
               "x_21" = birthrates$Year*(birthrates$Year >= myknots[1])*(birthrates$Year < myknots[2]),
               "x_31" = birthrates$Year*(birthrates$Year >= myknots[2])*(birthrates$Year < myknots[3]),
               "x_41" = birthrates$Year*(birthrates$Year >= myknots[3]))

line.fit <- lm(birthrates$Birthrate ~ .-1, data = data.frame(mybasis))
par(mar = c(2,3,2,0))
plot(birthrates, pch = 19, col = "blue")
abline(v = myknots, lty = 2)
title("Piecewise linear")

for (k in 1:4)
  points(c(bounds[k], bounds[k+1]), line.fit$coefficients[k] + c(bounds[k], bounds[k+1])*1,
        type = "l", lty = 1, col = "magenta", lwd = 4)
```



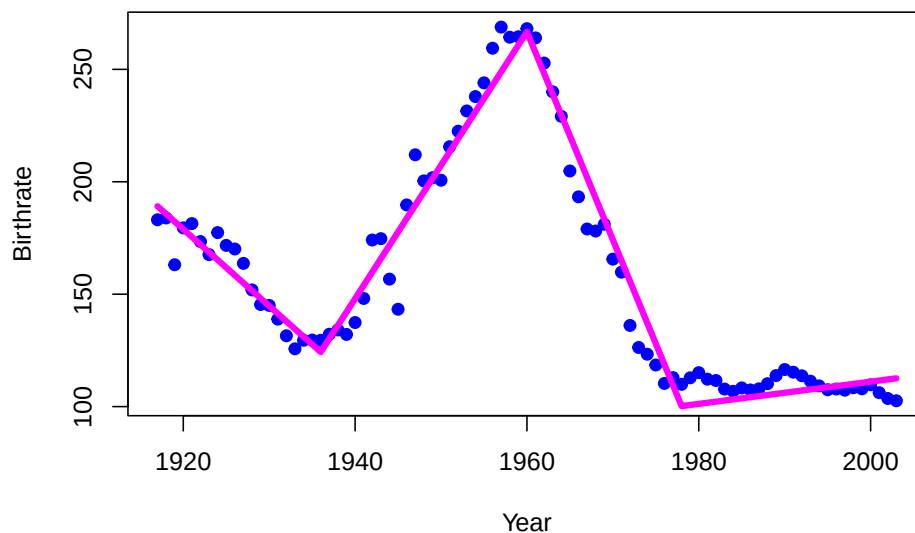
As we discussed in the lecture, it is a localized fit, but there are discontinuities and the overall fit is not good.

4.5.0.1 Splines Regression

Instead, we can use **splines** to fit the data. Let us first try to use the B-Splines. We will first try a **linear spline**

```
bsplines.lin.fit <- lm(Birthrate ~ splines::bs(Year, degree = 1, knots = myknots), data = birthrates)
plot(birthrates, pch = 19, col = "blue")
lines(birthrates$Year, bsplines.lin.fit$fitted.values, lty = 1, col = "magenta", lwd = 2)
title("Linear spline with the bs() function")
```

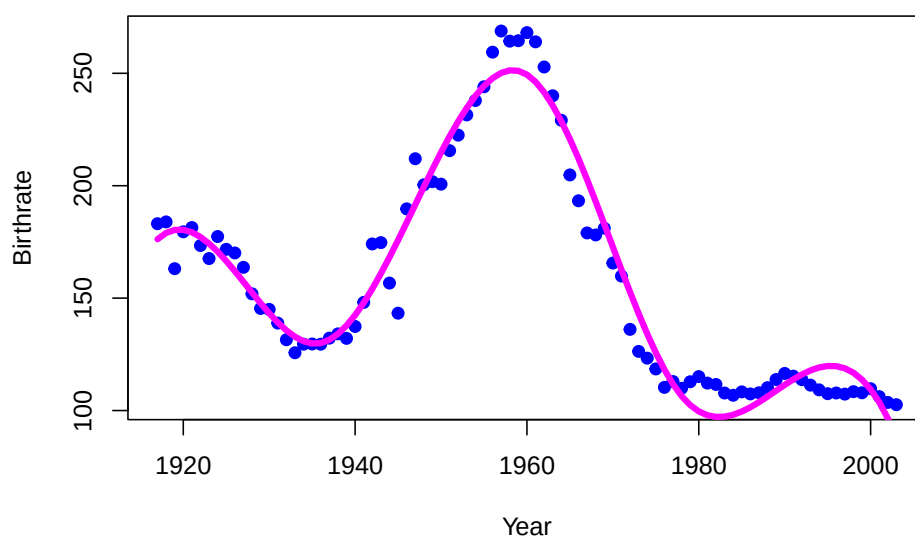
Linear pline with the bs() function



and then a **cubic spline** (as the ones discussed in class)

```
bsplines.3.fit <- lm(Birthrate ~ splines::bs(Year, degree = 3, knots = myknots), data = birthrates)
plot(birthrates, pch = 19, col = "blue")
lines(birthrates$Year, bsplines.3.fit$fitted.values, lty = 1, col = "magenta", lwd = 4)
title("Cubic spline with 3 knots")
```

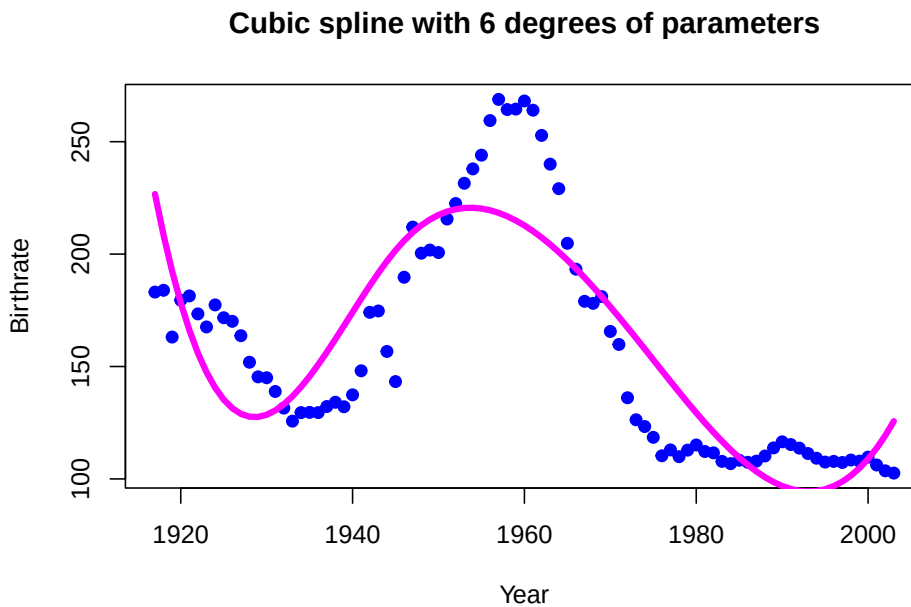
Cubic spline with 3 knots



In the plot below, we used our pre-defined knots. We can alternatively determine

the degrees of freedom in the function `bs`:

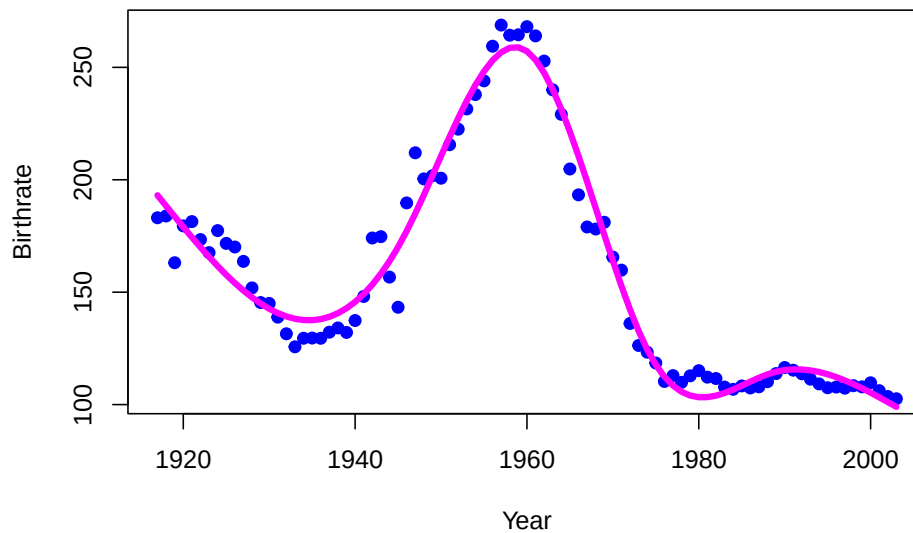
```
bsplines.3.fit.new <- lm(Birthrate ~ splines::bs(Year, df = 5), data = birthrates)
plot(birthrates, pch = 19, col = "blue")
lines(birthrates$Year, bsplines.3.fit.new$fitted.values, lty = 1, col = "magenta",
title("Cubic spline with 6 degrees of parameters"))
```



Let us now try the **Natural Cubic Splines**

```
library(splines)
ns.splines.fit = lm(Birthrate ~ ns(Year, df=6), data=birthrates)
plot(birthrates, pch = 19, col = "blue")
lines(birthrates$Year, ns.splines.fit$fitted.values, lty = 1, col = "magenta", lwd
title("Natural Cubic Splines with df=6"))
```

Natural Cubic Splines with df=6



4.5.0.2 A Simulated Example of Smoothing Splines

In this example, we are going to reproduce the example from the ESL book in section 5.5.2 in which the true function of the data is given by

$$f(x) = \frac{\sin(12(x + 0.2))}{x + 0.2}, x \in [0, 1]$$

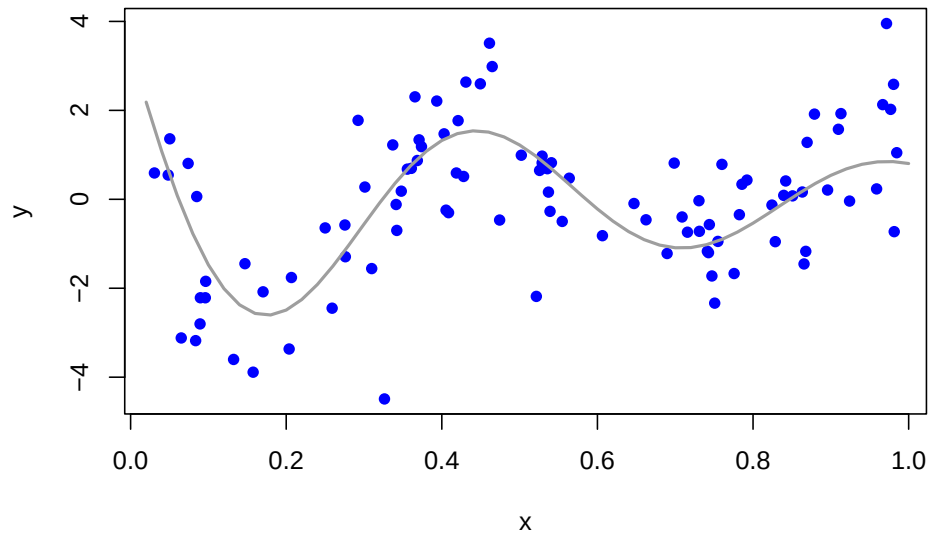
So, we are first going to simulate data from this curve:

```
set.seed(598)
n=100

x = sort(runif(n))
y = sin(12*(x+0.2))/(x+0.2) + rnorm(n, 0, 1)

plot(x, y, col="blue", pch=16 )

funf_x = 1:50/50
funf_y = sin(12*(funf_x+0.2))/(funf_x+0.2)
lines(funf_x, funf_y, col=8, lwd=2)
```



4.5.0.3 Fitting a Smoothing Spline model to the simulated data above

The function to use is `smooth.spline` and is part of the `splines` library in R.

```
library(splines)
?smooth.spline
```

The model is fitted as follows:

```
spline.model = smooth.spline(x, y, df=5)
spline.model
```

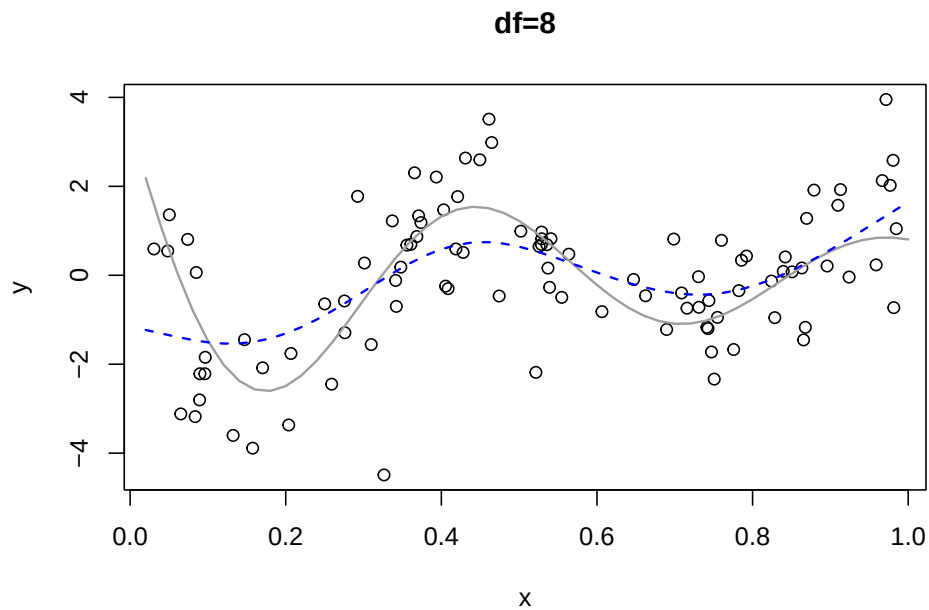
```
## Call:
## smooth.spline(x = x, y = y, df = 5)
##
## Smoothing Parameter spar= 0.9616029 lambda= 0.006048723 (12 iterations)
## Equivalent Degrees of Freedom (Df): 5.000634
## Penalized Criterion (RSS): 162.0406
## GCV: 1.795488
```

We can compute the fitted values as follows:

```
fitted.y = predict(spline.model, funf_x)
```

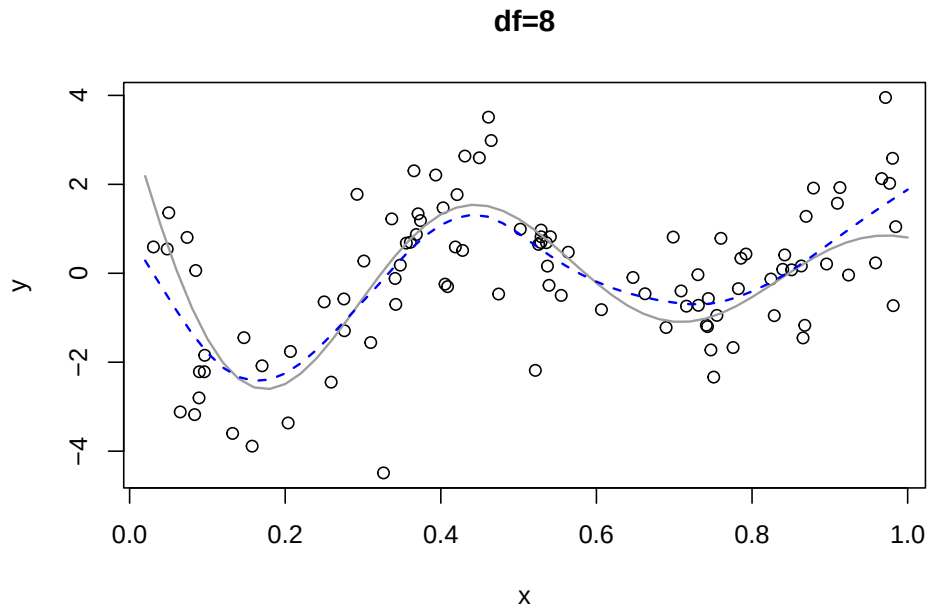
To illustrate the fit of the model, we can plot the fitted line on top of the data:

```
plot(x,y, xlab='x', ylab='y');
lines(funf_x, funf_y, col=8, lwd=1.5);
lines(fitted.y, lty=2, col='blue', lwd=1.5);
title('df=8');
```



The blue line is the true model, while the gray line is the fitted model with 5 degrees of freedom. As we discussed, if we change the degrees of freedom, e.g. $df=8$, then the line will be more/less sensitive to the data. For example for $df=8$ we have

```
plot(x,y, xlab='x', ylab='y');
lines(funf_x, funf_y, col=8, lwd=1.5);
lines(predict(smooth.spline(x, y, df=8), funf_x), lty=2, col='blue', lwd=1.5);
title('df=8');
```



4.5.0.4 Choice of Lambda

Many R packages come with an inbuilt option for determining lambda, primarily based on leave-one-out cross-validation (LOOCV) and generalized cross-validation (GCV).

There's no need to specify lambda directly. Instead, we indicate the desired degrees of freedom (df), ranging from 0 to n. R then determines the appropriate lambda. We can then consult both CV and GCV curves to decide the best lambda or df.

```
model.fit = smooth.spline(x, y, df=9);
```

When using the `smooth.spline` function with a degree of freedom set at 9, the returned value `model.fit$df` may have a slight deviation due to rounding errors.

The leverage output is equivalent to the diagonal entries of the smoothing matrix.

```
model.fit$df
```

```
## [1] 9.001417
```

```
model.fit$lev # leverage = diagonal entries of the smoother matrix
```

```
## [1] 0.30893986 0.16530188 0.15452511 0.10279673 0.08811473 0.08173143
## [7] 0.08148178 0.08160942 0.08169108 0.08413540 0.08441745 0.11256266
## [13] 0.12120898 0.12602066 0.13175493 0.13750648 0.13745737 0.11875674
```



```
## [19] 0.11154402 0.10015615 0.09976906 0.08986948 0.08508903 0.08020005
## [25] 0.07203883 0.06701137 0.06528799 0.06497591 0.06309976 0.06147779
## [31] 0.06087571 0.06065672 0.06070371 0.06080451 0.06105153 0.06369195
## [37] 0.06523756 0.06573835 0.06643670 0.06911934 0.06984682 0.07231683
## [43] 0.07347632 0.08032413 0.08330237 0.08402559 0.08427316 0.07806223
## [49] 0.07013561 0.06947727 0.06946921 0.06948635 0.06949033 0.07114094
## [55] 0.07177155 0.07287487 0.07400078 0.08729462 0.09996053 0.14342850
## [61] 0.14982974 0.13605679 0.10479241 0.09474069 0.08413352 0.07762692
## [67] 0.06748810 0.06720784 0.06346315 0.06318782 0.06301969 0.06275533
## [73] 0.06281727 0.06333844 0.06451613 0.06999234 0.07288373 0.07423106
## [79] 0.07646914 0.07952440 0.07869513 0.07674170 0.07634088 0.07528017
## [85] 0.07601450 0.07640336 0.07689899 0.07734393 0.08037947 0.08796840
## [91] 0.09303429 0.09391332 0.09611884 0.10022513 0.10988799 0.12007621
## [97] 0.13650369 0.15233032 0.15631587 0.17386379
```

```
sum(model.fit$lev)
```

```
## [1] 9.001417
```

By default, the function offers a GCV score, but you can alter it to LOOCV for comparison.

```
model.fit$cv # default: GCV
```

```
## [1] FALSE
```

```
sum((y-model.fit$y)^2)/(1-model.fit$df/n)^2/n
```

```
## [1] 1.577447
```

```
fit=smooth.spline(x, y, df=9, cv=T) # set 'cv=T' to return CV
fit$cv
```

```
## [1] TRUE
```

```
sum(((y-fit$y)/(1-fit$lev))^2)/n
```

```
## [1] 1.58123
```


Chapter 5

Regression Trees

In this section, we are going to discuss **tree-based** models for **regression**.

Framework

- Response Variable: $Y \in \mathbb{R}$ – continuous
- Features: $X = (X_1, X_2, \dots, X_p)$

Regression trees are a *non-linear* and *non-parametric* regression model. As we are going to discover, tree-based methods are non-parametric methods that *recursively* partition the feature space into hyper-rectangular subsets, and make predictions on each subset. At the same time, they are non-parametric since they do not assume a fixed functional relationship between the response and the set of predictors. Instead the complexity of the model is controlled by hyper-parameters which make them quite flexible as models.

Below we can see a sketch of a regression tree for a **wines** data set in which we want to predict the *quality* of the wine (a numerical outcome) versus *wine characteristics* (a set of predictors):

We will restrict our focus to *binary* partitions of the tree. In the example above, we first split the space in **two** regions, and model the response by the mean of Y in each region. *At each step*, we choose the variable and split-point to achieve the best fit. Then, one or both regions are split into two more regions, and this process is continued, until *some* stopping rule is applied. Specifically in our example, the first split rule is whether **alcohol** is less/greater than 10.85. Then, the region to the left for which $\text{alcohol} < 10.85$ is split based on **volatile.acidity** and value 0.2525, and the region to the right is split based on **alcohol** again and value 11.7417, and so on and so forth. When we stop, the predicted **quality** for the wine is the average of all the points in the respective region.

Terminology

- **Leaf Node:** Each leaf node corresponds to a rectangular-shaped region in the *feature space*.
- **Internal Node:** These nodes are parent nodes with two children – left and right nodes.
- A **split** involves a variable j and a particular split value s .

Due to their flexibility tree-based methods are easy to interpret. They are also *invariant* under any monotone transformation of predictors. The tree-based algorithms used to fit the trees also allow us to handle high-dimensional data without modifying the algorithm, and they are able to incorporate interactions between variables automatically and perform variable selection tasks.

Single-tree models do not have very high prediction accuracy which is why ensemble tree models have been introduced.

5.1 Single-Tree Building

As we discussed to build a single tree, one needs to recursively partition the predictor space. So, we start by **initializing the root** node and starting with all training data. Then, we choose a **splitting rule** consisting of a predictor $X^{(j)}$ and a constant c that we use to split the data in the node:

$$\mathbf{1}_{\{X^{(j)} \leq c\}}$$

Then, we recursively apply the procedure on each child node:

We denote a node by (t) , the children nodes by (t_L, t_R) . The split will consist of the pair (var j , value s), and in the end we have the leaves or terminal nodes. When we stop, we **predict** each terminal node using within-node data. Every leaf node (i.e. a rectangle region $R_m \in \mathcal{X}$) is assigned a constant for regression tree

$$\hat{f}(X) = \sum_m c_m \mathbf{1}_{\{X \in R_m\}}$$

So, the natural questions that arise are the following:

1. **Where** to split the tree?
2. **When** to stop?
3. **How** to predict the response at each leaf node?

5.1.1 Prediction at Leaf Nodes

As we saw before, each leaf node corresponds to region R_m that contains some samples. The **prediction** for a leaf node is *defined* to be the *average of the response variable Y* for the samples in R_m , $m = 1, \dots, M$:

$$\hat{f}(X) = \sum_{m=1}^M c_m \mathbf{1}_{\{X \in R_m\}}$$

In the expression above, we need to determine the constants c_m . So, we minimize the expression with respect to c_m :

$$\min_{c_m} \sum_{i=1, x_i \in R_m}^n (y_i - c_m)^2$$

Thus, we obtain c_m **to be the average of y_i s that corresponds to $x_i \in R_m$.**

5.1.2 Split Criterion

Finding the **best partition** in terms of minimum sum of squares is computationally infeasible which means that we need to use a greedy algorithm find the minimum. So, we start with all the data at the root, and the using a split variable j and a split point s , we obtain the following two half planes:

$$R_1(j, s) = \{X|X_j \leq s\} \quad \text{and} \quad R_2(j, s) = \{X|X_j > s\}$$

Then, we need to find the splitting variable j and split point s that solve

$$\min_{j, s} \left[\min_{c_1} \sum_{x_i \in R_1(j, s)} (y_i - c_1)^2 + \min_{c_2} \sum_{x_i \in R_2(j, s)} (y_i - c_2)^2 \right]$$

For any choice of j and s , the inner minimization is solved by:

$$\hat{c}_1 = \text{ave}(y_i | x_i \in R_1(j, s)) \quad \text{and} \quad \hat{c}_2 = \text{ave}(y_i | x_i \in R_2(j, s))$$

For each splitting variable, the determination of the split point s can be done quickly. So, the determination of the best pair is feasible.

The process we discuss above can be summarized in the following way: for each split, we define a *split criterion* $\Phi(j, s)$ e.g., we may consider the **deduction of RSS** for regression. By construction, regression trees are built in a *top-down greedy fashion*:

- Start with the *root* and consider
 - a. all possible variables $j = 1, \dots, p$ and
 - b. all possible split values,
 - c. and pick the best split, that is the one having the best Φ value.
- Now, the *data are divided into the left node and right node*.
- Repeat this procedure in each node.

Split Criterion $\Phi(j, s)$

Deduction in RSS, if we split samples at node t into t_R and t_L :

$$\Phi(j, s) = RSS(t) - [RSS(t_R) + RSS(t_L)]$$

where

$$RSS(t) = \sum_{x_i \in t} (y_i - c_t)^2, \quad \text{where} \quad c_t = \text{AVE}\{y_i : x_i \in t\}$$

Note that $\Phi(j, s)$ is always positive if we split the data into two groups (even randomly), unless the mean of the left node and the one of right node are the same.

Categorical Predictors

If we have a categorical predictor with m levels, then this can be naturally handled by the tree model. Having m levels implies that there will be $2^{m-1} - 1$ possible partitions of the m labels into *two* groups leading to a simplified computation for regression with a *squared error loss*:

- Order the m levels by their mean values of Y ,
- Split the categorical variable as if it were an ordered predictor, so there will only be $(m - 1)$ potential splits.

Missing Predictor Values

As we discussed in the beginning, tree-based model algorithms can also handle missing values automatically. In general, if we discard observations with missing values, this will lead to a serious depletion of the training set. However, we can start by ignoring missing value observations, and determine the split criteria using only non-missing observations. Once a split (j, s) is found, we can “fill-in” the missing values. This can be done, for example, using imputation techniques.

Alternatively, we can find *surrogate* variables that can predict the binary outcome “ $X_j < s$ ” and “ $X_j \geq s$ ” using a *one-split tree*. We rank those surrogate variables along with the blind rule “go with majority”. Any observation that is missing X_j is then classified with the first surrogate variable, or if missing that, the second surrogate variable (or the blind rule) is used, and etc.

5.1.3 Single-Tree Pruning

So far we discussed how to predict the response in the leaves of the tree and we developed a split criterion. Thus, it remains to decide *how and when we stop* which also determines the size of the tree.

To clarify, when we refer to the “*size of a tree*” we refer to the *number of splits* a tree has. A large tree (with many splits) can easily over-fit the data, but on the other hand it may capture important structures that a smaller tree might miss. So, intuitively balancing the tree size and accuracy resembles the “loss + penalty” framework.

One possible approach is to split tree nodes only if the decrease in the loss (e.g. RSS) exceeds certain threshold, however this can be short-sighted, since

what might seem as a useless split at this step might lead to a very good split below. So, we need to develop a *different strategy*: We construct a large tree which we then **prune**.

We start by growing a very large tree $T_{\max}$:

- until all terminal nodes are nearly pure, that is the variability of the points in that node is small.
- or when the number of data in each terminal node is less than certain threshold;
- or when the tree reaches a certain size.

The size of the initial tree is not critical as long as the tree is *sufficiently large*.

Notation

Subtree $T' \prec T$: any tree that can be obtained by pruning T , i.e. by collapsing all its internal nodes. Branch at node (t) : T_t

For any subtree $T \prec T_{\max}$ define the **complexity cost**

$$R_\alpha(T) = R(T) + \alpha|T|,$$

where $R(T)$ is the RSS for the regression tree T , $|T|$ is the tree size, i.e. the number of leaf nodes, and $\alpha > 0$ is the (**penalty**) cost of adding a split. Then, our goal is to find the value of α that minimizes the complexity cost. Or, in other words, we want to pick the best subtree that minimizes the cost

$$\begin{aligned} T(\alpha) &= \arg \min_{T \preceq T_{\max}} R_\alpha(T) \\ &= \arg \min_{T \preceq T_{\max}} [R(T) + \alpha|T|] \end{aligned}$$

The **Optimal subtree** $T^*(\alpha)$ is defined to be the smallest one among all $T(\alpha)$'s, that is

$$\begin{aligned} R_\alpha(T^*(\alpha)) &= \min_{T \preceq T_{\max}} R_\alpha(T) \\ T^*(\alpha) &\preceq \text{any } T(\alpha). \end{aligned}$$

where $T^*(\alpha)$ is unique.

Note that for a pair of leaf nodes (t_L, t_R) , there exists α^* , such that

- For any $\alpha \geq \alpha^*$, we would like to collapse them to just node t ;
- For any $\alpha \leq \alpha^*$, we prefer to keep the two leaf nodes.

That is, α^* is *the maximal price we would like to pay to keep that split*. In a similar way, we can extend this calculation to compute *the maximal price we would like to pay to keep a branch* T_t . For any non-leaf node t , we need to compute the maximal price we would like to pay for keeping the whole branch T_t , focusing only on samples at node t . So,

- The cost for keeping a branch T_t :

$$R_\alpha(T_t) = R(T_t) + \alpha|T_t|$$

- The cost for cutting a branch T_t :

$$R_\alpha(\{t\}) = R(\{t\}) + \alpha$$

So, we calculate the difference between the two costs:

$$\alpha^* = \frac{R(\{t\}) - R(T_t)}{|T_t| - 1}$$

This implies that if the given $\alpha > \alpha^*$, then it is too expensive to keep this branch and we would like to cut the whole branch and make t a leaf node.

The main disadvantage of this approach is that if we need to search all possible sub-trees of $T_{\max}$, then this is computationally intense. However, this is not necessary. Below we introduce the *Weakest Link Pruning* algorithm that finds and removes the least important branches of a large tree step-by-step to produce a sequence of smaller simpler trees. Specifically:

Weakest Link Pruning Algorithm

1. *Start* with $T_0 = T_{\max}$ and $\alpha_0 = 0$.
2. For any non-leaf node t ,

$$\alpha(t) = \text{the maximal price you would pay to keep } T_t$$

3. $\alpha_1 = \min_t \alpha(t)$ The corresponding non-terminal node t_1 is called the **weakest link**.
4. *Cut* the branch at t_1 .
5. *Update* the maximal price for each non-leaf node (we only need to recompute the maximal price for nodes that are the parents/grandparents of t_1).
6. *Find* α_2 and cut the branch at the 2nd weakest link.
7. *Keep* doing this until we get to the root.

This algorithm generates a **Solution Path** in the following sense:

$$T_{\max} = T_0 \succ T^*(\alpha_1) \succ T^*(\alpha_2) \succ \dots \succ \{\text{root node}\}$$

$$0 = \alpha_0 < \alpha_1 < \alpha_2 < \dots$$

All possible values of α are grouped into $(m + 1)$ intervals

$$\begin{aligned} I_0 &= [0, \alpha_1) \\ I_1 &= [\alpha_1, \alpha_2) \\ &\vdots \\ I_m &= [\alpha_m, \infty) \end{aligned}$$

where all $\alpha \in I_i$ share the same optimal subtree $T^*(\alpha_i)$.

Tuning of the hyper-parameter α

We can use K -Fold Cross validation to tune α – the hyper-parameter in the pruning algorithm above. Therefore, we can start by fitting a very large tree $T_{\max}$ and compute $I_0, I_1, \dots, I_m$ as above. Then, we can set

$$\begin{aligned} \beta_0 &= 0 \\ \beta_1 &= \sqrt{\alpha_1 \alpha_2} \\ &\vdots \\ \beta_{m-1} &= \sqrt{\alpha_{m-1} \alpha_m} \\ \beta_m &= \infty \end{aligned}$$

where each β_j is a typical value for its interval I_j .

To perform the K for cross-validation, we divide data into K groups and repeat $k = 1, \dots, K$. On each data set, except for the k -th group, we fit a full model and determine the optimal subtrees:

$$T_0 \succ T^*(\beta_1) \succ \dots \succ T^*(\beta_m) \succ \{\text{root node}\}$$

We then compute the prediction error on the k th group for each of the tree models. We finish by producing the CV plot over different α values and pick the optimal α via $\alpha_{\min}$ or α_{1se} .

5.2 Regression Forests

Regression forests naturally extend the idea of regression trees by combining many trees into an **ensemble**. Each tree is trained on a randomly sampled subset of data features, which reduced correlation among trees and mitigates the overfitting of a single large tree. The final prediction is obtained by averaging out the outputs of the individual trees, leading to a higher accuracy, stability and generalization performance.

5.2.1 Weak and Strong Learners

In our discussion of single-tree models, we highlighted their main drawbacks: they are often unstable – meaning that small changes in the data can lead

to very different trees – and they tend to lack accuracy. For this reason they are considered **weak learners**.

A natural idea, widely explored in the literature, is to combine many weak learners to form a **strong learner**. Examples of such ensemble methods include bagging, boosting, and random forests.

Bagging and random forests construct multiple trees under different *random perturbations*, and *average* their predictions, while boosting takes a different approach; it *progressively* trains simple models of small capacity, and then *adds* them together.

Conceptually, an **ensemble** of trees is as follows:

5.2.2 Bootstrap Aggregation (Bagging)

Bagging (i.e. for bootstrap aggregation) is an ensemble method designed to reduce the variance of a predictive model—particularly unstable ones. It works by repeatedly *resampling* the training data *with replacement*, fitting a model on each bootstrap sample, and then averaging their predictions (*for regression*) or taking a majority vote (*for classification*).

Consider having Training Data denoted by $\mathbf{Z} = \{(x_i, y_i)_{i=1}^n\}$. Then, we can obtain **bootstrap samples with replacement** $\mathbf{Z}^{*b} = \{(x_i^{*b}, y_i^{*b})_{i=1}^n\}$, where $b = 1, \dots, B$, that is

$$\begin{aligned} \mathbf{Z}^{*1} &: (x_1^{*1}, y_1^{*1}), (x_2^{*1}, y_2^{*1}), \dots, (x_n^{*1}, y_n^{*1}) \\ \mathbf{Z}^{*2} &: (x_1^{*2}, y_1^{*2}), (x_2^{*2}, y_2^{*2}), \dots, (x_n^{*2}, y_n^{*2}) \\ &\vdots \\ \mathbf{Z}^{*B} &: (x_1^{*B}, y_1^{*B}), (x_2^{*B}, y_2^{*B}), \dots, (x_n^{*B}, y_n^{*B}) \end{aligned}$$

Then, the regression (or classification) function, $\hat{f}^{*b}$, is trained on each sample $\mathbf{Z}^{*b}$ and the **Bagging Estimate** is defined to be

$$\hat{f}_{bag}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{*b}(x)$$

Because bagging relies on averaging the predictions of many individual trees, it can substantially reduce the variance of otherwise unstable learners like trees, thus improving predictive performance. However, the simple and interpretable structure of a single tree is lost once it is embedded in an ensemble. Moreover, if the individual trees are highly correlated, the variance reduction achieved through bagging may be limited.

5.2.3 Random Forests

The development of random forests can be seen as the convergence of several ideas that emerged in the 1990s. One major breakthrough was Breiman's introduction of bagging (bootstrap aggregation) that we discussed above, which demonstrated that averaging unstable predictors like trees could dramatically improve their performance by reducing variance. Around the same time, Ho proposed the random subspace method (1998), showing that further *diversity* among trees could be achieved by training each one on a randomly selected subset of features. This insight highlighted that injecting randomness into feature selection, not just the data, could strengthen ensembles of trees. Breiman built on both of these advances when he introduced **random forests**: a method that combines *bagging* with *random feature selection* at every split. By doing so, random forests create a large collection of de-correlated trees whose averaged predictions are *both accurate and stable*. This synthesis of bagging and random subspaces represents a turning point in ensemble learning, establishing random forests as one of the most powerful and widely used machine learning methods.

The random forest algorithm can be described in a few key steps:

1. Bootstrap Sampling (Bagging): For each tree in the forest, a bootstrap sample of the training data is drawn – this introduces variability among the trees.
2. Tree Construction with Random Feature Selection: Each tree is grown on its bootstrap sample, but with an important twist: at each split in the tree, only a randomly chosen subset of features is considered as candidates for the split – this further decorrelates the trees and enhances the variance reduction achieved by averaging.
3. Tree Growth: Typically, the trees in a random forest are grown deep (without pruning), allowing them to capture complex patterns in the data. Overfitting of individual trees is not a concern, since averaging across many trees stabilizes the final model.
4. Prediction: For regression, the forest's prediction is the average of the predictions from all trees.
5. Out-of-Bag (OOB) Error Estimation: Since each bootstrap sample leaves out about one-third of the training data, these “out-of-bag” observations can be used to estimate prediction error, providing a built-in form of cross-validation.

Sketch of the Algorithm

For $b = 1, \dots, B$

1. Draw a bootstrap sample of size n $\mathbf{Z}^{*b}$ from the training data.
2. Grow a BIG tree T_b by recursively repeating the following steps for each terminal node of the tree, until the minimum node size is reached:

1. Select m variables at random from the p variables.
2. Pick the best variable/split-point among the m .
3. Split the node into two children nodes.

Output the forest, i.e. *ensemble* of trees: $\{T_b\}_{b=1}^B$

Predict at a new point x (Regression case)

$$\hat{f}^B(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

Restrictions

When constructing each tree in the forest, at every split, we consider a subset of m features is chosen at random from the total of p available features. The algorithm then determines the best split using only this subset of m features. A common rule of thumb is to set $m = \sqrt{p}$ for classification and $m = p/3$ for regression.

By restricting each split to a random subset of features, the trees are less likely to all rely on the same strong predictors, thereby reducing correlation among trees and increasing the overall effectiveness of the ensemble.

5.2.4 Out-of-Bag (OOB) Samples

Out-of-bag (OOB) samples are the observations that do not appear in the bootstrap dataset Z^{*b} and are therefore not used to build the corresponding tree T_b . These left-out samples provide a natural test set for evaluating T_b . In practice, the OOB error is computed by aggregating predictions for each observation using only the trees where that observation was excluded from training. Implementations such as `randomForest` report predictions and error rates based on this OOB evaluation. The resulting error estimate is typically very close to what would be obtained through cross-validation.

5.2.4.1 Variable Importance

Variable Importance in Random Forests can be assessed in two main ways. Based on:

1. Reduction in RSS (or Impurity): At each split, the algorithm records the improvement in the residual sum of squares (RSS) that results from splitting on a given variable. This improvement is attributed to the splitting variable. For each variable, these improvements are accumulated over all splits in a tree, and then averaged across all trees in the forest. The result reflects how much each variable contributes to reducing prediction error through the splitting process.

2. Permutation of Predictors: For each tree T_b , compute its prediction error on the out-of-bag (OOB) samples (mean squared error for regression, classification error for classification). Next, randomly permute the values of the j th predictor within the OOB data and recalculate the prediction error. The increase in error caused by this permutation indicates the importance of that variable. This difference is averaged across all trees and, to account for variability, further normalized by the corresponding standard deviation.

5.2.5 Boosting Trees

Boosting is an ensemble learning technique that combines multiple weak learners, such as trees, into a single, strong predictive model. Unlike bagging or random forests, which grow trees independently, boosting builds trees **sequentially**, with each new tree focused on correcting the errors of the previous ensemble.

For example, if we consider an additive model,

$$F(x) = f_1(x) + f_2(x) + \dots + f_{T-1}(x) + f_T(x)$$

it is difficult to solve for all f_t s. Instead we solve it using a forward stagewise greedy algorithm.

Boosting leads to very high predictive accuracy, often outperforming bagging methods, and it is flexible and capable of capturing complex nonlinear relationships. However, it is more sensitive to noise and outliers compared to bagging methods, and it requires careful tuning of hyperparameters to avoid overfitting.

5.3 Tree-Based Regression Models in R

```
library(ggplot2)
library(rpart)

##
## Attaching package: 'rpart'

## The following object is masked from 'package:faraway':
##
##      solder

library(rpart.plot)
library(tree)
```

The `whitewines.csv` data set contains information related to white variants of the Portuguese "Vinho Verde" wine.

 Specifically, we have recorded the following information: (a)fixed acidity, (b)volatile acidity, (c)citric acid, (d)residual sugar, (e)chlorides, (f)free sulfur dioxide, (g)total sulfur dioxide, (h)density, (i)pH, (j)sulphates,

(k)alcohol, (l)quality' (score between 0 and 10) The data set can be downloaded here: [whitewines.csv](#)

```
whitewines = read.table("data/week6/whitewines.csv", sep=";", header=TRUE)
head(whitewines)

##   fixed.acidity volatile.acidity citric.acid residual.sugar chlorides
## 1           7.0           0.27         0.36           20.7       0.045
## 2           6.3           0.30         0.34            1.6       0.049
## 3           8.1           0.28         0.40            6.9       0.050
## 4           7.2           0.23         0.32            8.5       0.058
## 5           7.2           0.23         0.32            8.5       0.058
## 6           8.1           0.28         0.40            6.9       0.050
##   free.sulfur.dioxide total.sulfur.dioxide density    pH sulphates alcohol
## 1                  45                  170  1.0010  3.00      0.45      8.8
## 2                  14                  132  0.9940  3.30      0.49      9.5
## 3                  30                   97  0.9951  3.26      0.44     10.1
## 4                  47                  186  0.9956  3.19      0.40      9.9
## 5                  47                  186  0.9956  3.19      0.40      9.9
## 6                  30                   97  0.9951  3.26      0.44     10.1
##   quality
## 1        6
## 2        6
## 3        6
## 4        6
## 5        6
## 6        6
dim(whitewines)

## [1] 4898   12
```

In R there are two packages that fit **tree** models: **tree** and **rpart**. The first package we only used it to generate the first figures in the slides via its function **partition.tree**. Below is the code that generates the plots in the lectures.

The tree Package

The syntax for fitting a regression tree with **tree** is similar to that used for linear regression models. For illustration purposes, let's focus on just two predictors: **alcohol** and **volatile.acidity** to predict **quality**.

We start by fitting the tree:

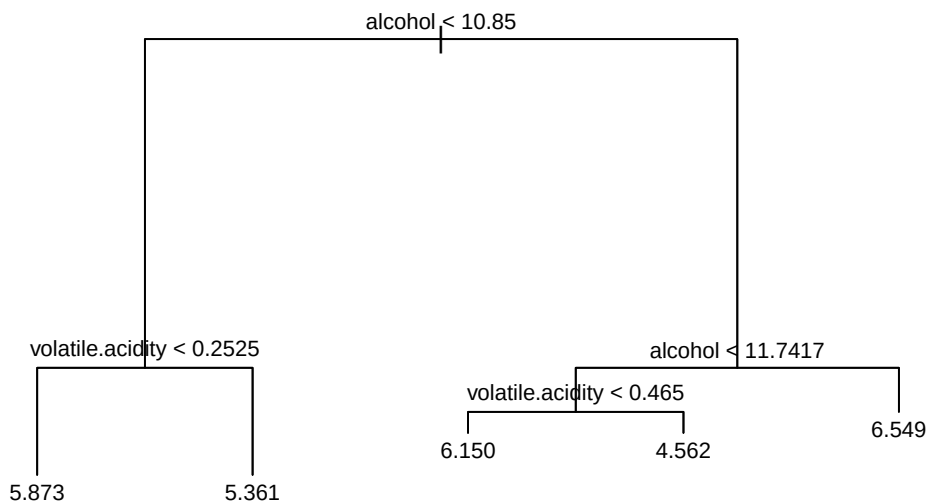
```
quality.tree = tree(quality ~ alcohol + volatile.acidity, data=whitewines)
quality.tree

## node), split, n, deviance, yval
##      * denotes terminal node
```

```
##
## 1) root 4898 3841.00 5.878
##    2) alcohol < 10.85 3085 1845.00 5.606
##      4) volatile.acidity < 0.2525 1475 868.00 5.873 *
##      5) volatile.acidity > 0.2525 1610 775.30 5.361 *
##    3) alcohol > 10.85 1813 1378.00 6.341
##      6) alcohol < 11.7417 881 670.00 6.121
##      12) volatile.acidity < 0.465 865 616.50 6.150 *
##      13) volatile.acidity > 0.465 16 13.94 4.562 *
##      7) alcohol > 11.7417 932 624.70 6.549 *
```

To illustrate the tree structure, we plot the fitted tree model:

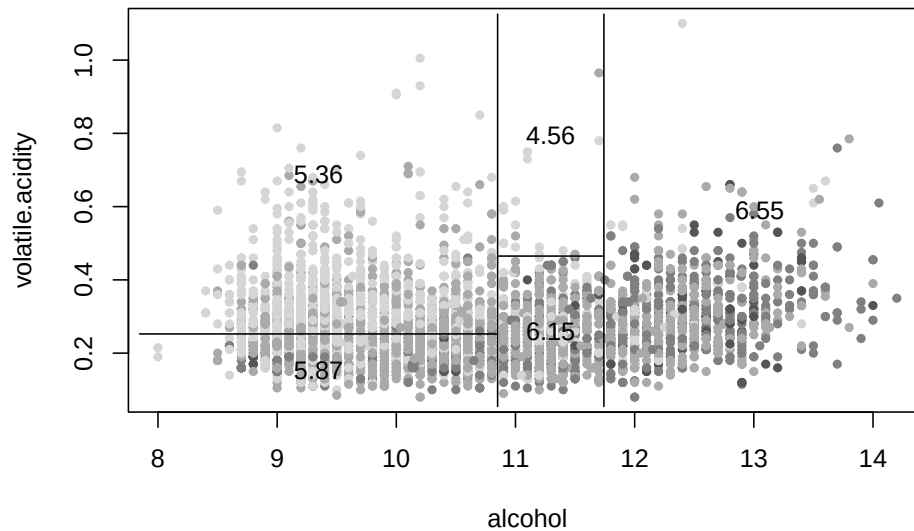
```
plot(quality.tree)
text(quality.tree, cex = .75)
```



and create the rectangular regions (as in the lecture), we have:

```
quality.quantiles = cut(whitewines$quality, unique(quantile(whitewines$quality, 0: 5 / 5)),
  include.lowest = TRUE)

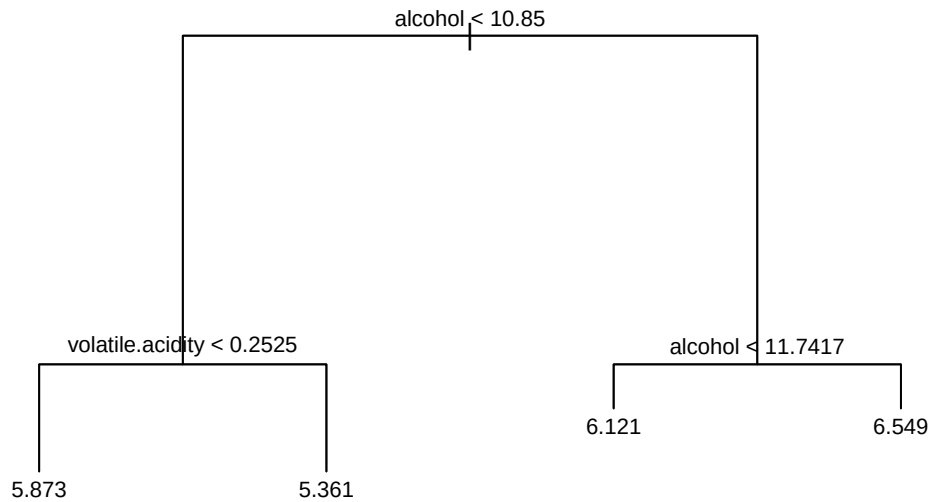
plot(whitewines$alcohol, whitewines$volatile.acidity, col = grey(5: 1 / 6)[quality.quantiles],
  pch = 20, ylab = "volatile.acidity", xlab = "alcohol")
partition.tree(quality.tree, ordvars = c("alcohol", "volatile.acidity"), add = TRUE)
```



If we want, we can also prune the tree using the `prune.tree` function:

```
smaller.tree = prune.tree(quality.tree, best=4)
smaller.tree
```

```
## node), split, n, deviance, yval
##      * denotes terminal node
##
## 1) root 4898 3841.0 5.878
##    2) alcohol < 10.85 3085 1845.0 5.606
##      4) volatile.acidity < 0.2525 1475 868.0 5.873 *
##      5) volatile.acidity > 0.2525 1610 775.3 5.361 *
##    3) alcohol > 10.85 1813 1378.0 6.341
##      6) alcohol < 11.7417 881 670.0 6.121 *
##      7) alcohol > 11.7417 932 624.7 6.549 *
##
plot(smaller.tree)
text(smaller.tree, cex = .75)
```

We now detach the `tree` package and switch to the `rpart` package.

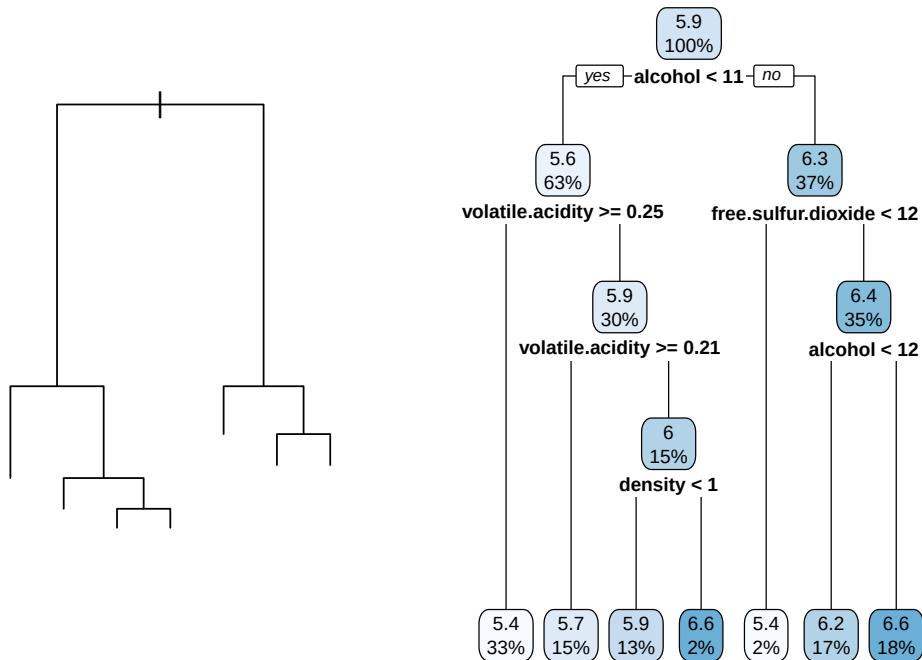
```
detach("package:tree")
```

The `rpart` Package

Fitting a Regression Tree

The syntax for fitting a regression tree with `rpart` to the one we use for linear regression models. The default tree plot may not be visually appealing, so for better visualization, we can use the `rpart.plot` package.

```
set.seed(598)
wine.tree1 = rpart(quality ~ ., data = whitewines)
par(mfrow = c(1, 2))
plot(wine.tree1)
rpart.plot(wine.tree1)
```



Note that in this example (unlike the previous one), we use all predictors not just two.

Understanding the Fitted Tree

Starting at the root node, the average value for all observations is **5.9**, representing 100% of the data points. The *first split* occurs based on the variable **alcohol**. The left node contains roughly 63% of the samples with an average of **5.6**, while the right node contains the remaining 37% with an average of **6.3**. Leaf nodes appear at the bottom, and their color shading indicates the magnitude of their predictions.

Cross-Validation

Cross-validation is automatically performed when using the `rpart` command. To inspect the cross-validation for pruning, use the `printcp` command, which provides a table.

CP stands for **Complexity Parameter** which is a scaled version of α – the *price* or *penalty* we pay for keeping a split.

```
printcp(wine.tree1)
```

```
##
## Regression tree:
## rpart(formula = quality ~ ., data = whitewines)
##
```

```
## Variables actually used in tree construction:
## [1] alcohol          density          free.sulfur.dioxide
## [4] volatile.acidity
##
## Root node error: 3841/4898 = 0.7842
##
## n= 4898
##
##          CP nsplit rel error  xerror    xstd
## 1 0.161007      0   1.00000 1.00030 0.021275
## 2 0.052469      1   0.83899 0.84260 0.020094
## 3 0.027342      2   0.78652 0.79453 0.019601
## 4 0.017711      3   0.75918 0.76429 0.018478
## 5 0.010355      4   0.74147 0.75134 0.018065
## 6 0.010000      6   0.72076 0.74656 0.017995
```

In the output above: `nsplit` means the number of splits, which implies that the number of leaf nodes (or size of a tree) is equal to `nsplit + 1`. `rel error` and `xerror` are the training error (RSS) and CV error, but these values are relative to a reference training error at the root node. Recall that the connection between α and CP is

$$RSS(T) = \alpha|T|, \frac{RSS(T)}{RSS(root)} + CP \cdot |T|$$

Pruning

We can employ the `prune` command to obtain the optimal subtree by providing specific Complexity Parameter (CP) values. As an example:

```
prune(wine.tree1, cp = 0.3)
```

```
## n= 4898
##
## node), split, n, deviance, yval
##      * denotes terminal node
##
## 1) root 4898 3840.99 5.877909 *
```

```
prune(wine.tree1, cp = 0.2)
```

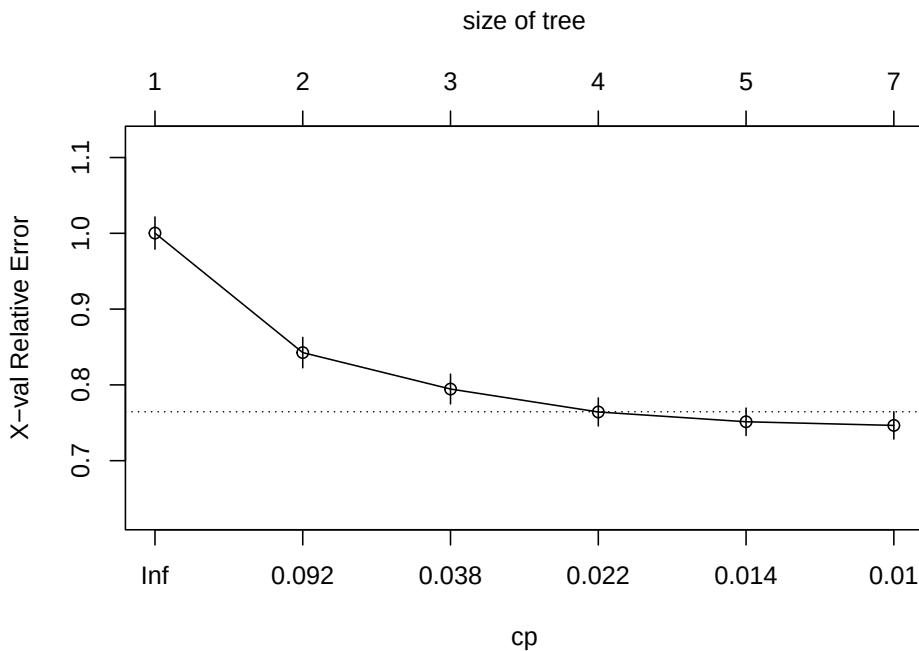
```
## n= 4898
##
## node), split, n, deviance, yval
##      * denotes terminal node
##
## 1) root 4898 3840.99 5.877909 *
```

```
prune(wine.tree1, cp = 0.156)
```

```
## n= 4898
##
## node), split, n, deviance, yval
##      * denotes terminal node
##
## 1) root 4898 3840.990 5.877909
##   2) alcohol< 10.85 3085 1844.906 5.605511 *
##   3) alcohol>=10.85 1813 1377.659 6.341423 *
```

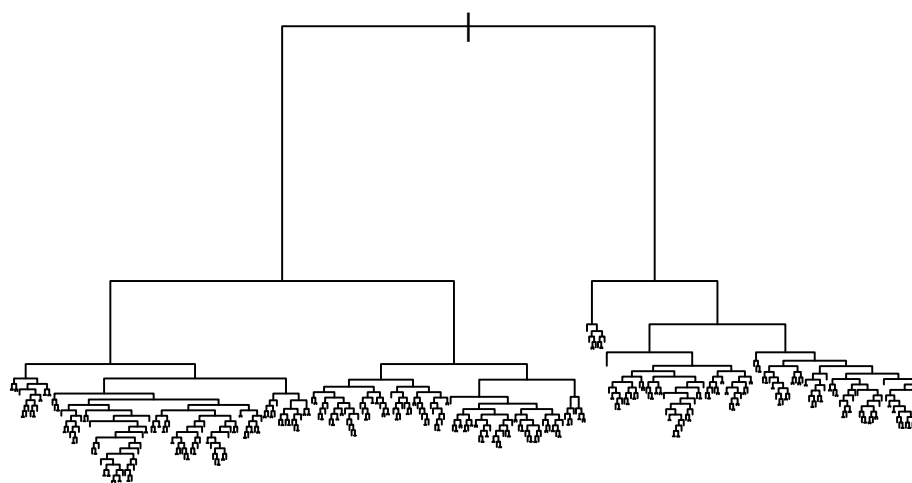
A CV path plot

```
plotcp(wine.tree1)
```

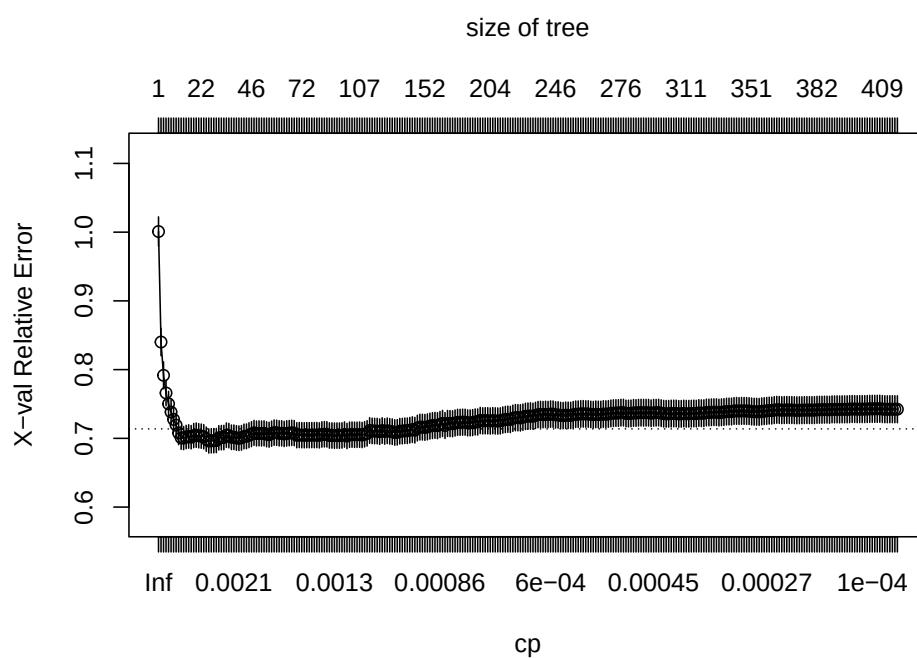


If you are uncertain whether the cross-validation error has reached its minimum, you can start with a larger tree:

```
wine.tree2 = rpart(quality~., data = whitewines, control = list(cp = 0, xval = 10))
#printcp(wine.tree2)
plot(wine.tree2)
```



```
plotcp(wine.tree2)
```



Optimal CP Value

We can use either the *Minimum Cross-Validation error* or the *One Standard Error (1-SE)* rule to choose the **optimal CP** value for pruning.

```
# Index of CP with lowest xerror
optimal.cp = which.min(wine.tree2$cptable[, "xerror"])
```

```

# The optimal CP value
wine.tree2$cptable[optimal.cp, 4]

## [1] 0.6961474

# upper bound for equivalent optimal xerror
wine.tree2$cptable[optimal.cp, 4] + wine.tree2$cptable[optimal.cp, 5]

## [1] 0.713746

# Row IDs for CPs whose xerror are equivalent to min(xerror)
tmp.id = which(wine.tree2$cptable[, 4] <= wine.tree2$cptable[optimal.cp, 4] + wine.tree2$cptable[optimal.cp, 5])
tmp.id = min(tmp.id)

# CP.1se = any value between row(tmp.id) and (tmp.id - 1)
CP.1se = (wine.tree2$cptable[tmp.id - 1, 1] + wine.tree2$cptable[tmp.id, 1]) / 2

# Prune tree with CP.1se
wine.tree3 = prune(wine.tree2, cp = CP.1se)

```

Let us now try to understand the relationship between the *1st column* and the *3rd column* of the CP table. The difference between adjacent rel error in the table is equivalent to the corresponding CP values. This difference signifies the **improvement gained in RSS by adding an additional split**, effectively setting the *cost* of that split.

```
#cbind(wine.tree2$cptable[, 1], c(-diff(wine.tree2$cptable[, 3]), 0))
```

After the tree is fit, the predict function `predict.rpart` from the `rpart` package can be used for making predictions on new or existing data.

Random Forests in R

We start by loading the random forest package in R. We also split the data in training and testing data sets.

```

library(randomForest)

## randomForest 4.7-1.2

## Type rfNews() to see new features/changes/bug fixes.

##
## Attaching package: 'randomForest'

## The following object is masked from 'package:ggplot2':
##
##     margin

```

```
n = nrow(whitewines)
ntest = round(n*0.3)
set.seed(598)
test.id = sample(1:n, ntest)
```

Model Fitting

To fit the model, we use the `randomForest` function. Our goal is to predict the quality using all the other columns in the data matrix as potential predictors.

```
rf.wines = randomForest(quality ~ ., data = whitewines[-test.id, ], importance = T, ntree=400);
```

The `randomForest` function outputs several quantities. Let us focus on `mtry` which is the number of variables selected for each split.

```
rf.wines$mtry
```

```
## [1] 3
```

In our example, `mtry` is set 3, aligning with the recommended value for regression-based Random Forest models: $p/3$, where p is the number of predictors.

We can use the fitted model to make predictions, and we can calculate the average test error:

```
yhat.test = predict(rf.wines, whitewines[test.id, ])

sum((whitewines$quality[test.id] - yhat.test)^2)/length(test.id)
```

```
## [1] 0.411837
```

If we think about the *training* error, then the random forest approach leads us to two types of training errors: one based on the prediction function (using the training data) and the other one based on the out-of-bag samples.

```
yhat.train = predict(rf.wines, whitewines[-test.id, ])
sum((whitewines$quality[-test.id] - yhat.train)^2)/(n - ntest)
```

```
## [1] 0.07441902
```

We can see that the training error is significantly smaller compared to the prediction error.

```
sum((whitewines$quality[-test.id] - rf.wines$predicted)^2)/(n - ntest)
```

```
## [1] 0.3726565
```

The error based on the out-of-bag sample is closer to the testing error and a more reliable estimator to the model's generalization performance. We can also

see how many times an observation was in an out-of-bag sample (below we only output the first 10 observations for illustration purposes):

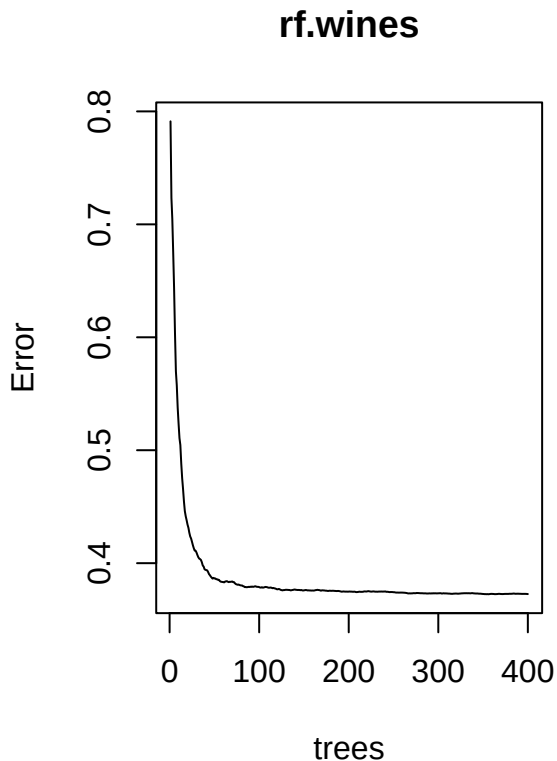
```
rf.wines$oob.times[1:10]
```

```
## [1] 132 141 166 131 151 150 142 157 143 137
```

Performance Plots

We can visualize the performance of the random forest fitted model

```
par(mfrow=c(1, 2))  
plot(rf.wines)
```



The function is plotting the MSE values derived from out-of-bag predictions. The plot shows that the model's performance improves rapidly with the initial addition of trees but starts to plateau beyond a certain point. This suggests that adding more trees does not lead to overfitting, although one could potentially achieve similar performance with fewer trees.

Variable Importance

In the `randomForest` function we specified `importance` as `'TRUE'` which means that we can extract an importance table for the variables.


```
rf.wines$importance
```

##	%IncMSE	IncNodePurity
## fixed.acidity	0.06555382	160.5398
## volatile.acidity	0.16982629	269.3357
## citric.acid	0.08526387	181.1612
## residual.sugar	0.14416479	193.1863
## chlorides	0.11123063	209.8724
## free.sulfur.dioxide	0.13832758	267.2291
## total.sulfur.dioxide	0.09535885	192.1764
## density	0.24062906	295.2742
## pH	0.08382337	174.5245
## sulphates	0.04954697	155.4755
## alcohol	0.36331159	413.3186

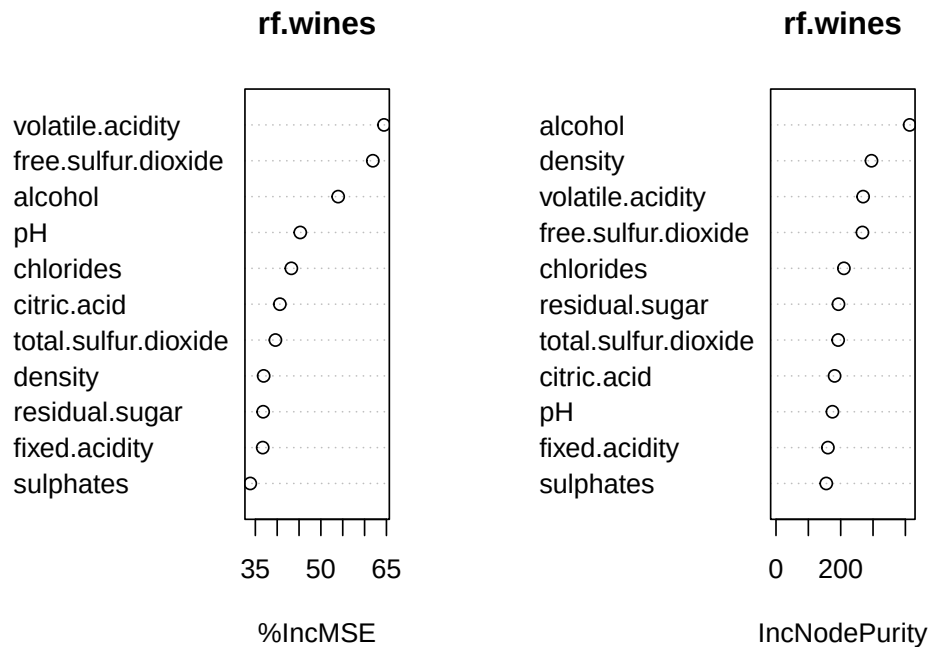
The first column `%IncMSE` shows importance based on accuracy gain before/after permutation. The second column `IncNodePurity` shows importance based on RSS gain. Both scaled and unscaled versions of these measures are available, along with their standard errors.

```
importance(rf.wines, scale = F)
```

##	%IncMSE	IncNodePurity
## fixed.acidity	36.68890	160.5398
## volatile.acidity	64.44043	269.3357
## citric.acid	40.61625	181.1612
## residual.sugar	36.78927	193.1863
## chlorides	43.23887	209.8724
## free.sulfur.dioxide	61.88785	267.2291
## total.sulfur.dioxide	39.60061	192.1764
## density	36.90432	295.2742
## pH	45.28066	174.5245
## sulphates	33.84465	155.4755
## alcohol	53.95691	413.3186

We can also visualize the variable importance using the `varImpPlot`:

```
par(mfrow = c(1,2))
varImpPlot(rf.wines, type=1)
varImpPlot(rf.wines, type=2)
```



Gradient Boosting Machines in R

Before running a GBM model, we need to tune the following hyperparameters:

- **Loss Function:** For regression tasks (like here) we use the squared error
 - **n.trees:** number of trees (default = 100)
- **shrinkage:** shrinkage factor or learning rate (default = 0.1)
- **bag.fraction:** fraction of the training data used for learning (default = 0.5)
- **cv.folds:** number of folds for cross-validation (default = 0 - i.e. no cv error returned)
- **interaction.depth:** depth of individual trees (default = 1)

```
library(gbm)
```

```
## Loaded gbm 2.2.2
```

```
## This version of gbm is no longer under development. Consider transitioning to gbm3,
```

Fitting a GBM:

```
gbm.wines = gbm(quality ~ ., data=whitewines[-test.id, ], distribution = "gaussian",
  n.trees = 100, shrinkage = 1, interaction.depth = 3, bag.fraction = 1,
```

```

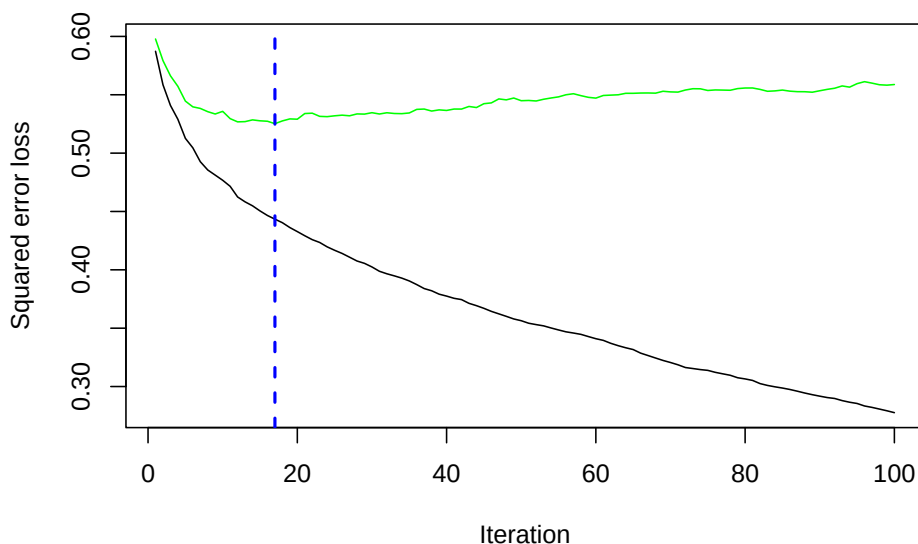
                                cv.folds = 5)
gbm.wines

## gbm(formula = quality ~ ., distribution = "gaussian", data = whitewines[-test.id,
##      ], n.trees = 100, interaction.depth = 3, shrinkage = 1, bag.fraction = 1,
##      cv.folds = 5)
## A gradient boosted model with gaussian loss function.
## 100 iterations were performed.
## The best cross-validation iteration was 17.
## There were 11 predictors of which 11 had non-zero influence.

```

We plot the CV error to find the optimal number of trees to prevent overfitting:

```
opt.size = gbm.perf(gbm.wines)
```

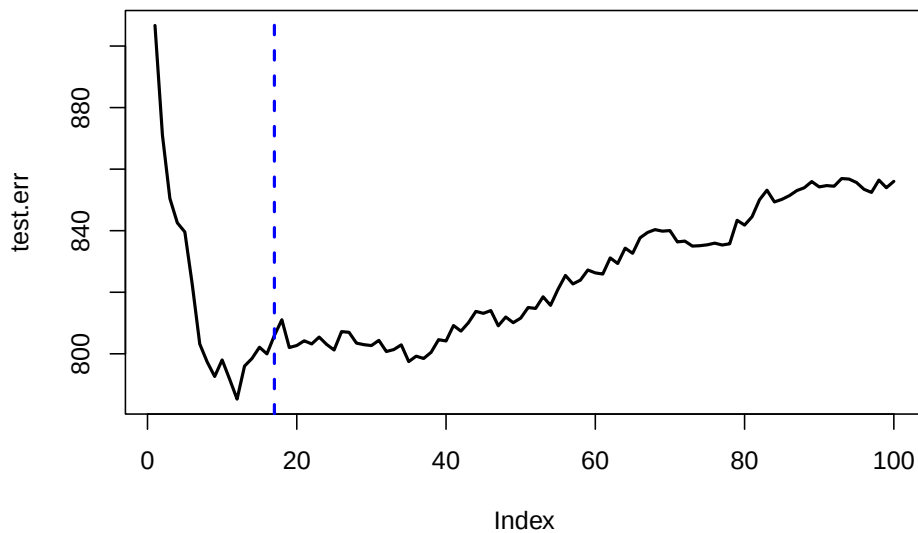


We can also evaluate the model's performance on the test data against the number of trees.

```

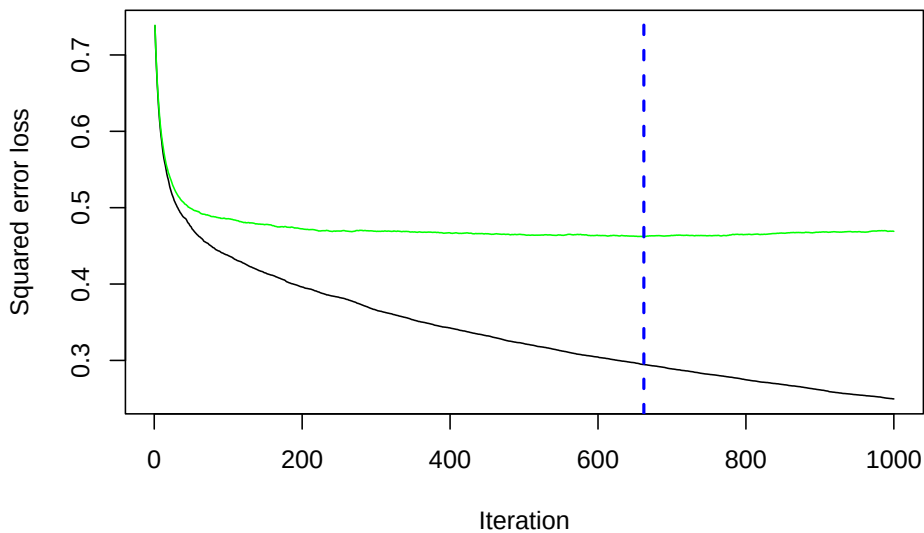
size = 1:gbm.wines$n.trees
test.err = rep(0, length(size))
for(i in 1:length(size)){
  y.pred = predict(gbm.wines, whitewines[test.id, ], n.trees = size[i])
  test.err[i] = sum((whitewines$quality[test.id] - y.pred)^2)
}
plot(test.err, type = "n")
lines(size, test.err, lwd = 2)
abline(v = opt.size, lwd = 2, lty = 2, col = "blue")

```



We can change the tuning parameters and fit a new GBM:

```
gbm.wines.2 = gbm(quality ~ ., data=whitewines[-test.id, ], distribution = "gaussian",
                  n.trees = 1000, shrinkage = 0.1 , interaction.depth = 3, bag.fraction = 0.5)
opt.size.2 = gbm.perf(gbm.wines.2)
```

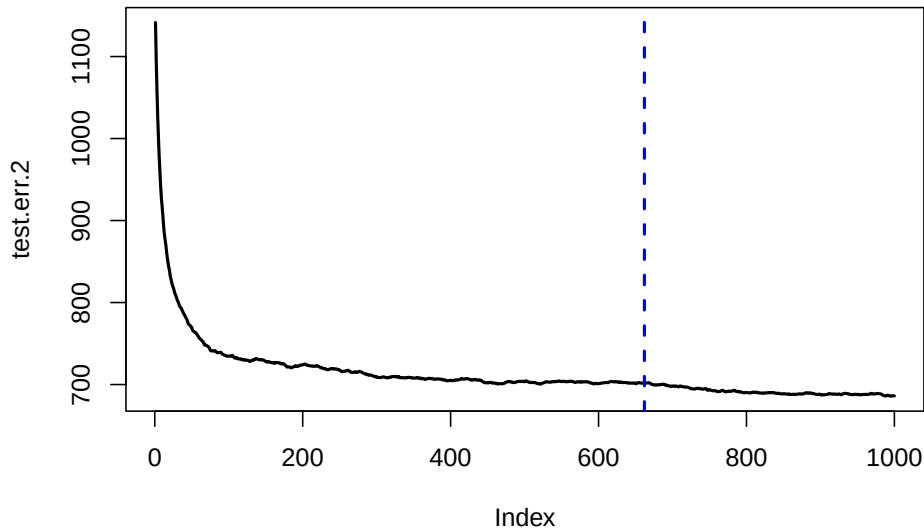


```
size.2 = 1:gbm.wines.2$n.trees
test.err.2 = rep(0, length(size.2))
for(i in 1:length(size.2)){
  y.pred.2 = predict(gbm.wines.2, whitewines[test.id, ], n.trees = size.2[i])
  test.err.2[i] = sum((whitewines$quality[test.id] - y.pred.2)^2)
```

```

}
plot(test.err.2, type = "n")
lines(size.2, test.err.2, lwd = 2)
abline(v = opt.size.2, lwd = 2, lty = 2, col = "blue")

```



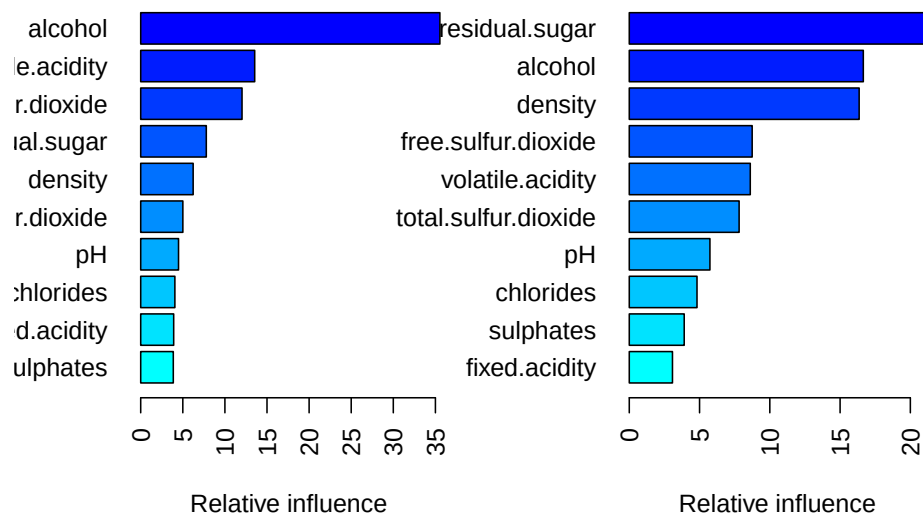
We can still quantify the variable importance:

```

par(mfrow=c(1, 2))
summary(gbm.wines, cBars = 10, method = relative.influence, las = 2)

##              var    rel.inf
## alcohol          alcohol 35.531879
## volatile.acidity  volatile.acidity 13.540395
## free.sulfur.dioxide free.sulfur.dioxide 12.034661
## residual.sugar     residual.sugar  7.791508
## density            density  6.230189
## total.sulfur.dioxide total.sulfur.dioxide 5.006459
## pH                pH  4.494230
## chlorides          chlorides  4.058416
## fixed.acidity      fixed.acidity  3.918535
## sulphates          sulphates  3.871500
## citric.acid        citric.acid  3.522228
summary(gbm.wines, cBars = 10, method = permutation.test.gbm, las = 2)

```



```
##          var    rel.inf
## 1  residual.sugar 21.288079
## 2      alcohol 16.652574
## 3      density 16.356976
## 4 free.sulfur.dioxide 8.744650
## 5  volatile.acidity 8.609951
## 6 total.sulfur.dioxide 7.815605
## 7          pH 5.738623
## 8      chlorides 4.820094
## 9      sulphates 3.908482
## 10 fixed.acidity 3.073419
## 11      citric.acid 2.991547
```

Chapter 6

Introduction to Classification

Classification problems are a fundamental supervised learning task in which the goal is to *predict a categorical outcome* based on a set of input features. Unlike regression, where the response is continuous, classification involves discrete labels—such as “spam” vs. “not spam” or “diseased” vs. “healthy”. The challenge lies in learning a function that accurately maps the input variables to the correct *class*, often by estimating probabilities or decision boundaries. Techniques range from simple models like logistic regression to more complex methods such as decision trees, support vector machines, and ensemble approaches. As in all statistical learning, the key is balancing model flexibility with the risk of overfitting, ensuring reliable predictions on new, unseen data.

Classification Framework

In the classification framework, we start with **training data** $\mathcal{D}_n = \{x_i, y_i\}_{i=1}^n$, where $x_i \in \mathbb{R}^p$ are the p measurements/features. In our discussion here, we are going to focus the **binary** case first in which the *response* $y_i \in \{0, 1\}$, $i = 1, \dots, n$ observations. The specific class coding is not as important, so it could be “0/1” or “-1/1” etc.

The main purpose in classification problems is to find a **classifier** $f : \mathbb{R}^p \rightarrow \{0, 1\}$, that is a **map** between the features’ space and the classes. To quantify the performance of the classifier, at any target point x with outcome y , we typically use the 0 – 1 loss, that is the misclassification error

$$L(f(x), y) = \begin{cases} 0, & \text{if } y = f(x) \\ 1, & \text{otherwise} \end{cases}$$

Ideally, we would like to construct a *prediction rule* that makes small errors not

only on the n samples, but also on future observations. So, to find an optimal classifier we need to minimize:

$$\min_f \frac{1}{n} \sum_{i=1}^n L(f(x_i), y_i)$$

The Optimal Classifier

Assume we know the **true** model (how the data were generated), or that we have infinitely many samples, then

$$\frac{1}{n} \sum_{i=1}^n L(f(x_i), y_i) \longrightarrow E_{X,Y} L(f(X), Y)$$

where the expectation is taken with respect to the true data generating process $P(x, y)$.

Optimal Classifier

The optimal classifier is obtained by minimizing a risk function. Specifically:

$$f^* = \arg \min_f Risk[f]$$

where $Risk[f] = E_{X,Y} L(f(X), Y)$

Suppose we adopt the **0-1 loss**, and are allowed to use any function f . *What is the optimal f^* ?*

If we denote by $p(x)$ the *marginal* distribution function of X , $p(y|x)$ is the *conditional* distribution function of Y given $X = x$, and $p(x, y) = p(x)p(y|x)$ is the *joint* distribution function of (X, Y) , then

$$\begin{aligned} Risk[f] &= E_{X,Y} L(f(X), Y) \\ &= \int_{\mathcal{X}} \int_{\mathcal{Y}} L(f(x), y) p(x, y) dy dx \\ &= \int_{\mathcal{X}} \int_{\mathcal{Y}} L(f(x), y) p(y|x) p(x) dy dx \\ &= \int_{\mathcal{X}} \left[\int_{\mathcal{Y}} L(f(x), y) p(y|x) dy \right] p(x) dx \end{aligned}$$

The problem of finding f that minimizes $Risk[f]$ is reduced to a series of *sub-problems*:

For each given x , we need find the optimal value $f(x)$ to *minimize the inside integral*, i.e.

$$f^*(x) = \arg \min_f \int_{\mathcal{Y}} L(f(x), y) p(y|x) dy$$

If we do this for every x , then the resulting f^* minimizes $Risk[f]$.

The expression above turns out to be not difficult to solve. Note that Y is just a (discrete) Bernoulli random variable taking two possible values $\{0, 1\}$ with pmf

$$\begin{cases} P(Y = 1|X = x) = \eta(x) \\ P(Y = 0|X = x) = 1 - \eta(x) \end{cases}$$

Given x , the integral over y is given by

$$\begin{aligned} \int_y L(f(x), y) p(y|x) dy &= L(f(x), 1) P(Y = 1|x) + L(f(x), 0)P(Y = 0|x) \\ &= L(f(x), 1) \eta(x) + L(f(x), 0)(1 - \eta(x)) \\ &= \\ &= \begin{cases} 1 - \eta(x), & \text{if } f(x) = 1 \\ \eta(x), & \text{if } f(x) = 0 \end{cases} \end{aligned}$$

This leads to

$$f^*(x) = \arg \min_f R[f] = \begin{cases} 1, & \text{if } \eta(x) := P(Y = 1|x) \geq 0.5 \\ 0, & \text{if } \eta(x) := P(Y = 1|x) < 0.5 \end{cases}$$

The optimal classifier f^* is called the **Bayes rule** and the corresponding risk $R[f^*]$ is referred to as the **Bayes risk** or **Bayes error**.

How about Multi-class Classification?

The previous calculation can be easily extended to multi-class classification. In this case, the response will take values in a discrete set $y \in \{1, \dots, K\}$ and the **optimal rule** will be obtained by

$$f^*(x) = \arg \max_k P(Y = k|X = x)$$

0/1 classifiers partition the X -space into regions, each assigned a label of 0 or 1. In many cases, we often describe the decision boundary rather than the function f itself, which is where the term linear classifiers arises - these are classifiers whose decision boundaries are linear

6.1 Discriminant Analysis

Discriminant analysis is a classification technique that models the *difference* between classes by assuming that the data for each class comes from a specific probability distribution. In other words, it tries to discriminate between classes

by estimating the **boundaries** that separate them. We have three main types of discriminant analysis more-or-less based on the decision boundary that we obtain as a result:

1. Quadratic Discriminant Analysis (QDA)
2. Linear Discriminant Analysis (LDA) and its connection to Fisher Discriminant Analysis (FDA)
3. Naive Bayes

All these variations of discriminant analysis are based on **Bayes Theorem**. In the calculations above, we discussed that it is crucial to estimate the joint distribution $P(x, y) = P(x|y)p(y)$ which means that we need to be able to estimate

- $p(x|y)$ the conditional distribution of X given Y in each class
- $p(y)$ the marginal frequency of Y in each class.

and then use the Bayes theorem to flip things and obtain the desired $P(y|x)$, that is the optimal classifier (i.e. the Bayes rule) does classification based on the conditional probability which by the Bayes Theorem takes the following form:

Bayes Theorem for Classification

$$\begin{aligned} P(Y = k|X = x) &= \frac{P(X = x, Y = k)}{P(X = x)} = \frac{P(X = x|Y = k)P(Y = k)}{P(X = x)} \\ &= \frac{\pi_k f_k(x)}{P(X = x)} \propto \pi_k f_k(x) \end{aligned}$$

- $f_k(x)$: **conditional** density function of $X|Y = k$
- $\pi_k = P(Y = k)$: the **marginal** probability or prior probability for class k .

Using Bayes theorem, we can express the decision function as

$$f(x) = \arg \max_k \pi_k f_k(x) = \arg \max_k [\log \pi_k + \log f_k(x)]$$

which implies that in order to construct the optimal classifier, we need to estimate π_k and f_k .

6.1.1 Quadratic Discriminant Analysis

Given $Y = k$, the p -dim feature $\mathbf{x} \sim MVN(\mu_k, \Sigma_k)$ is modeled by a multivariate normal distribution with mean μ_k and covariance matrix Σ_k , that is

$$\mu_k = \begin{pmatrix} \mu_{k,1} \\ \mu_{k,2} \\ \vdots \\ \mu_{k,p} \end{pmatrix}_{p \times 1} \quad \text{and} \quad \Sigma_k^{-1} = \begin{pmatrix} \theta_{k,11} & \cdots & \theta_{k,1p} \\ \theta_{k,21} & \cdots & \theta_{k,2p} \\ \vdots & \ddots & \vdots \\ \theta_{k,p1} & \cdots & \theta_{k,pp} \end{pmatrix}_{p \times p}$$

Then, the pdf of $\mathbf{x} \sim MVN(\mu_k, \Sigma_k)$, is given by

$$f_k(\mathbf{x}) = \frac{1}{(\sqrt{2\pi})^p} \frac{1}{|\Sigma_k|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k) \right\}$$

A few calculations on the *exponent* lead to

$$(\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k) = \sum_{j=1}^p \sum_{\ell=1}^p \theta_{k,j\ell} (\mathbf{x}_j - \mu_{k,j})^T (\mathbf{x}_j - \mu_{k,\ell})$$

Therefore, we can write that we have that

$$P(Y = k | \mathbf{x}) \propto \pi_k f_k(x) \propto e^{-\frac{d_k(x)}{2}}$$

where

$$\begin{aligned} d_k(x) &= 2 \left[-\log f_k(x) - \log(\pi_k) \right] \\ &= (\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k) + \log |\Sigma_k| - 2 \log \pi_k \end{aligned}$$

The first term of the **discriminant function/exponent** is the so-called Mahalanobis distance between $\mathbf{x}$ and μ_k .

So, we predict $\mathbf{x}$ to class k , if d_k achieves the minimum among $(d_1(\mathbf{x}), \dots, d_K(\mathbf{x}))$. The classification rule is called **Quadratic Discriminant Analysis (QDA)*** since it leads to **quadratic decision boundaries**.

To apply this method in practice, we need to be able to estimate π_k , μ_k and Σ_k . Indeed,

- *Sample Frequency & Sample Mean* for each class:

$$\hat{\pi}_k = \frac{n_k}{n}, \quad \hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i = k} \mathbf{x}_i, \quad k = 1 : K$$

- *Sample Covariance matrix* for each class:

$$\hat{\Sigma}_k = \frac{1}{n_k - 1} \sum_{i: y_i = k} (\mathbf{x} - \hat{\mu}_k)_{p \times 1} (\mathbf{x} - \hat{\mu}_k)_{1 \times p}^T$$

Remark: In the case that $\hat{\Sigma}_k^{-1}$ does not exist, we replace it by $(\hat{\Sigma}_k + \eta I_p)^{-1}$ where η is a small number e.g. $\eta = 0.01$.

6.1.2 Linear Discriminant Analysis

Assume now that the entries of the covariance matrix Σ are not dependent on k , i.e. $\Sigma_k = \Sigma$. Then, the discriminant function simplifies to

$$d_k(\mathbf{x}) = (\mathbf{x} - \mu_k)^T \Sigma^{-1} (\mathbf{x} - \mu_k) + \log |\Sigma| - 2 \log \pi_k$$

which is a **linear function** of $\mathbf{x}$:

$$(\mathbf{x} - \mu_k)^T \Sigma^{-1} (\mathbf{x} - \mu_k) = \mathbf{x}^T \Sigma^{-1} \mathbf{x} - 2\mathbf{x}^T \Sigma^{-1} \mu_k + \mu_k^T \Sigma^{-1} \mu_k$$

In this case, the covariance matrix Σ can be estimated by the *pooled sample covariance matrix*

$$\hat{\Sigma} = \frac{1}{n - K} \sum_{i=1}^K \sum_{i: y_i = k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^T$$

Remark: If $\hat{\Sigma}^{-1}$ does not exist, then we can replace it by $(\hat{\Sigma} + \eta I_p)^{-1}$ where η is a small number to compute $\hat{\Sigma}^{-1}$. Alternatively, we can compute a generalized inverse as follows (assume $p = 3$):

$$\hat{\Sigma} = U_{p \times 3} \begin{pmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & 0 \end{pmatrix} U^T \Rightarrow \hat{\Sigma}^{-1} = U \begin{pmatrix} 1/d_1 & 0 & 0 \\ 0 & 1/d_2 & 0 \\ 0 & 0 & 0 \end{pmatrix} U^T$$

6.1.3 Reduced Rank LDA

Let us further simplify the covariance matrix Σ by assuming that it is an **identity** matrix $\mathbf{I}_p$. The discriminant function for LDA will now become

$$d_k(\mathbf{x}) = \|\mathbf{x} - \mu_k\|^2 - 2 \log \pi_k$$

Note that the feature vector x only appears in the 1st term which is $\|\mathbf{x} - \mu_k\|^2$ is the *squared distance from $\mathbf{x}$ to μ_k* , the center of the k th class.

Next, we will show that we can replace the squared distance $\|\mathbf{x} - \mu_k\|^2$ in the original p -dim space by a square distance in a $(K - 1)$ -dim subspace showing that LDA naturally leads to dimension reduction from p to $K - 1$.

Without loss of generality assume that $\bar{\mu} = 0$. Indeed, we care about the *relative* distance among points so we can always move the origin to $\bar{\mu}$ without affecting the distance.

In this case $(\mu_1, \dots, \mu_K)$ form a $(K - 1)$ -dim subspace in $\mathbb{R}^p$, denoted by $\mathcal{A}$. For any vector $\mathbf{x}$ in $\mathbb{R}^p$

$$\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2, \quad \mathbf{x}_1 \in \mathcal{A}, \mathbf{x}_2 \in \mathcal{A}^c$$

In the expression above

- $\mathbf{x}_1$: projection of $\mathbf{x}$ onto the $(K-1)$ -dim subspace $\mathcal{A}$
- $\mathbf{x}_2$: the projection of $\mathbf{x}$ onto the $(p-K+1)$ -dim subspace that is *orthogonal* to $\mathcal{A}$.

In practice, we can get this decomposition by running a regression of $\mathbf{x}$ against $B = (\mu_1, \dots, \mu_{K-1})_{p \times (K-1)}$ and

$$\begin{aligned}\mathbf{x}_1 &= H\mathbf{x} = \text{the fitted vector,} \\ \mathbf{x}_2 &= \text{the residual vector, where } H = B(B^T B)^{-1} B^T\end{aligned}$$

Due to orthogonality of $\mathbf{x}_1$ and $\mathbf{x}_2$, we have

$$\begin{aligned}\|\mathbf{x}\|^2 &= \|\mathbf{x}_1 + \mathbf{x}_2\|^2 = (\mathbf{x}_1 + \mathbf{x}_2)^T (\mathbf{x}_1 + \mathbf{x}_2) \\ &= \mathbf{x}_1^T \mathbf{x}_1 + \mathbf{x}_1^T \mathbf{x}_2 + \mathbf{x}_2^T \mathbf{x}_1 + \mathbf{x}_2^T \mathbf{x}_2 = \|\mathbf{x}_1\|^2 + \|\mathbf{x}_2\|^2\end{aligned}$$

[norm-square (length-square) of a vector] = [norm-square of its projection onto $\mathcal{A}$] + [norm-square of its projection onto $\mathcal{A}^c$]

Square distance from a point $\mathbf{x}$ to the center of the k th class:

$$\begin{aligned}\|\mathbf{x} - \mu_k\|^2 &= \|(\mathbf{x}_1 - \mu_k) + (\mathbf{x}_2 - 0)\|^2 \\ &= \|\mathbf{x}_1 - \mu_k\|^2 + C, k = 1, \dots, K\end{aligned}$$

where the constant $C = \|\mathbf{x}_2\|^2$ is the same for all k . Note that the projection of μ_k onto $\mathcal{A}^c$ is zero. For classification, we *only* want to know the **magnitude** of $d_k(x)$:

$$\begin{aligned}d_k(x) &= \|\mathbf{x} - \mu_k\|^2 - 2 \log \pi_k \\ &= \|\mathbf{x}_1 - \mu_k\|^2 - 2 \log \pi_k + \text{constant}\end{aligned}$$

Operate LDA on the reduced $(K-1)$ -dim space $\mathcal{A}$: replace data $\mathbf{x}$ by $\mathbf{x}_1$ (its projection onto $\mathcal{A}$) and compute

$$d_k(\mathbf{x}) = \|\mathbf{x}_1 - \mu_k\|^2 - 2 \log \pi_k$$

which is the same as running LDA in the original p -dimensional space.

In practice, Σ will **not** be the identity matrix. So, we need to consider the SVD (singular value decomposition) of Σ

$$\Sigma = U D^2 U^T$$

where

- D is diagonal with non-negative entries
- U is an orthogonal (i.e. rotation) matrix with $U U^T = I_p$

Start by transforming $\mathbf{x}$ to $\tilde{\mathbf{x}}$:

$$\mathbf{x} \in \mathbb{R}^p \Rightarrow \tilde{\mathbf{x}} = \Sigma^{-1/2} \mathbf{x} \in \mathbb{R}^p$$

with $\Sigma^{-1/2} := U D^{-1} U^T$. Then the covariance matrix of $\tilde{\mathbf{x}}$ is identity. Indeed:

$$\begin{aligned} \text{Cov}(\tilde{\mathbf{x}}) &= \text{Cov}(\Sigma^{-1/2} \mathbf{x}) \\ &= \Sigma^{-1/2} \text{Cov}(\mathbf{x}) (\Sigma^{-1/2})^T \\ &= U D^{-1} U^T \Sigma U D^{-1} U^T \\ &= U D^{-1} U^T U D^2 U^T U D^{-1} U^T = I_p \end{aligned}$$

Remark: This has the same flavor as the re-scaling of an one-dimensional variable to have unit variance. In the one-dim case, we multiply the variable by $\frac{1}{\sqrt{\sigma^2}}$, while here we multiply by $\Sigma^{-1/2}$, i.e. the inverse of the square-root of Σ in the multi-dimensional setting. Note that $\Sigma^{-1/2}$ is symmetric and $\Sigma^{-1/2} \Sigma^{-1/2} = \Sigma^{-1}$

Summary of Reduced Rank LDA

Training: Input $(x_i, y_i)_{i=1}^n$ / Output: $(\pi_k, \mu_k)_{k=1}^K$ and Σ

Make Prediction:

Input: test point $\mathbf{x}^*$ and output from Training

Compute SVD of $\Sigma = U D^2 U^T$ and $A = U D^{-1} U^T$

Compute the projection matrix H based on $(A\mu_1, \dots, A\mu_K)$

$$\begin{aligned} B_{p \times (K-1)} &= \begin{bmatrix} A(\mu_1 - \bar{\mu}), \dots, A(\mu_{K-1} - \bar{\mu}) \end{bmatrix} \\ H &= B(B^T B)^{-1} B \end{aligned}$$

Dimension reduction: $\tilde{\mathbf{x}}^* = H A \mathbf{x}^*$

For $k = 1 : K$, compute

$$d_k(\mathbf{x}^*) = \|\tilde{\mathbf{x}}^* - \mu_k\|^2 - 2 \log \pi_k$$

Output: $\arg \min_k d_k(\mathbf{x}^*)$.

<>br

When $p \geq K$, this means that for LDA, one can project the data onto a lower-dimensional subspace ($\dim = K - 1$), e.g. just one dimension for binary classification.

The same subspace also arises in a dimension reduction method called Fisher Discriminant Analysis (FDA), though FDA is motivated from a slightly different aspect.

Risk of Overfitting

Each of the $(K - 1)$ directions are linear combinations of the original p dimensions, and the weights are all learned from the data so overfitting can occur. %good separation on “just” the $(K - 1)$ (or even less) directions on the training data, but the same good result cannot be reproduced on the test data.

6.2 Fisher Discriminant Analysis (FDA)

Supervised dimension reduction

We want to find a direction such that the projection of data (of multiple classes) onto this direction is well separated, that is the group means of μ_i s are far apart from each other, and within each group, the variation/spread is small, i.e., maximize the following ratio

$$\frac{\text{Between group variation}}{\text{Within group variation}} = \frac{a^T B a}{a^T W a}$$

The matrices B and W are the **within-class** & **between-class** sample covariance matrices, defined by

$$W_{p \times p} = \frac{1}{n - K} \sum_{k=1}^K \sum_{i: y_i = k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^T$$

$$B_{p \times p} = \frac{1}{K - 1} \sum_{k=1}^K n_k (\hat{\mu}_k - \bar{\hat{\mu}})(\hat{\mu}_k - \bar{\hat{\mu}})^T$$

If we re-write x_i s as follows

$$\begin{aligned} x_1 &= x_1 - \hat{\mu}_{y_1} + \hat{\mu}_{y_1} \\ x_2 &= x_2 - \hat{\mu}_{y_2} + \hat{\mu}_{y_2} \\ &\dots = \dots \\ x_n &= x_n - \hat{\mu}_{y_n} + \hat{\mu}_{y_n} \end{aligned}$$

we can think of W as the sample covariance matrix over $x_i - \hat{\mu}_{y_i}$ (same as the pooled sample covariance matrix $\hat{\Sigma}$ in LDA), and B as the sample covariance matrix over the K class centers $\hat{\mu}_{y_i}$.

Generalized Eigenvalue Problem

$$\max_a \frac{a^T B a}{a^T W a} \Rightarrow \max_a (a^T B a) \quad \text{subject to } a^T W a = 1$$

Assume $W = U D^2 U^T$, and define $b = W^{1/2} a$, where $W^{1/2} := U D U^T$ is symmetric. The inverse of $W^{1/2}$ computes as $W^{-1/2} = U D^{-1} U^T$. Therefore,

$$\begin{aligned}
a^T B a &= a^T W^{1/2} W^{-1/2} B W^{-1/2} W^{1/2} a \\
&= (W^{1/2} a)^T W^{-1/2} B W^{-1/2} (W^{1/2} a) \\
&= b^T W^{-1/2} B W^{-1/2} b
\end{aligned}$$

subject to $\|b\|^2 = 1$. The optimization above is a classical eigenvalue problem! In this problem, we can solve the directions *sequentially* as follows:

b_1 = the 1st eigenvector of matrix $W^{-1/2} B W^{-1/2}$, then solve $a_1 = W^{-1/2} b_1$.

b_2 = the 2nd eigenvector of matrix $W^{-1/2} B W^{-1/2}$, then solve $a_2 = W^{-1/2} b_2$.

Although b_j s are orthonormal, the a_j s are **not**:

$$\begin{aligned}
b_j^T b_j &= 1, & a_j^T W a_j &= 1 \\
b_j^T b_\ell &= 0, & a_j^T W a_\ell &= 0
\end{aligned}$$

We can extract at most $(K - 1)$ directions, since the rank of B is $(K - 1)$. The $(K - 1)$ directions span exactly the same space as the one from the reduced rank LDA.

LDA vs. FDA

Linear Discriminant Analysis (LDA) is primarily a classifier, whereas Fisher Discriminant Analysis (FDA) is a dimension reduction method: it identifies a set of directions rather than providing a classification rule. Although FDA does not explicitly assume that the data are normally distributed, the subspace it produces often resembles the reduced space obtained from LDA. This similarity arises because FDA implicitly assumes that the data in each group are approximately normally distributed with the same covariance matrix. FDA is a supervised dimension reduction technique, in contrast to Principal Component Analysis (PCA), which is unsupervised. Moreover, FDA can be applied to regression problems as well. However, care must be taken to avoid overfitting, particularly in high-dimensional settings.

Risk of Overfitting

Consider an example where you only have $n = 6$: 3 of the observations are from group 1 (blue), and the other 3 from group 2 (orange). In this case, the projection results in a clear separation between the two classes:

- Class one points project to 1.
- Class zero points project to 0.

This separation arises not due to meaningful features but because the feature dimension matches the sample size n .

When the feature dimension equals the sample size, we can generate an $n \times n$ design matrix and find a direction in the n -dimensional space that effectively separates the classes, even if the features are random noise.

This can be remedied if we use:

- **L1 Penalty:** Add L1 regularization to the optimization problem when estimating covariance matrices or their inverses and mean vectors. This encourages sparsity in the estimated parameters.
- **Assume Independence:** Assume that the p -dimensional density function is composed of independent components. This implies that the covariance matrix (sigma) becomes diagonal. This approach is known as Naive Bayes and can help address high-dimensional overfitting.

6.3 Naive Bayes Classifiers

For multi-class problems the optimal decision rule is

$$\arg \max_k P(Y = k | X = \mathbf{x}) = \arg \max_k \pi_k f_k(\mathbf{x})$$

We require $f_k(x)$ to be

$$f_k(\mathbf{x}) = f_{k1}(\mathbf{x}_1) \times f_{k2}(\mathbf{x}_2) \times \dots \times f_{kp}(\mathbf{x}_p)$$

i.e. each dim of $\mathbf{x}$ is independent. (You can view independence as regularization for high-dimensional problems.)}

Each density f_{kj} ($j = 1 : p, k = 1 : K$) is estimated separately within each class, e.g., discrete features via histograms; numerical features via kernel density estimates (non-parametric NB) or normal densities (parametric NB).

Summary of Parametric Naive Bayes

Training:

Input: $(x_i, y_i)_{i=1}^n$ Output: $(\pi_k, (\mu_{jk}, \sigma_{kj}^2)_{j=1:p})_{k=1}^K$

Make Prediction:

Input: test point $\mathbf{x}^*$ and output from Training

For $k = 1 : K$, compute

$$\begin{aligned} d_k(\mathbf{x}^*) &= -2 \log \left[\pi_k \frac{1}{\sqrt{\sigma_{k1}^2}} e^{-\frac{(x_1^* - \mu_{k1})^2}{2\sigma_{k1}^2}} \dots \frac{1}{\sqrt{\sigma_{kp}^2}} e^{-\frac{(x_p^* - \mu_{kp})^2}{2\sigma_{kp}^2}} \right] \\ &= -2 \log \pi_k + \sum_{j=1}^p \left[\log \sigma_{kj}^2 + \frac{(x_j^* - \mu_{kj})^2}{\sigma_{kj}^2} \right] \end{aligned}$$

Output: $\arg \min_k d_k(\mathbf{x}^*)$

Summary

In Discriminant Analysis (DA), we estimate the joint distribution}

$$\begin{aligned} P(X = x, Y = k) &= P(Y = k|X = x) P(X = x) \\ &= P(X = x|Y = k) P(Y = k) \end{aligned}$$

and then obtain $P(Y = k|X = x)$.

It is conceptually simple and works for some low-dimensional problems, but not an effective way of building classifiers.

6.4 The Digits Recognition Example

Discriminant Analysis in R

To illustrate the concepts presented in this section, we are going to work with the handwritten digits data set from the ESL book. The data can be found in the `ElemStatLearn` library (which is not compatible with newer versions of R) or in the `tensorBSS` library. The **goal** in this example is to classify the digits correctly. To that end, we are going to use QDA, LDA, FDA and Naive Bayes techniques.

```
library(tensorBSS) ## Needed for the ESL data sets (if you have a newer R)
```

```
## Loading required package: JADE
```

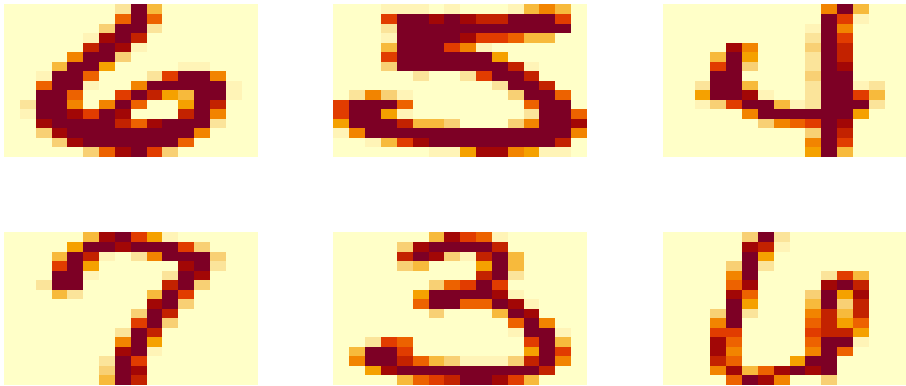
```
## Loading required package: fICA
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
## as.zoo.data.frame zoo
```

```
library(MASS) ## Needed for the DA functions
library(e1071)
data(zip.train)
```

In this data set, the response is in the **first** column. If we want (for illustration purposes), we can extract the first six images and plot them:

```
par(mfrow=c(3,3), mai=c(0.25,0.25, 0.25, 0.25))
for (i in 1:6){
  digit.img = matrix(zip.train[i, -1], nrow = 16, byrow = TRUE)
  digit.img = t(digit.img)[,16:1]
  image(1:16,1:16,digit.img, axes = FALSE)
}
```



There is no pre-loaded testing data set in this library, hence we split the give data in testing and training:

```
set.seed(598)
n <- nrow(zip.train)

train.index <- sample(1:n, 800)
remaining.index = setdiff(1:n, train.index)
test.index = sample(remaining.index, 200)

train.zip <- zip.train[train.index, ]
test.zip <- zip.train[test.index, ]

X.train <- train.zip[, 2:257]
Y.train <- train.zip[, 1]

X.test <- test.zip[, 2:257]
Y.test <- test.zip[, 1]
```

Linear Discriminant Analysis (LDA)

We start by running the Linear Discriminant Analysis (LDA):

```
digits.lda = lda(X.train, Y.train)
```

Then, we predict and compute a prediction table. We also compute the accuracy of the method.

```
Y.predicted = predict(digits.lda, X.test)$class
# table(Y.test, Y.predicted) ## Remove comment to see output
1 - sum(Y.test != Y.predicted) / length(Y.test)
```

```
## [1] 0.845
```

The accuracy here is approximately 84.5%!

Quadratic Discriminant Analysis (QDA)

If we try to run a Quadratic Discriminant Analysis (QDA), then we obtain an error message due to the rank deficiency of the sigma matrix for each group. Unfortunately, the MASS package is not able to handle this issue.

```
## digits.qda = qda(X.train,Y.train)
```

Fisher Discriminant Analysis (FDA)

Let us try to see if FDA works. Project the data onto FDA directions (returned by the `lda` function).

```
FDA.digits.directions = digits.lda$scaling
dim(FDA.digits.directions)
```

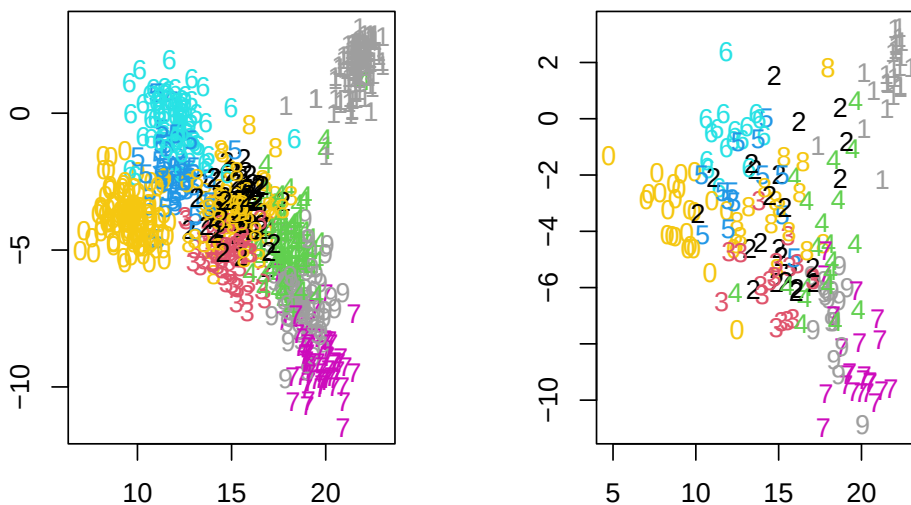
```
## [1] 256 9
```

These directions turned out to be 9 in our case!

```
F = X.train%%FDA.digits.directions
```

```
par(mfrow=c(1,2))
plot(F[,1],F[,2], type="n", xlab="", ylab="")
text(F[,1], F[,2], Y.train, col=Y.train+7)
```

```
F.test = X.test%%digits.lda$scaling;
plot(F.test[,1], F.test[,2], type="n", xlab="", ylab="")
text(F.test[,1], F.test[,2], Y.test, col=Y.test+7)
```



For illustration purposes, we display the projection of the data onto the first two dimensions. Projecting in other dimensions will give a different result, so feel free to change the dimensions and see the difference.

On the left, you can observe that the training data is well-separated into ten classes when using these two directions. However, replicating the same result with the test data on the right is not possible. This discrepancy arises because of overfitting to the training data.

Computational Remarks

Let's discuss how to compute the FDA directions. We start by computing the within group covariance matrix W :

```
tmp.lm = lm(X.train ~ as.factor(Y.train) - 1); # each x_i - its group mean
W = cov(tmp.lm$res)
eigenW = eigen(W)
summary(eigenW)
```

```
##           Length Class  Mode
## values      256  -none- numeric
## vectors 65536  -none- numeric
```

```
A = eigenW$ve %%% diag(1/sqrt(eigenW$va)) %%% t(eigenW$ve)
```

Then, we compute B the between-group covariance matrix:

```
B = cov(tmp.lm$fitted)
```

So, now we can extract the FDA directions:

```
FDA.eigen = eigen(A %%% B %%% t(A))
FDA.eigen.dir = A %%% FDA.eigen$ve[, 1:9]

for(j in 1:9)
  print(cor(FDA.eigen.dir[, j], FDA.digits.directions[, j]))
```

```
## [1] 1
## [1] 1
## [1] 1
## [1] 1
## [1] -1
## [1] 1
## [1] 1
## [1] 1
## [1] 1
```

FDA vs PCA

Obtain the top 100 PCs:

```
Sigma = cov(X.train)
dig.eigen=eigen(Sigma);
PCA.dir = dig.eigen$ve[, 1:100] # pick top 100 PCA directions
dim(PCA.dir)
```

```
## [1] 256 100
```

Note that directions from PCA are orthogonal:

```
round(t(PCA.dir[, 1:10]) %*% PCA.dir[, 1:10], dig=2)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [1,]    1    0    0    0    0    0    0    0    0    0
## [2,]    0    1    0    0    0    0    0    0    0    0
## [3,]    0    0    1    0    0    0    0    0    0    0
## [4,]    0    0    0    1    0    0    0    0    0    0
## [5,]    0    0    0    0    1    0    0    0    0    0
## [6,]    0    0    0    0    0    1    0    0    0    0
## [7,]    0    0    0    0    0    0    1    0    0    0
## [8,]    0    0    0    0    0    0    0    1    0    0
## [9,]    0    0    0    0    0    0    0    0    1    0
## [10,]   0    0    0    0    0    0    0    0    0    1
```

but directions from FDA are not:

```
round(t(FDA.eigen.dir) %*% FDA.eigen.dir, dig=2)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9]
## [1,] 112.81 -36.31 -1.72 -74.75 21.61 -23.03 28.10 84.45 25.98
## [2,] -36.31 76.33 10.78 36.52 -16.64 0.05 -63.48 -22.84 -3.74
## [3,] -1.72 10.78 75.85 5.00 4.98 2.03 -18.44 -8.86 1.40
## [4,] -74.75 36.52 5.00 221.53 -15.22 20.63 -0.99 -78.55 -47.91
## [5,] 21.61 -16.64 4.98 -15.22 113.25 -24.92 -8.64 -0.74 -12.94
## [6,] -23.03 0.05 2.03 20.63 -24.92 85.31 36.17 -22.76 -2.01
## [7,] 28.10 -63.48 -18.44 -0.99 -8.64 36.17 217.71 6.06 -2.48
## [8,] 84.45 -22.84 -8.86 -78.55 -0.74 -22.76 6.06 165.63 6.32
## [9,] 25.98 -3.74 1.40 -47.91 -12.94 -2.01 -2.48 6.32 101.53
```

But, FDA directions are orthogonal with respect to the within-class covariance matrix W :

```
round(t(FDA.eigen.dir) %*% W %*% FDA.eigen.dir, dig=2)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9]
## [1,]    1    0    0    0    0    0    0    0    0
## [2,]    0    1    0    0    0    0    0    0    0
## [3,]    0    0    1    0    0    0    0    0    0
## [4,]    0    0    0    1    0    0    0    0    0
## [5,]    0    0    0    0    1    0    0    0    0
```

```
## [6,] 0 0 0 0 0 1 0 0 0
## [7,] 0 0 0 0 0 0 1 0 0
## [8,] 0 0 0 0 0 0 0 1 0
## [9,] 0 0 0 0 0 0 0 0 1
```

Another approach would be to apply PCA first to reduce the dimensionality of the problem, and then apply FDA. As an example:

```
newX.train = X.train %%% PCA.dir[, 1:100];
new.digits.lda = lda(newX.train, Y.train);
new.X.test = X.test %%% PCA.dir[, 1:100]
Y.test.pred = predict(new.digits.lda, new.X.test)$class
table(Y.test, Y.test.pred)
```

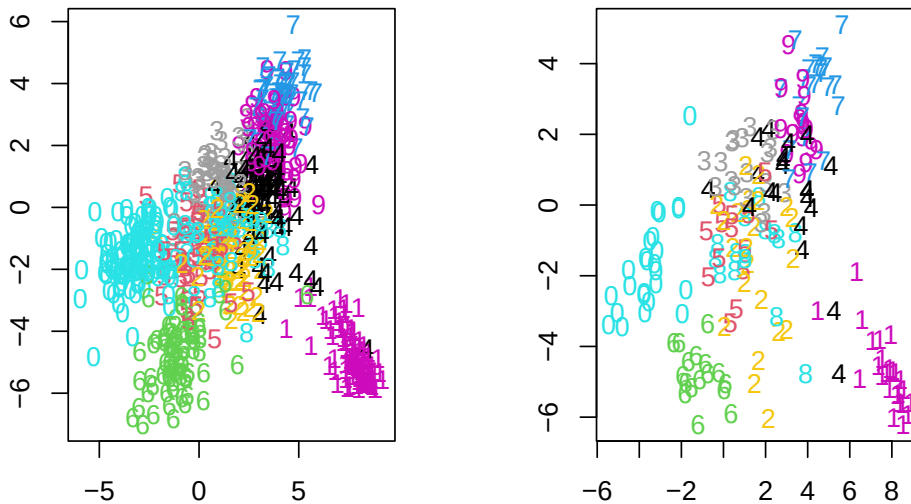
```
##      Y.test.pred
## Y.test 0 1 2 3 4 5 6 7 8 9
##      0 25 0 0 0 0 0 0 0 0
##      1 0 25 0 0 0 0 0 0 0
##      2 0 0 23 0 0 0 0 0 1
##      3 0 0 0 16 0 0 0 1 1 0
##      4 0 2 0 0 18 0 0 0 0 1
##      5 0 0 0 4 0 9 0 0 0 1
##      6 0 0 0 0 0 0 17 0 0 0
##      7 0 0 0 0 0 0 0 20 0 3
##      8 0 1 1 0 2 0 0 0 12 0
##      9 0 0 0 0 0 0 0 2 0 15
```

```
sum(Y.test != Y.test.pred)
```

```
## [1] 20
```

```
newF=newX.train%%new.digits.lda$scaling;
newFtest=X.test%%dig.eigen$ve[,1:100]%%new.digits.lda$scaling

par(mfrow=c(1,2))
plot(newF[,1],newF[,2], type="n", xlab="", ylab="")
text(newF[,1], newF[,2], Y.train, col= Y.train+5)
plot(newFtest[,1], newFtest[,2], type="n", xlab="", ylab="")
text(newFtest[,1], newFtest[,2], Y.test, col=Y.test+5)
```



In this case, we observe that there is not as clear a separation observed in training data as before. This reduction in separation is expected due to the substantial dimension reduction from 256 to 100. While the training data's separation may not be as good as before, this approach produces more consistent results between the training and test datasets, which can help mitigate overfitting issues.

Naive Bayes Prediction

Implementation of the Naive Bayes predictor. We use the `naiveBayes` function in the `e1071` library.

```
zip17 = train.zip[train.zip[,1] %in% c(1,7),]
zip17.test = test.zip[test.zip[,1] %in% c(1,7),]
Y.17 = as.factor(zip17[,1])
X.17 = zip17[,-1]

NBfit = naiveBayes(X.17, Y.17)

y_pred = predict(NBfit, zip17.test)
accuracy = mean(y_pred == zip17.test[,1])

table(Predicted = y_pred, Actual = zip17.test[,1])
```

##		Actual
## Predicted	1	7
##	1	22 0
##	7	3 23

Chapter 7

Logistic Regression

In the previous section, we discussed how discriminant analysis models the joint distribution of predictors and the class variable. In this section, we are going to focus on logistic regression that takes a different approach: it **directly** models the conditional probability of class membership as a function of the predictors. For *binary outcomes*, this is achieved through the *logistic function*, which maps a linear combination of predictors to values between 0 and 1.

Logistic Regression has the advantage of making fewer assumptions about the distribution of the predictors, provides coefficients that can be interpreted in terms of odds ratios, and can be extended to handle multiple classes through multinomial formulations. Because of its simplicity, interpretability, and solid probabilistic foundation, logistic regression is one of the most widely used methods for classification.

7.1 Fitting a Logistic Regression model

The logistic regression starts by directly modeling the *conditional* probability

$$\eta(x) = P(Y = 1|X = x)$$

where $\eta(x)$ is a function bounded in $[0, 1]$.

Specifically, consider a **link** function g that maps $\eta(x)$ into $(-\infty, \infty)$, i.e.

$$g(\eta(x)) = \mathbf{x}^T \beta$$

Based on this formulation, we can see that the response Y conditional on X follows a Bernoulli distribution:

$$P(Y = y_i|X = x_i) = \eta(x_i)^{y_i} (1 - \eta(x_i))^{1-y_i}$$

where we defined as

$$\eta(x) = P(Y = 1|X = x) \quad \text{and} \quad 1 - \eta(x) = P(Y = 0|X = x)$$

For the Logistic Regression, we use the **logit link** function that is defined as

$$\log \frac{\eta(x)}{1 - \eta(x)} = x^T \beta \quad \Leftrightarrow \quad \eta(x) = \frac{\exp(x^T \beta)}{1 + \exp(x^T \beta)}$$

The factor $\log \frac{\eta(x)}{1 - \eta(x)}$ is called **log-odds**, and we model it as a **linear** function of x .

7.1.1 Maximum Likelihood Approach

At each data point we have observations $(x_i, y_i, \eta(x_i))$ that we want to use to estimate $\eta(x)$ or equivalently β . Instead of starting by minimizing a squared error loss (for a given loss function), as we did in the regression with continuous response case, we follow a maximum likelihood approach.

Maximum likelihood estimation is a method that provides the *most likely value* for a parameter, given the data we observed. In a parametric framework, the likelihood function is a function of the model parameters and the data, and the maximum likelihood estimator is obtained by maximizing the likelihood function with respect to the parameters.

Recall, that conditionally on X the response Y follows a Bernoulli distribution with probability $\eta(x)$. Therefore, the likelihood function is written as

$$\begin{aligned} \mathcal{L}(\beta|\mathbf{y}, \mathbf{x}) &= \prod_{i=1}^n p(y_i|x_i\beta) \\ &= \prod_{i=1}^n \eta(x_i)^{y_i} (1 - \eta(x_i))^{1-y_i} \end{aligned}$$

To fit the Logistic Regression model, we need to maximize the likelihood function with respect to the parameters β :

$$\max_{\beta} \mathcal{L}(\beta|\mathbf{y}, \mathbf{x})$$

This is equivalent to maximizing the log of the likelihood, the so-called *log-likelihood function*:

$$\begin{aligned} \ell(\beta|\mathbf{y}, \mathbf{x}) &= \sum_{i=1}^n \log \left\{ \eta(x_i)^{y_i} (1 - \eta(x_i))^{1-y_i} \right\} \\ &= \left\{ \sum_{i=1}^n y_i \log \frac{\eta(x_i)}{1 - \eta(x_i)} + \log(1 - \eta(x_i)) \right\} \end{aligned}$$

Recall that we defined

$$\log \frac{\eta(x)}{1 - \eta(x)} = x^T \beta$$

Combining all the above, after a couple of lines of algebra, the log-likelihood function becomes:

$$\ell(\beta|\mathbf{y}, \mathbf{x}) = \sum_{i=1}^n \left\{ y_i x_i^T \beta - \log(1 + \exp(x_i^T \beta)) \right\}$$

and the Maximum Likelihood Estimator (MLE) $\hat{\beta}$ is obtained by *maximizing*

$$\max_{\beta} \ell(\beta|\mathbf{y}, \mathbf{x}) = \max_{\beta} \sum_{i=1}^n \left\{ y_i x_i^T \beta - \log(1 + \exp(x_i^T \beta)) \right\}$$

Note here that β here is a vector, so we need to compute the gradient vector $\frac{\partial \ell(\beta|\mathbf{y}, \mathbf{x})}{\partial \beta}$, set it equal to 0, solve with respect to β , and check the Hessian matrix $\frac{\partial^2 \ell(\beta|\mathbf{y}, \mathbf{x})}{\partial \beta \partial \beta^T}$ to ensure that our solution is indeed a maximum.

Solving for the MLE

Taking the derivative with respect to β , and using chain rule along the way, the gradient computes as

$$\begin{aligned} \frac{\partial \ell(\beta|\mathbf{y}, \mathbf{x})}{\partial \beta} &= \sum_{i=1}^n \frac{\partial}{\partial \beta} (y_i x_i^T \beta) - \frac{\partial}{\partial \beta} (\log(1 + \exp(x_i^T \beta))) \\ &= \sum_{i=1}^n y_i x_i^T - \sum_{i=1}^n \frac{1}{1 + \exp(x_i^T \beta)} \cdot \frac{\partial}{\partial \beta} (1 + \exp(x_i^T \beta)) \\ &= \sum_{i=1}^n y_i x_i^T - \sum_{i=1}^n \frac{1}{1 + \exp(x_i^T \beta)} \cdot \exp(x_i^T \beta) x_i^T \\ &= \sum_{i=1}^n y_i x_i^T - \sum_{i=1}^n \frac{\exp(x_i^T \beta) x_i^T}{1 + \exp(x_i^T \beta)} \end{aligned}$$

To compute the Hessian, we need to take the second derivative with respect to β , using chain rule in a similar way:

$$\begin{aligned}
\frac{\partial^2 \ell(\beta|\mathbf{y}, \mathbf{x})}{\partial \beta \partial \beta^T} &= \sum_{i=1}^n \frac{\partial}{\partial \beta} y_i x_i^T - \frac{\partial}{\partial \beta} \left(\frac{\exp(x_i^T \beta) x_i^T}{1 + \exp(x_i^T \beta)} \right) \\
&= - \sum_{i=1}^n \frac{\partial}{\partial \beta} \left(\frac{\exp(x_i^T \beta) x_i^T}{1 + \exp(x_i^T \beta)} \right) \\
&= - \sum_{i=1}^n \frac{\partial}{\partial (\exp(x_i^T \beta))} \left(\frac{\exp(x_i^T \beta) x_i^T}{1 + \exp(x_i^T \beta)} \right) \frac{\partial (\exp(x_i^T \beta))}{\partial \beta} \\
&= - \sum_{i=1}^n x_i \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \frac{1}{1 + \exp(x_i^T \beta)} x_i^T \\
&= - \sum_{i=1}^n x_i \eta(x_i) (1 - \eta(x_i)) x_i^T
\end{aligned}$$

Finally, we have

- **Gradient Vector**

$$\frac{\partial \ell(\beta|\mathbf{y}, \mathbf{x})}{\partial \beta} = \sum_{i=1}^n y_i x_i^T - \sum_{i=1}^n \frac{\exp(x_i^T \beta) x_i^T}{1 + \exp(x_i^T \beta)} = \mathbf{X}^T (\mathbf{y} - \mathbf{p}),$$

where

$$p_i = \eta(x_i) = P(Y = 1|X = x_i)$$

- **Hessian Matrix**

$$\frac{\partial^2 \ell(\beta|\mathbf{y}, \mathbf{x})}{\partial \beta \partial \beta^T} = - \sum_{i=1}^n x_i \eta(x_i) (1 - \eta(x_i)) x_i^T = -\mathbf{X}^T \mathbf{W} \mathbf{X}$$

The Hessian matrix is *negative semi-definite*, which means that the log-likelihood function is *concave*, and as a consequence *any local maximum is global maximum*.

We do not have an explicit solution for the MLE in the logistic regression, but it can be easily obtained using an iterative Newton-Raphson approach.

Newton-Raphson for Logistic MLE

$$\begin{aligned}
f(x) &\approx f(x_0) + f'(x_0)(x - x_0) \\
\Rightarrow x &= x_0 - \frac{f(x_0)}{f'(x_0)}
\end{aligned}$$

In our case:

$$\begin{aligned}
\beta &= \beta_0 + (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{y} - \mathbf{p}) \\
&= (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{X} \beta_0 \\
&\quad + \mathbf{W}^{-1} (\mathbf{y} - \mathbf{p})
\end{aligned}$$

We can also obtain the Logistic Regression MLE via a *Reweighted LS Algorithm*:

1. Start with some initial values β^0 .
2. Calculate the corresponding $p_i^0 = \eta^0(x_i)$ for $i = 1, \dots, n$.
3. Define $W = \text{diag}(p_i^0(1 - p_i^0))_{i=1}^n$.
4. Calculate

$$\mathbf{z} = \mathbf{X}\beta^0 + \mathbf{W}^{-1}(\mathbf{y} - \mathbf{p}^0)$$

5. Update $\beta^0 \leftarrow \beta^1$ with

$$\beta^1 = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{z}$$

6. Iterate the above steps until convergence.

Interpreting Parameters

What is the effect of β ? Each unit increase of X_i increases the log-odds of Y by β_j .

7.2 Logistic Regression with Separable Data

Consider the following scenario with two classes, each containing two points. Assume that the intercept is zero for simplicity.

In the log-likelihood, the *first term* expression corresponds to the **orange points** while the *second term* corresponds to the **blue points**:

$$\frac{e^{\beta_1 x_1 + \beta_2 x_2}}{1 + e^{\beta_1 x_1 + \beta_2 x_2}} = \frac{1}{1 + e^{-(\beta_1 x_1 + \beta_2 x_2)}} - \frac{1}{1 + e^{\beta_1 x_1 + \beta_2 x_2}}$$

and the only difference between the two expressions is a “*minus*” sign.

In such a scenario, one can observe that as the β s increase, the probabilities will approach 1, and as we continue increasing the coefficients, the likelihood increases - giving us an error that does not converge. On the other hand, the decision boundary will remain constant, so as a method it is a valid approach to use.

Remark: One could ask whether regularization can help in such a scenario. Unfortunately, ridge or lasso will not resolve the convergence issues. The problem is not related to overfitting and cannot be mitigated by adjusting the regularization strength. The root cause of this issue is that when the classification rule is flawless, i.e. $Y = 1$ with probability 1 when $x^T \beta > \beta_0$, the model parameters (β_0, β) become undefined tending toward positive or negative infinity.

7.3 Retrospective Sampling

So far we assumed that the data consist of i.i.d. samples drawn from a population of interest. **But**, in some cases, such as studying a certain disease, if we want to sample the whole population, we should select an very large sample -in the thousands or millions $\rightarrow$ Highly Inefficient!

Therefore, we obtain a balanced sample from a group of patients having the disease and from a group of healthy patients (control group); say we have 50 individuals in each group. When we assess the model's performance on a test sample, there is an inherent difference between the training and the testing data, since there will not have the same 50-50 split (balance) in the test data.

To address this situation in our model, we define a new random variable Z indicating whether an individual has been sampled. We then can estimate the probability

$$P(Y = 1|Z = 1, X = x)$$

instead of $P(Y = 1|X = x)$, without any reliance on Z . So, in the framework of the logistic regression, consider the following probabilities

$$r_0 = P(Z = 1|Y = 0), \quad r_1 = P(Z = 1|Y = 1)$$

Then,

$$\begin{aligned} P(Y = 1|Z = 1, X = x) &= \frac{P(Z = 1|Y = 1, x)P(Y = 1|x)}{P(Z = 1, Y = 0|x) + P(Z = 1, Y = 1|x)} \\ &= \frac{r_1 \exp(\alpha + x^T \beta)}{r_0 + r_1 \exp(\alpha + x^T \beta)} \\ &= \frac{\exp(\alpha^* + x^T \beta)}{1 + \exp(\alpha^* + x^T \beta)} \end{aligned}$$

where $\alpha^* = \alpha + \log \frac{r_1}{r_0}$. Although the data have been sampled retrospectively, the logistic model continues to apply with the same coefficients β but a different intercept.

7.4 Logistic Regression for the Heart dataset in R

In this example we are going to work with the South Africa heart data that can be found here.

```
library(readr)
SAheart <- read_csv("https://web.stanford.edu/~hastie/ElemStatLearn/datasets/SAheart.d")

## Rows: 462 Columns: 11
## -- Column specification -----
```

```
## Delimiter: ","
## chr (1): famhist
## dbl (10): row.names, sbp, tobacco, ldl, adiposity, typea, obesity, alcohol, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
SAheart = as.data.frame(SAheart)
```

The SAheart data set is the outcome of a study of heart disease risk factors among South African men of mixed ancestry. It contains 462 observations with variables that describe lifestyle, medical history, and physiological measures—such as age, alcohol intake, tobacco use, blood pressure, cholesterol levels, and family history of heart disease.

Our goal in this example is to estimate the probability that an individual had coronary heart disease (CHD).

The data set consists of the following variables:

- `sbp` - systolic blood pressure
- `tobacco` - cumulative tobacco (kg)
- `ldl` - low density lipoprotein cholesterol
- `adiposity` - numeric measure of a person's relative body fat.
- `famhist` - family history of heart disease (Present, Absent)
- `typea` - type-A behavior
- `obesity`
- `alcohol` - current alcohol consumption
- `age` - age at onset

```
heart.logistic = glm(chd ~ ., data=SAheart, family=binomial)
summary(heart.logistic)
```

```
##
## Call:
## glm(formula = chd ~ ., family = binomial, data = SAheart)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -5.9207616  1.3265724  -4.463 8.07e-06 ***
## row.names    -0.0008844  0.0008950  -0.988 0.323042
## sbp          0.0076602  0.0058574   1.308 0.190942
## tobacco      0.0777962  0.0266602   2.918 0.003522 **
## ldl          0.1701708  0.0597998   2.846 0.004432 **
## adiposity    0.0209609  0.0294496   0.712 0.476617
```

```
## famhistPresent 0.9385467 0.2287202 4.103 4.07e-05 ***
## typea         0.0376529 0.0124706 3.019 0.002533 **
## obesity       -0.0661926 0.0443180 -1.494 0.135285
## alcohol       0.0004222 0.0045053 0.094 0.925346
## age           0.0441808 0.0121784 3.628 0.000286 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 596.11  on 461  degrees of freedom
## Residual deviance: 471.16  on 451  degrees of freedom
## AIC: 493.16
##
## Number of Fisher Scoring iterations: 5
```

We obtain the fitted values for the regression model, and we report in a table:

```
yhat = (heart.logistic$fitted.values>0.5)
table(yhat, SAheart$chd)
```

```
##
## yhat      0    1
## FALSE 255  79
## TRUE   47  81
```

7.4.1 Goodness-of-Fit

In logistic regression, we do not have a residual sum of squares (RSS), so we need alternative metrics to assess the fit of the model. Traditionally, deviance measures serve as good candidates to quantify the fit and they are also based on the likelihood function. So, when all the x_i s are unique, the deviance is defined as

$$Deviance = -2 \left(\sum_{i:y_i=1} \log \hat{p}_i + \sum_{i:y_i=0} \log(1 - \hat{p}_i) \right)$$

where $\hat{p}_i$ represents the estimated probability of $P(Y = 1|X = x_i)$.

In the end of the summary output in R, we see two types of deviances:

- **Null Deviance:** The deviance associated with a logistic regression model that includes only the intercept term (i.e., it doesn't consider any predictor variables). In this case, the estimated probability $P(Y = 1|X = x_i)$ is determined solely by the sample mean of the observed values, irrespective of the feature vector x_i .
- **Residual Deviance:** This deviance corresponds to the current logistic regression model under consideration. For example, for the logistic regression model we just fit, and the estimated probability of $P(Y = 1|X = x_i)$

obtained using the model fitted values.

We can also compute them here

```
p.hat=heart.logistic$fitted.values;

-2*(sum(log(p.hat[SAheart$chd==1])) + sum(log(1-p.hat[SAheart$chd==0])))

## [1] 471.1591

p0.hat = rep(mean(SAheart$chd), length(SAheart$chd));

-2*(sum(log(p0.hat[SAheart$chd==1])) + sum(log(1-p0.hat[SAheart$chd==0])))

## [1] 596.1084
```

Solving this problem using numerical optimization -without the use of the glm function

```
## Computation of the negative log-lik
logistic.loglik <- function(b, x, y)
{
  bm = as.matrix(b)
  xb = x %*% bm
  # this returns the negative loglikelihood
  return(sum(y*xb) - sum(log(1 + exp(xb))))
}

## Gradient
logistic.gradient <- function(b, x, y)
{
  bm = as.matrix(b)
  expxb = exp(x %*% bm)
  return(t(x) %*% (y - expxb/(1+expxb)))
}

## We can now use the `optim` function to solve the problem:

SAheart$famhist = ifelse(SAheart$famhist=="Present",1,0)
SAheart$famhist

## [1] 1 0 1 1 1 1 0 1 1 1 0 1 0 0 1 1 0 1 1 1 0 1 1 0 0 1 0 0 0 1 0 0 0 0
## [38] 0 1 1 1 0 0 1 0 0 1 1 0 0 1 0 0 1 0 0 0 0 0 1 0 1 1 0 0 1 1 0 1 0 0 0 0 1
## [75] 0 0 0 1 0 1 1 0 1 0 1 0 0 1 1 0 0 0 0 0 0 0 1 1 1 0 1 0 1 0 0 0 0 1 0 0 0
## [112] 1 0 1 0 1 0 0 1 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 1 0 1 0 1 0 1 0 0 0 0 1 1
## [149] 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 0 0 0 1 1 1 0 0 0 0 1 1 1 1
## [186] 1 0 0 1 1 0 1 1 0 0 1 0 0 1 0 0 1 0 0 1 0 1 1 0 1 0 1 0 0 0 0 0 0 0 0 0 1
## [223] 0 1 0 0 1 0 1 1 1 0 0 0 0 0 0 1 0 0 0 1 0 0 1 0 1 1 0 1 0 0 0 1 1 1 1 1 0
```

```
## [260] 1 0 1 1 1 1 0 0 0 1 1 1 0 1 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0
## [297] 0 0 0 1 1 0 0 1 0 1 1 0 0 1 1 0 1 0 0 0 1 0 1 0 1 1 0 1 0 0 1 0 0 0 0 1 1
## [334] 0 0 1 1 0 1 0 0 1 1 0 0 1 1 0 1 0 1 1 1 1 1 1 0 0 1 0 1 0 1 1 1 0 0 1 1 0
## [371] 1 1 0 1 1 0 0 1 0 0 0 1 1 0 0 1 1 0 0 1 1 1 0 0 0 0 1 0 1 0 0 0 1 1 0 1 1
## [408] 1 0 0 0 0 0 1 1 1 0 1 1 0 0 0 1 1 0 0 1 0 1 0 0 0 0 1 0 1 0 0 0 1 1 0 1 1
## [445] 0 1 0 1 1 1 0 0 1 1 0 0 1 0 0 0 0 0 1

x = as.matrix(cbind("intercept" = 1, SAheart[, 2:10]))
y = as.matrix(SAheart[,11])
b = rep(0, ncol(x))

# Because we defined the negative of the log-likelihood, we do
# mazimization instead of minimization:

optim(b, fn = logistic.loglik, gr = logistic.gradient, method = "BFGS", x = x, y = y, c

## $par
## [1] -6.150733305 0.006504017 0.079376464 0.173923988 0.018586578
## [6] 0.925372019 0.039595096 -0.062909867 0.000121675 0.045225500
##
## $value
## [1] -236.07
##
## $counts
## function gradient
##      74      16
##
## $convergence
## [1] 0
##
## $message
## NULL
```

Our result here matches the `glm()` solution. If a general solver function is not readily available, we can always use Newton-Raphson. For that, you will need to write a function to calculate the Hessian matrix and then proceed to optimization.

7.4.2 Variable Selection

Variable selection might also be necessary when working in a logistic regression framework. So, we will explore which methods/criteria that we used in regression can seamlessly transfer to logistic regression.

AIC/BIC

We can employ criteria such as AIC and BIC because the logistic regression model has a likelihood. However, unlike linear regression, we are not able to employ the leaps-and-bounds algorithm to explore all possible sub-models since the likelihood in logistic regression differs from residual sum of squares. Instead, we rely on a greedy algorithm, such as stepwise, backward, or forward.

As an example

```
step.logistic = step(heart.logistic, scope=list(upper=~., lower=~1))

## Start:  AIC=493.16
## chd ~ row.names + sbp + tobacco + ldl + adiposity + famhist +
##      typea + obesity + alcohol + age
##
##           Df Deviance    AIC
## - alcohol   1   471.17 491.17
## - adiposity  1   471.67 491.67
## - row.names  1   472.14 492.14
## - sbp        1   472.88 492.88
## <none>       471.16 493.16
## - obesity    1   473.47 493.47
## - ldl        1   479.65 499.65
## - tobacco    1   480.27 500.27
## - typea      1   480.75 500.75
## - age        1   484.76 504.76
## - famhist    1   488.29 508.29
##
## Step:  AIC=491.17
## chd ~ row.names + sbp + tobacco + ldl + adiposity + famhist +
##      typea + obesity + age
##
##           Df Deviance    AIC
## - adiposity  1   471.69 489.69
## - row.names  1   472.14 490.14
## - sbp        1   472.94 490.94
## <none>       471.17 491.17
## - obesity    1   473.49 491.49
## + alcohol    1   471.16 493.16
## - ldl        1   479.68 497.68
## - tobacco    1   480.73 498.73
## - typea      1   480.80 498.80
## - age        1   484.82 502.82
## - famhist    1   488.43 506.43
##
## Step:  AIC=489.69
```

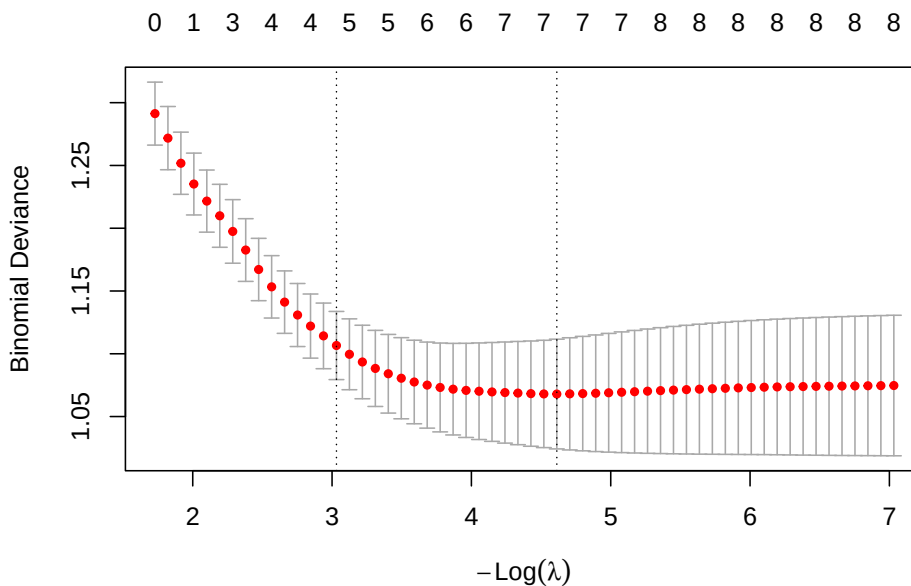
```
## chd ~ row.names + sbp + tobacco + ldl + famhist + typea + obesity +
##      age
##
##              Df Deviance    AIC
## - row.names  1   472.55 488.55
## - sbp        1   473.55 489.55
## <none>       471.69 489.69
## - obesity    1   473.86 489.86
## + adiposity  1   471.17 491.17
## + alcohol    1   471.67 491.67
## - typea      1   481.05 497.05
## - tobacco    1   481.35 497.35
## - ldl        1   481.68 497.68
## - famhist    1   488.89 504.89
## - age        1   493.97 509.97
##
## Step:  AIC=488.55
## chd ~ sbp + tobacco + ldl + famhist + typea + obesity + age
##
##              Df Deviance    AIC
## - sbp        1   473.98 487.98
## <none>       472.55 488.55
## - obesity    1   474.65 488.65
## + row.names  1   471.69 489.69
## + adiposity  1   472.14 490.14
## + alcohol    1   472.55 490.55
## - tobacco    1   482.54 496.54
## - ldl        1   482.95 496.95
## - typea      1   483.19 497.19
## - famhist    1   489.38 503.38
## - age        1   495.48 509.48
##
## Step:  AIC=487.98
## chd ~ tobacco + ldl + famhist + typea + obesity + age
##
##              Df Deviance    AIC
## - obesity    1   475.69 487.69
## <none>       473.98 487.98
## + sbp        1   472.55 488.55
## + adiposity  1   473.47 489.47
## + row.names  1   473.55 489.55
## + alcohol    1   473.94 489.94
## - tobacco    1   484.18 496.18
## - typea      1   484.30 496.30
## - ldl        1   484.53 496.53
## - famhist    1   490.58 502.58
```

```
## - age          1   502.11 514.11
##
## Step:  AIC=487.69
## chd ~ tobacco + ldl + famhist + typea + age
##
##           Df Deviance    AIC
## <none>          475.69 487.69
## + obesity      1   473.98 487.98
## + sbp          1   474.65 488.65
## + row.names    1   475.26 489.26
## + adiposity    1   475.44 489.44
## + alcohol      1   475.65 489.65
## - ldl          1   484.71 494.71
## - typea        1   485.44 495.44
## - tobacco      1   486.03 496.03
## - famhist      1   492.09 502.09
## - age          1   502.38 512.38
```

7.4.3 Penalized Logistic Regression

Similar to regular regression, we can apply penalization techniques in a logistic regression framework to address collinearity or high-dimensional settings. This can be done in R again using the `glmnet` package.

```
library(glmnet)
lasso.fit = cv.glmnet(x = data.matrix(SAheart[, 2:10]), y = SAheart[,11], nfold = 10, family = "b")
plot(lasso.fit)
```



```
coef(lasso.fit, s="lambda.min")

## 10 x 1 sparse Matrix of class "dgCMatrix"
##           lambda.min
## (Intercept) -5.73717333
## sbp         0.00416794
## tobacco     0.07056290
## ldl         0.14789200
## adiposity   .
## famhist     0.81077289
## typea       0.02967640
## obesity     -0.01618876
## alcohol     .
## age         0.04396417
```

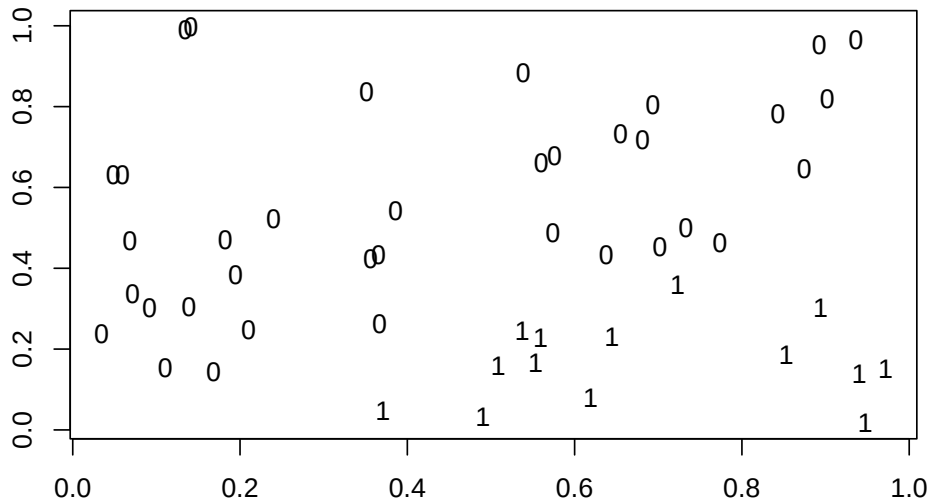
7.4.4 Illustration of divergence of ML estimators

Consider a very simple case in which

$$y = \begin{cases} 1, & \text{if } x_1 - 2x_2 > 0 \\ 0 & \text{otherwise} \end{cases}$$

Simulate such data:

```
x = matrix(runif(100), 50, 2)
y = ifelse(x[,1] - 2*x[,2]>0, 1, 0);
plot(x[,1], x[,2], type="n", xlab="", ylab="");
text(x[,1], x[,2], y)
```



```
logistic.fit = glm(y~x, family = binomial)
```

```
## Warning: glm.fit: algorithm did not converge
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

The warning message shows exactly what we discussed in the lectures. The MLE of the coefficients tend towards $\pm$ infinity and the algorithm does not converge.

However, despite that fact, we can see that the separation like is well-defined allowing us to use the fitted model for predictions.

```
round(logistic.fit$fitted.values, 0)
```

```
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
##  0  0  0  1  1  0  1  0  1  0  1  1  0  0  0  0  0  0  1  0  0  0  1  0  0  0
## 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
##  0  0  0  0  0  1  0  1  1  1  0  1  0  0  0  0  0  0  0  0  0  1  0  0
```